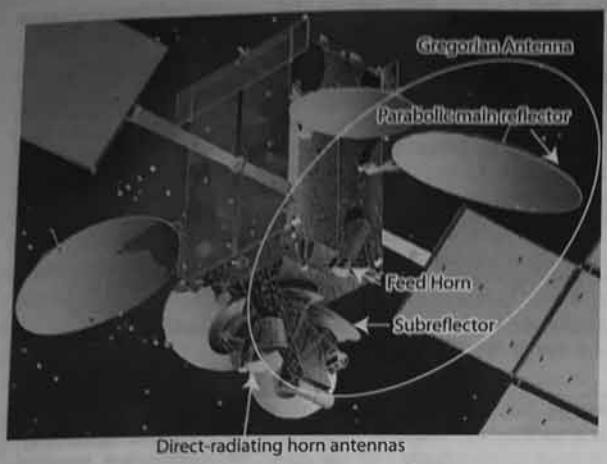


Fig. 16-19. Modern Geosynchronous Satellite that uses both and Direct-radiating Earth-coverage Horn Antennas and High-gain Gregorian Antennas with Parabolic Main Reflectors and Offset Subreflectors.

16.3.2 Communications Payload Block Diagram Design

Once preliminary antenna designs and performance predictions are available, the communications payload design team has sufficient information to begin designing the heart of the communications payload. The *payload block diagram* details all of the components and component connectivity required to implement the channelization and the corresponding amplification, filtering, frequency conversion, switching, and routing that takes place between payload receive and transmit apertures. The payload block diagram specifies not only the types of components that will be required, but also describes the number of each component required for the complete payload, the way that the components are connected to one another, and key characteristics about the components, such as part identification codes, nominal power, operating band, and channel assignments of certain active and passive components will be useful throughout the remainder of the design process and into later testing and troubleshooting.

Developing the payload block diagram requires a variety of initial payload budgets to be completed. Like the link analysis discussed in Sec. 16.2, these budgets track various aspects of payload performance, and compliance to performance specifications. In addition, they are invaluable tools for defining unit specifications, supporting trades during the design process, and supporting troubleshooting during integration and test phases. Like the link budgets, many of these payload budgets are maintained and updated throughout the design, integration, and test process, with initial performance estimates being replaced with improved performance predictions as the design matures, and with measured data as it becomes available.

Table 16-16. Antenna Types for Satellite Systems. (Formulas from Jasik [1961]) In these equations, C, D, L, and λ are in meters and f is in GHz.

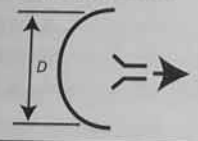
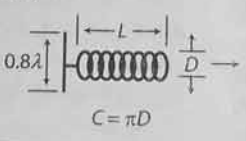
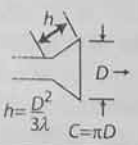
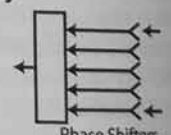
Antenna Type	*Parabolic Reflector*	*Helix*	*Horn*	*Array*
				
Beam Shape	Conical	Conical	Conical	Conical (scanning)
Typical Max Gain (dBI)	15–65	5–20	5–20	5–20
Peak Gain (dBI)	$17.8 + 20 \log D + 20 \log f$ ($\eta = 0.55$)	$10.3 + 10 \log (C^2 L / \lambda^3)$ $0.8 \le C / \lambda \le 1.2$ ($\eta = 0.70$)	$20 \log (C/\lambda) - 2.8$ ($\eta = 0.52$)	$10 \log (A/\lambda^2) + 8$
Half-Power Beamwidth (degs)	$\frac{21}{fD}$	$\frac{52}{\sqrt{C^2 L / \lambda^3}}$	$\approx \frac{225}{(C/\lambda)}$	
Size, m	D	DL	D	A
Mass, kg	10–30	10–15	1–2	20–40
Features	• Folded Optics, parabolic offset feed typically employed for satellite application • Mechanical steering • Simple lightweight structure • High *EIRP*	• Lower frequency applications • Deployed helix, increased complexity	• Ideal for global coverage, low gain applications with broad coverage area	• Flat packaging • Electronic beam steering • High aperture efficiency • Multiple independent steerable beams • High losses in feed distribution • High *EIRP* • High cost and weight

Table 16-16, Fig. 16-19, Eq. 16-33◀

Figure 16-20 illustrates how the link budget is used to represent the allocation of *EIRP* contributions between elements in the output section of the power-amplification chain and transmit-antenna contribution. Similar allocations are made in a variety of ways with other link budgets listed in Table 16-17, providing a basis for the development of key components of unit-performance specifications, unit, and higher level testing, and for troubleshooting and understanding various measures of payload performance sensitivity.

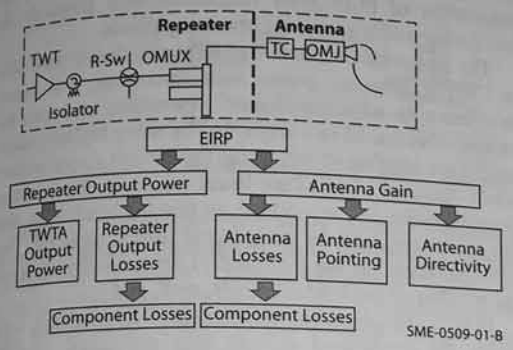


Fig. 16-20. Allocation of *EIRP* Contributions Between the Power-amplification Chain and Transmit-antenna Contributions.

Table 16-17. Payload Budgets Required as Part of the Communications Payload Design Process

EIRP and G/T	Phase Shift
Gain/Loss	AM/PM Transfer and Conversion
Noise Figure	Overdrive
Input Losses	Amplitude Linearity
G/T Stability	*G/T* Degradation
SFD Variation over Frequency	Coherent Self Interference
Output Power and Losses	Transmit Power Loading
EIRP Stability	General Spurs
Frequency Response and Rejection	Specific Spurs
Gain Slope	Passive Intermodulation
Group Delay	Peak Power for Multipactor
Group Delay Ripple	RF Susceptibility
Group Delay Stability	Antenna Isolations

Because of their great cost, designers and manufacturers of satellite communications payloads are constantly challenged to maximize the performance, and hence the potential revenue generating capacity of each communications payload. As a result, satellite communications payloads are highly optimized to support the unique mission requirements of each application. Payload budgets provide a structured way of predicting, tracking, balancing, and allocating many aspects of communications payload performance between different parts of the system.

The most common communications mission today are for geosynchronous FSS and BSS satellite communications applications [Elbert, 2008]. To facilitate the description of key components that are used to build and populate the payload block diagram, we will refer to the highly simplified description of the key FSS payload functional blocks illustrated in Fig. 16-21.

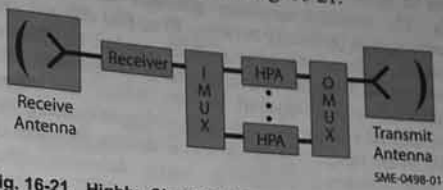


Fig. 16-21. Highly Simplified Schematic Illustrating Key Functional Blocks of a Typical FSS Payload.

Receive Antenna: Uplinked signals (and noise) are captured by the receive antenna, which is designed to efficiently couple energy from free space into coaxial transmission lines, where they pass through prefilters to remove out-of-band signals before reaching the receiver. The receive antenna plays a pivotal role in determining the sensitivity of the satellite to uplinked signals, as well as the geographic distribution of users that can be served.
Receiver: The receiver amplifies the uplinked signal and converts it to the downlink frequency. The receiver comprises a *low noise amplifier* (LNA), a *down converter* (D/C), a *local oscillator* (LO) and a *power supply*, and may be realized either as individual components, or an integrated assembly. Receivers are typically required to have low noise generation, high gain, high linearity, and zero to low spurious signal generation. The noise figure of the LNA is the primary contributor to the payload *G/T* performance, and is a function of frequency, bandwidth, and operating temperature. Figure 16-22 includes a simple block diagram that illustrates how the components of a receiver relate to one another and to the receiver inputs and outputs. A photograph of a satellite communications payload receiver is also shown in Fig. 16-22.

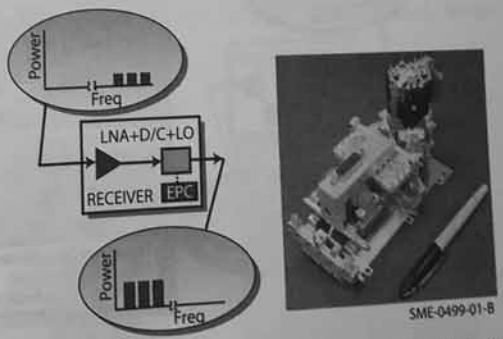


Fig. 16-22. Schematic Diagram of Receiver Functionality and an Image of an Integrated Receiver Unit. High-frequency, low-power signals at the receiver input are amplified and converted to lower frequency.

Table 16-17, Fig. 16-22, Eq. 16-33◀

HPA: The *high-power amplifier* (HPA) provides commandable amplification on the downlink signal path within the communications payload in a compact form factor, using very mature technology that offers high reliability. The HPA usually consists of a *channel amplifier* (CAMP), a *traveling wave tube* (TWT) amplifier, and an *electronic power conditioner* (EPC) power supply. The channel amplifier provides medium-power amplification (typically between 40 and 60 dB) and gain control. A linearizer is sometimes needed to improve transponder linearity when the HPA is operated near maximum (or *saturated*) power, where its output becomes non-linear. A final stage of amplification (typically in the range of 40 to 60 dB) is usually provided by a high-efficiency *traveling wave tube amplifier* (TWTA), which comprises a TWT and an EPC. In a travelling wave tube, an electron beam interacts with an RF signal, and energy transferred from the electron beam increases the energy of the RF signal, typically providing RF output power between 60 and 150W. The principal metrics of TWTA performance are the RF power output, and the overall efficiency (typically in the range of 50 to 65%) for converting DC bus current into RF output power. Figure 16-23 illustrates the relationship between the HPA and HPA inputs and outputs, and also shows a photograph of a flight-qualified HPA.

Solid State Power Amplifiers (SSPAs) are sometimes used in place of TWTAs at lower L-band and S-band frequencies, serving the same function as TWTs, but without need for high-voltage power supplies. This component savings translates directly to reducing payload mass, but does so at the expense of reduced RF output power and lower efficiency (25–55%). Where SSPAs are a viable alternative, the specifics of the application requirements determine which choice is most appropriate [Brown, 2002].

The HPA is the main contributor to payload nonlinearity (an important metric of communications payload performance), and is therefore the principal source of signal distortion when operated in its non-linear range, near saturation. Non-linearity is characterized by the ratio of the carrier power to the power in the third-order intermodulation product, an interfering signal that is generated in the amplifier at a frequency that is offset from that of the carrier. Figure 16-24 depicts the non-linear input-output characteristic of a TWTA operated near saturation, AM-PM signal modulation, and variation in the generation of third-order intermodulation products at low power and high power operating points.

The increased production of intermodulation products at TWTA operating points near saturation, will cause interfence to adjacent-channel operation, and can be mitigated in one of two ways: (A) by using higher-power amplifiers capable of delivering the desired output power when operated within their linear range, or (B) increase channelization of the payload, so that the bandwidth and the power required for each signal path is reduced. The latter strategy has the potential to dramatically increase the number of payload channels, cost, and mass.

IMUX: The *input multiplexer* (IMUX) assembly separates the broadband signals into individual channel frequencies. This function may also be realized through the use of hybrid splitting or channel dropping through integrated circulators (as shown). An IMUX is necessary to minimize unwanted intermodulation products generated by subsequent high power amplifiers operated in their non-linear region. IMUX passband response is typically a contributor to linear impairment, such as amplitude and group delay distortion. Figure 16-25 (Left) depicts the function of an IMUX, and includes a photograph of an IMUX representative of those that might be part of a satellite communications payload.

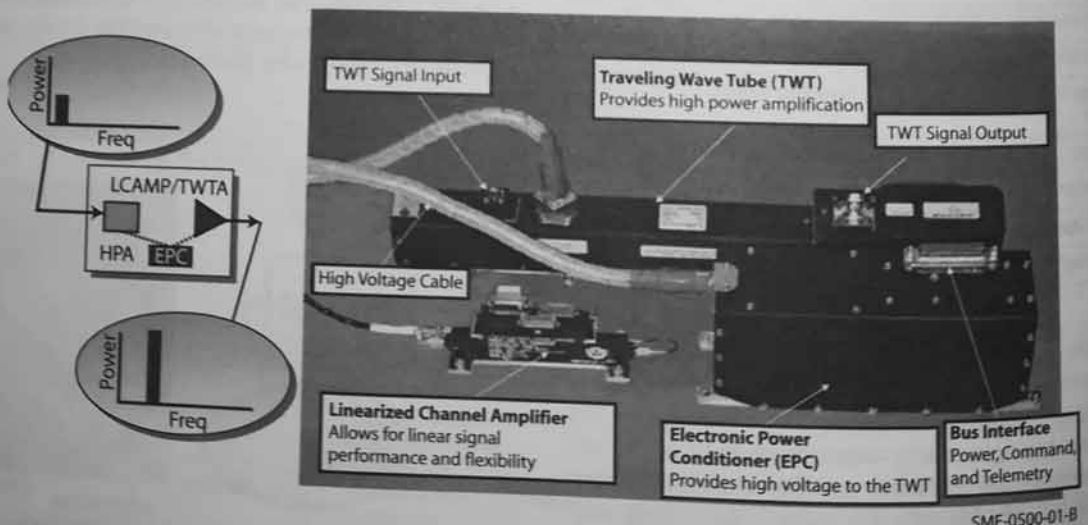


Fig. 16-23. Schematic Diagram of High-power Amplifier Functionality, and an Image of a High-Power Amplifier Comprising a Linearized Channel Amplifier (LCAMP), a Traveling Wave Tube (TWT) Amplifier, and an Electronic Power Conditioner (EPC).

Table 16-17◀, Fig. 16-23, Eq. 16-33◀

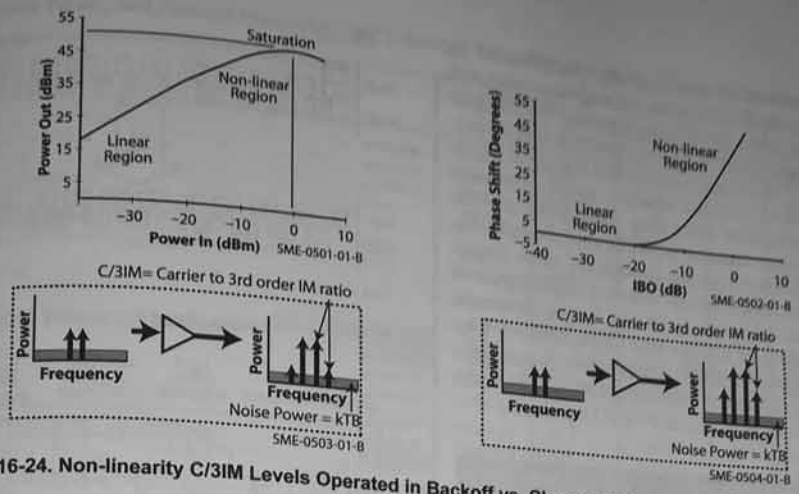


Fig. 16-24. Non-linearity C/3IM Levels Operated in Backoff vs. Close to Saturation.

OMUX: The *output multiplexer* (OMUX) assembly combines narrow-band signals from individual amplifier chains onto a single common broadband signal. An OMUX is necessary for minimizing the number of ports that are required on the transmit antenna. OMUX passband response is typically a contributor to linear impairment such as amplitude and group delay distortion. High power-handling capability is a critical requirement in OMUX design. Figure 16-25 (Right) depicts the function of an OMUX, and includes a photograph of an 8-channel OMUX.

Transmit Antenna: The *transmit antenna* efficiently couples energy from waveguide (or coaxial, in some instances) transmission lines to free space. The transmit antenna determines the coverage area that can be supported by the communications payload downlink, and in conjunction with the HPAs, determines the *EIRP* of the communications payload.

Reflector Surface Reuse: Although transmit and receive antennas are shown separately in Fig. 16-17, transmit and receive functions are frequently supported on *orthogonal polarizations* of a single antenna, allowing overall payload mass to be reduced, and space to be made available for adding antennas to serve other payloads or functions, or for other components. As a result, it is common for transmit and receive functions to be combined in a single antenna when uplink and downlink frequencies are within about 30% of one another. Because they operate at different frequencies, combining transmit and receive functions into a single antenna generally prevents either from being fully optimized, with performance being degraded further as the frequency difference grows. In most applications where transmit and receive functions are combined and performance degradation is split evenly between the transmit and receive functions, gain is degraded approximately 0.2 dB, as compared with the performance that might have been obtained for separately optimized antennas.

Payload Reliability and Robustness: Many factors contribute to the overall *reliability* of a system design. The reliability of the payload is strongly influenced by both the required mission life, and by failure rates for components used in the design. Failure rate is often expressed as either the *failure in time* (FIT) rate, which expresses the number of failures per billion operating hours, or in terms of the *mean time between failures* (MTBF), which is the measure used in the overall *probability of success (Ps)* analyses of the payload design over the mission life.

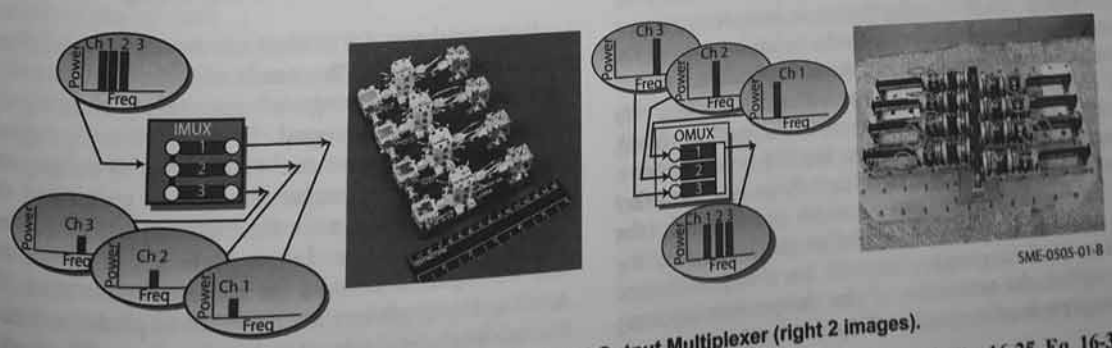


Fig. 16-25. IMUX Input Multiplexer (left 2 images) and OMUX Output Multiplexer (right 2 images).

Table 16-17◀, Fig. 16-25, Eq. 16-33◀

The communications payload *Ps* performance is affected most by the FIT rates for active components and their electronics, and high voltage supplies. To mitigate against these types of failures and potential loss of service, switching circuits are implemented at several points in the payload block diagram to allow failed units to be replaced with redundant units (Fig. 16-26). To provide the most appropriate balance between cost, complexity, and reliability, these switching circuits are often implemented as *switch rings,* which afford a high degree of flexibility in terms of the choice of which redundant units can be used in place of a given unit that has failed. Payload block diagrams typically incorporate redundancy switch rings and redundant units in Receiver (LNA plus downconverter) and HPA (CAMP+TWTA) blocks.

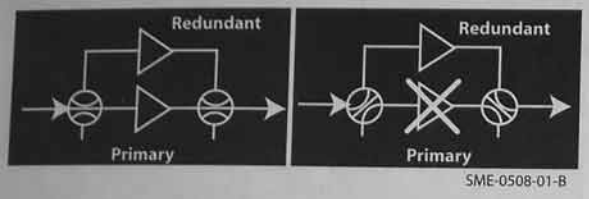


Fig. 16-26. Redundancy Switch Rings.

This detailed payload block diagram also describes *ancillary units* and hardware, including *passive devices* such as isolators, loads, hybrids, and even barrel connectors, which depicts the communication payload architecture for each beam or coverage area defined by the mission need. In this example, we have identified the major components of the payload, the ancillary components have not been defined at this stage of the block diagram, but would be at subsequent stages of more detailed block diagram release. The next level detailed block diagram would include spur filters, isolators, splitters, test couplers, detailed redundancy ring topology, and defined interconnect coax, transitions, waveguide.

Figure 16-27 shows the channelized frequency plan and payload block diagram for a typical FSS payload. Though more detailed than the diagram shown in Fig. 16-21, which reduced the diagram to its most fundamental functional blocks, the payload block diagram shown here is still greatly simplified in comparison with working block diagrams for even the simplest of modern communications payloads, which provide identifying information for each individual unit, including passive hardware such as isolators, loads, hybrids, waveguide and cable segments, and even barrel connectors.

The details of the payload's channelization begin to emerge early in the development of the payload block diagram and associated budgets, and from this comes a detailed picture of the channelized frequency plan that will be used by the communications payload. Like the payload block diagram and the payload budgets, the channelized frequency plan will be a key reference throughout the remainder of the design development, through payload integration, test, and on-orbit operations

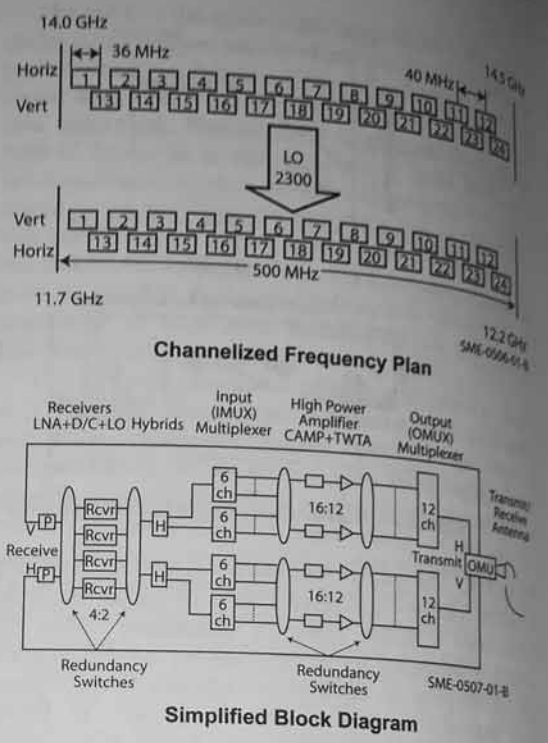


Fig. 16-27. A Representative FSS Payload Channelized Frequency Plan and Simplified Block Diagram.

Figure 16-27 shows a representative FSS payload channelized frequency plan and simplified payload block diagram.

The frequency plan shows the local oscillator frequency that defines the frequency conversion, as well as channel spacing that is staggered between polarizations to minimize cross-talk interference. The frequency plan and the payload block diagram shows that the design has been developed to include a total of 24 channels. The block diagram indicates that transmit and receive functions are supported on orthogonal polarizations by a single antenna, as well as redundancy for critical units. For example, the notation 4:2 under the receiver block means that 4 receivers are provided at launch to support 2 active receivers corresponding to "four-for-two" redundancy.

16.3.3 Communications Payload Sizing

Parts databases for modern communications satellites catalog hundreds of thousands of individual parts that must be procured, integrated into successively higher-level assemblies, and tested. The payload block diagram is a critically important tool that is used by design, integration, and test personnel alike to understand the hardware requirements for the communications payload and how they are intended to relate to one another and operate together. Despite the enormous parts count, the satellite bus engineers are expected to predict and track the satellite's total size, mass, power, thermal dissipation requirements, and center of gravity to within very tight

Table 16-17◀, Fig. 16-27, Eq. 16-33◀

Table 16-18. Mass, Power, and Thermal Dissipation (MPT) Budget Table, Based on Build History for Multiple Satellites.

	Payload MPT				Quantities			Unit			Total		
					Unit	Total	Active	Mass (kg)	Power (W)	Thermal (W)	Mass (kg)	Power (W)	Thermal (W)
1	Input Ancilliary (Test Coupler, Preselect Filter)			waveguide	# uplinks	4		0.30			1.20		
2	RCVR Input Red Ring Switches			waveguide	#RCVRs	8		0.20			1.60		
3	RCVR, LNA+D/C+LO			waveguide	#RCVRs	8	4	1.20	10.0	10.0	9.60	40.0	40.0
4	RCVR Output Red Ring Switches			coax	#RCVRs	8		0.10			0.80		
5	Input Ancilliary (Hybrid, SatPads, ...)			coax	#RCVRs	8					0.00		
6	IMUX, single channel			coax	#channels	40		0.15			1.20		
7	TWTA Input Red Ring			coax	#TWTs	64		0.26			12.48		
8	LCAMP				#TWTs	64	48	0.10			6.40		
9	TWT RAD	150.0	65%	92%	#TWTs	64	48	0.20	3.4		12.80	163.2	
10	EPC SINGLE		OBO:	0.0	#TWTs	64	48	1.00	230.8	40.4	64.00	11078.9	1938.5
11	TWTA Output Red Ring			waveguide	#TWTs	64	48	2.00	24.3	24.3	128.00	1167.2	1167.2
12	Termination Loads			waveguide	#TWTs	64		0.30			19.20		
13	OMUX, single channel			waveguide	#channels	48		0.90			43.20		
14	Output Ancilliary (Harmonic Filter, Test Coupler)			waveguide	#downlinks	4		1.00			4.00		
15	Output Loss			2.0			48						2657.1
16	Interconnects (coax, waveguide, brackets,harness ...)			46%							121.8		
17	Antenna Subsystem (feed, subreflector, main reflector)			waveguide	# up/downlinks	2		20.00			40.00		
18	Total										466.3	12447.3	5802.8

tolerances, and to refine those estimates and retire their small contingency margins as the satellite design matures, and further still as integration proceeds.

Because of its role in specifying components and component counts, completion of the payload block diagram represents a major milestone toward developing precise predictions of the payload mass, power, and thermal dissipation requirements. Once the payload block diagram has been defined, the payload equipment list can be generated, allowing the satellite bus engineering team to assess the communication payload and overall satellite *mass, power, and thermal dissipation* (MPT) requirements. Whereas early MPT estimates are top-down estimates that are based on previous satellite-build experience, the first reasonably precise estimates are only possible when a bottoms-up accounting can be performed.

Likewise, with the component specifications and connections described by the payload block diagram, bus and payload engineers can work together to define and optimize mechanical layouts. This is the first step toward verifying that all components can be physically accommodated within the satellite bus with the connectivity that is required, and the first step toward predicting the satellite center of mass.

In Table 16-18 we see the MPT budget for a typical FSS satellite communications payload, showing allocations that have been made to each type of payload component. Allocations for any given satellite can be compared with final allocations that were made on earlier satellites as a means of judging whether allocations are realistic, or whether further review or design optimizations may be required.

Lines 1 through 14 of the table contain entries that correspond with functional blocks of the payload block diagram. The mass allocation for these items is derived from historical data, and provided for reference. The unit column under quantity provides a guideline for interpreting each value listed in the table. For many units, component size is proportional to the signal wavelength so that mass values vary as a function of operating frequency, with greater mass allocations being required as the operating frequency decreases. Allocations for interconnects, such as coaxial and waveguide connectors, are made on line 16. This mass is not trivial, and generally exhibits a dependence on operating frequency.

DC power consumption occurs only in active units (Receiver and HPA), and is captured on line 3 and on lines 8 through 10. Power consumption for active units increases as operating frequency increases, with the largest differences occurring in TWTs (line 9). This systematic effect is mainly associated with a decrease in efficiency as units progress toward higher-frequency operation. Note that in certain applications, it is worthwhile to consider the trade between efficiency and the increased risk of single point failure vs. the advantage of improved power use efficiency that may be achieved through the use of power supplies that support multiple units (such as dual EPCs, that may supply two TWTs, or master converters that may power multiple LNAs).

Thermal dissipation is also dominated by TWT power levels, and increase with operating frequency for the very same reasons as described above for DC power consumption. An important factor that offsets this pattern is the availability of radiation-cooled TWT configurations for K_u-band and K_a-band applications, which offer greater efficiency than the conduction-cooled tubes available for C-band applications. Radiation-cooled tubes are able to dissipate approximately 50% less heat to the satellite bus than their conduction-cooled counterparts, radiating the difference directly to space.

Like the link and payload budgets, the MPT budget provides a systematic way to estimate and track the communications payload MPT requirements, and to verify

Table 16-18, Fig. 16-27◀, Eq. 16-33◀

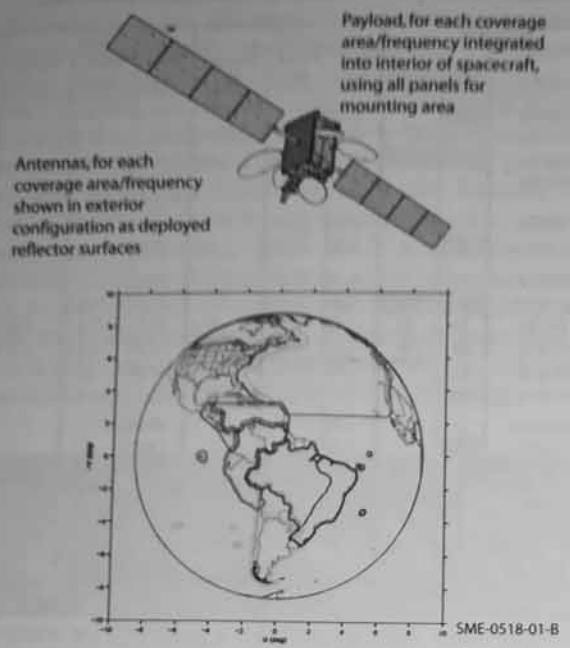


Fig. 16-28. FSS C/K_u-Band Satellite Concept and Coverage Requirement.

compatibility of the payload with both the satellite bus and the launch vehicle.

Sample MPT Budget for an FSS Payload

Here we consider a hybrid C/K_u-band FSS payload, rather than the pure K_u-band payloads that have been used as examples earlier in this section, with 24-channel coverage being divided between two polarizations for one C-band coverage area (South America) and for a second non-overlapping K_u-band service area (CONUS), as illustrated in Fig. 16-28. In this example, both receive and transmit antennas use 2.8 m shaped parabolic reflectors to optimize coverage contours and average link margins within the regions to be served. Because the uplink and downlink frequencies are within 16% of one another for the C-band payload, and within 30% for the K_u-band payload, transmit and receive functions are supported by single antennas for both payloads. As in the previous examples, channelizaton is achieved with 12 36 MHz wide channels on each polarization, with channels spaced at 40 MHz intervals so that 500 MHz spectrum allocated for this mission is almost completely used (Fig. 16-29). The payload design yields a block diagram four times that of the detailed block diagrams shown in Fig. 16-30. The communications payload MPT budget for this communications satellite is shown in Table 16-19.

The MPT budget for this communications payload predicts a total payload mass of 756 kg, a DC power requirement of 17 kW, and a thermal dissipation requirement of 8.4 kW, each of which is within the limits that can be accommodated by modern satellite buses routinely placed in geosynchronous orbit.

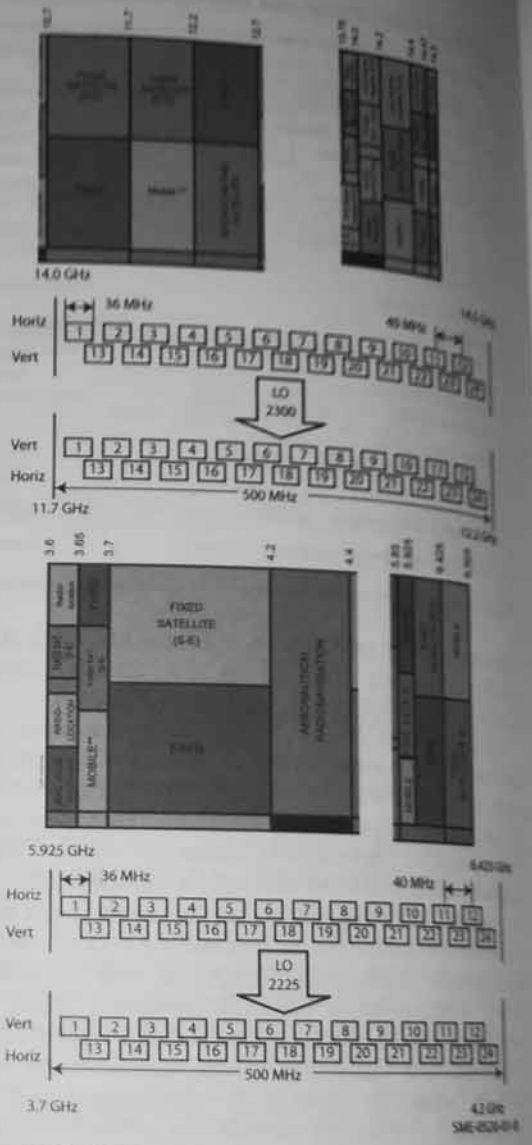


Fig. 16-29. FSS C/K_u-Band Satellite Concept and Coverage Requirement.

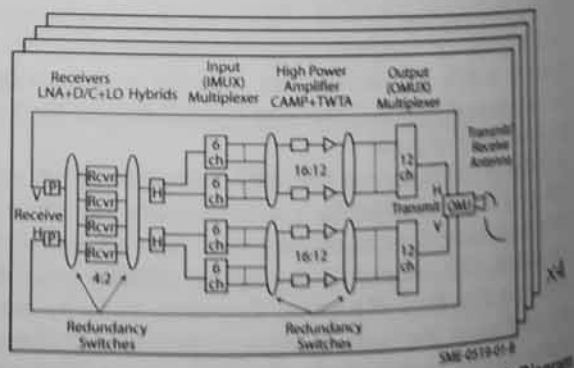


Fig. 16-30. FSS C/K_u-Band Satellite Payload Block Diagram (2 coverage areas and 2 frequencies payloads scaled from Fig. 16-27.

Table 16-18◀, Fig. 16-30, Eq. 16-33◀

Table 16-19. FSS C-Band and K_u-Band Mass, Power, and Thermal Dissipation (MPT) Budget Table.

C-Band MPT				Quantities			Unit			Total		
				Unit	Total	Active	Mass (kg)	Power (W)	Thermal (W)	Mass (kg)	Power (W)	Thermal (W)
1 Input Ancilliary (Test Coupler, Preselect Filter)		C-Band	WR137	# uplinks	4		0.70			2.80		
2 RCVR Input Red Ring Switches		C-Band	Coax	#RCVRs	8		0.15			1.20		
3 RCVR, LNA+D/C+LO		C-Band	Coax	#RCVRs	8	4	1.00	12.0	12.0	8.00	48.0	48.0
4 RCVR Output Red Ring Switches		C-Band	Coax	#RCVRs	8		0.15			1.20		
5 Input Ancilliary (Hybrid, SatPads, ...)				#RCVRs	8					0.00		
6 IMUX, single channel		C-Band	Coax	#channels	48		0.15			1.20		
7 TWTA Input Red Ring		C-Band	Coax	#TWTs	64		0.35			16.80		
8 **LCAMP**		C-Band	Coax	#TWTs	64	48	0.15			9.60		
9 **TWT CC**				#TWTs	64	48	0.30	3.4		19.20	163.2	
10 **EPC Single**	65.0	65%	91%	#TWTs	64	48	0.80	100.0	35.0	51.20	4800.0	1680.0
11 TWTA Output Red Ring		OBO:	0.0	#TWTs	64	48	1.30	14.1	14.1	83.20	678.7	678.7
12 Termination Loads		C-Band	Coax	#TWTs	64		0.30			19.20		
13 OMUX, single channel		C-Band	Coax	#channels	48							
14 Output Ancilliary (Harmonic Filter, Test Coupler)		C-Band	WR229	#downlinks	4		0.90			43.20		
15 Output Loss		C-Band	WR229			48	1.00			4.00		911.2
			1.5									
16 Interconnects (coax, waveguide, brackets, ...)		C-Band	30%									
17 Antenna Subsystem (feed, subreflector, main reflector)			waveguide	# up/downlinks	2		20.00			78.2		
18 **Total**										40.00		
										379.8	5689.5	3317.8

K_u-Band MPT				Quantities			Unit			Total		
				Unit	Total	Active	Mass (kg)	Power (W)	Thermal (W)	Mass (kg)	Power (W)	Thermal (W)
1 Input Ancilliary (Test Coupler, Preselect Filter)		K_u-Band	WR75	# uplinks	4		0.30			1.20		
2 RCVR Input Red Ring Switches		K_u-Band	WR75	#RCVRs	8		0.20			1.60		
3 RCVR, LNA+D/C+LO		K_u-Band	WR75	#RCVRs	8	4	1.20	10.0	10.0	9.60	40.0	40.0
4 RCVR Output Red Ring Switches		K_u-Band	Coax	#RCVRs	8		0.10			0.80		
5 Input Ancilliary (Hybrid, SatPads, ...)				#RCVRs	8					0.00		
6 IMUX, single channel		K_u-Band	Coax	#channels	48		0.15			1.20		
7 TWTA Input Red Ring		K_u-Band	Coax	#TWTs	64		0.26			12.48		
8 **LCAMP**				#TWTs	64	48	0.10			6.40		
9 **TWT RAD**	136.0	65%	92%	#TWTs	64	48	0.20	3.4		12.80	163.2	
10 **EPC DUAL**		OBO:	0.0	#TWTs	64	48	1.00	209.2	36.6	64.00	10043.1	1757.5
11 TWTA Output Red Ring		K_u-Band	WR75	#TWTs	32	24	2.00	44.9	44.9	64.00	1077.3	1077.3
12 Termination Loads		K_u-Band	WR75	#TWTs	64		0.30			19.20		
13 OMUX, single channel				#TWTs	64							
14 Output Ancilliary (Harmonic Filter, Test Coupler)		K_u-Band	WR75	#channels	48		0.90			43.20		
15 Output Loss		K_u-Band	WR75	#downlinks	4		1.00			4.00		
			1.8			48						2215.0
16 Interconnects (coax, waveguide, brackets, ...)		K_u-Band	40%							96.2		
17 Antenna Subsystem (feed, subreflector, main reflector)			waveguide	# up/downlinks	2		20.00			40.00		
18 **Total**										376.7	11323.6	5089.8

16.4 Sample Missions

Get More

We can now apply this approach to the SCS and FireSat II missions. See the book website for this expanded discussion along with additional MPT tables for each mission.

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Get Ref

For annotated bibliography of communications payloads, see the book website.

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Table 16-19◀, Fig. 16-30◀, Eq. 16-33◀

17 Observation Payloads

Jeffery J. Puschell, *Raytheon*

17.1 Observation Payload Design
Electro-optical Observation Payloads; Radio Frequency Observational Payloads
17.2 Observation Payload Sizing
Payload Operations Concepts; Required Payload Capability; Spatial Sampling; Signal Processing; Radiometric Sensitivity Performance; Payload Size, Mass and Power; Figures of Merit for Evaluating Candidate Payloads; Example of Observation Payload Sizing; Life-cycle Cost and Operability of the Payload
17.3 Sample Mission—VIIRS
17.4 The Evolution of Observation Payloads

Space-based observation payload missions have dramatically changed our perception of the world by delivering vitally important imagery and remote sensing data on life-threatening hurricanes and other severe weather, surveillance and reconnaissance data of critical importance to military planners, national decision makers, stunning imagery, and measurements of distant astrophysical objects at wavelengths not accessible from the Earth's surface. Future space-based observation missions are expected to provide continuous video surveillance of the Earth, improve weather forecasting by constraining forecast models with more accurate and detailed measurements, contribute to more complete understanding of climate change, and help resolve some of the deepest cosmological mysteries of our time related to dark matter and dark energy.

Observation payloads can be divided into two broad categories defined by whether the observed light is from active illumination of the scene by the payload itself or a cooperative illumination source, or from passive illumination by the Sun or by other uncooperative sources including the scene itself. Radars and laser-based sensors like DLR's TerraSAR-X and NASA's ICESAT are examples of active systems. Passive sensor examples include visible wavelength imagers like the commercial GeoEye and DigitalGlobe imagers that rely on reflected sunlight along with infrared remote sensors like NASA's Atmospheric InfraRed Sounder (AIRS) and passive microwave radiometers like the US DOD Special Sensor Microwave Imager Sounder (SSMIS) that detect thermal emission from the scene.

Earth observations by both active and passive payloads occur in three primary spectral regions known as solar reflectance, thermal infrared and microwave or radio frequency (RF). Space astronomy payloads like NASA's Chandrasekhar X-Ray Observatory and Compton Gamma Ray Observatory have observed objects at X-ray and gamma ray wavelengths, in addition to these four primary spectral regions. Schowengerdt [2006] and Schott [2007] are excellent sources for detailed discussion of solar reflectance and thermal infrared systems. Sharkov [2003] presents a comprehensive treatment of passive microwave observation systems. Cantafio [1989] provides extended discussions of active space-based RF systems.

Solar reflectance systems operate from the near ultraviolet (–0.35 μm) to the shortwave infrared (~2.5 μm) by observing reflected sunlight. These systems offer the potential for higher spatial resolution than other spectral regions because of their relatively short wavelengths, but can only operate in daylight, unless a laser or other source of active scene illumination is used [Nischan, et al. 2003].

Thermal infrared or emissive systems operate in the spectral region dominated by thermal emission from commonly observed objects on Earth (~3–300 μm). Objects of interest include the surface of the Earth (~300 K), the upper atmosphere (~220 K), biomass fires (~600–1,200 K), and missile launches (~300–1,600 K). The thermal infrared region is affected by absorption of light by molecules in the atmosphere. Observations of the Earth's surface are possible only within certain wavelength ranges known as atmospheric windows, which occur in between strong molecular absorption bands. These strong absorption bands provide a means for probing temperature structure of the atmosphere along with the 3-D distribution of trace gases such as water vapor and ozone. Thermal infrared sensors can operate both day and night because the detected signal is from thermal emission by the scene itself and does not rely on reflected sunlight.

Microwave/RF systems operate at millimeter and centimeter wavelengths. Passive microwave radiometers operate chiefly at millimeter wavelengths (20–200 GHz). Spatial resolution of these systems is three to five orders of magnitude worse than for visible wavelength sensors with the same aperture size, but they are capable of collecting unique information over large areas. Active RF

Table 17-1. Advantages of Various Observation Paylod Types.

Observation Payload Type	Advantages
Passive EO—Solar Reflectance Spectral Region	Mature technology enabling high sensitivity measurements, high spatial resolution even with relatively small wavelengths, rich phenomenology
Passive EO—Emissive	Mature technology enabling high sensitivity measurements, high spatial resolution even with relatively small wavelengths, rich phenomenology, effective day or night
Passive Microwave	Mature technology enabling high sensitivity measurements, rich phenomenology, effective day or night and through virtually all weather conditions
Passive X-Ray and Gamma-Ray	Enables detection of high energy events on Earth and throughout the universe
Active EO	Day/night operation even at Solar reflectance wavelengths
Active RF—Real Aperture Radar	Day/night, all weather capability for detecting precipitation, mapping clouds, characterizing large scale features and altimetry
Active RF—Synthetic Aperture Radar	Day/night, all weather performance with spatial resolutions comparable to much shorter wavelength systems because of the large synthetic aperture

systems also known as radars provide their own illumination of the scene in the centimeter to millimeter bands. The reflected signals can be processed to identify physical features in the scene. Radar systems can be designed to penetrate most atmospheric disturbances, such as clouds, because only larger features can reflect signals at radar wavelengths.

Regardless of spectral region or mode of operation, sensitivity of an observational payload depends on the observation geometry, effective sensor aperture size, integration time, detector sensitivity, spectral bandwidth, transmission through the atmosphere, spatial sample size and the spectral radiance from the scene itself. Sections 17.1 and 17.2 discuss these design considerations in detail.

17.1 Observation Payload Design

Observation payloads collect data (e.g., imagery, spectral radiance, distance) on remote objects such as the surface of the Earth, relatively nearby objects such as another satellite or distant astrophysical objects that can be processed into information (e.g., temperature, reflectance, chemical composition, topography) to provide insight into physical characteristics of these remote objects for scientific, weather forecasting, military and policy making purposes—just to name a few. Observation payloads operate across the entire electromagnetic spectrum from gamma rays and X-rays to RF frequencies. In addition, observation payloads include cosmic ray and other directional particle sensors used to probe the Sun, supernovae, active galactic nuclei, and other unknown astrophysical objects. Table 17-1 lists the advantages of various observation payload types. For specific mission applications for observation payloads missions, e.g., missile warning, environmental, and remote sensing, see the book website.

Two basic types of observation payloads exist: so-called passive payloads such as visible-infrared and microwave imagers that observe intrinsic emission from the scene, whether it be reflected sunlight, thermal emission or gamma rays and active payloads such as lidars and radars that supply their own source of light to enable specific types of measurement such as ranging or to enhance signal-to-noise ratio and spatial resolution.

Space-based observation payloads include the following major categories.

Passive solar reflectance systems observe that part of the electromagnetic spectrum dominated by reflected sunlight—the ultraviolet (~0.3 μm) through the shortwave infrared (SWIR) at ~2.5 μm. These systems offer the potential for relatively high spatial resolution from space, because their shorter operating wavelengths diffract less than longer wavelengths. However, because they rely on reflected sunlight for Earth observations, most of these systems operate effectively only during daytime. Observation payloads in this category include:

- Largely panchromatic imagers like GeoEye and DigitalGlobe
- Commercial imagers that specialize in collecting high special resolution imagery, multispectral (~10 spectral band with wavelength, λ, and spectral bandwidth, $\delta\lambda$, such that $\lambda/\delta\lambda \sim 10$ typically)
- Synoptic imagers like the GEOS imager
- Polar orbiting environmental imagers like MODIS that provide routine global-scale observations
- Hyperspectral (~100 spectral band with $\lambda/\delta\lambda$~100 across a broad contiguous spectral region) imagers, such as Hyperion that provide more detailed spectral measurements over limited spatial regions.
- Low light level imagers like the OLS onboard the DMSP satellite that provide panchromatic imagery of moonlit or airglow lit scenes on Earth at night
- Polarimeters such as POLDER that provide measurements of linear polarization of spectral radiance in different spectral bands across the entire solar reflectance spectral region

See the book website for more color photos from various types of observation payloads.

Emissive or thermal infrared systems observe that part of the electromagnetic spectrum dominated by thermal emission from the scene itself in atmospheric transmission windows from ~3 μm to ~300 μm in wavelength. In some cases, these systems operate outside atmospheric transmission windows to enable less cluttered observations high above the surface of the Earth. These systems operate effectively day or night, because they do not rely on sunlight to create an observable signal. Like solar reflectance systems, emissive systems include:

- Multispectral imagers like the GOES imager and MODIS
- Hyperspectral systems like the Thermal Emission Spectrometer (TES) that flew on the Mars Global Surveyor to map mineralogy on Mars and search for evidence of water
- Ultraspectral (~1,000 spectral bands with $\lambda/\delta\lambda$ ~1,000 across a broad contiguous spectral region)
- Systems like the AIRS sensor on board the NASA Aqua satellite and the IASI on board the Eumetsat MetOp satellite that provide detailed spectra of the Earth's atmosphere and surface to enable 3-D instantaneous maps of temperature distribution on the surface and in the atmosphere along with the maps of water vapor and other trace gases

Passive microwave systems observe thermal emission at much longer wavelengths than emissive infrared systems. The microwave part of the electromagnetic spectrum extends from ~1 mm to ~1 m in wavelength. Normally, microwave radiation is referenced to frequency rather than wavelength. In frequency, the microwave spectral region ranges from ~0.3 GHz to ~300 GHz. Like emissive infrared systems, passive microwave systems operate effectively day or night. At frequencies less than 10 GHz, microwave radiation is relatively unaffected by clouds, so that these longer wavelength systems can operate under virtually all weather conditions. For higher frequency microwave systems, clouds and fog along with absorption by water and oxygen molecules in the atmosphere can influence the signal from the surface. Well known passive microwave systems include:

- AMSU is used primarily for mapping temperature and water vapor structure in the atmosphere. The system flying today is the most recent version of systems that have operated on board the TIROS/POES satellites since 1978. AMSU is divided into two parts called A (used mostly for temperature sounding) and B (used mostly for water vapor sounding), which together operate in 20 spectral channels ranging from 23.8 GHz to 183.3 GHz in frequency and with different spectral resolutions for some frequencies. AMSU-A has a spatial sample size at nadir of about 45 km. The higher frequency AMSU-B has a spatial sample size of about 15 km at nadir
- SSM/I is used primarily for near-surface wind speed, total column water vapor, total column cloud liquid water and precipitation. It has been on board DMSP satellites since 1987, but has heritage back to the Nimbus 7 and Seasat missions launched in 1978. SSM/I operates in seven channels ranging from 19 GHz to 89 GHz in frequency. SSM/I operates with spatial resolutions ranging from 13 km to 69 km, depending on frequency
- The SSMI/S instrument is the follow-on instrument to SSM/I and measures microwave energy at 24 discrete frequencies from 19 to 183 GHz with a swath width of 1,700 km at 12.5 km to 75 km spatial resolution at nadir. The synoptic solar reflectance and emissive infrared systems used in conjunction with these passive microwave systems operate with ~1 km spatial resolution using much smaller aperture sizes, because of the diffraction performance advantage of shorter wavelengths

X-ray imagers observe thermal and non-thermal emission from astrophysical objects including the Sun, neutron stars, supernovae, the center of our Galaxy and distant unknown objects at wavelengths ranging from less than 3 pm to 10 nm, corresponding to frequencies ranging from above 10 EHz (1 EHz is 10^{18} Hz) down to 30 PHz (1 PHz is 10^{15} Hz). X-rays are often referenced in units of electron Volts (eV) rather than wavelength or frequency. In eV, the X-ray spectral region extends from about 0.1 KeV to 400 KeV. The first X-ray observations from space were made with geiger counters that flew on board converted V-2 rockets starting in 1949. UHURU also known as SAS-1 launched in 1970 was the first satellite with an X-ray observation payload. The Chandra X-ray observatory, launched in 1999, observes the 0.1 KeV to 10 KeV spectral region, with 2.4 μrad (0.5 arcsec) angular resolution using a 1.2 m diameter X-ray telescope. The satellite is in a highly elliptical orbit that enables the observatory to make X-ray measurements from well out the Earth's Van Allen belts. The effective collection area for the Chandra observatory is only 0.04 m^2 at 1 KeV, despite the much larger telescope diameter. Conventional optical approaches are not effective at X-ray wavelengths, because no suitable refractive material is available to build X-ray lenses and standard reflector telescopes do not work because X-rays are either absorbed or transmitted at near normal incidence to reflecting mirrors. Therefore, X-ray instruments use either a coded aperture method or a grazing incidence telescope, also known as a Wolter telescope, to collect and focus X-rays. In Wolter telescopes, the angle of reflection from the mirror is very low—typically 10 arc-minutes to 2 deg. Chandra uses a Wolter telescope that consists of nested cylindrical paraboloid and hyperboloid surfaces coated with iridium or gold. Chandra uses four pairs of nested mirrors. The relatively

thick substrate (2 cm) and very careful polishing allowed a very precise optical surface, which is largely responsible for Chandra's unprecedented and unmatched angular resolution. However, thickness of the substrates limits the fill factor of the aperture, leading to the low effective collection area.

Gamma ray payloads observe emission of the highest energy photons from astrophysical objects including the Sun, supernovae, the center of our Galaxy and distant unknown objects. Gamma rays do not penetrate Earth's atmosphere very well so direct astronomical observations of gamma rays must occur from high in the atmosphere or from space. Wavelengths for gamma rays are shorter than 3 pm, corresponding to frequencies greater than 10 EHz (1 EHz is 10^{18} Hz). The first gamma ray observations of the Sun and distant galactic and extragalactic objects were made by OSO-3 launched in 1967. Shortly thereafter, mysterious gamma ray bursts from very distant, but still only partially identified objects were discovered by the Vela satellites designed to detect gamma rays from nuclear explosions on Earth. Recent gamma ray observation payloads include the instruments on board the Fermi Gamma Ray Telescope, a successor to the Compton Gamma Ray Observatory. These instruments detect gamma rays in two broad spectral ranges using scintillation detectors that detect small flashes of light created by gamma rays passing through a crystal and pair production techniques at the highest energies that create electron-positron pairs created by incoming gamma rays.

Active electro-optical systems like lidars, ladars and altimeters use active illumination of the scene by a laser to create a measurable signal that can be read out rapidly over time to remotely sense backscattered light from water droplets, aerosols and trace gases in the atmosphere as a function of distance from the system and for range measurements and altimetry, when looking at the Earth, Moon, and other space objects.

- The first lidar in space was the Lidar In-space Technology Experiment (LITE) carried onboard the Space Shuttle in 1994. LITE measured clouds with a backscatter lidar based on a Nd:YAG laser operating at three different wavelengths: 1,064 nm, 532 nm (second harmonic) and 355 nm (third harmonic)

- CALIPSO, launched as part of the A-train of Earth Science satellites in 2006, is also a backscatter lidar based on a Nd:YAG laser. CALIPSO transmits both the fundamental (1,064 nm) and second harmonic (532 nm) of the Nd:YAG laser in a nadir beam. The vertical resolution is 30 m and horizontal resolution 333 km

- ICESAT launched in 2003 carried a Nd:YAG based laser altimeter used primarily for measuring thickness of Arctic ice sheets, but also for providing global topography and vegetation data. The main challenge for building a laser-based observational space payload is producing a reliable laser for extended duration use in space—previous instruments like ICESAT experienced failure of some laser modules after only a few months of operation

Lessons learned from ICESAT benefit future missions such as the upcoming AEOLUS mission with the ALADIN lidar, planned for the Atmospheric Dynamics Mission (ADM). ALADIN is a wind-measuring lidar, based on the third harmonic of a Nd:YAG laser. ALADIN will measure wind speed along the lidar line of sight by measuring the Doppler shift of the backscattered light using two methods: coherent (heterodyne) detection by mixing the return signal with a local oscillator and a direct direction method using a narrowband etalon. These techniques will be applied to measuring Doppler shifts of light returned to a receiver from Mie (aerosol) and Rayleigh (molecular) scattering respectively. While useful as a pathfinder for future winds lidar missions by providing information not available to today to numerical weather forecast models, AEOLUS will only measure a single component of the 3-D wind vector.

Active RF systems like radars and synthetic aperture radars (SARs) use active illumination of the scene by a coherent RF transmitter to create a signal that can be detected and read out rapidly in time to produce detailed range resolved RF imagery of precipitation in clouds, surface wakes on the ocean and detailed topography of Earth and other planetary solid surfaces. Like passive microwave systems, active RF systems operate at much longer wavelengths than visible and infrared systems that are routinely used for imagery—typically at centimeter wavelengths. Consequently, for practical payload sizes, active RF systems tend to operate at much coarser spatial resolution than visible and infrared systems—but they can operate through almost any possible weather condition. For example:

- Precipitation radar (PR) onboard the TRMM satellite. The TRMM PR launched in 1997 was the first spaceborne radar designed to measure vertical structure of rainfall precipitation in the lower atmosphere. It operated at 13.8 GHz (2 cm in wavelength) using a 128-element electronically scanned array to allow fast and complex cross-track scanning over a swath width of 215 km with a cross-range spatial resolution of about 4.3 km and a range resolution of 250 m. As shown in Fig. 17-1, the TRMM PR operated in conjunction with other observational payload systems—a passive microwave imager and a visible-infrared imager

- Systems (like TRMM PR) are very useful, but they cannot provide spatial resolution approaching visible-infrared systems with feasible collector sizes. A different type of radar takes advantage of

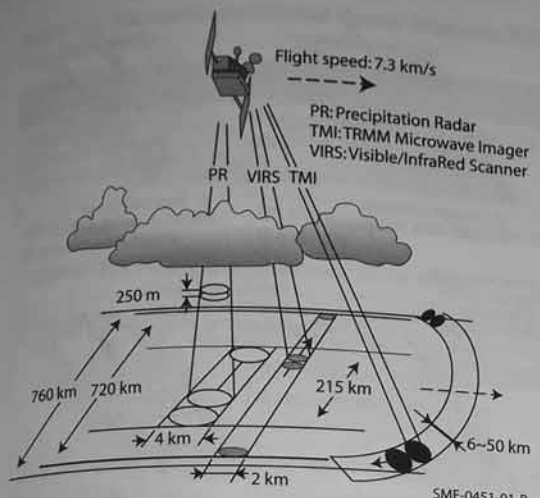


Fig. 17-1. The TRMM PR observed rainfall observation in conjunction with a passive microwave imager, the TMI, and a visible-infrared imager called VIRS. While the PR and VIRS made cross-track scans, the TMI made a conical scan, similar to the SSM/I and SSMI/S.

the relatively relaxed tolerances at these long RF wavelengths for coherent phase and Doppler shift measurements of a radar return signal

- Synthetic Aperture Radar (SAR) processes coherent measurements of phase and Doppler shift across the wide beamwidth of a RF system to synthesize a virtual aperture with a size equivalent to the distance the physical antenna moves while the scene location remains illuminated by the beam. The corresponding spatial resolution in the along spacecraft track direction approaches resolutions achievable with visible and infrared systems.

- SEASAT launched in 1978 was the first known SAR in space. It operated in L-band at a frequency of 1.275 GHz or 23.5 cm in wavelength. SEASAT operated for only about 105 days, failing when a short circuit in the satellite's electrical system ended the mission. Nevertheless, the mission was very successful and laid the processing foundation for following systems

- Since SEASAT, SARs have been developed and flown by Canada, Japan, Europe, and Russia. TerraSAR-X, an X-band (9.65 GHz) SAR developed by the German Aerospace Center (DLR) and launched in 2007 has produced remarkable imagery that is especially useful for urban landscapes. Along with its partner satellite, TanDEM-X, launched in 2010, high resolution digital elevation maps are being produced for flood plain maps and traffic monitoring among many other applications

To determine the radiometric performance of an optical instrument, we must consider the observation geometry, effective aperture, integration time, detector sensitivity, spectral bandwidth, ground pixel size, and the transmission through the atmosphere. For passive Earth observations, we define three basic categories, depending on the spectral range

Wavelengths covered in the first case range from 0.3 to 2.5 μm, and they include the ultraviolet, visible, near and short wavelength infrared. The instrument receives radiation reflected from the atmosphere, as well as from the surface of the Earth, when illuminated by the Sun. We will normally neglect thermal radiation emitted by the Earth in this case

The second case applies to wavelengths ranging from the long infrared region (8–14 μm) through the microwave and RF. In these wavelengths, instruments receive thermal radiation emitted from the atmosphere and the surface of the Earth when reflected radiation from the Sun is negligible

The mid-wavelength infrared spectral region (3–8 μm) is our final case, where contributions from both emissive and reflected sources must be considered. The radiance from emissive and reflected sources is shown in Fig. 17-2. It can be seen that the thermal emitted radiance of the Earth increases with wavelength, corresponding to Planck's law, and that the reflected radiance from the Earth's surface decreases with wavelength

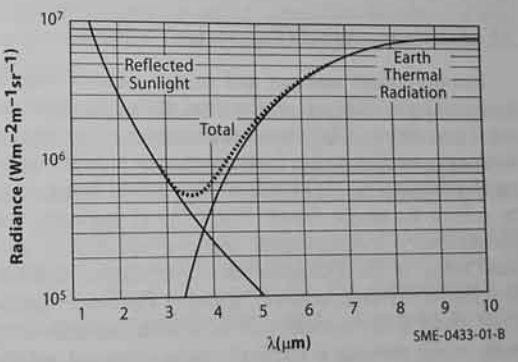


Fig. 17-2. Radiance from Direct and Reflected Sources. Radiance contribution in Watts per square meter per meter (wavelength) per unit solid angle of reflected sunlight from the Earth and emitted radiation from the Earth as a function of wavelength. The sum is shown as a dashed line. The Sun is modeled as a blackbody with a temperature of 6,000 K, the reflection coefficient of the Earth's surface and the transmission of the atmosphere are modeled as constants for clarity.

In the ultraviolet through SWIR bands, we resolve images produced by energy (chiefly from the Sun) reflected from the target scene rather than energy from the relatively low self-emission that occurs in the reflectance bands. But in the infrared, we see things almost entirely by their self-emission, with very little energy being reflected, particularly at night

We may use the same optical train elements —lenses, prisms, mirrors, and filters—to collect

Table 17-1◀, Fig. 17-2, Eq. 17-0

infrared energy as for visible and UV, but we must apply them differently. For example, ordinary glass is opaque to IR beyond 3 μm, whereas germanium, which is opaque in the visible band, is transparent in the 1.8 to 25 μm region

We must consider atmospheric scattering caused by molecules, aerosols and other particles in the air. The amount of scattered radiation from atmospheric molecules is a function of the inverse fourth power of the wavelength

For haze, the situation is more complex, but the scattering extinction coefficient goes roughly as the inverse first to second power of the wavelength. Thus, IR penetrates haze and dust much better than visible radiation because the IR wavelengths are four or more times those in the visible spectrum. The same phenomena explain the reddish color of the sky near dawn and sunset. At these times, shorter green, blue, indigo, and violet wavelength signals are greatly attenuated as they travel farther through the atmosphere than when the Sun is overhead

The following sections provide more detailed information on the best known observation payloads— those that use electro-optical techniques in the solar reflectance and emissive spectral regions to observe and measure physical properties of scenes in the ultraviolet through submillimeter spectral region, and those that use radio frequency technology.

17.1.1 Electro-optical Observation Payloads

Electro-optical imagers and remote sensors scan a scene using mechanical or electrical means. Geostationary Earth Orbit (GEO) spacecraft appear to be almost stationary relative to the Earth, so that an optical instrument operating in GEO needs to scan in two dimensions to collect an image frame. The original approach for geostationary environmental imaging spacecraft, still used today in the European Meteosat Second Generation (MSG) system, involves spinning the entire spacecraft about a north-south axis to enable the east-west scan, while moving a relatively large elliptical mirror (~80 cm along the major axis) in front of the imager optics to perform the north-south scan. Three axis stabilized spacecraft in geosynchronous orbit like the US GOES environmental satellite system have imagers that use a two axis scanner in front of the imager collection optics to scan the scene in two dimensions. Alternatively, large format two dimensional focal plane array and optical technologies for advanced missile warning sensors have been used to create a full-Earth staring imager that requires no mechanical scanner to sample the entire Earth scene.

The sub-satellite point for a spacecraft in low Earth orbit (LEO) moves with respect to the scene on the ground at a speed of approximately 7 km/s (See Chap. 8). In most LEO systems, the motion of the spacecraft is used to eliminate scanning in the *ground track* direction (except for spacecraft motion and image rotation corrections), so that the scanning system of the optical instrument need move only in the *cross-track* direction. Several methods for doing so are described below. Illustrated in Fig. 17-3A, *whiskbroom sensors* scan a single detector element on the ground in the cross-track direction, thereby creating a single scanline of pixels in the final image or other data product. Whiskbroom scanners can also use multiple detectors, to create multiple scanlines of pixels simultaneously (Fig. 17-3B). This technique reduces the required scan rate compared to a system with a single detector. The required scan rate is reduced so that the dwell time per detector sample is multiplied by the number of detector elements used. *Push broom scanners* cover a full swath width using a linear arrangement of detector elements called a linear array, line array, or scanning array. As illustrated in Fig. 17-3C, the read-out process delivers one line after another in the ground track direction, like pushing a broom, hence the name. The integration time is defined by the velocity of the sub-satellite point and the size of the pixels on the ground. To improve signal-to-noise ratio of the image, at the expense of area coverage rate, push broom imagers or their host spacecraft can compensate for motion of the sub-spacecraft point along the Earth's surface to increase effective dwell time.

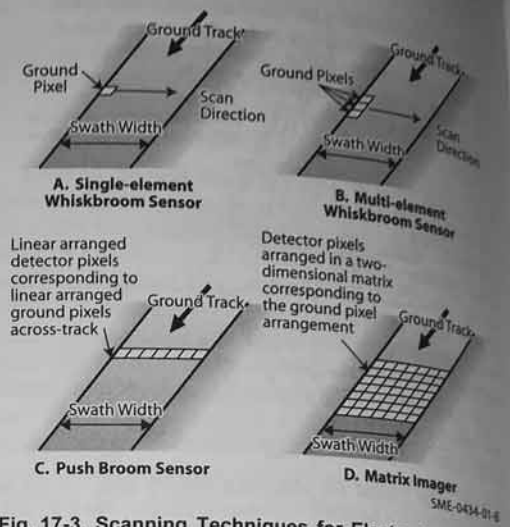


Fig. 17-3. Scanning Techniques for Electro-Optical Instruments. See text for discussion.

Step-and-stare scanners use a 2-dimensional array of detector elements to collect data over an entire region of interest by staring at part of the scene and then stepping to another location in the scene in a predetermined pattern of steps and staring again, until the region is completely covered. Each detector element corresponds to a spatial resolution element on the ground. These systems operate in two basic modes in LEO. In the first mode, integration times are determined by velocity of the sub-satellite point and size of the ground pixel, as in the pushbroom sensor. No advantage is achieved with

Table 17-2. Comparison of Optical Sensor Scanning Methods. We list relative advantages and disadvantages of different scanning mechanisms

Scanning Technique	Advantages	Disadvantages
Whiskbroom Scanner—Single Detector Element	High uniformity of the response function over the scene Relatively simple optics	Short dwell time per pixel High bandwidth requirement and time response of the detector
Whiskbroom Scanner—Multiple Detector Elements	Uniformity of the response function over the swath Relatively simple optics	Relatively high bandwidth and time response of the detector
Push Broom Sensor	Uniform response function in the along-track direction No mechanical scanner required Relatively long dwell time (equal to integration time)	High number of pixels per line imager required Relatively complex optics
Step-and-Stare Imager with Detector Array	Well defined geometry within the image Long integration time (if motion compensation is performed)	High number of pixels per array imager required Complex optics required to cover the full image size Calibration of fixed pattern noise for each pixel Highly complex scanner required if motion compensation is performed

respect to the integration time in the simplest case. A short integration time is needed to avoid image smear during read-out. The second mode allows a longer integration time. If we shift the image array in the focal plane or move the line of sight of the instrument, the image motion can be compensated to very low speeds relative to the ground. The instrument line of sight can be adjusted by moving a scan mirror or by tilting the spacecraft itself to compensate for the movement of the sub-spacecraft point. Step-and-stare sensors require relatively complex scanner optics or spacecraft operations to cover large areas at high signal to noise ratio from low Earth orbit, especially. The fixed pattern noise must be removed from the image because each pixel has a somewhat different responsiveness and dark signal, creating additional complexity. Table 17-2 summarizes the distinguishing features of optical scanning methods.

In GEO and other high altitude orbits, another approach to capturing a 2 dimensional scene can be used that involves positioning the scene with respect to the instrument. In this case, the instrument moves to the next part of the scene between integration periods either along-track or across track to image the entire scene. This scan motion can be continuous across a large swath, as in the geosynchronous GOES and MTSAT imagers. If this shift occurs in a series of discrete steps, this approach is referred to as a step-and-stare imager. This technique is used in geosynchronous orbit in the GOES Sounder, for example. If the instrument field of view only covers a part of the required scene, then moderately complex (and also moderately sized) optics are required. We can use highly agile and accurate pointing mirrors in front of the instrument's optics to adjust the line of sight.

For space missions, optical instruments often rely on existing designs. Custom tailoring of current optical and detector design approaches are common in optimizing the performance and cost of the instrument. Let's follow a scene photon through an observational payload by considering first the optical subsystem involving a telescope and a spectral separation approach and then the detectors.

Telescopes. This discussion considers the telescope optics used to collect and transform light from the scene onto a focal plane along with approaches used to separate the incoming light into different wavelength bands.

We treat reflective telescopes (telescopes using mirrors) and refractive telescopes (telescopes using lenses) together. There are advantages and disadvantages to both these telescopes that will be discussed in detail later.

There are several ways to describe a telescope. Parallel rays of light falling on a perfect lens will all converge at the focal point, whose distance from the lens is called the *focal length, f*. The focal length largely determines the length of the optical collection system and, for a single lens, it is related to the lens surface's radius of curvature. The focal length of a spherical reflecting surface is one-half its radius. For a parabolic reflector whose y-z surface is defined by the equation $z^2 = 4fy$, the quantity f is the focal length; it equals the distance from the focus to the nearest point on the reflecting surface.

In design practice, we normally determine the required focal length based on instantaneous field of view or angular spatial sampling size and the physical size of detector elements in the image plane. The plate scale, s, or length per field-of-view angle is given by:

Table 17-2, Fig. 17-3◀, Eq. 17-0

$$s = f \text{ (unit length/rad)}$$
$$= \left(\frac{\pi}{180}\right) f \qquad (17\text{-}1)$$
$$\approx 0.01745 f \text{ (unit length/rad)}$$

where s and f are in the same units. The image size is a function of s and the size of the detector— ranging from a single element to a large array—employed at the focal plane. As Fig. 17-4 shows, the focal length needed to record an object or scene of radius R, is given by the magnification, M,

$$M = \frac{f}{h} = \frac{r_d}{R} \qquad (17\text{-}2)$$

where h is the distance from the spacecraft to the object, r_d is the radius of the detector array in the image plane, and R is the radius of the object, with the image and object measured perpendicular to the line of sight. The magnification or scale, r_d/R, is the ratio of the image size to the object size. It is ordinarily a very small number for space-based observation payloads. We express the scale on the image plane as "1 μm equals × km on the ground."

We can also describe an optical element or system by its so-called infinity F-number or F-stop, often written as f/, F, F No., or F#. It is defined as f/D, where D is the effective entrance aperture, which is the effective diameter of the lens or primary telescope mirror. Image brightness is proportional to F^{-2}, so an $f/4$ lens gives an image four times brighter than an $f/8$ lens.

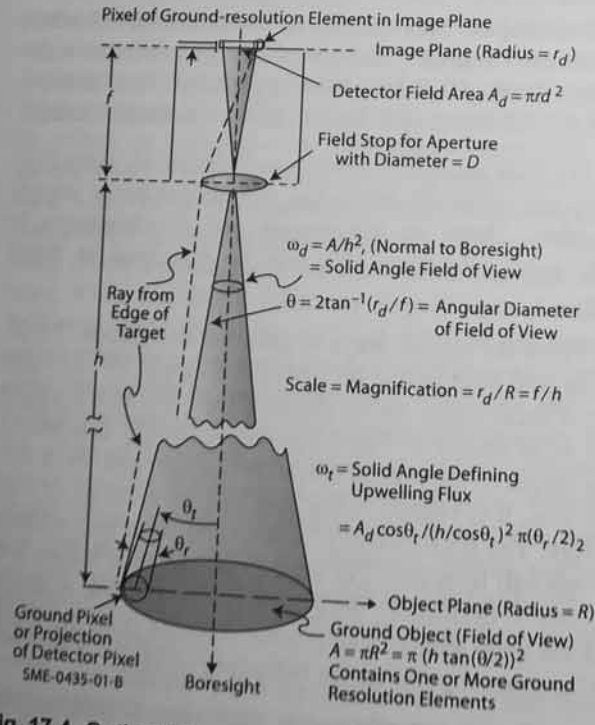


Fig. 17-4. Optical Characteristics of a Refractive System. Note one-to-one correspondence of the ground-resolution element's size to the pixel size at the image plane. The operating wavelength is λ. As resolution elements move away from nadir, flat-Earth approximations become less precise.

Table 17-3, Fig. 17-4, Eq. 17-4

The numerical aperture, NA, gives the same information in another way:

$$NA \equiv \frac{1}{2F\#} = \frac{D}{2f} \qquad (17\text{-}3)$$

or

$$F\# = \frac{1}{2NA} = \frac{f}{D} \qquad (17\text{-}4)$$

The largest numerical aperture for optics used in air is 1. Thus, the smallest F# is 0.5.

Table 17-3. Principal Aberrations in Optical Systems. See Table 17-4 for which of these are mitigated in various optical systems.

Chromatic Aberration is dispersion of the light due to the refractive index of a lens being a function of the wavelength. Causes different colors to focus at different distances.
Spherical Aberration is dispersion in which light from the periphery of a spherical lens or mirror is focused nearer the element than light from the center. Can be eliminated by making the optical surface parabolic, rather than spherical.
Coma is dispersion of off-axis portions of the image. (So named because in a telescope, off-axis star images look liketear drops, or the coma of a comet pointing toward the center of the image.)
Astigmatism is aberration in which the distorted image is asymmetric such as when light in a horizontal plane comes to a slightly different focus than light in a vertical plane. A common problem in human vision.
Distortion is when an otherwise sharp image is distorted in shape, such as when straight light on the surface being viewed appear curved on the focal plane. An uncorrectable distortion occurs when trying to image the celestial sphere onto a flat focal plane. (See Sec. 8.1.)
Curvature of Field is when a sharp image is formed on a focal surface which isn't flat. Can be corrected in film systems by using a slightly curved focal plane.

All telescopes experience *aberrations*, or imperfections in the quality of the image, in addition to diffraction which limits system resolution as discussed below. Principal optical aberrations are listed in Table 17-3. *Chromatic aberration* is wavelength or color dependent. This imperfection is due to various wavelengths being bent by different amounts when passing through the lens. As a result, only systems with at least one refractive element will suffer chromatic aberration. This is due to the fact that reflective surfaces treat all wavelengths the same, which is not absolutely true since some surfaces will reflect visible light, for example, but not X-rays. However, when reflection does occur, it is independent of wavelength to the first order.

Three basic types of telescopes are shown in Fig. 17-5. In each of these telescope approaches, there is a corresponding refractive and reflective design. Table 17-4 shows the aberrations that can be corrected in these telescopes. The lens doublet is the classic refractive telescope lens and can be designed to elimi-

Table 17-4. Aberrations that can be Corrected by the Three Basic Optical Systems. Checks indicate errors that are fully correctable and parenthetical checks indicates that corrections are possible only for dedicated design parameters. (See Table 17-3 for definitions.)

Image Error	Doublet/ Schmidt	TeleOptic/ Cassegrain	Triplet/ TMA
Lateral Chromatic Aberration	✓	✓	✓
Length Chromatic Aberration	✓	✓	✓
Spherical Aberration	✓	✓	✓
Coma	✓	✓	✓
Astigmatism		(✓)	✓
Distortion	(✓)	(✓)	✓
Curvature of Field		(✓)	✓

nate spherical aberrations, coma, distortion and chromatic aberrations. In tele-optic lens systems, the distance between the optical element and the focal plane is shorter than the focal length. Spherical aberrations, coma, astigmatism, curvature of field effects, and even chromatic aberrations can all be eliminated by tele-optic lenses. The simplest refractive (spherical) optical system that theoretically allows for correction of all distortions is the lens triplet. While this approach offers many important advantages, it can be relatively challenging to implement because each of the optical elements is highly sensitive to displacement or tilt. The ray traces in Fig. 17-5 show different locations of the image in the focal plane relating to various viewing angles. All five of the third-order aberrations as compensated by the lens triplet include: spherical aberrations, coma, astigmatism, curvature of field, and distortion. Similar to tele-optics, it is also free from chromatic aberrations. The same behaviors exist in the corresponding reflective systems as well. The Cassegrain telescope is a reflective implementation of a tele-optic lens, and the Schmidt Mirror System is an all-reflective doublet. The Three-Mirror Anastigmatic system is similar to the lens triplet with respect to all the aberration corrections, but with an all-reflective design. Generally, reflective optical systems are free from chromatic aberrations. However, reflective systems typically have a much smaller field of view than their refractive counterparts.

Space remote sensing telescopes tend to be more complex than ones used on the ground. This is due in part to effects such as thermal distortions and radiation which can alter performance of instruments and cause space-based telescopes to fall short of meeting performance requirements over the intended lifetime of the instrument. These environmental effects and the challenging image quality and spectral coverage requirements for many systems limit the technologies for manufacturing the lenses and mirrors. Even if operating temperature ranges are controlled to within a few deg, thermal distortions can still limit the performance of the optical system. Cosmic radiation effects can degrade the transparency of most optical glass over time. Figure 17-6 shows the cross sectional view of the high resolution optical lens system, a German-built Modular Optoelectronic Multispectral Scanner (MOMS 2P) designed to achieve a resolution of 6 meters on the ground.

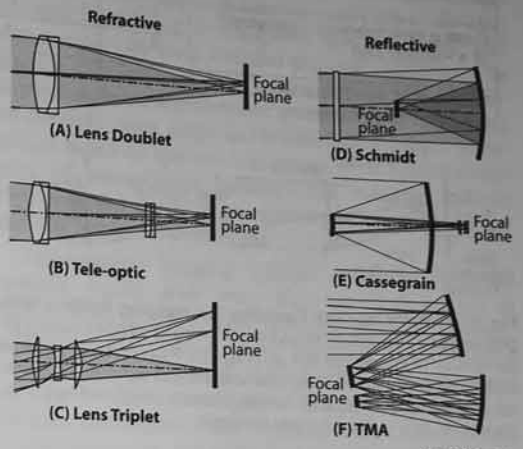


Fig. 17-5. Basic Configurations for Refractive and Reflective Optical Systems. Each of the reflective systems on the right is analogous to the corresponding refractive system on the left. TMA = Three-mirror anastigmatic.

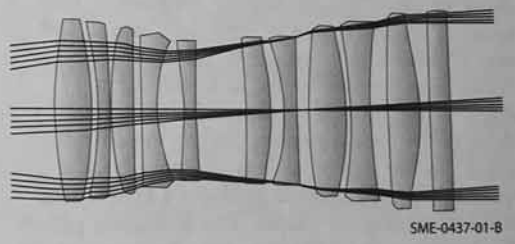


Fig. 17-6. Lens Cross Section of the Panchromatic Objective of the MOMS 2P Instrument. The sensor has a focal length of 0.66 m and an aperture size of 0.15 m. The complexity of this optical system is representative of sophisticated remote sensing payloads.

The angular resolution of an optical system determines its ability to enable recognition of fine spatial detail in Earth observation data. A telescope that can just barely resolve two close stars is said to have a resolving power equal to the angular separation of those two stars. We are more interested in the ability to distinguish fine detail on the surface for Earth observing systems. Thus, for these systems, spatial resolution is normally expressed as the smallest physical separation of two objects on the surface that can be resolved by the instrument.

Table 17-4, Fig. 17-6, Eq. 17-4◀

Live Calc

Table 17-5. Diffraction-Limited Resolution. Note that the Synthetic Aperture Radar provides resolutions similar to visual or IR systems, independent of range and wavelength for ranges up to the maximum signal-to-noise limit by synthesizing the required aperture.

		Ground Resolution = $h\lambda/D$		
	Aperture Size, D	Visible (λ = 0.5 μm)	IR (λ = 3 μm)	Passive Microwave (f = 10 GHz; λ = 3 cm)
From an orbiting spacecraft (h = 900 km)	1 m	0.45 m	2.70 m	27.0 km
	3 m	0.15 m	0.9 m	9 km
From a synchronous spacecraft (h = ~35,800 km)	1 m	17.9 m	107 m	1,074 km
	3 m	6.0 m	35.8 m	358 km
From SR-71 (h = 20 km or 70,000 ft)	0.3 m	0.033 m	0.200 m	2.00 km

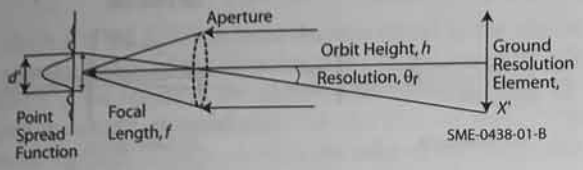


Fig. 17-7. Point Spread Function for Imaging System with Diffraction. The optical wave front from an ideal point source on the ground is imaged as the point spread function by the optical system. The diameter of the aperture and the wavelength determine the extent of the point spread function measured by the diameter, d', of the first intensity minimum.

For a perfectly designed and fabricated optical system, the theoretical limit to angular resolution performance is determined by diffraction, which is the apparent bending of light occurring whenever light encounters an obstacle, such as an element in a telescope or other optical system. Diffraction blurs the image of a point source of light, such as a distant star, into a pattern with elements that get progressively fainter away from the center, as shown in Fig. 17-7. The shape of the diffraction pattern is determined by the shape of the optical element causing the diffraction, so that a circular mirror or lens produces circular rings, whereas a square aperture produces a pattern with 2-D series of rectangular shapes. The diffraction pattern is also known as the diffraction disk, the Airy disk (for a circular aperture) or the point spread function. For a circular aperture, the angular distance, θ_r, from the maximum at the center of a point source image to the first dark interference ring, is called the Rayleigh limit and is given by

$$\theta_r = \frac{1.22\lambda}{D} = \frac{1.22}{f\,F_{cutoff}} \tag{17-5}$$

where λ is the wavelength, D is the aperture diameter of the optical instrument, f remains the focal length, and F_{cutoff} is the optical cutoff spatial frequency in units of inverse length (say m^{-1}) that limits the resolution of the image in the focal plane. θ_r is expressed in radians. The bright lobe at the center of this Airy disk diffraction pattern contains 84% of the total energy arriving at the focal plane from a point source, from its center out to the first minimum. For a satellite at altitude, h, the linear resolution or ground resolution limit, X', at nadir as defined by the cutoff frequency is

$$X' = \frac{h\lambda}{D} \tag{17-6}$$

A more general expression is to replace h with the slant range, R_s, from Eq. (8-31), to determine the resolution away from nadir (R_s here = D in Chap. 8). Keep in mind that this is the resolution perpendicular to the observation system line of sight and that actual spatial resolution on the surface of the Earth is degraded by the effect of geometric projection or 1/sin ε, where ε is the elevation angle at the specific spacecraft and Earth locations of interest, obtained from Eq. (8-28). Ground resolution at nadir for several representative system configurations is given in Table 17-5. Most space-based optical systems have the ability to produce images with angular resolution as good as the theoretical limit and are therefore described as being diffraction limited.

When we implement an optical system using a detector array, we add an additional design parameter, the quality factor, Q, defined as the ratio of the physical size in the focal plane corresponding to the spatial frequency cutoff to the detector element size, d, that is

$$Q = \frac{d'}{d} = \frac{X'}{X} \tag{17-7}$$

where d' is given by

$$d' = \frac{\lambda f}{D} \tag{17-8}$$

X is the ground pixel size, and X' is the ground resolution corresponding to d' in the focal plane (See Fig. 17-8). Q typically ranges from 0.4 to 2. For $Q > 1$, the detectors are smaller than the cutoff and resolution is limited by diffraction in the optics. This gives the best possible image resolution for a given aperture. For $Q < 1$, the resolution is limited by detector size. This design point may be chosen if image quality is less important than aperture size, as would be the case, for example, when increased light gathering power is required. A good starting point is $Q = 1$, which allows good image quality.

From the definition of the magnification, Eq. (17-2), we have:

$$\frac{d}{X} = \frac{d'}{X'} = \frac{f}{h} \tag{17-9}$$

Combining Eqs. (17-7) to (17-9), we obtain expressions for the detector size, d, in terms of the other basic system parameters:

$$d = \frac{d'X}{X'} = \frac{d'}{Q} = \frac{\left(\frac{\lambda f}{D}\right)}{Q} \tag{17-10}$$

where the parameters are defined above and, as usual, λ is the wavelength, f is the focal length, and D is the aperture diameter.

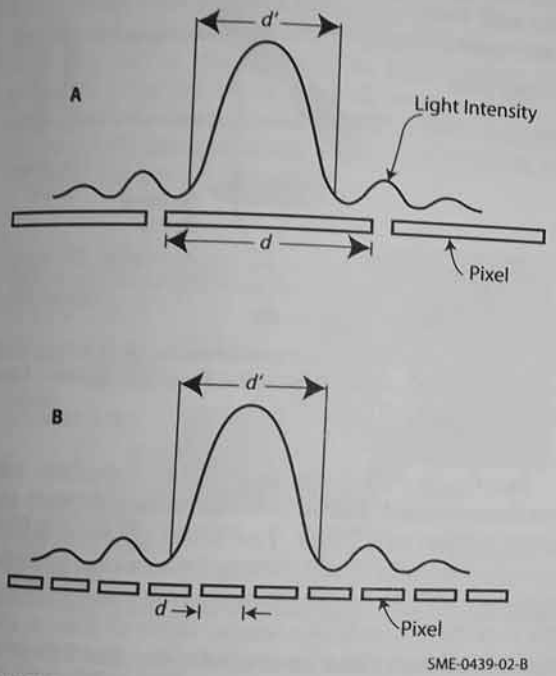


Fig. 17-8. Effect of Varying Quality Factor. Different sizing of the detector pixel with respect to the point spread function is shown by varying the Quality Factor, Q. A low quality factor results in the relative sizing in diagram (A) on top and a large quality factor results in the relative sizing in diagram (B) on bottom.

Spectral separation approaches. As noted above, space-based observation payloads operate with different spectral sampling approaches, depending on intended mission. These spectral sampling approaches fall into four broad categories summarized below called panchromatic, multispectral, hyperspectral and ultraspectral.

Panchromatic or PAN spectral sampling involves one band per broad spectral region (e.g., one band covering the entire visible part of the spectrum). PAN imaging is useful for collecting high SNR, high spatial resolution imagery where spectral content of a scene is considered less important than image sharpness and clarity. Well known systems using PAN imaging include the commercial imaging GeoEye and DigitalGlobe systems.

Multispectral Imaging (MSI) uses ~3–50 relatively wide ($\lambda/\delta\lambda$~10 typically) spectral bands to identify major features of a scene (rocks, trees, crops, water). Well known MSI examples include Landsat, AVHRR, GOES imager and sounder and MODIS.

Hyperspectral Imaging (HSI) systems collect ~30–300 moderately high spectral resolution ($\lambda/\delta\lambda$~100) spectral bands over a contiguous spectral region. HSI systems provide more detailed physical information on a scene than PAN and MSI system and are used to identify components within major scene features (e.g., types of minerals, trees, crops and health, bathymetry, composition) based on spectral signatures. Well known examples of space-based hyperspectral systems include, Hyperion and the TES instrument that mapped Mars at thermal infrared wavelengths.

Ultraspectral Imaging systems collect ~300–3,000 spectral bands with very high spectral resolution ($\lambda/\delta\lambda$~1,000) across a broad contiguous spectral region. Ultraspectral systems are used primarily in the infrared to collect detailed spectra of the Earth's atmosphere and surface for physical analysis of gas absorption features for trace gas detection and mapping and retrieval of vertical temperature structure and distribution of water vapor in the atmosphere. Well known ultraspectral systems include AIRS, MIPAS and IASI.

Spectral separation occurs in a device called a spectrometer that can use filters, dispersive elements or interferometers to isolate spectral ranges of light for remote sensing and imaging. PAN and MSI systems most often use filters to separate out desired ranges of wavelength for measurement whereas HSI and ultraspectral systems almost always use dispersive or interferometric approaches but can use filter spectrometers. Some systems sample the Earth with a hyperspectral approach and then process data onboard the spacecraft to MSI bands for transmission to Earth, as illustrated in Fig. 17-9.

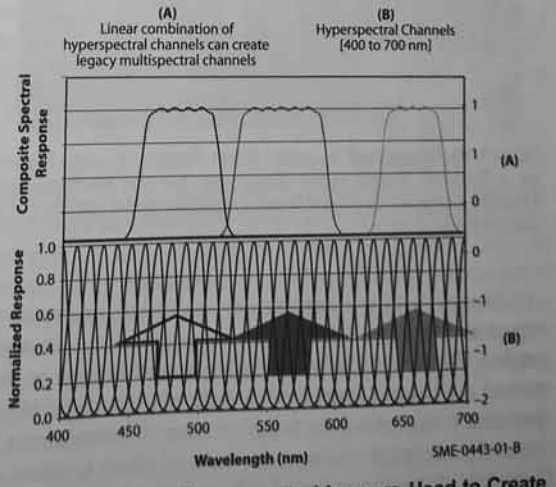


Fig. 17-9. Data from Hyperspectral Imagers Used to Create Virtual Multispectral Channels. (Courtesy of NASA)

Table 17-5◀, Fig. 17-9, Eq. 17-10

Filter spectrometers use optical bandpass filters to transmit narrow wavelength bands from the scene spectrum to an individual detector, or a 1-D or 2-D focal plane detector array. Early optical filters were based on stacks of colored glass that defined spectral bandpasses. Virtually all optical filters used today are based on an interferometric device called an etalon to define high tailored spectral responses precisely. Multispectral filter spectrometers can use discrete filters (one or more per focal plane), filter wheel mechanisms to rotate different filters into a detector line of sight or integrated spectral filter assemblies also known as butcher block assemblies (Fig. 17-10) to cover ~10 spectral bands in a single focal plane. Hyperspectral filter spectrometer approaches include tunable filters and spatially variable filters such as the wedge filter that can collect data as a pushbroom imager. Advantages of filter spectrometers include relatively high optical throughput and relatively easy implementation for multispectral applications, especially. One of the disadvantages of filter spectrometers is that high spectral resolution filters with predictable characteristics can be very expensive and difficult to produce consistently. Filter spectrometer examples include all of the best known MSI systems (e.g., MODIS, GOES, AVHRR) and the hyperspectral Linear Etalon Imaging Spectral Array (LEISA) on the New Horizons to Pluto.

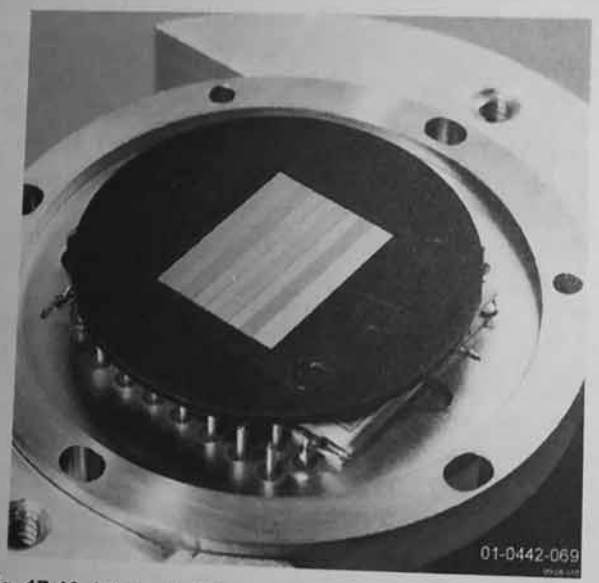


Fig. 17-10. Integrated Spectral Filter Assembly also Known as a Butcher Block Filter Assembly for the Multispectral THEMIS Imager On-board Mars Odyssey 2001. (Photo Courtesy of Raytheon)

Dispersive spectrometers use a grating or prism to separate light at different wavelengths into distinct angles. The spectrum of a single ground sample is dispersed and then focused onto different detectors along one dimension of a 2-D detector array. The spectrum is typically sampled at Nyquist rate so that number of detectors along the spectral axis is twice the number of points in reported spectrum. Advantages of dispersive spectrometers relative to Michelson interferometer approaches discussed in the next paragraph include superior performance in the SWIR and in that part of the MWIR where noise is photon limited and no moving parts in the spectrometer except possibly for a mechanical cooler. Disadvantages include sensitivity to temperature changes that can affect the focal length and therefore spectral calibration and radiometric accuracy and a tendency toward relatively large complex instruments especially for missions requiring high spectral resolution and wide field of view Earth coverage. Well known examples of space-based dispersive spectrometers include Galileo-NIMS, AIRS, and Hyperion.

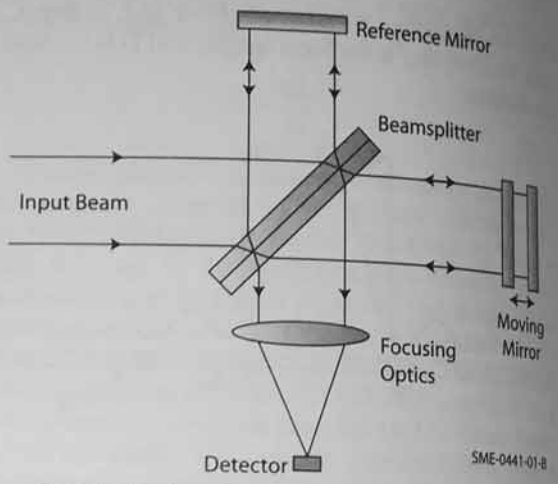


Fig. 17-11. Michelson Interferometer, the Most Widely Used Form for a Temporally Modulated Fourier Transform Spectrometer.

Interferometers achieve spectral separation using constructive and destructive interference of light from the same part of a scene. Two broad classes of interferometers flown in space include Fourier transform spectrometers (FTS) in which the optical path difference between two interfering beams varies in time or space and multibeam interferometers like the Fabry-Perot interferometer in which multiple beams from the same scene element interfere to select a series of narrowband transmission ranges from which one is selected using another etalon or blocking filter. Nearly all optical filters used in MSI systems are based on a special type of Fabry-Perot interferometer (FPI) called an etalon. The best known interferometer used in space is the temporally modulated FTS (TFTS) implemented frequently with a design approach similar to what is shown in Fig. 17-11. In principle, a temporally modulated FTS can measure all optical frequencies simultaneously. In reality, practical limits such as detector sensitivity cutoffs, optical coatings and electronics frequency ranges restrict spectral range. The moving element in the TFTS and the resulting variation in optical path difference modulates intensity of optical frequencies across a broad spectral range in front of a single detector for each pixel that converts received optical signal into an elec-

trical signal. The resulting plot of detected irradiance versus time (corresponding to changes in optical path difference) is called an interferogram. The interferogram is converted to an optical spectrum using a Fourier transform. General advantages of interferometer approaches include simplest and most compact instrument approach for high spectral resolution and coverage applications (especially for the FPI) and superior performance for detector noise limited TFTS instruments, because detector noise is multiplexed across the spectrum. Disadvantages for the TFTS include reliability concerns because of the moving elements in the interferometer and the so-called multiplex disadvantage for photon noise limited instruments in which photon noise from the broad spectral band observed by the TFTS in every interferogram sample is multiplexed into each spectral channel by the Fourier transform. FPI disadvantages include relatively low optical throughput, especially at high spectral resolution, which leads to larger collection optics than other spectrometers and limited spectral range for each etalon which can lead to requirements for many etalons in a system, to cover the broad spectral range required for some missions. Space-based interferometer examples include TFTS instruments like IASI, TE,S MIPAS, and FPI-based instruments like HRDI.

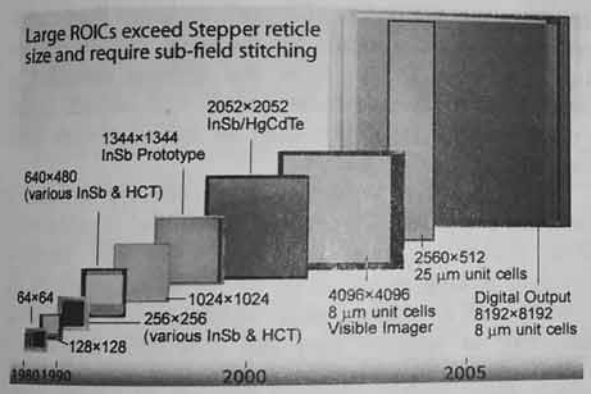


Fig. 17-12. Improvement in Detector Array Size and Readout Rate from 1980 to 2005.

Detectors. We make a distinction between discrete detectors consisting of one or a few light-sensitive elements with discrete electronics and detector arrays, which consist of a large number (hundreds to tens of millions) of tightly packed light-sensitive elements combined with a readout integrated circuit (ROIC) to recover the measured signal from each detector element with minimum complexity. As shown in Fig. 17-12, since the 1980s, detector technology has taken advantage of improvements in integrated circuit design and fabrication capabilities and advanced rapidly from relatively small format ~64 × 64 detector array designs to much larger format (~8,192 × 8,192) digital output approaches.

We must select the materials used for detector elements depending on the spectral range of the instrument being designed. The ability of photon detector elements to absorb photons relates directly to the energy of the incident photons (and consequently to the wavelength of the radiation as well) and the effective band gap of the material. Detector material generates random noise that reduces sensitivity of the instrument. This noise increases with operating temperature of the detector. Therefore, frequently, we must lower the temperature of the detector elements so that the self-generated noise does not degrade the signal-to-noise ratio of the instrument. This requirement becomes more stringent as the wavelength of the radiation being detected increases. With few exceptions, detectors have to be cooled for wavelengths in the short wave infrared (SWIR) and longer. All matter, at temperatures greater than absolute zero, generates thermal photons. Thus, frequently, for infrared imagers, the optics and interior structures of the instrument have to be cooled to reduce interference from light radiated by the instrument itself.

For the spectral range between 400 nm and 1,100 nm, silicon detectors and imagers are used most frequently. Silicon is attractive because it is possible to combine the detector elements and read-out electronics in a single monolithic chip. Furthermore, silicon substrates are available in larger size than other materials, which enables larger format devices. We can produce 1-dimensional line arrays or 2-D area arrays with a very large number of elements (e.g., 10,000 × 10,000 and greater) with silicon. Incident photons are converted to an electrical output signal in the array. For charge-coupled device (CCD) arrays with read-out electronics, the process begins when incident photons are converted by each detector element into electrons according to a conversion efficiency dictated by the characteristic spectral response. The electrons are then collected for the entire integration time for each detector element. The read-out of the charge generated—up to millions of electrons per detector element—is performed via an analog shift register across the arrays into a serial output port. 2-D CCDs may have more image smear relative to 1-D arrays during frame transfer. Also, in 2-D monolithic array designs, detector fill factor is generally lower than in 1-D arrays or in so-called hybrid arrays in which a detector array is bonded to a separate ROIC. Figure 17-13 shows a typical spectral response function for a silicon imager.

When radiometric performance of the optical instrument is paramount, we use time delay integration (TDI) methods. TDI describes an imaging principle that uses the image motion along the rows of a 2-D detector array to extend the integration time. Integration time is extended by electronically shifting the integrating pixel cell synchronously to the movement along the row. The signal-to-noise ratio of this concept is improved by the square root of the number of TDI stages. The primary advantage of TDI imager systems compared to line imagers is the improved signal-to-noise ratio. The disadvantage is the increased requirement for spacecraft attitude and orbit stability (due to the required synchronization of the shifting pixel). Some TDI sys-

Table 17-5◀, Fig. 17-12, Eq. 17-10◀

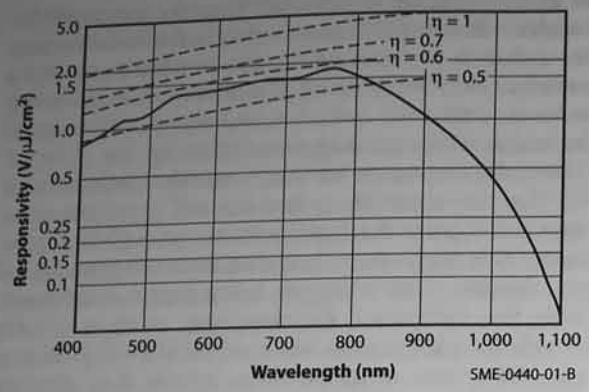


Fig. 17-13. Spectral Response Function of a Silicon Line Imager. The spectral response is shown in terms of the output voltage resulting from illumination by energy density vs. the wavelength from 400 to 1,100 nm. The imager reaches a quantum efficiency (generated electrons per incident photon), denoted by η, above 50% between 400–800 nm. Outside this wavelength region the quantum efficiency drops to small values making the imager less suitable above 900 nm.

tems perform the integration in software by adding together measurements from the different detectors that view the same part of the scene.

We classify and select infrared detectors according to their spectral band of operation and a figure of merit called specific detectivity for photon detectors. The operating temperature of the detector dictates the cooling requirements for the sensor focal plane. Infrared sensors often have non negligible time constants for response with respect to integration time. The size of infrared detector arrays is ultimately limited by the size of substrate material on which the detectors can be grown. Recent work at Raytheon has produced infrared arrays with format sizes up to 4,096 × 4,096. We detect infrared wavelengths with thermal detectors or photon detectors. Thermal detectors exploit the fact that absorbed heat raises the temperature of the detector, which changes its electrical characteristics. The advantage of thermal detectors is relatively uniform response with respect to wavelength. Thermal detectors can also be operated at ambient temperatures, although they have lower sensitivity and much slower response times, which can limit their applicability in whiskbroom scanners, for example. Photon detectors use absorbed photons to generate charge carriers. These systems offer the advantages of higher sensitivity and shorter time response, but they must be operated at low temperatures.

Infrared detectors are often rated by the specific detectivity, D^*, given by

$$D^* = \frac{\sqrt{A\Delta f}}{NEP} \qquad (17\text{-}11)$$

where A is the detector area, Δf is the noise equivalent bandwidth, and NEP is the noise equivalent power of the detector. The factor D^* is strongly wavelength dependent showing its peak value at the cut-off frequency. See the book website for a list of IR FPA manufacturers.

Get More

The selection of a detector or detector array is driven by several factors. The primary design issues center on maximizing detectivity in the spectral band of interest while operating at the highest possible temperature and a sufficiently small time constant. In addition, we must consider the geometry of the detector and the array as well as associated calibration issues.

17.1.2 Radio Frequency Observational Payloads

Radio frequency observation payloads can scan the Earth in ways that are analogous to electro-optical payloads. Likewise, systems for collecting or transmitting optical and RF signals operate exactly the same in theory with collecting, reflecting and focusing light—only the physical hardware is different. Thus, the primary mirror in a visible and infrared system and the large reflector of a dish antenna in passive microwave radiometry and radar, for example, are equivalent.

However, because of the much longer wavelengths at which RF systems operate compared with optical systems, the equations shown in the previous section for diffraction limited angular resolution and corresponding horizontal resolution on the surface lead to RF system apertures that are orders of magnitude larger than corresponding optical systems for the same surface resolution. Likewise, for the same real aperture diameter, a passive microwave or radar system will provide much coarser surface resolution than an optical system with the same aperture diameter.

Nearly all RF systems detect collected irradiance coherent using the superheterodyne method. Superheterodyne detection, invented in the early 20th century, involves mixing an incoming RF signal collected by an antenna with a coherent, phase stable and low noise local oscillator to produce sum and difference frequencies called intermediate frequencies (IFs) that are much lower in frequency than the incoming signal and therefore easier to process. The IF signals contain the same amplitude and phase information as the RF signal. The amplitude is scaled by a constant, measureable amount and the phase difference is a constant that can also be measured.

This ability of RF systems to measure both amplitude and phase of the received signal with a relatively high degree of accuracy enables a technique called synthetic aperture radar (SAR) that can provide radar imagery with spatial resolution in one dimension that is comparable to spatial resolutions obtained with high performance optical systems.

To understand the importance of SAR, consider first how real aperture radar works. Radar illuminates an area with pulses of microwave radiation. These pulses are reflected from the illuminated scene and collected and processed by a receiver. Radar determines the distance of the reflecting object by precisely measuring the time difference between transmission time of the outgoing

pulse and time the return pulse is received. This distance or range is the time difference times the speed of light divided by two. Spatial resolution along the radar line of sight is not range or directly wavelength dependent, as it is for horizontal resolution perpendicular to the line of sight. Instead, it is determined by the effective pulse-width, δt multiplied by the speed of light, c, divided by two, which is equivalent to the reciprocal of the effective pulse bandwidth, B times the speed of light. That is

$$\text{Range Resolution} = \frac{c\delta t}{2} = \frac{c}{B} \tag{17-12}$$

Pulse bandwidth and therefore data rate inside the instrument grow, as the range resolution improves. Many radars, including SARs, transmit a pulse using a linear frequency modulated (FM) "chirp" in which the frequency of the transmitted radar pulse varies linearly over a selected frequency range. This variation in frequency determines the RF bandwidth of the system. Chirp length and slope are constrained by range resolution for a system and depend on RF pulse power, pulse repetition frequency (PRF) and analog-to-digital converter (ADC) sampling rate and dynamic range. For example, the German TerraSAR-X system launched in 2007 and described by Werninghaus and Buckreuss, [2010], operates with a transmit bandwidth ranging from 5 to 300 MHz (with 300 MHz being used in an experimental mode). Consequently, TerraSAR-X is capable of operating with range resolutions extending from 1 m to 60 m.

In the direction perpendicular to the radar beam, the so-called cross range or azimuthal direction, spatial resolution of real aperture radar is limited by its physical size, just like a passive optical system. In this case, cross range spatial resolution is given by $1.22\,\lambda R_s/D$. From space, good range resolution can be achieved, but poor cross range horizontal spatial resolution, typically on the order of a kilometer even for practical but very large systems, cannot be avoided by real aperture radar at frequencies low enough to penetrate all weather conditions.

Interestingly enough, this significant disadvantage for radar, of large beam width or angular field of view, can be turned into an advantage, if the system has the ability to measure precisely, record and process the phase and Doppler shift of the radar return signal. A radar pointing toward the Earth's surface with a wide beam illuminates a scene location for an extended period of time (possibly up to seconds in duration) as the satellite passes overhead. If, during this time, the radar measures the phase and Doppler shift for each return, an aperture can be synthesized that is equivalent to the distance the real aperture antenna moves while the scene location is in the beam. This synthesized aperture can be orders of magnitude larger than the real radar antenna, which makes it possible to produce radar imagery with a meter of resolution. Longer wavelengths and smaller aperture diameters result in wider beam divergences, which can illuminate a scene location longer and therefore provide a longer baseline for creating a synthetic aperture.

Figure 17-14 illustrates a typical space-based SAR with co-located transmitter and receiver. This transmitter-receiver configuration is known as a monostatic configuration. As the satellite passes by a target location of interest, it maps out a strip along the Earth's surface. The dots in the figure represent a series of positions at which the SAR transmitter emitted a pulse. Each pulse travels to the Earth's surface and illuminates a scene location. A small fraction of that outgoing pulse returns to the receiver and is collected by the same antenna. The SAR records the phase histories of the responses at each scene location as the transmitter beam moves across the scene and then phase shifts and adds them together with appropriate weighting to minimize sidelobes in the synthesized beam to determine the range and scene brightness of one spatial resolution element at a time until the complete scene is described. SAR achieves a very high signal processing gain because this summation of the range correlated radar responses is done coherently with both amplitude and phase. In many cases, thousands of pulses are summed for each spatial resolution element, resulting in a signal processing gain of up to 10,000 or more for the TerraSAR-X compared with a single pulse.

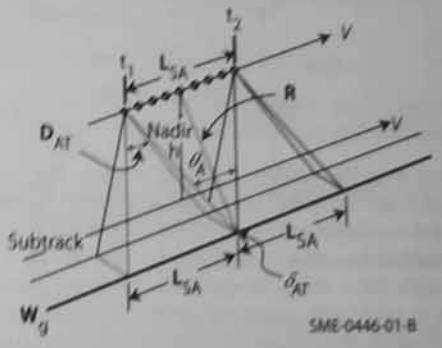


Fig. 17-14. Basic Principles of Synthetic Aperture Radar.

SAR performance is constrained by ambiguity relationships associated with the 2-D scene sampling that limit PRF and real aperture collector size. The basic idea is to collect data in the along track and in the range direction that can be processed into a unique image. In Fig. 17-14, L_{SA} is synthetic aperture length, and is the distance the satellite moves to illuminate the target. V is the satellite velocity, R is the target range, θ_A and D_{AT} are respectively the aperture azimuth beamwidth and the along-track antenna length of the real SAR antenna, δ_{AT} is the Earth target resolution element along-track dimension, and W_g is the satellite ground swath. The dwell or integration time, t_I, is given by:

$$t_I = \frac{L_{SA}}{V} = \frac{\lambda R}{V D_{AT}} \tag{17-13}$$

Table 17-5◀, Fig. 17-14, Eq. 17-13

The aperture azimuth beamwidth is given by:

$$\theta_A = \frac{\lambda}{D_{AT}} \tag{17-14}$$

Additionally, the following relations apply:

$$\mathbf{L_{SA}} = \theta_A \mathbf{R} = \frac{\lambda \mathbf{R}}{D_{AT}} \tag{17-15}$$

$$\delta_{AT} = \frac{\lambda}{2\mathbf{L_{SA}}}\mathbf{R} = \frac{\lambda \mathbf{R}}{2\lambda \mathbf{R}/\mathbf{D_{AT}}} = \frac{\mathbf{D_{AT}}}{2} \tag{17-16}$$

The achievable azimuthal (i.e., along track) resolution of a SAR is approximately one half the along track radar antenna size and this resolution is independent of radar frequency and range to first order and improves with smaller real antenna aperture. Based on that one equation alone, it appears that along track resolution could be improved simply by making physical size of the antenna length very small. However, achieving a given along track resolution also requires that the radar send out a pulse every time the radar moves half of the along track antenna size. That is

$$2 \times \frac{V}{D_{AT}} < \mathrm{PRF} \tag{17-17}$$

or

$$\mathrm{PRF} > \frac{V}{R_A} \tag{17-18}$$

where V is the satellite velocity, D_{AT} is the antenna length, and R_A is the azimuthal resolution. Better azimuthal resolution requires a higher transmitter PRF. This means that less time is available between pulses to collect data. Only one pulse can be in the target space at a time to avoid ambiguities in range that would prevent construction of a unique image solution. This condition sets another limit on PRF which is a function of swath width ($S = R_f - R_n$) as shown in Fig. 17-15 for the SEASAT SAR imaging geometry, and the pulse duration, T, resulting in

$$\mathrm{PRF} < \left(2T + \frac{2S}{c}\right)^{-1} \tag{17-19}$$

Not surprisingly, more spacing between pulses is required as the swath width or nadir angle increase. Thus, ultimately, a trade must be made in the design or specific system configuration between increased spatial coverage and improved along track spatial resolution, δ_{AT}. All of this reduces to a relationship between swath width, S, and along track spatial resolution for the single look case [Raney, 1998].

$$2 \times \frac{S}{\delta_{AT}} < \frac{\mathrm{c}}{V} \tag{17-20}$$

Table 17-5◀, Fig. 17-15, Eq. 17-23

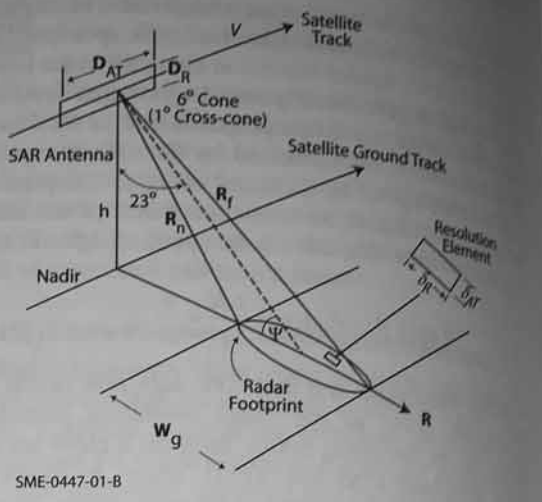


Fig. 17-15. SEASAT SAR Imaging Geometry.

In Fig. 17-15, the velocity, V is 7,450 m/s, the altitude, h is 800 km, and the ground swath, W_g is 100 km. $\mathbf{R}$ is the cross-track or range direction, Ψ is the grazing angle, and the satellite ground track is in the along track (AT) or azimuth direction. The following relations apply:

$$\delta_{AT} \geq \frac{\mathbf{D_{AT}}}{2} \tag{17-21}$$

$$\delta_R \geq \frac{\mathrm{c}}{2\beta} \tag{17-22}$$

$$\mathrm{W_g} = \frac{\mathbf{R_f} - \mathbf{R_n}}{\cos\Psi} = \frac{\lambda \mathbf{R}}{\mathbf{D_R} \sin\Psi} \tag{17-23}$$

Where c is the speed of light, and β is the transmitted pulse bandwidth.

17.2 Observation Payload Sizing

The sizing of an observational payload is an iterative process that involves trading and optimizing different payload approaches, frequently across a large trade space of candidate possibilities. In the early stages of this process, it is important to be able to examine different options without designing each one in detail. Initial assessment of design alternatives requires estimates of size, mass, power, data rate and constraints on spacecraft pointing control and stability among other interface characteristics to narrow down the possibilities for more thorough consideration. These assessments flow from detailed description and understanding of the mission to the operations concept and initial specifications for the observation payload that determine payload size, mass, and power, as illustrated in Table 17-6.

High-resolution wide field-of-view optical instruments, especially those with hyperspectral capability across a broad spectral region, typically generate data

Table 17-6. The Observation Payload Sizing Process. Note that Steps 2 and 3 are highly iterative and may well drive both the mission and payload design.

Step	Where Discussed
1. Define Mission Requirements and Objectives	Chaps. 3, 4, 6
2. Determine Mission Geometry and Orbit Parameters	Chaps. 8, 10
3. Define Payload Requirements (ITERATE WITH STEP 2)	Sec. 17.2.2
4. Define Payload Operations Concept	Sec. 17.2.1
5. Determine Spatial Sampling	Sec. 17.2.3
6. Determine On-board Signal Processing	Sec. 17.2.4
7. Determine Radiometric Sensitivity Performance	Sec. 17.2.5
8. Estimate Payload Size, Mass, and Power	Sec. 17.2.6
9. Determine Payload Figures of Merit	Sec. 17.2.7
10. Document and Iterate	

rates on the order of several hundred Mbps and above. To send this data stream to a ground station in real time, the system may need several parallel channels with capacities up to 300 Mbps each. If data collection is short compared to the available downlink time, the data can be stored on-board and transmitted to the ground over time to reduce the required downlink data rate.

Trends for future systems indicate a growing tendency toward more on-board processing to decrease downlink data rate requirements and to enable broadcast of information derived from observational payload data directly to users. Continuing improvements in powerful processing hardware and associated algorithms enable sophisticated on-board processing capability comparable to ground-based processing. Given the significant impact that high speed processing systems can have on payload size, mass, and power, selection of an on-board data processing approach is becoming an ever more important element in the observation payload design trade space. To maintain the flexibility required for continuous improvements in system performance, software uploads to the hardware must be possible to update and modify operational algorithms from the user community. The need to upload large quantities of software relatively quickly must be considered, when specifying uplink data rate requirements for an observation system.

This section discusses each of the steps in the observation payload sizing process. In addition, figures of merit for comparing different design approaches are considered along with the description of an observation payload that fulfills requirements for the FireSat II mission.

17.2.1 Payload Operations Concepts

The operations concept for using the observation payload must be considered early in the preliminary design process. The end-to-end mission need must be understood along with the physics, associated engineering and phenomenology of the collection and interpretation of data. The payload operations approach as driven by end data user requirements needs to be a realistic and cost-effective way to achieve mission objectives.

Specific collection operations involving tasking and scheduling of the sensor, processing of the mission data, and distribution of the data can dramatically increase complexity of the systems engineering challenge and decrease the final accuracy of the system, if not selected carefully and within context of quantitative figures of merit. Not only can physical effects such as atmospheric correction, calibration, and rectification degrade system performance, but technical effects such as quantization and data compression errors can decrease the resolution of the system from the perspective of the end user. For additional information about technical aspects of the end-to-end throughput or image chain problem for remote sensing and imagery collected from space, see Schott [2007].

The operations concept for a system like FireSat II must consider all aspects of the operational mission, including different mission phases and alternative operating modes (See Chap. 29). In preliminary mission planning, we should be aware of the expected sequence of events during each mission phase. For normal FireSat II mission operations, a sample timeline includes the following steps:

1. Fire starts somewhere on the Earth
2. Sensor field-of-view passes over the fire and collects data, as part of a global fire reconnaissance mission
3. Measurements of the fire are transmitted through the sensor data stream
4. Data is analyzed at the mission ground station or processed on board
5. Fire detection algorithm determines the probability of a fire at the analyzed location—this could be a multistep process with a rapid, but crude fire detection algorithm that triggers a more accurate algorithm or set of measurements
6. If fire is detected, system generates appropriate messages to Customer that indicates presence of fire at specific location
7. Customer organization reports fire and notifies responsible authorities and appropriate researchers
8. System continues to monitor the fire and region near the fire, in conjunction with its global reconnaissance mission to track progress of the detected fire and monitor nearby areas for new fires

To ensure mission success, an operations concept for each mission phase and each spacecraft operating mode—including contingency and failure modes—must be created within operating environment limitations (See Chap. 29).

Table 17-6, Fig. 17-15◀, Eq. 17-23◀

Table 17-7. Characteristics of Typical Payloads.

Purpose	Instrument Name	Size L × W × D (m)	Mass (kg)	Avg. Pwr. at 28 V (W)	Data Rate (Mbps)	Aperture (m)	Pointing Accuracy (deg)
Solar Physics	Lyman—Alpha Coronograph	2.8 × 0.88 × 0.73	250	87	13.5	—	0.003
	X-ray Telescope Spectrom.	2.7 × 1 dia.	465	30	0.4	—	0.003
	Solar Optical Telescope	7.3 × 3.8 dia.	6,600	2,000	50+	1.25	—
	Solar Magnetic Velocity Field	2 × 0.4 × 0.4	183	322	2+	—	0.003
	100 m Pinhole Camera	1 × 1 × 2	1,000	500	0.5	—	—
	Extreme UV Telescope	2.75 × 0.86 × 0.254	128	164	1.28	—	—
	Solar Gamma Ray Spectrom.	1 × 1 × 13	2,000	500	0.1	140 cm²	0.003
Space Plasma Physics	Ion Mass Spectrometer	0.5 × 0.5 × 0.4	80	334	0.01	—	1
	Beam Plasma	0.6 × 0.7 × 0.7+ two 0.7 dia. ant.	17	38	0.016	—	5
	Plasma Diagnostics	—	2,000	250	50	—	—
	Doppler Imaging Interferom.	$(0.25)^3$	100	620	0.2	—	—
	Proton (Ion) Accelerators	6.7 × 3.4 × 3.10	500	1,500	0.256 (4.2 TV)	—	—
High Energy Astrophysics	Gamma Ray Burst	2 × 4 dia.	1,000	120	0.01	3	1
	Cosmic Ray Transition	3.7 × 2.7 dia.	1,500	230	0.1	2.7	—
	X-ray Spectrom./Polarimeter	1.6 × 1.6 × 3	2,000	300	0.03	—	—
	Short X-ray	1 × 1 × 3	1,000	300	0.025	1 × 3	0.1
	Hi Energy Gamma Ray Tele.	3 dia. × 4	10,000	100	0.003	3	0.1
Resources	Gravity Gradiometer	0.23 m sphere	10	1			0.1
	Synthetic Aperture Radar	2.8 × 3.7 × 1.4	808	3,000	120	8 × 2.8	1–2
	Multi-Spectral Mid-IR	1.4 × 1.5 × 0.9	270	170	8	0.18	2.5
	Enhanced Thematic Mapper	2 × 0.7 × 1.1	400	600	75	0.406	0.1
Materials Processing	Materials Experiment Assem.	1 × 1 × 2	900	500	0	—	0.005
	Solidification Experiment		1,100	3,000	0.02	—	—
Environmental	Limb Scanning Radiometer	4.8 × 1.9 dia.	~800	125	0.52	—	—
	Microwave Radiometer	4 × 4 × 4	325	470	0.20	4	±0.1
	Dual Frequency Scatterom.	4.6 × 1.5 × 0.3	150	200	0.01	4.6 × 0.3	1
	Ocean SAR	20 × 2 × 0.2	250	300	120	20 × 2	0.1
	Solar Spectrum	0.4 × 0.3 × 0.6	16	60	Low	—	±3
	Doppler Imager	1.25 × 0.6 × 0.8	191	165	20	—	±3
	Imaging Spectoral	1.0 × 1.6 × 1.0	250	225	11	0.18	0.04
Planetary Science	Multi-Spectral Imager	0.54 × 0.37 × 0.29	13	12	2	0.12	—

17.2.2 Required Payload Capability

Often, mission requirements can be met in a number of different ways. It's not always obvious how to go through the various candidate approaches and select the best one for any given mission. The general approach outlined here is a simple, repeatable method for selecting a payload to satisfy an observation mission. After we select a physical phenomenology (for example, measuring thermal infrared radiance to detect the high brightness temperature that is characteristic of forest fires), three system characteristics need to be determined. The first is the spatial sampling approach needed to satisfy the mission information need, followed by the radiometric sensitivity levels needed to achieve fire detection and false alarm requirements. The third and frequently most important system characteristic from a mission implementation and cost perspective are the implications of spatial sampling and radiometric sensitivity characteristic on payload size, mass, and power.

Table 17-7, Fig. 17-15◀, Eq. 17-23◀

Table 17-7 provides a small sampling of remote sensing payloads and corresponding spacecraft missions. Categorizing remote sensing missions is complicated by the fact that sensors usually have multiple uses, and they can be categorized according to any number of different aspects, such as measurement technique (active or passive), event measured (such as fire or deforestation), and measurement resolution (spatial, spectral, radiometric, temporal). Frequently, these multiple uses lead to conflicting requirements that further complicate payload design and development. Remote sensing payload examples include imaging spectroradiometers that passively measure the intensity of scene spectral radiance across a broad range in wavelength and polarimeters that measure polarization state of a transmitted, scattered or reflected wave. Scatterometers are active RF instruments (radars) that measure surface roughness of the ocean by measuring radar return signals that depend on wind speed and direction.

Measurement procedures used in observation payloads and associated processing algorithms are customized to deliver information that can be retrieved from the measurements to understand the subject. We describe the fundamental measurement content provided by passive Earth observation instruments in terms of spatial and spectral resolution and radiometric sensitivity. We continue by discussing the basic types of detectors and collection techniques that are available in the design of a remote sensing instrument.

We frequently use imaging sensors, based on electro-optical devices, to measure and analyze spatial features. Visible wavelength imaging is a common approach for providing data needed for topographic mapping and photogrammetry. Altimeters are also especially valuable in providing fundamental spatial measurements. Kramer [2001] lists examples of these types of sensors.

As noted above, spatial resolution depends on the range from the sensor to the target, the aperture and focal length of the collection optics, detector element size, signal processing approach and the wavelength of the incident energy. We can characterize the spatial resolution by the smallest angular distance the sensor can resolve or directly as the ground range in units of length for the size of the smallest object that can be discriminated. Effective spatial resolution is a function of these many different design parameters in a remote sensing system and more, including the scene itself. Optical system and focal plane array characteristics determine theoretical limits to spatial resolution for any system geometry. In addition, these same characteristics define the ground sample distance (GSD), the distance on the ground (usually referenced to the sub-spacecraft point) at which the sensor samples the scene. Frequently, the effective spatial resolution is different from the GSD. Normally, the effective spatial resolution is about twice the GSD. However, for low contrast targets or target detections with low signal-to-noise ratio, effective spatial resolution can be coarser than twice the GSD. The smallest physical features that can be discriminated for specific target characteristics using any candidate sensor measurement approach determine the limits of spatial resolution in a remote sensing system. Information from spectral content of scenes can be used within context of plausible assumptions to derive spatial information that has finer resolution than the GSD.

Spectral content is provided by spectrometers that collect data that can be used to identify different components in a scene (e.g., rocks, trees, and water) and sometimes chemical composition of objects being measured. The spectral radiance received by a sensor is a composite of the spectral radiances emitted by all objects in the instantaneous field-of-view. The radiance reaching a possible multispectral FireSat II imager, for example, will include spectra from soil, vegetation, and cities in addition to any fire that might be there. The combined spectrum of all these scene elements may be very different from the spectrum of a forest fire by itself. Combining information from different spectral bands has been used often and successfully to isolate key scene elements from each other and improve the usefulness of a sensor.

Space-based spectrometers are frequently grouped into one of three categories: multispectral, hyperspectral and ultraspectral. Multispectral systems typically have tens of discrete spectral bands with relatively low ($\lambda/d\lambda$~10) spectral resolution. Hyperspectral and ultraspectral systems employ hundreds and thousands of spectral bands, respectively, across a contiguous spectral range with spectral resolutions that are typically moderate ($\lambda/d\lambda$~100) for hyperspectral systems and high ($\lambda/d\lambda$~1,000) for ultraspectral systems. Following Slater [1980], we consider a multispectral image of a scene with concrete, asphalt, soil, and grass. Figure 17-16 shows typical spectral reflectance for these four scene elements. If a candidate imager has only three bits of radiometric sensitivity (low, medium, and high), for example, then no single image, whether panchromatic or filtered to a particular band region could distinguish these four scene features. However, even a simple, but well calibrated two-band multispectral system can uniquely resolve these four scene elements using bands in the 600–700 nm and 700–900 nm wavelengths. Returns for each of these materials are shown in Table 17-8.

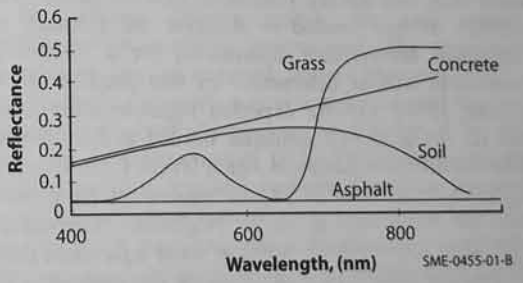


Fig. 17-16. Typical Spectral Reflectance Curves for Grass, Concrete, Soil, and Asphalt. A single-band low-resolution sensor can not distinguish all 4 types, but a 2-band, low-resolution sensor can, as shown in Table 17-8 [Slater, 1980].

Table 17-8. Spectral Resolution for Sample Two-Band Sensor. Combining low-resolution sensor returns for the two-band sensor can uniquely identify the four materials from Fig. 17-16.

Material	Sensor Reading 600–700 nm	Sensor Reading 700–900 nm
Concrete	Medium	High
Asphalt	Low	Low
Soil	Medium	Low
Grass	Low	High

Multispectral data with more spectral bands and especially hyperspectral and ultraspectral data contain much more physically meaningful detail, which can be used to recognize and quantify characteristics of com-

Table 17-8, Fig. 17-16, Eq. 17-23◀

plex scenes. However, the higher spectral resolution of hyperspectral and ultraspectral systems, especially, reduces measured signal-to-noise ratio (SNR)—all other system characteristics being equal—which can impact system design in many ways ranging from reducing spatial resolution to recover SNR to increasing aperture size to improve light gathering capability.

We can also use the measured spectral radiance level to extract useful information. For example, if we have an imager with four or more bits of sensitivity operating in the 700–900 nm wavelength, then a single spectral radiance measurement could be used to distinguish among all four candidate scene types, if we assume a pixel contains only one scene type and we understand the details of the solar illumination and viewing geometry precisely. The ability to measure spectral radiance precisely goes beyond detecting either the presence or absence of energy at a given wavelength. For example, if a FireSat II sensor were designed to detect fires based on measuring brightness temperatures of Earth's thermal emission, then a high enough brightness temperature would indicate presence of fire. But, the radiometric resolution of the instrument needs to be good enough to allow the system to discriminate between a forest fire and a campfire.

Thus, merely detecting the presence of thermal energy in a spectral band characteristic of burning biomass does not satisfy FireSat II mission needs. The number of bits needed to describe the intensity or brightness temperature information for a FireSat II instrument will be determined by two practical limits [Slater, 1980]: (1) the expected signal-to-noise ratio, and (2) the level of confidence needed to discriminate between different threshold signal levels. Consequently, signal-to-noise ratio (SNR) or equivalent parameters such as difference in scene brightness temperature equivalent to the system noise or noise equivalent delta temperature (NEDT) are performance discriminators for candidate observation payload designs.

17.2.3 Spatial Sampling

The spatial sampling of a scene by an optical instrument is determined by the scanning method used as described in Table 17-2, and by spacing of the detector elements in the focal plane, which can be constrained by optical system design characteristics such as operating wavelength, aperture diameter, and focal length. If the spatial sampling method is not selected with care, obvious image artifacts and not so obvious radiometric errors may be introduced into the data. Figure 17-17 shows the effect of sampling by a detector array observing three scenes with different spatial frequency structures. The response of the detectors to each scene pattern is shown as a gray level intensity in the three diagrams. Spatial frequency is defined as the inverse of the width of a black-and-white line pair, and is expressed in line pairs per meter or simply in units of inverse meters. The same detector array, shown to the right of each arrow, samples

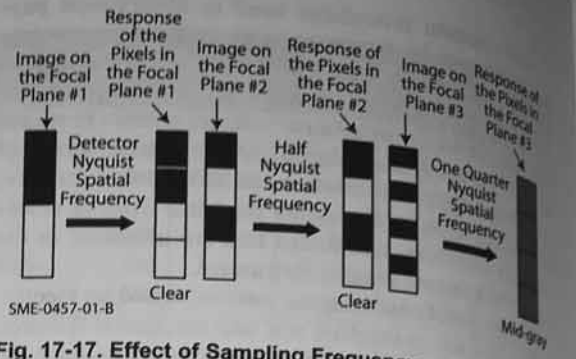


Fig. 17-17. Effect of Sampling Frequency on Image Quality. The three diagrams show illumination patterns having different spatial frequencies present in the scene (left of the arrows) sampled at the same sampling frequency, that is, the same detector array (right of the arrows). The response of the pixels is shown as intensity (gray level) in the three diagrams. From left to right: Sampling an illumination pattern with Nyquist frequency (the highest possible frequency that can be reconstructed without error). Sampling an illumination pattern with half the Nyquist frequency. Sampling an illumination pattern with one quarter the Nyquist frequency.

the scene in all three cases. That is, in these simple examples, the different scene structures are sampled at the same spatial sampling frequency, defined as the inverse of the distance between centers of the sampling detectors. The maximum spatial frequency which can be reconstructed without error after the sampling/restoration process is half of the sampling frequency or twice the sample distance in the focal plane. The left diagram in Fig. 17-17 illustrates this so-called detector Nyquist sampling case, where the scene feature as seen in the focal plane is twice the size of the detector. In the center, sampling of an illumination pattern occurs at half the Nyquist frequency of the scene structure, showing that reconstruction of the original illumination pattern is feasible without error even at half the Nyquist rate. However, if the sampling pattern is shifted by half a pixel with respect to the illumination pattern, then reconstruction fails. And on the right, sampling an illumination pattern with one quarter the Nyquist frequency of the scene structure results in a pixel response of mid-gray since every detector pixel is illuminated over half its area in the example. All samples in the third case show the same constant value which corresponds to a spatial frequency of zero, which was not present in the original scene. This creation of signal power at new frequencies is known as aliasing. Such frequency components cannot be removed by additional processing of the reconstructed image. To avoid these errors completely in an actual system, the scene must be sampled at twice the optical cutoff frequency described in Eq. (17-5). This Nyquist sampling of the optical cutoff frequency defines the lowest rate at which this band limited data must be sampled to avoid errors in scene reconstruction. Many systems do not sample the scene at the Nyquist rate of the cutoff frequency, because normally, very little power exists in the higher frequencies being aliased into the measurements and because radiometric sensitivity requirements typically determine aperture

Table 17-8◀, Fig. 17-17, Eq. 17-23◀

diameter and spatial sampling needs. Furthermore, it is not usually cost effective to sample an optical system at the Nyquist rate of the optical cutoff frequency because this approach results in a much larger number of detectors than would otherwise be required by alternative approaches, such as sampling at twice the Rayleigh criterion or at the spatial resolution required by the mission, which can significantly increase system cost relative to other approaches. The result is that we usually have a certain amount of unavoidable aliasing which degrades image quality and radiometric accuracy of such systems.

The Modulation Transfer Function (MTF) is the ratio of the measured contrast variation in the image to the actual contrast variation in the ground scene as a function of spatial frequency. MTF is zero at the optical cutoff frequency described in Eq. (17-5). Figure 17-18 shows the theoretical MTF of an optical system with and without a central obscuration such as an on-axis secondary mirror and support structure. The theoretical MTF can be approximated by a line starting at 1 when the spatial frequency is 0 and falling to 0 at the cut-off frequency, $F_{cut\text{-}off} = D/\lambda f$, where D is the aperture diameter, λ is the wavelength, and f is the focal length of the optical system. The MTF of a telescope is the autocorrelation function of the entrance pupil function. In optical terms, the MTF is the absolute value of the complex Optical Transfer Function (OTF) which describes how the complex amplitudes of the optical wave front are transferred by an optical system at different spatial frequencies.

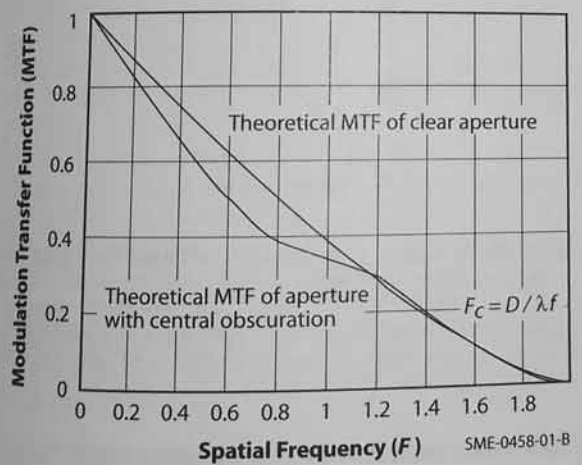


Fig. 17-18. Modulation Transfer Function of Circular Diffraction-Limited Optical System vs. Spatial Frequency. MTF curves for a clear circular optical system is compared with one having a central obscuration (as found in on-axis reflective telescopes). The MTF can be thought of as a function dependent on the spatial frequency, F, which describes the modulation (contrast) function through the optical system (analogous to the frequency dependent gain of an electrical transmission block). The MTF starts at 1 for spatial frequencies near 0 and drops to 0 at the cutoff frequency, $F_c = D/\lambda f$.

The MTF describes the ability of an imaging system to transfer contrast from a scene to an image in the focal plane as function of spatial frequency. It is a measure of the quality of an optical system. The point-spread function illustrated in Fig. 17-8 describes exactly the same properties by showing the two-dimensional intensity distribution in the focus of the optical system. The two functions are related by Fourier transform. System level MTF is also affected by spatial sampling in the focal plane by detector arrays and by signal processing of data collected with 2-D imaging detector arrays.

17.2.4 Signal Processing

Analog signal processing for visible wavelength detector arrays based on CCDs or Si:PIN diode arrays is very similar to processing used in infrared detector arrays with an integrated multiplexer. In all cases, weak analog signals need to be amplified and conditioned, maintaining high dynamic range and high processing speed. High speed electronic signal processing with high accuracy can become a cost and schedule driver in development of high-resolution wide field of view instruments. Concerns about electrical noise and other electromagnetic interference dictate that this processing be conducted as close to the focal plane assembly as possible—frequently on the chip itself. Simultaneously, careful thermal isolation between the detector arrays and the heat dissipating electronics must maintain a stable minimum operating temperature for the detector arrays.

For infrared detector arrays, the dark current increases with operating temperature. The exact rate depends on detector material, bias voltage and fabrication methods. However, across a wide range of designs, the dark noise typically goes up by a factor of 10 with each rise in temperature of 10 K. Infrared imagers are usually cooled to low temperatures; therefore, we must minimize each heat leak to their cryostat to keep the cooling power low. Frequently, the cooling requirement becomes more stringent as the wavelength of the radiation to be measured becomes longer. For wavelengths in the 100 μm range, detectors are typically cooled to 4 K. Even small heat leaks, such as those from the necessary electrical wiring and mechanical connection to the spacecraft, transport excess heat that increase required cooling power.

Figure 17-19 shows the typical functional blocks found in the electronics of an optical instrument. The signal flow through the electronics begins with a detector in the focal plane of the instrument. This detector converts photons to analog electrical signals, which are then amplified, conditioned and converted to a digital signal stream. Any additional signal processing is performed in the digital data processing block. The analog and digital signal processing blocks operate synchronously with the readout of the detector arrays. In most observation systems, a central timing subsystem supplies appropriate timing signals to all relevant electronic blocks to guarantee synchronous operation. Given electrical signals processed in the instrument electronics are weak, special attention to power conditioning and

Table 17-8◀, Fig. 17-18, Eq. 17-23◀

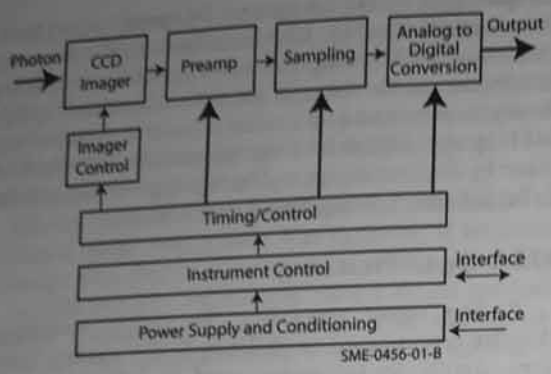


Fig. 17-19. Block Diagram of an Optical Instrument. Photons are collected by the Detector Array, then amplified and processed. The primary supporting functions of power and control are also depicted. The sampling process for charge coupled device signal processing typically uses correlated double sampling to eliminate reset noise.

grounding is required. Careful filtering and clean electrical grounding across a wide electrical bandwidth must be implemented to decouple digital and analog signals.

The instrument control processor manages the signal processing and timing functions performed by the instrument and it interfaces to the main spacecraft processor. Depending on the design, the spacecraft processor may transmit time-tagged commands that are executed by the instrument processor.

Analog signal processing for detector arrays frequently involves correlated double sampling. This technique takes two slightly time-shifted samples of the analog signal and subtracts one from the other to extract their image-related video signal. The adaptation of this video signal to the input range of the analog-to-digital converter requires setting gain and offset parameters for every detector in the system. Any non-uniformity in photo response and dark signal must be removed from each pixel. This task is frequently conducted on board the spacecraft because some of the most straightforward data compression algorithms become invalid if the detector samples are not normalized in this way. Furthermore, normalizing detector samples is required for most effective use of on-board calibration methods in the sensor.

The process of normalizing pixels is based on characteristics of the detector arrays. For arrays with a highly linear response, one-or two-point correction is generally sufficient. Highly nonlinear detectors, like some infrared detectors, require n-point correction techniques. Figure 17-20 illustrates an example of two-point correction. The offset (dark signal) and gain factor (response) are corrected by first subtracting the individual offset value and then by dividing by the individual response for each detector.

High-resolution wide field of view optical instruments typically generate data rates on the order of several hundred Mbps and above. To send this data stream to a ground station in real time, the system may need several parallel channels with capacities up to 300 Mbps each. If data collection is short compared to the available downlink time, a buffer memory can reduce the downlink data rate over a longer transmission time.

Trends for future systems indicate a growing tendency toward more on-board information processing to decrease communication requirements. Continuing improvements in powerful processing hardware and associated algorithms enable sophisticated on-board processing capability comparable to ground-based processing. To maintain the flexibility required for continuous improvements in system performance, software uploads to the hardware must be possible to update and modify operational algorithms from the user community. The need to upload relatively large quantities of software must be considered, when specifying uplink data rate requirements for an observation system.

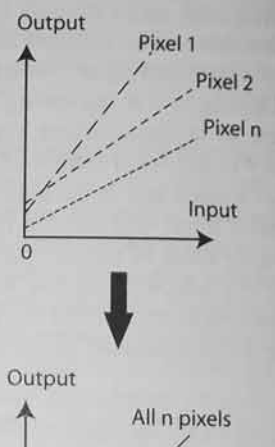


Fig. 17-20. Two-point Normalization of Pixel Response. The top diagram shows the original response functions of individual pixels. The bottom diagram shows the normalized response function of the same pixels, where the effects of the variations in offset values and responsivity values have been removed from the individual pixels.

17.2.5 Radiometric Sensitivity Performance

In order to describe the radiometric sensitivity performance of an observation payload, we must follow a scene photon through the entire end-to-end system. For a solar reflective system operating in the visible or near infrared, that means we start with the source of the reflected light, the Sun. Figure 17-21 shows the radiance distribution of the Sun versus wavelength.

The measured power density in a spectral band of interest is given by integration of the different components of spectral radiance input at the scene over solid angle and across the spectral bandwidth of the sensor. To first-order we assume that lambertian (ideal or dif-

Table 17-8◀, Fig. 17-20, Eq. 17-23◀

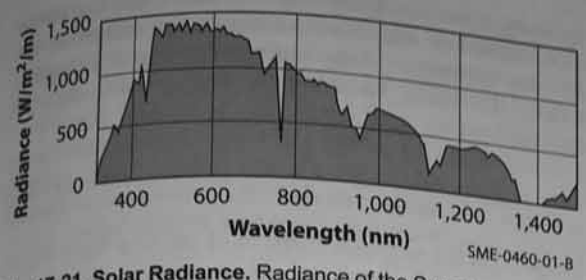


Fig. 17-21. Solar Radiance. Radiance of the Sun at sea level as a function of the wavelength.

fuse) reflection with a constant reflection coefficient occurs at the target scene (reasonably good approximation for small spectral bandwidth). The area of the ground sample radiating power to the instrument determines the power density per solid angle. The atmosphere attenuates this radiation by a constant transmission factor (again, invoking an approximation for small spectral bandwidth). Scattered light in the atmosphere contributes significantly to upwelling signal levels, especially over relatively low reflectance areas like the ocean. The entrance aperture of the payload at orbital altitude collects a very small fraction of the total power radiated by the scene. The collected signal power is attenuated further by transmission through the optics, during the integration time some amount of energy (power times integration time) is collected by each detector in the array. This collected energy divided by the photon energy (which is wavelength dependent, of course) leads to the number of available signal photons per detector. These signal photons are converted into electrons or pairs of electrons and holes inside the detector, depending on the detector material. The quantum efficiency of the detector multiplied by the number of photons is equal to the number of available electrons or electron/hole pairs. These charge carriers are collected by the detector junction and correspond to the output signal of the detector. This signal is expressed in units of electrons, whether electrons or holes are collected by the junction. For convenience, all noise components in the signal measurement process such as dark current, readout noise or instrument background for emissive infrared systems are stated in electrons.

Radiometric performance of an instrument is determined by the signal-to-noise ratio and dynamic range. The signal-to-noise ratio describes the image quality for any given set of measurement conditions, including sensor aperture diameter and instantaneous field of view and scene intensity. The number of noise electrons associated with the signal follows Poisson statistics and equals the square root of the number of signal electrons. Readout of the detector arrays results in additional noise electrons. The signal noise is added to the readout noise since they are statistically independent values, resulting in a total number of noise electrons in the denominator of the signal-to-noise ratio. For emissive infrared systems especially, background from structures inside the instrument that radiate spurious thermal photons create additional noise in the detector that has to be considered in the total number of noise electrons, too.

The required maximum dynamic range of the instrument is the ratio of the sum of the signal, instrument background, dark noise, and readout noise electrons the sensor sees between maximum possible brightness and completely dark scenes. In the solar reflectance part of the spectrum, the brightest scene is typically reflection from thick clouds or snow, and the darkest scene is deep space.

In order to estimate radiometric performance of instruments operating in the mid and long-wavelength emissive infrared spectral region, spectral emission of the surface and atmosphere of the Earth must be modeled. A reasonable approximation for much of the Earth's surface is a blackbody with an equivalent temperature of 290 K. The atmospheric transmission as a function of the wavelength is well understood for any given path orientation and set of atmospheric characteristics up to high zenith angles from the surface. The multiplication of the spectral radiance with the atmospheric transmission results in the upwelling radiance from the surface measured at the sensor. The atmosphere emits significant amounts of thermal radiation and can be the largest single source of emission, depending on wavelength. The integration of upwelling spectral radiance over the selected bandwidth and area of the spatial resolution element results in the power per solid angle. Input power at the sensor's entrance aperture is calculated, based on sensor instantaneous field of view and effective aperture of the receiving optics. This power is converted via the optical transmission factor to input power at detector (usually in the picoWatt range). As with the solar reflective sensors, during the integration time, a certain amount of energy is accumulated per detector. The division of that energy by the energy of one photon gives the number of photons available per detector which is converted by the quantum efficiency to the available number of electrons or electron/hole pairs per detector which corresponds to its output signal.

To characterize the radiometric performance of an instrument that measures thermal emission, a more commonly used figure of merit for radiometric sensitivity is the noise-equivalent delta temperature (NEDT) for the instrument at a given scene temperature. NEDT is given by the temperature difference at a specified scene temperature that is normally relevant to the mission, which generates a signal equivalent to the total number of noise electrons at that scene temperature. NEDT characterizes the instrument's ability to resolve temperature variations for a specified scene temperature.

17.2.6 Payload Size, Mass and Power

We need to be able to estimate the size, mass, power and other system level characteristics of an observation payload before completing a detailed design. We want to be able to compare system options without necessar-

Table 17-8◀, Fig. 17-21, Eq. 17-23◀

ily designing each in detail. This section describes methods to estimate the overall size and key parameters, including data rates. In Sec. 17.2.8, we will apply these methods to FireSat II.

Regardless of the nature of any particular payload, we would like to estimate its size, weight, and power even before we have done a detailed design. To do so, we can use three basic methods:

- Analogy with existing systems
- Scaling from existing systems
- Budgeting by components

The most straightforward approach is to use an analogy with existing systems. To do this, we turn to the list of existing payloads in Table 17-7 or other payloads that we may be aware of which have characteristics matching the mission we have in mind. Kramer [2001] offers a very thorough list of existing sensors. We look for payloads whose performance and complexity match what we are trying to achieve and make a first order estimate that our payload will have characteristics comparable to the previously designed, existing payload. While this approach is rough, it does provide a first estimate and some bounds to decide whether the approach we have in mind is reasonable.

A second approach, described in more detail below, is scaling the payload estimate from existing systems. This can provide moderately accurate estimates of reasonable accuracy if the scale of the proposed payload does not differ too greatly from current payloads. In addition, scaling provides an excellent check. Most existing payloads have been carefully designed and optimized. If our new payload is either too large or too small relative to prior ones, there should be some reason for this change in characteristics. If more detailed estimates based on detailed budgets don't scale from existing systems, we must understand why.

The most accurate process for first-order payload sizing is budgeting by components. Here, we develop a list of payload components such as detectors, optics, optical bench, and electronics. We then estimate the weight, power, and number of each. This is the best and most accurate approach but may be very difficult to apply at early mission stages because we simply don't have enough initial information. Ultimately, we will size the payload with budgeting by components. We will develop budgets as outlined in Chap. 14 for each payload instrument for weight, power, and any critical payload parameters. These budgets will then help us monitor the ongoing payload development. However, even with a detailed budget estimate, it is valuable to use scaling as a check on component budgeting. Again, we wish to understand whether the components scale from existing payloads and, if not, why not.

Scaling from Existing Systems

An excellent approach for early design concept work is to adjust the parameters in Table 17-7 to match the instrument we are designing. We will scale the instruments based on aperture—a main design parameter that we can determine from preliminary mission requirements. To scale, we compute the aperture ratio, R, defined by

$$R = \frac{A_i}{A_o} \tag{17-24}$$

where A_i is the required aperture of our new instrument, and A_o is the aperture of a similar instrument (Table 17-7). We then estimate the size, weight, and power based on ratios with the selected instrument from Table 17-7, using the following:

$$L_i \sim RL_o \quad L = \text{linear dimensions} \tag{17-25}$$

$$S_i \sim L_i^2 \quad S = \text{surface area} \tag{17-26}$$

$$V_i \sim L_i^3 \quad V = \text{volume} \tag{17-27}$$

$$W_i \sim KR^3 W_o \quad W = \text{weight} \tag{17-28}$$

$$P_i \sim KR^3 P_o \quad P = \text{power} \tag{17-29}$$

The factor K should be 2 when R is in the range of 0.5 or less and 1 otherwise. This reflects an additional factor of 2 in weight and power for increased margin when scaling the system down by a factor of more than 2. When the system grows, the R^3 term will directly add a level of margin. For instruments more than a factor of five smaller than those listed in Table 17-7, scaling becomes unreliable. We recommend assuming a mass density of 1 gm/cm^3 and power density of 0.005 W/m^3 for small instruments. An example of these computations for FireSat II is in Sec. 17.3.8.

17.2.7 Figures of Merit for Evaluating Candidate Payloads

Multi-parameter performance indices can be defined for comparing optical instruments with similar performance characteristics. For high spatial resolution imagers three basic figures of merit describe the quality or expected information content in the image. The three defining figures of merit are signal-to-noise ratio at zero spatial frequency, MTF of an instrument at the detector Nyquist frequency (corresponding to the leftmost diagram in Fig. 17-17), and the ground sample distance, GSD. We define a relative quality index (RQI) to allow relatively straightforward quantitative comparisons with a reference instrument denoted by the subscript ref.

$$\text{RQI} = \frac{\text{SNR}}{\text{SNR}_{ref}} \times \frac{\text{MTF}}{\text{SNR}_{ref}} \times \frac{\text{GSD}_{ref}}{\text{GSD}} \tag{17-30}$$

This relative quality index allows the designer to trade requirements with respect to each other. For example, a higher SNR can compensate for a lower MTF at the detector Nyquist frequency for a given GSD. Such

comparisons enable first-order insights into the relationships between complexity, performance, and cost of candidate imagers. For example, suppose we consider a well understood reference instrument with SNR of 512, MTF of 0.5 and GSD of 25 m. If we then compute design parameters for some specific mission, we can calculate a relative quality index, for this candidate design with respect to the known reference instrument. For instance, if our design choices lead to an instrument with SNR of 705.2, MTF of 0.47 and GSD of 30 m, then the RQI for this system will be 108%, suggesting the candidate instrument is slightly better than the reference instrument. This type of index offers a straightforward method for comparing several competing sensors across three (or potentially more) key performance figures of merit.

17.2.8 Example of Observation Payload Sizing

To illustrate the preliminary design process for payloads, we will estimate the basic parameters for the FireSat II payload developed throughout Chaps. 1–10. We cannot expect to carry out a detailed design without substantial input from an IR payload designer. Still, we would at least like to know whether the FireSat II payload is the size of a shoebox or the size of a truck.

Table 17-9 contains the details of the design analysis process for visible and infrared systems. We start with basic system parameters such as the orbital height, minimum observation angle and ground resolution. We then calculate quantities that describe instrument performance. In particular, we determine the image processing parameters and system data rate, size of the optics for a given spatial resolution, and the radiometric sensitivity performance of the sensor. Sample computations for the FireSat II imager are presented in the third column.

The FireSat II altitude trade led to a preliminary altitude, h = 700 km. From this, we can determine the apparent angular radius of the Earth, ρ:

$$\rho = \sin^{-1}\left(\frac{R_E}{R_E + h}\right) = 64.3\,\text{deg} \qquad (17\text{-}31)$$

A key parameter in the system design is the minimum elevation angle, ε, at which the system can work. We do not have an estimate of that yet, but we do know that IR payloads do not work well at small elevation angles. Therefore, we will tentatively assume a minimum elevation angle of 20 deg, recognizing that this may be a very critical trade at a later stage. With this assumption, we can compute the nadir angle range, η, the maximum ground-track angle or swath width, λ, and the maximum range to the target, D, from the formulas in Sec. 8.3:

$$\sin\eta = \cos\varepsilon \sin\rho = 57.9\,\text{deg} \qquad (17\text{-}32)$$

$$\pi = 90 - \eta - \varepsilon = 12.1\,\text{deg} \qquad (17\text{-}33)$$

$$D = R_E\left(\frac{\sin\lambda}{\sin\eta}\right) = 1{,}578\ \text{km} \qquad (17\text{-}34)$$

These equations imply that the sensor on board the spacecraft will have to swing back and forth through an angle of ±57.9 deg to cover the swath. The swath width on the ground will be 2 × 12.1 = 24.2 deg wide in Earth-central angle, with a maximum distance to the far edge of the swath of 1,578 km. Had we been able to work all the way to the true horizon (ε = 0), the maximum Earth central angle would be 90 − ρ = 27.5 deg, and the swath width would be 55 deg. Increasing the minimum elevation angle to 20 deg has very dramatically reduced the size of the available swath.

We next find the orbit period, P, and longitude shift per orbit, ΔL, (Secs. 10.2.1 and 10.2.1.6 web):

$$P = 1.659 \times 10^{-4} \times (6{,}378 + h)^{3/2} = 98.8\ \text{min} \qquad (17\text{-}35)$$

$$\Delta L = 98.8 \times (360/1440) = 24.7\,\text{deg} \qquad (17\text{-}36)$$

Therefore, at the equator, successive node crossings are 24.7 deg apart. Notice that this is slightly larger but very close to the 24.2 deg swath width which we computed above. This is an important characteristic for FireSat II. It would be extremely valuable to have the swaths overlap so that every FireSat II spacecraft can cover all locations on the Earth twice per day. Therefore, in designing the payload, we should work hard to maintain either the altitude or the minimum elevation angle to provide some swath overlap. Doing so could dramatically reduce the number of spacecraft required and therefore the cost of the system.



Fig. 17-22. Computation of FireSat II Ground-Track Parameters.

As Fig. 17-22 shows, the swath width does not need to be quite as large as the spacing between nodes along the equator. Even at the equator, it is enough to have a swath width equal to S, the perpendicular separation between the ground tracks. In Chap. 10, we selected an inclination for FireSat II of 55 deg to cover up to 65 deg latitude. Consequently, we can use the spherical triangle ABC shown in the figure to compute S as follows:

Table 17-8◀, Fig. 17-22, Eq. 17-36

Table 17-9. Parameter Calculations for a Passive Optical Sensor. IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
A. Orbit and Viewing Geometry			
Select altitude, h, and eccentricity, e	Table 10-1; Sec. 10.4	h =700 km, e = 0	See Sec. 10.6 for constellations
Compute orbit period, P	Tables 8-11, 8-12, IRC	P = 98.8 min	
Compute ground track velocity, V_g		V_g = 6.76 km/s	Varies with true anomaly
Compute node shift per orbit, ΔL	IRC	ΔL = 24.7 deg	= longitude shift at all latitudes
Compute Earth angular radius, ρ	Tables 8-11, 8-12, IRC	ρ = 64.3 deg	
Determine min working elevation angle, ε_{min}		ε_{min} = 20 deg	Major system trade; small elev angle provides more coverage, but makes it harder for payload
Compute max nadir angle, η_{max}	Tables 8-11, 8-12, IRC	η_{max} = 57.9 deg	Only slightly less than Earth ang. radius
Compute max Earth central angle, λ_{max}	Tables 8-11, 8-12, IRC	λ_{max} = 12.1 deg	
Compute slant range, R_s	Tables 8-11, 8-12, IRC	R_s = 1,584 km	min distance = altitude
Compute swath width, SW	= 2 λ_{max}	SW = 24.3 deg	
B. Pixel Parameters and Data Rate			
Specify maximum along-track ground sampling distance, Y_{max}	Design parameter	Y_{max} = 1.12 km	Based on spatial resolution requirements at λ_{max}
Determine along-track instantaneous field of view, IFOV	$IFOV_Y = \frac{Y_{max}}{R_s} \cdot \frac{180\text{deg}}{\pi}$	$IFOV_Y$ = 0.041 deg	One pixel width in along-track direction
Specify max cross-track ground sampling distance, X_{max}	Design parameter	X_{max} = 1.12 km	Based on spatial resolution requirements at λ_{max}
Specify number of cross-track detector samples at nadir in one pixel	Design parameter	N_{samp} = 3	Makes pixel same size in along-track and cross-track directions
Determine cross-track IFOV	$IFOV_X = \frac{X_{max}}{R_s} \cdot \frac{180\text{deg}}{\pi N_{samp}}$	$IFOV_X$ = 0.014 deg	
Determine cross-track ground pixel resolution, X, at nadir	$X = IFOV_X \cdot N_{samp} \cdot h\left(\frac{\pi}{180\text{deg}}\right)$	X = 495 m	Best cross-track resolution for this instrument
Determine along-track pixel resolution, Y, at nadir	$Y = IFOV_Y \cdot h\left(\frac{\pi}{180\text{deg}}\right)$	Y = 495 m	Best along-track resolution for this instrument
Determine number of cross-track pixels, Z_c	$Z_c = \left(\frac{2 \cdot \eta_{max}}{IFOV_X}\right)$	Z_c = 8,569	Ground pixel size varies along the swath
Find number of swaths recorded along-track in 1 sec, Z_a	$Z_a = \frac{V_g \cdot 1\,\text{sec}}{Y}$	Z_a = 13.7	Number of successive swaths without gaps at nadir
Find number of pixels recorded in 1 sec, Z	$Z = Z_c \cdot Z_a$	$Z = 1.12 \times 10^5$	
Specify number of bits used to encode each pixel, B	Design parameter	B = 14 bits	Based on radiometric resolution requirement and dynamic range
Specify number of spectral bands, N_{bands}	Design parameter	N_{bands} = 5	
Estimate data rate, DR	$DR = Z \cdot B \cdot N_{bands}$	DR = 8.2 Mbps	Large number may challenge downlink capacity
C. Define Sensor Integration Parameters			
Specify number of pixels for whiskbroom inst. N_m	Design parameter	N_m = 16	Must be large enough to allow sufficient integration time
Specify swath overlap, O	Design parameter	O = 1.185	
Find pixel integration period, T_i	$T_i = \frac{Y}{V_g} \cdot \frac{N_m}{O} \cdot \frac{X}{2\pi h}$	T_i = 111 μs	Integration time of each detector pixel at nadir
Find resulting pixel read-out frequency, F_p	$F_p = 1/T_i$	F_p = 9 kHz	

Table 17-9, Fig. 17-22◀, Eq. 17-36◀

Table 17-9. Parameter Calculations for a Passive Optical Sensor. (Continued) IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
D. Define Sensor Optics			
Specify width for cross-track detectors, d_x	Design parameter	$d_x = 50\ \mu m$	Typical for available infrared detectors
Specify quality factor for imaging, Q	Design parameter	$Q = 0.4$	$0.4 < Q < 2$ ($Q = 1.1$ for good image quality)
Specify operating wavelength, λ	Design parameter	$\lambda = 10.8\ \mu m$	Based on subject trades
Define focal length, f	$f = \frac{h \cdot d_x}{(X/N_{samp})}$	$f = 0.2$ m	Use altitude and Eq. (17-8)
Find diffraction-limited aperture, D	$D = \frac{\lambda f}{Q d_x}$	$D = 0.12$ m	Eq. 17-10
Compute F-number of optics, F#	$F\# = f/D$	$F\# = 1.85$	Typical range = 4–6
Compute along-track field of view of optical system, FOV	$FOV = IFOV_Y \cdot N_m$	$FOV = 0.65$ deg	FOV for the array of pixels
Determine cut-off frequency, F_c	$F_c = D/(\lambda f)$	$F_c = 51{,}000$/m	Referred to nadir
Determine cross-track Nyquist frequency, F_{nc}	$F_{nc} = 1/(2d_x)$	$F_{nc} = 10{,}000$/m	Referred to ground sample resolution at nadir
Determine along-track Nyquist frequency, F_{na}	$F_{na} = 1/(2d_x N_{samp})$	$F_{na} = 3{,}333$/m	Referred to ground sample resolution at nadir
Compute relative Nyquist frequencies, F_{qc} and F_{qa}	$F_{qc} = \frac{F_{nc}}{F_c}$; $F_{qa} = \frac{F_{na}}{F_c}$	$F_{qc} = 20\%$ $F_{qa} = 6.7\%$	% of the cutoff frequency used for this case
Find optics PSF as a function of distance, r, from center of detector	$PSF(r) = [\lambda J_1(Z)/Z]^2$ $Z \equiv \pi r D/\lambda f$		Use $-2d < r < 2d$ J_1 is the Bessel function of order 1
Find optical modulation transfer function (MTF_0) for clear circular optics	$MTF_0(F) = \frac{2C}{\pi}\left[\frac{\pi}{2} - \frac{C}{F_c\sqrt{1-C^2}} - a\sin C\right]$		Use $0 < F < F_c$ $C \equiv F / F_c$
Compute detector MTF cross-track, MTF_X, and along-track, MTF_Y	$MTF_X = [\sin(F_X)/F_X]^2$ $MTF_Y = \lvert\sin(F_Y)/F_Y\rvert$		Use $0 < F < F_c$ $F_X \equiv \pi X F$ $F_Y \equiv \pi Y F$
Compute system MTF cross-track, MTF_S	$MTF_S(F) = MTF_0(F) \cdot MTF_x(F)$		Let F range: $0 < F < F_c$
E. Estimate Sensor Radiometry (for Nadir Viewing)			
Define equivalent blackbody temperature, T	Design parameter	T = 290 K	Blackbody temperature of the Earth
Define the operating bandwidth, $\Delta\lambda$	Design parameter	$\Delta\lambda = 0.4$ μm	Based on subject trades
Determine blackbody spectral radiance, L_λ	$L_\lambda = E(\lambda)/\pi$ $E(\lambda)$ from Eq. (15-2)		Use range $\lambda \pm \Delta\lambda/2$
Look up transmissivity, $\tau(\lambda)$ of the atmosphere	See Fig. 15-5	See Fig 15.5	Evaluate operating bandwidth
Compute upwelling radiance, L_{upi}	$L_{upi}(\lambda) = L_\lambda\, \tau(\lambda)$		Total input radiance as a function of wavelength
Compute integrated upwelling radiance, L_{int}	$L_{int} = \sum L_{upi}(\lambda_i - \lambda_{i+1})$	$L_{int} = 3.31$ W/m²/sr	Evaluate over operating bandwidth
Compute radiated power, L, from a single detector sample at nadir	$L = L_{int} \cdot X \cdot Y/N_{samp}$	$L = 2.7 \times 10^5$ W/sr	Total power from the ground scene that arrives at the instrument
Compute input power, P_{in}, at sensor	$P_{in} = \frac{L}{h^2} \cdot \left(\frac{D}{2}\right)^2 \cdot \pi$	$P_{in} = 5.7 \times 10^{-9}$	P_{in} is power at the entrance to the optics
Define optical transmission factor τ_o	Design parameter	$\tau_0 = 0.80$	Typical value for optical systems
Find input power, P_D, at the detector pixel	$P_D = P_{in} \cdot \tau_0$	$P_D = 4.5 \times 10^{-9}$ W	Very little power arrives at each pixel
Determine available energy, E, after integration time at nadir	$E = P_D \cdot T_i / N_{samp}$	$E = 1.7 \times 10^{-13}$ J	Radiometric design challenge for FireSat II

Table 17-9. Parameter Calculations for a Passive Optical Sensor. (Continued) IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
E. Estimate Sensor Radiometry (for Nadir Viewing) (Continued)			
Find number of available photons, N_p	$N_P = E\,\lambda / hc$	$N_P = 9.2 \times 10^6$	h is Planck's constant, c is speed of light
Define quantum efficiency, QE, of detector at λ	Design parameter	$QE = 0.80$	Typical physical property of detector material
Compute number of photocarriers available, N_e	$N_e = N_p \cdot QE$	$N_e = 7.3 \times 10^6$	Evaluate for an ideal detector
Determine number of noise electrons, N_n	$N_n = \sqrt{N_e}$	$N_n = 2{,}708$	Considers only Shott noise
Determine number of read-out noise electrons, N_r	Design parameter	$N_r = 5{,}000$	Typical value
Determine total number of noise electrons, N_t	$N_t = \sqrt{N_e + N_r}$	$N_t = 2{,}709$	Assumes uncorrelated noise processes
Find signal-to-noise ratio of the image, SNR	$SNR = N_e / N_t$	$SNR = 2{,}707$	Assuming signal dominates background
Determine sensor dynamic range, DR	$DR = N_e / N_r$	$DR = 1{,}467$	With respect to cold space
F. Find the Noise-Equivalent Temperature Difference			
Compute noise-equivalent temperature difference, NEΔT	$NE\Delta T = \frac{\lambda T^2}{W\,SNR}\left[1 - \exp\left(\frac{W\Delta\lambda}{\lambda^2}\right)\right]$	$NE\Delta T = 0.02$ K	Temperature limit the instrument can resolve $W = 1.4388 \times 10^{-2}$

$$S = \sin^{-1}[\sin(24.8\text{ deg})\sin(55\text{ deg})] = 20.1\text{ deg} \quad (17\text{-}37)$$

The perpendicular separation between the orbits at the equator is 20.1 deg. Because the swath width is 24.2 deg, we now have some overlap margin even at the equator and substantial margin at higher latitudes, which are the primary areas of interest. We could, therefore, increase the minimum elevation angle to 25 deg. This would be a reasonable option. At present, we instead choose to hold ε at 20 deg and to provide some margin on altitude and elevation angle for later payload trades. We next compute the required resolution and data rates for FireSat II. From Table 3-4, we initially estimated the needed ground resolution as 500 m. Thus, the required resolution at nadir, θ_r from an altitude of 700 km is:

$$\theta_r = 0.500/700 = 7.1\times10^{-5}\text{ rad} = 0.041\text{ deg} \quad (17\text{-}38)$$

However, as shown in Fig. 17-23, the projected IFOV or Ground IFOV (GIFOV) at end of scan for a maximum slant range of 1,580 km grows to about 3.0 km in the cross track direction and 1.1 km in the along track dimension. This growth in GIFOV would cause the sensor to fall a factor of six short of the needed 500 m ground resolution in the worst case direction, and would probably prevent any further development of this system. We could mitigate this effect by designing in zoom optics that would reduce the IFOV with scan angle. However, we seek to develop a reliable, low cost system and introducing a complex zoom optics system would likely increase system cost while also reducing its reliability. A lower cost approach is to increase the format size of the focal plane assembly to sample at a higher spatial frequency to enable better ground resolution at end of scan. For example, increasing the spatial sampling rate by a factor of three in the cross track direction leads to the ground resolution versus nadir angle curve shown in the lower part of Fig. 17-23. In this case, GIFOV grows from 500 m × 500 m at nadir to 1.1 km × 1.1 km at end of scan. The price we pay in system performance for this approach is reduced sensitivity at end of scan. In the 3.8 μm band, NEDT at 300 K for this single +0.041deg^2 IFOV is close to 0.5 K. However, this is a fire detection and mapping system and while it might be useful to have better sensitivity at end of scan at room temperature, we are most interested in performance for fire signatures at temperatures close to 450 K and above. NEDT at 450 K for this system concept is about 0.2 K, which is more than adequate for distinguishing fire from other scene features.

Using this resolution, we can follow the procedure outlined in Table 17-9 to compute the data rate for FireSat II as 8 Mbps. This data rate is relatively high, but well within capabilities of existing technology. Nevertheless, to reduce cost, we should look at ways to reduce the downlink data rate. One approach is to process the data on board or more simply, turn off the imager over the oceans or other areas where fire detection is of limited utility or use well known and well understood data compression techniques such as Rice compression to reduce data rate by roughly a factor of two. For now, we will leave the value as computed, recognizing that this may need to be reduced later in the system design.

We next compute mapping and pointing budgets for FireSat II. We do not have a firm mapping requirement, but we do have some broad sense of what is needed. We begin, therefore, with a rough estimate of performance

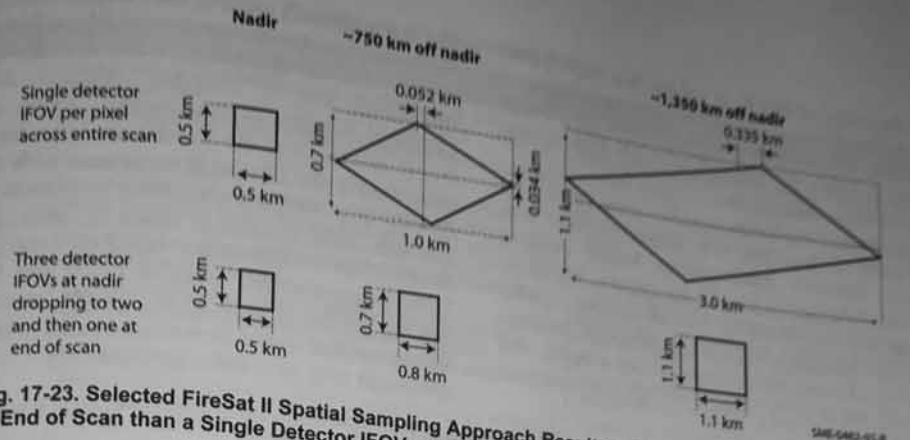
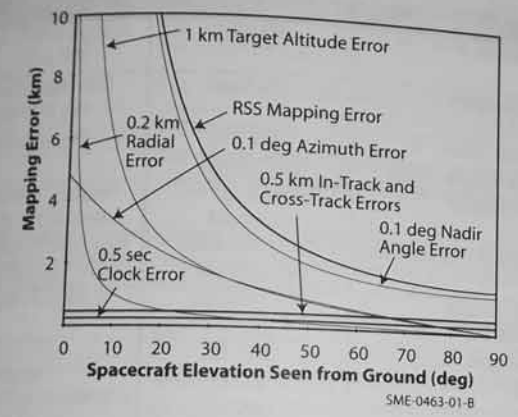
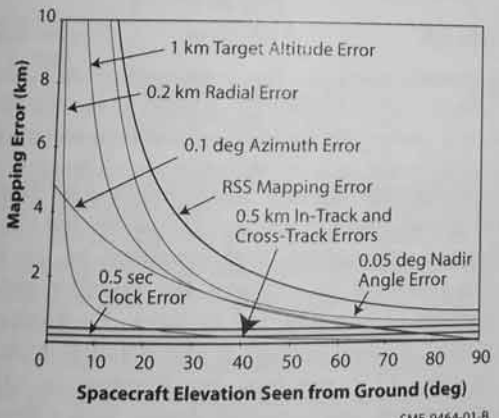


Fig. 17-23. Selected FireSat II Spatial Sampling Approach Results in Much Better Ground Resolution at End of Scan than a Single Detector IFOV per Pixel.



A. 0.1 deg Nadir Angle Error



B. 0.05 deg Nadir Angle Error

Fig. 17-24. FireSat II Mapping Budget. Reducing the nadir angle error below 0.05 deg will have relatively little impact on the overall mapping error because of the 1 km target altitude error. Compare with Fig. 8-50.

parameters and create the mapping error as a function of the elevation angle shown in Fig. 17-24. In this figure, we have used a 0.1 deg nadir angle and azimuth errors corresponding to a relatively inexpensive pointing system based on an Earth sensor. We know we can go to a more expensive system if necessary. In looking at Fig. 17-24A, we see that the mapping error at our chosen minimum elevation angle of 20 deg is between 6 and 8 km. While we are not certain what our mapping requirement is, we are reasonably sure that it is smaller than 6 km. We need to locate fires more accurately than this. Note also that the accuracy has been set almost entirely by our crude attitude number of 0.1 deg.

The next most critical parameter is the 1 km error in target altitude. This means that we assume we can determine the altitude of the fire above the Earth to 1 km—a reasonable accuracy with an oblate Earth model. But significantly improving this accuracy would require carrying a map of the altitudes of all of the regions of interest. That could be very difficult, particularly in mountainous areas, and would cost a lot more money. Therefore, it is of little value to drive the error in nadir angle down below approximately 0.05 deg because it would no longer be the dominant error source. Fig. 17-24B shows the curves that we would achieve with the error in nadir angle reduced to 0.05 deg and all of the other error sources remaining the same. Now the contribution of the errors in nadir angle and target altitude are comparable, so we will use this budget to establish a preliminary mapping requirement of 5.5 km at a 20 deg elevation angle, and 3.5 km at a 30 deg elevation angle. This may still be considerably cruder than we would like, so we may need to revisit this issue.

Now that we understand the mission geometry, we evaluate the optical, signal processing, and radiometric performance of the instrument using the process described in Table 17-9. The third column in that table summarizes results of computations for a whiskbroom sensor design for the FireSat II mission.

The example FireSat II design considers only initial feasibility of the instrument. Several challenges remain with this design and additional iterations are required within context of mission requirements and constraints to optimize the instrument design. For example, the computed data rate of 25.5 Mbps will present some design challenge for a very low cost system. A second challenge with the design is the relatively poor SNR for some of the bands at end of scan for lower temperature

Table 17-9◀, Fig. 17-24, Eq. 17-38◀

scenes in the range of 300 K or colder. We're prepared to accept this performance limitation, however, because our mission is to detect fire, not measure sea surface temperature or cloud top temperature with high spatial resolution across a wide swath.

We now select a "similar instrument" from Table 17-7 for our FireSat II example. We appear to have two options: the Planetary Multispectral Visible-Infrared Imager or the Multispectral Mid-IR instrument. We tentatively select the Multispectral Mid-IR as our similar instrument and will scale from its fundamental parameters of 1.4 m × 1.5 m × 0.9 m size, 270 kg weight, and 170 W power, for its 0.18 m aperture. We first compute the aperture ratio, R

$$R = 0.11/0.18 = 0.61 \quad (17\text{-}39)$$

The value 0.61 is close enough to 0.5 that we select K to be 2 to reflect our assessment that we are dealing with a significantly scaled down system that requires increased margin. With this fundamental ratio, we now estimate the FireSat II payload parameters as

$$\text{Size} = 0.9 \text{ m} \times 0.9 \text{ m} \times 0.5 \text{ m} \quad (17\text{-}40)$$

$$\text{Weight} = 2 \times 270 \times 0.61^3 = 123 \text{ kg} \quad (17\text{-}41)$$

$$\text{Power} = 2 \times 170 \times 0.61^3 = 77 \text{ W} \quad (17\text{-}42)$$

If we now scale from the Planetary Multispectral Visible-Infrared Imager, we derive a very different estimate for the FireSat II payload parameters. In this case,

$$R = 0.11/0.12 = 0.92 \quad (17\text{-}43)$$

We now estimate the FireSat II payload parameters as

$$\text{Size} = 0.50 \text{ m} \times 0.34 \text{ m} \times 0.27 \text{ m} \quad (17\text{-}44)$$

$$\text{Weight} = 12.8 \times 1.00^3 = 10.0 \text{ kg} \quad (17\text{-}45)$$

$$\text{Power} = 12 \times 1.00^3 = 9.3 \text{ W} \quad (17\text{-}46)$$

As described in Sec. 17.2.6, we have incorporated a factor of 2 to provide margin against scaling down the payload size for the Multispectral Mid-IR instrument significantly. The aperture diameter for the planetary instrument is the same as for this FireSat II payload. Therefore, we have not built margin into this scaling calculation.

Clearly, we find that simple scaling laws can give widely divergent estimates for size, mass, and power for concept payloads, depending on the reference payload. To resolve this difference, we need to look in more detail at the closest reference instrument, namely the Planetary Multispectral Visible-Infrared instrument and use a combination of the other methods described in Sec. 17.2.5, analogy with existing systems and budgeting by components to derive a higher confidence estimate.

The planetary instrument, while identical in aperture size with the FireSat II payload, is different at the next level of detail. It has two focal planes, versus the single focal plane for FireSat II. That difference allows us to reduce the dimensions of the instrument by a few centimeters in width and depth. Likewise, the planetary instrument has a long sunshield that is not needed for FireSat II, which further reduces the instrument length by tens of centimeters. However, FireSat II requires a scanner that adds back most of that length. A rough estimate of the scanner size is that it would be a few centimeters shorter than the sunshield so that a small reduction in instrument size is expected. Another difference is FireSat II requires active cooling, which adds to the power and mass of the instrument. Likewise, the higher data rate for the FireSat II instrument adds a substantial amount of power to the nominal instrument description. The size, mass, and power values in Table 17-10 account for all of these differences, based on existing hardware. Thus, as summarized in Table 17-10, our first guess is that the FireSat II payload is a simple, moderately sized instrument that would fit well on a small to medium-sized spacecraft with high data rate capability.

Table 17-10. Space Summary of FireSat II Initial Parameter Estimates.

Parameter	Nominal Value	Comments
Altitude, h	700 km	Range = 600 to 800 km
Inclination, i	55 deg	Coverage to 65 deg latitude
Swath width, $2\lambda_{max}$	24.2 deg	20.1 deg required
Nadir angle range, η	±57.9 deg	
Instrument	mid-range IR scanner	Payload needs expert input
Ground resolution	501 m at nadir (along track)	Key parameter—needs trade study
Instrument resolution	2.714 mrad	0.00409 deg
Aperture, A	0.11 m	
Size	0.48 × 0.22 × 0.18 cm^3	
Weight, W	20 kg	
Power, P	50 W	
Data rate, DR	8 Mbps/sec	May be limiting feature
Mapping	3.5 km @ ε=30 deg 5.5 km @ ε=20 deg	

Our initial analysis of a small, lightweight FireSat II payload shows that the mission is feasible but challenging. Several refinements and iterations on the design have the potential to improve the viability and cost-effectiveness of the payload concept. To illustrate the end point of a more complete and detailed design process, in Sec. 17.3, we turn our attention to VIIRS, a large instrument and a mature design with a fire detection mission among many other missions.

17.2.9 Life-cycle Cost and Operability of the Payload

Spacecraft designs are heavily driven by the need to minimize cost, while simultaneously meeting an acceptable level of performance. Over the years, a number of approaches have been proposed and implemented to treat cost as an independent variable. For systems where cost and performance may be traded, it is no longer sufficient to simply state mission requirements clearly and realistically. Rather, mission requirements may now become involved in the iterative process of design

Table 17-11. Impact of Remote Sensing Payloads on the Spacecraft Design.

Impact Area	Typical Requirements to Support Payload	Additional Considerations
Structure	Mount the optical instruments isostatically to the spacecraft bus	Carefully analyze aging of material (e.g., stress release in metal parts), humidity release, transition to microgravity, and acceleration forces
	Do not apply excessive forces or torques to the payload instrument	Typical stability requirements at critical locations within the optical instrument housing are in the μm and mdeg range
	Make the mounting structure or base plate for optical components stiff enough to prevent any misalignment when subjected to the forces and vibrations of launch	
Thermal	Make large opto-mechanical assemblies temperature stabilized or isothermal	Large reflective systems (which use Zerodur, Aluminum, or Beryllium or newly developed materials such as SiC or CsiC as materials for the mirrors) and mounting structures (which use composite materials) are temperature sensitive and may require semi-active temperature control of structure and mirrors
	Operate refractive optical systems typically within a specific temperature range to achieve required performance (frequently they employ semi-active temperature control)	Temperature gradients in optical components can severely degrade performance
	Make reflective systems entirely from the same material which leads to a compensation of thermal effects (typically done for cryogenic optical systems)	
External Alignment	Align the optical axis of the instrument and the line of sight of the pointing device with an external reference on the spacecraft. External alignments may need to be on the order of 1 arc sec	User reference cubes to achieve alignment
		External alignment requires a calibrated optical bench
Pointing	For monocular optical instruments, make the pointing requirements on the order of 0.1 to 0.01 of the swath width, typically	Mount attitude determination sensors (e.g., star sensors) to the instrument (not the bus) to minimize the effects of thermoelasticity
	For stereoscopic instruments, automated digital terrain mapping requires pointing knowledge of 1/5 of a pixel	Do pointing by maneuvering the spacecraft or by pointing devices (such as pointing mirrors for the front of the instrument or gimbals for the entire instrument)
Assembly Integration and Verification	Optical instruments require clean rooms and clean laminar air flow benches for all integration and verification activities	Cleaning optical surfaces is generally not possible
	Clean room requirements typically range from 100 to 100,000 ppm	During exposure to the environment, use cleanliness samples to verify the level of contamination
System Accommodation	Sensor must have an unobstructed field-of-view	Orient radiators and passive coolers for infrared systems to prevent interference with optical devices
	Sensor must have a guard cone to prevent performance degradation due to stray light	Calibration devices impose geometric constraints with respect to the optics of the system and the orbit
	Avoid pointing toward the Sun	

[Fleeter, 2000]. Several excellent examples of how to include cost as a system parameter rather than a given are described by Shishko, et al. [1996].

Working through trade-offs associated with cost, performance, and requirements in early stages of payload definition keeps payload designers focused on the best sensor characteristics to maximize mission performance and minimize cost. Designers sometimes want to perform a purely analytical evaluation of the costs and benefits of various design options. Unfortunately, relative benefits of different design features are difficult if not impossible to quantify in a clear and universally accepted way. Well understood and clearly stated analysis methods can be useful for providing a common framework for evaluating different designs. Ultimately, however, judgments about meeting mission goals within cost and schedule limitations rely on the insight of experienced designers and other analysts, adding to the challenges and significance of this portion of the payload definition process.

After we establish the final payload type and basic payload performance requirements, final design of the payload can begin. The final payload design could be as simple as an evaluation of existing payloads that are available or it could involve detailed design, fabrication, and testing of an entirely new instrument. The final step in the payload definition and sizing process is the decision by the responsible organization to buy or build the payload.

Integrating a payload into a spacecraft introduces a number of practical considerations for the spacecraft and other payload subsystems. These derived requirements can have a significant impact on other parts of the spacecraft. Table 17-11 contains an overview of some of the spacecraft accommodation considerations for a payload. Resolving the impact of these requirements on the spacecraft and other parts of the payload means we must assess the performance, cost, and technical risk for each subsystem to accommodate the payload.

17.3 Sample Mission—VIIRS

Spaceflight instruments that can automatically detect fires already exist. The remarkably versatile MODIS (Moderate-resolution Imaging Spectroradiometer), has been flying on board NASA Earth Observing System satellites since 1999. MODIS has provided an abundance of data in support of scientific investigations into Earth's atmosphere, oceans and land use that goes well beyond

Table 17-11, Fig. 17-24◀, Eq. 17-46◀

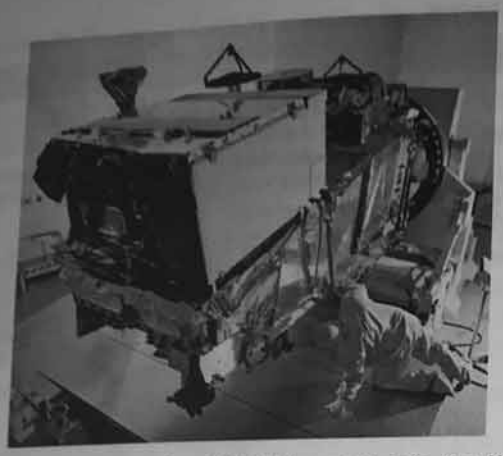


Fig. 17-25. Photograph of FU1 Integrated onto the NPP Spacecraft at Ball Aerospace. (Photo courtesy of Ball Aerospace)

fundamental needs for the FireSat II mission. MODIS is being replaced by the Visible/Infrared Imager Radiometer Suite (VIIRS), the next-generation imaging spectroradiometer for the NOAA/NASA Joint Polar Satellite System (JPSS) and the US Department of Defense Weather Satellite System (DWSS). VIIRS, described in Fig. 17-25, replaces and improves upon three different operational and research sensors with a single instrument. These three sensors are the MODIS, the Advanced Very High Resolution Radiometer (AVHRR) operating on board NOAA POES and EUMETSAT Polar System Metop since 1978 and the Operational Line Scanner (OLS) on board DMSP since 1976.

VIIRS is a whiskbroom scanner that monitors Earth from a 833 km polar Sun-synchronous orbits at visible-infrared wavelengths (0.4–12 μm) and provides well calibrated moderate (~km) spatial resolution measurements of light upwelling from Earth in support of a large number of high priority applications ranging from weather prediction for civilian and military needs to climate change monitoring, land usage, public health alerts and predictions of electrical power usage. Figure 17-26 shows that VIIRS offers significant improvements over the systems it replaces by providing 22 spectral bands with 4 times better spectral coverage than AVHRR, thereby enabling new agricultural, climate, disaster monitoring, public health and weather data products, 3 times better spatial resolution than AVHRR and MODIS at end of scan enabling sharper imagery over a much greater area and a fully calibrated Day/Night Band (DNB) that improves nighttime weather forecasting and military applications compared with OLS. VIIRS benefits from substantial US R&D investment in MODIS and other NASA Earth observing systems that led to a wide range of new environmental data products, including fire products, from well characterized instruments in polar sun synchronous orbits.

A key to the VIIRS architecture shown in Fig. 17-27 is a rotating telescope assembly that enables a flexible design needed to meet a remarkably diverse set of requirements for multispectral imaging spectroradiometry and low light level day/night imaging. Advantages of the rotating telescope design relative to scan mirror based systems like AVHRR and MODIS include better control of straylight, smaller range in angle of incidence of light on the fore optics to reduce striping and polarization sensitivity and better protection from contamination and degradation over time because all of the optical elements are placed deep inside the instrument housing,

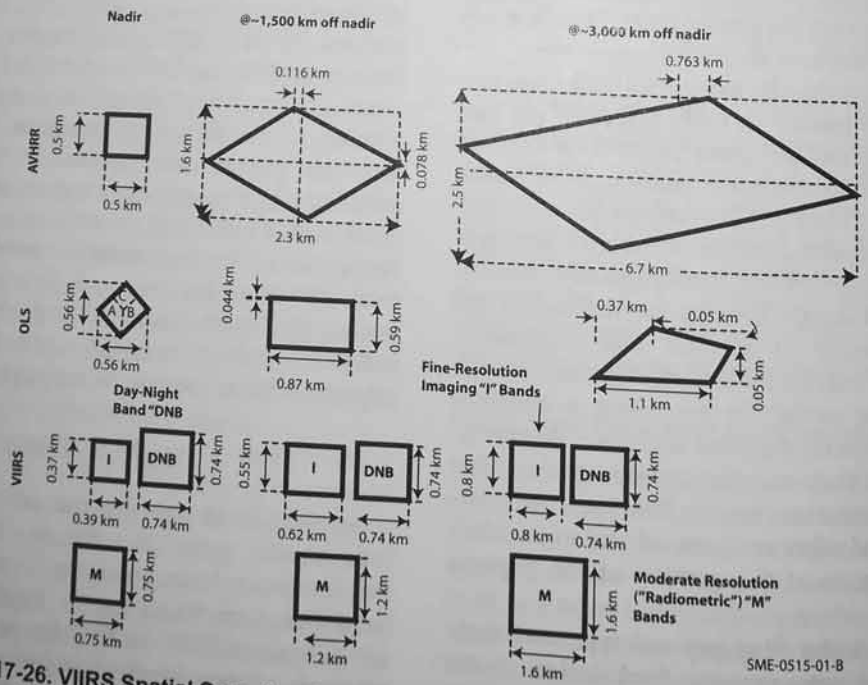


Fig. 17-26. VIIRS Spatial Sampling Approach Compared with the Legacy AVHRR and OLS Sensors. (Graphic courtesy of NASA)

Table 17-11◀, Fig. 17-26, Eq. 17-46◀

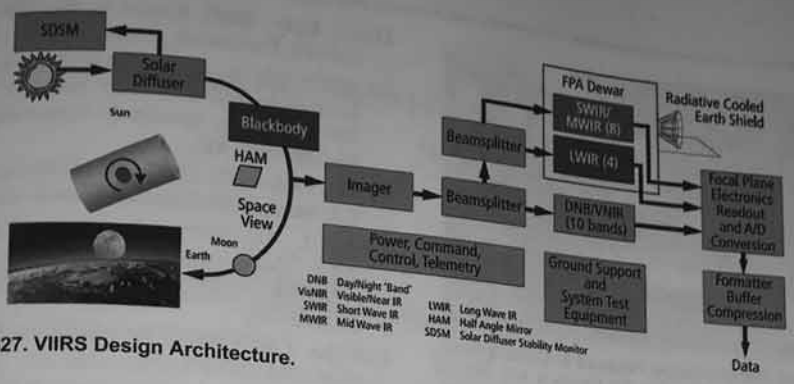


Fig. 17-27. VIIRS Design Architecture.

Many challenges were overcome in design and fabrication of the rotating telescope assembly to produce a well understood high performance subsystem. Significant US investment in the VIIRS architecture has resulted in an excellent imager that is ready to provide high fidelity data for the international science, weather and other environmental data product communities with unprecedented spatial resolution at end of scan.

The VIIRS fire detection algorithm is derived from the MODIS algorithm described by Kaufman and Justice [1998] and Giglio, et al. [2003]. The algorithm uses calibrated and geolocated data from five VIIRS spectral bands: M5 (0.672 μm), M7 (0.865 μm), M11 (2.25 μm), M13 (4.05 μm), M15 (10.763 μm), M16 (12.013 μm). In addition, the algorithm makes use of solar zenith and azimuth angles along with a global database describing whether a pixel is land, water or sometimes water. The algorithm follows these steps.

Identify potential fire pixels. The algorithm prescreens pixels by selecting only those clear sky pixels on land that have valid measurements. Then, the algorithm selects potential fire pixels by identifying pixels with spectral characteristics consistent with fire, based on temperature thresholds and differences in apparent brightness temperature between bands.

Generate statistics on pixel characteristics near potential fire pixels and separate potential fire pixels from the background. In this part of the process, the algorithm uses neighboring pixels to estimate the radiometric signal of a potential fire pixel in the absence of fire. Then, a series of threshold tests are used to perform relative fire detection. These tests look for the characteristic signature of an active fire in which both the M13 brightness and the M13 and M15 brightness temperature difference depart substantially from that of the non-fire background.

Make tentative fire detection based on absolute brightness temperature in M13. If the M13 brightness temperature is greater than 320 K at night, then fire detection is confirmed. During the day, the M13 brightness temperature in a pixel must be greater than 360 K for tentative fire detection.

Reject false alarms. Tentative fire detections made during the day are susceptible to false alarms because brightness temperature measurement can be contaminated with sunglint and other false signals resulting from background characterization errors in regions with abrupt boundaries between land and water or along the boundaries of deserts. These false alarms can be rejected with good reliability by using information derived from the solar reflective M5, M7 and M11 bands.

Export fire pixel locations with quality control information.

The VIIRS sensor and algorithm system illustrates many of the features of an automated fire detection system. In the context of the FireSat II mission, this example provides a point design with finalized tradeoffs in size, mass, power, resolution, and data processing that has already been built and tested.

17.4 The Evolution of Observation Payloads

In less than 60 years, space-based observation payloads have evolved from simple 35-mm film movie cameras on board converted World War II missiles providing imagery for just a few minutes to aggregations of widely different instruments covering much of the electromagnetic spectrum simultaneously from Earth orbit for more than 10 years. The very first space-based observation payload flew on board a German V-2 rocket captured by the United States near the end of World War II. This rocket was launched from White Sands Missile Range on October 24, 1946 and produced the image shown in Fig. 17-28. Clyde Holliday from Johns Hopkins University/Applied Physics Laboratory, developer of this first space-based imager, wrote: "Results of these tests now are pointing to a time when cameras may be mounted on guided missiles for scouting enemy territory in war, mapping inaccessible regions of the earth in peacetime, and even photographing cloud formations, storm fronts, and overcast areas over an entire continent in a few hours…the entire land area of the globe might be mapped in this way" [Holliday and Hamilton, 1950]. See the book website for a more extended discussion of how these payloads have evolved.



As new observational payloads are developed with ever increasing data collection rates, the driving technology for these systems is expected to shift from focal

Table 17-11◀, Fig. 17-27, Eq. 17-46◀



Fig. 17-28. The First Observational Payload to Collect Images from Space (>100 km alt.). Did So Using a V-2 Rocket Launched from White Sands Missile Range on October 24, 1946.

plane technology for visible-infrared systems toward on board and ground processing to handle the vast amounts of data generated by ever more capable systems. Future systems based on distributed architectures that are refreshed routinely may be able to keep pace with availability of improved technology better than today's systems which are refreshed on decade time scales.

Space-based observation systems are most valuable when they're effective, ubiquitous, affordable, reliable, flexible and rapidly replaceable. Readers should not confuse low system cost with low performance. The original Iridium system proved that it is possible to achieve both low cost and unprecedented performance. Unfortunately, the failure of the original Iridium business model prevented this production approach from being sustained in near term upgrades and in the building of similar expected systems. Any future observation system must meet its operational or research performance requirements. For instance, regardless of satellite constellation, space imagers must meet operational performance requirements for parameters such as spatial resolution, spectral coverage, area coverage rate, sensitivity and geolocation. Likewise, systems in any architecture must meet system level hardness, reliability and operational lifetime needed to fulfill customer mission needs. System-level trade studies early in program development need to be broadened to consider alternative system architectures that might be built at lower overall cost using highly engineered and reliable systems designed for affordable, efficient manufacturing and operationally effective performance and built using well understood and continuously improving processes. These early studies would trade satellite design life, technology improvement schedule, constellation refresh rate, survivability, and cost within context of firm system performance parameters.

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17 Observation Payloads

Jeffery J. Puschell, *Raytheon*

17.1 Observation Payload Design
Electro-optical Observation Payloads; Radio Frequency Observational Payloads
17.2 Observation Payload Sizing
Payload Operations Concepts; Required Payload Capability; Spatial Sampling; Signal Processing; Radiometric Sensitivity Performance; Payload Size, Mass and Power; Figures of Merit for Evaluating Candidate Payloads; Example of Observation Payload Sizing; Life-cycle Cost and Operability of the Payload
17.3 Sample Mission—VIIRS
17.4 The Evolution of Observation Payloads

Space-based observation payload missions have dramatically changed our perception of the world by delivering vitally important imagery and remote sensing data on life-threatening hurricanes and other severe weather, surveillance and reconnaissance data of critical importance to military planners, national decision makers, stunning imagery, and measurements of distant astrophysical objects at wavelengths not accessible from the Earth's surface. Future space-based observation missions are expected to provide continuous video surveillance of the Earth, improve weather forecasting by constraining forecast models with more accurate and detailed measurements, contribute to more complete understanding of climate change, and help resolve some of the deepest cosmological mysteries of our time related to dark matter and dark energy.

Observation payloads can be divided into two broad categories defined by whether the observed light is from active illumination of the scene by the payload itself or a cooperative illumination source, or from passive illumination by the Sun or by other uncooperative sources including the scene itself. Radars and laser-based sensors like DLR's TerraSAR-X and NASA's ICESAT are examples of active systems. Passive sensor examples include visible wavelength imagers like the commercial GeoEye and DigitalGlobe imagers that rely on reflected sunlight along with infrared remote sensors like NASA's Atmospheric InfraRed Sounder (AIRS) and passive microwave radiometers like the US DOD Special Sensor Microwave Imager Sounder (SSMIS) that detect thermal emission from the scene.

Earth observations by both active and passive payloads occur in three primary spectral regions known as solar reflectance, thermal infrared and microwave or radio frequency (RF). Space astronomy payloads like NASA's Chandrasekhar X-Ray Observatory and Compton Gamma Ray Observatory have observed objects at X-ray and gamma ray wavelengths, in addition to these four primary spectral regions. Schowengerdt [2006] and Schott [2007] are excellent sources for detailed discussion of solar reflectance and thermal infrared systems. Sharkov [2003] presents a comprehensive treatment of passive microwave observation systems. Cantafio [1989] provides extended discussions of active space-based RF systems.

Solar reflectance systems operate from the near ultraviolet (–0.35 μm) to the shortwave infrared (~2.5 μm) by observing reflected sunlight. These systems offer the potential for higher spatial resolution than other spectral regions because of their relatively short wavelengths, but can only operate in daylight, unless a laser or other source of active scene illumination is used [Nischan, et al. 2003].

Thermal infrared or emissive systems operate in the spectral region dominated by thermal emission from commonly observed objects on Earth (~3–300 μm). Objects of interest include the surface of the Earth (~300 K), the upper atmosphere (~220 K), biomass fires (~600–1,200 K), and missile launches (~300–1,600 K). The thermal infrared region is affected by absorption of light by molecules in the atmosphere. Observations of the Earth's surface are possible only within certain wavelength ranges known as atmospheric windows, which occur in between strong molecular absorption bands. These strong absorption bands provide a means for probing temperature structure of the atmosphere along with the 3-D distribution of trace gases such as water vapor and ozone. Thermal infrared sensors can operate both day and night because the detected signal is from thermal emission by the scene itself and does not rely on reflected sunlight.

Microwave/RF systems operate at millimeter and centimeter wavelengths. Passive microwave radiometers operate chiefly at millimeter wavelengths (20–200 GHz). Spatial resolution of these systems is three to five orders of magnitude worse than for visible wavelength sensors with the same aperture size, but they are capable of collecting unique information over large areas. Active RF

Table 17-1. Advantages of Various Observation Paylod Types.

Observation Payload Type	Advantages
Passive EO—Solar Reflectance Spectral Region	Mature technology enabling high sensitivity measurements, high spatial resolution even with relatively small wavelengths, rich phenomenology
Passive EO—Emissive	Mature technology enabling high sensitivity measurements, high spatial resolution even with relatively small wavelengths, rich phenomenology, effective day or night
Passive Microwave	Mature technology enabling high sensitivity measurements, rich phenomenology, effective day or night and through virtually all weather conditions
Passive X-Ray and Gamma-Ray	Enables detection of high energy events on Earth and throughout the universe
Active EO	Day/night operation even at Solar reflectance wavelengths
Active RF—Real Aperture Radar	Day/night, all weather capability for detecting precipitation, mapping clouds, characterizing large scale features and altimetry
Active RF—Synthetic Aperture Radar	Day/night, all weather performance with spatial resolutions comparable to much shorter wavelength systems because of the large synthetic aperture

systems also known as radars provide their own illumination of the scene in the centimeter to millimeter bands. The reflected signals can be processed to identify physical features in the scene. Radar systems can be designed to penetrate most atmospheric disturbances, such as clouds, because only larger features can reflect signals at radar wavelengths.

Regardless of spectral region or mode of operation, sensitivity of an observational payload depends on the observation geometry, effective sensor aperture size, integration time, detector sensitivity, spectral bandwidth, transmission through the atmosphere, spatial sample size and the spectral radiance from the scene itself. Sections 17.1 and 17.2 discuss these design considerations in detail.

17.1 Observation Payload Design

Observation payloads collect data (e.g., imagery, spectral radiance, distance) on remote objects such as the surface of the Earth, relatively nearby objects such as another satellite or distant astrophysical objects that can be processed into information (e.g., temperature, reflectance, chemical composition, topography) to provide insight into physical characteristics of these remote objects for scientific, weather forecasting, military and policy making purposes—just to name a few. Observation payloads operate across the entire electromagnetic spectrum from gamma rays and X-rays to RF frequencies. In addition, observation payloads include cosmic ray and other directional particle sensors used to probe the Sun, supernovae, active galactic nuclei, and other unknown astrophysical objects. Table 17-1 lists the advantages of various observation payload types. For specific mission applications for observation payloads missions, e.g., missile warning, environmental, and remote sensing, see the book website.

Two basic types of observation payloads exist: so-called passive payloads such as visible-infrared and microwave imagers that observe intrinsic emission from the scene, whether it be reflected sunlight, thermal emission or gamma rays and active payloads such as lidars and radars that supply their own source of light to enable specific types of measurement such as ranging or to enhance signal-to-noise ratio and spatial resolution.

Space-based observation payloads include the following major categories.

Passive solar reflectance systems observe that part of the electromagnetic spectrum dominated by reflected sunlight—the ultraviolet (~0.3 μm) through the shortwave infrared (SWIR) at ~2.5 μm. These systems offer the potential for relatively high spatial resolution from space, because their shorter operating wavelengths diffract less than longer wavelengths. However, because they rely on reflected sunlight for Earth observations, most of these systems operate effectively only during daytime. Observation payloads in this category include:

- Largely panchromatic imagers like GeoEye and DigitalGlobe
- Commercial imagers that specialize in collecting high special resolution imagery, multispectral (~10 spectral band with wavelength, λ, and spectral bandwidth, $\delta\lambda$, such that $\lambda/\delta\lambda \sim 10$ typically)
- Synoptic imagers like the GEOS imager
- Polar orbiting environmental imagers like MODIS that provide routine global-scale observations
- Hyperspectral (~100 spectral band with $\lambda/\delta\lambda$~100 across a broad contiguous spectral region) imagers, such as Hyperion that provide more detailed spectral measurements over limited spatial regions.
- Low light level imagers like the OLS onboard the DMSP satellite that provide panchromatic imagery of moonlit or airglow lit scenes on Earth at night
- Polarimeters such as POLDER that provide measurements of linear polarization of spectral radiance in different spectral bands across the entire solar reflectance spectral region

See the book website for more color photos from various types of observation payloads.

Emissive or thermal infrared systems observe that part of the electromagnetic spectrum dominated by thermal emission from the scene itself in atmospheric transmission windows from ~3 μm to ~300 μm in wavelength. In some cases, these systems operate outside atmospheric transmission windows to enable less cluttered observations high above the surface of the Earth. These systems operate effectively day or night, because they do not rely on sunlight to create an observable signal. Like solar reflectance systems, emissive systems include:

- Multispectral imagers like the GOES imager and MODIS
- Hyperspectral systems like the Thermal Emission Spectrometer (TES) that flew on the Mars Global Surveyor to map mineralogy on Mars and search for evidence of water
- Ultraspectral (~1,000 spectral bands with $\lambda/\delta\lambda$ ~1,000 across a broad contiguous spectral region)
- Systems like the AIRS sensor on board the NASA Aqua satellite and the IASI on board the Eumetsat MetOp satellite that provide detailed spectra of the Earth's atmosphere and surface to enable 3-D instantaneous maps of temperature distribution on the surface and in the atmosphere along with the maps of water vapor and other trace gases

Passive microwave systems observe thermal emission at much longer wavelengths than emissive infrared systems. The microwave part of the electromagnetic spectrum extends from ~1 mm to ~1 m in wavelength. Normally, microwave radiation is referenced to frequency rather than wavelength. In frequency, the microwave spectral region ranges from ~0.3 GHz to ~300 GHz. Like emissive infrared systems, passive microwave systems operate effectively day or night. At frequencies less than 10 GHz, microwave radiation is relatively unaffected by clouds, so that these longer wavelength systems can operate under virtually all weather conditions. For higher frequency microwave systems, clouds and fog along with absorption by water and oxygen molecules in the atmosphere can influence the signal from the surface. Well known passive microwave systems include:

- AMSU is used primarily for mapping temperature and water vapor structure in the atmosphere. The system flying today is the most recent version of systems that have operated on board the TIROS/POES satellites since 1978. AMSU is divided into two parts called A (used mostly for temperature sounding) and B (used mostly for water vapor sounding), which together operate in 20 spectral channels ranging from 23.8 GHz to 183.3 GHz in frequency and with different spectral resolutions for some frequencies. AMSU-A has a spatial sample size at nadir of about 45 km. The higher frequency AMSU-B has a spatial sample size of about 15 km at nadir
- SSM/I is used primarily for near-surface wind speed, total column water vapor, total column cloud liquid water and precipitation. It has been on board DMSP satellites since 1987, but has heritage back to the Nimbus 7 and Seasat missions launched in 1978. SSM/I operates in seven channels ranging from 19 GHz to 89 GHz in frequency. SSM/I operates with spatial resolutions ranging from 13 km to 69 km, depending on frequency
- The SSMI/S instrument is the follow-on instrument to SSM/I and measures microwave energy at 24 discrete frequencies from 19 to 183 GHz with a swath width of 1,700 km at 12.5 km to 75 km spatial resolution at nadir. The synoptic solar reflectance and emissive infrared systems used in conjunction with these passive microwave systems operate with ~1 km spatial resolution using much smaller aperture sizes, because of the diffraction performance advantage of shorter wavelengths

X-ray imagers observe thermal and non-thermal emission from astrophysical objects including the Sun, neutron stars, supernovae, the center of our Galaxy and distant unknown objects at wavelengths ranging from less than 3 pm to 10 nm, corresponding to frequencies ranging from above 10 EHz (1 EHz is 10^{18} Hz) down to 30 PHz (1 PHz is 10^{15} Hz). X-rays are often referenced in units of electron Volts (eV) rather than wavelength or frequency. In eV, the X-ray spectral region extends from about 0.1 KeV to 400 KeV. The first X-ray observations from space were made with geiger counters that flew on board converted V-2 rockets starting in 1949. UHURU also known as SAS-1 launched in 1970 was the first satellite with an X-ray observation payload. The Chandra X-ray observatory, launched in 1999, observes the 0.1 KeV to 10 KeV spectral region, with 2.4 μrad (0.5 arcsec) angular resolution using a 1.2 m diameter X-ray telescope. The satellite is in a highly elliptical orbit that enables the observatory to make X-ray measurements from well out the Earth's Van Allen belts. The effective collection area for the Chandra observatory is only 0.04 m^2 at 1 KeV, despite the much larger telescope diameter. Conventional optical approaches are not effective at X-ray wavelengths, because no suitable refractive material is available to build X-ray lenses and standard reflector telescopes do not work because X-rays are either absorbed or transmitted at near normal incidence to reflecting mirrors. Therefore, X-ray instruments use either a coded aperture method or a grazing incidence telescope, also known as a Wolter telescope, to collect and focus X-rays. In Wolter telescopes, the angle of reflection from the mirror is very low—typically 10 arc-minutes to 2 deg. Chandra uses a Wolter telescope that consists of nested cylindrical paraboloid and hyperboloid surfaces coated with iridium or gold. Chandra uses four pairs of nested mirrors. The relatively

thick substrate (2 cm) and very careful polishing allowed a very precise optical surface, which is largely responsible for Chandra's unprecedented and unmatched angular resolution. However, thickness of the substrates limits the fill factor of the aperture, leading to the low effective collection area.

Gamma ray payloads observe emission of the highest energy photons from astrophysical objects including the Sun, supernovae, the center of our Galaxy and distant unknown objects. Gamma rays do not penetrate Earth's atmosphere very well so direct astronomical observations of gamma rays must occur from high in the atmosphere or from space. Wavelengths for gamma rays are shorter than 3 pm, corresponding to frequencies greater than 10 EHz (1 EHz is 10^{18} Hz). The first gamma ray observations of the Sun and distant galactic and extragalactic objects were made by OSO-3 launched in 1967. Shortly thereafter, mysterious gamma ray bursts from very distant, but still only partially identified objects were discovered by the Vela satellites designed to detect gamma rays from nuclear explosions on Earth. Recent gamma ray observation payloads include the instruments on board the Fermi Gamma Ray Telescope, a successor to the Compton Gamma Ray Observatory. These instruments detect gamma rays in two broad spectral ranges using scintillation detectors that detect small flashes of light created by gamma rays passing through a crystal and pair production techniques at the highest energies that create electron-positron pairs created by incoming gamma rays.

Active electro-optical systems like lidars, ladars and altimeters use active illumination of the scene by a laser to create a measurable signal that can be read out rapidly over time to remotely sense backscattered light from water droplets, aerosols and trace gases in the atmosphere as a function of distance from the system and for range measurements and altimetry, when looking at the Earth, Moon, and other space objects.

- The first lidar in space was the Lidar In-space Technology Experiment (LITE) carried onboard the Space Shuttle in 1994. LITE measured clouds with a backscatter lidar based on a Nd:YAG laser operating at three different wavelengths: 1,064 nm, 532 nm (second harmonic) and 355 nm (third harmonic)

- CALIPSO, launched as part of the A-train of Earth Science satellites in 2006, is also a backscatter lidar based on a Nd:YAG laser. CALIPSO transmits both the fundamental (1,064 nm) and second harmonic (532 nm) of the Nd:YAG laser in a nadir beam. The vertical resolution is 30 m and horizontal resolution 333 km

- ICESAT launched in 2003 carried a Nd:YAG based laser altimeter used primarily for measuring thickness of Arctic ice sheets, but also for providing global topography and vegetation data. The main challenge for building a laser-based observational space payload is producing a reliable laser for extended duration use in space—previous instruments like ICESAT experienced failure of some laser modules after only a few months of operation

Lessons learned from ICESAT benefit future missions such as the upcoming AEOLUS mission with the ALADIN lidar, planned for the Atmospheric Dynamics Mission (ADM). ALADIN is a wind-measuring lidar, based on the third harmonic of a Nd:YAG laser. ALADIN will measure wind speed along the lidar line of sight by measuring the Doppler shift of the backscattered light using two methods: coherent (heterodyne) detection by mixing the return signal with a local oscillator and a direct direction method using a narrowband etalon. These techniques will be applied to measuring Doppler shifts of light returned to a receiver from Mie (aerosol) and Rayleigh (molecular) scattering respectively. While useful as a pathfinder for future winds lidar missions by providing information not available to today to numerical weather forecast models, AEOLUS will only measure a single component of the 3-D wind vector.

Active RF systems like radars and synthetic aperture radars (SARs) use active illumination of the scene by a coherent RF transmitter to create a signal that can be detected and read out rapidly in time to produce detailed range resolved RF imagery of precipitation in clouds, surface wakes on the ocean and detailed topography of Earth and other planetary solid surfaces. Like passive microwave systems, active RF systems operate at much longer wavelengths than visible and infrared systems that are routinely used for imagery—typically at centimeter wavelengths. Consequently, for practical payload sizes, active RF systems tend to operate at much coarser spatial resolution than visible and infrared systems—but they can operate through almost any possible weather condition. For example:

- Precipitation radar (PR) onboard the TRMM satellite. The TRMM PR launched in 1997 was the first spaceborne radar designed to measure vertical structure of rainfall precipitation in the lower atmosphere. It operated at 13.8 GHz (2 cm in wavelength) using a 128-element electronically scanned array to allow fast and complex cross-track scanning over a swath width of 215 km with a cross-range spatial resolution of about 4.3 km and a range resolution of 250 m. As shown in Fig. 17-1, the TRMM PR operated in conjunction with other observational payload systems—a passive microwave imager and a visible-infrared imager

- Systems (like TRMM PR) are very useful, but they cannot provide spatial resolution approaching visible-infrared systems with feasible collector sizes. A different type of radar takes advantage of

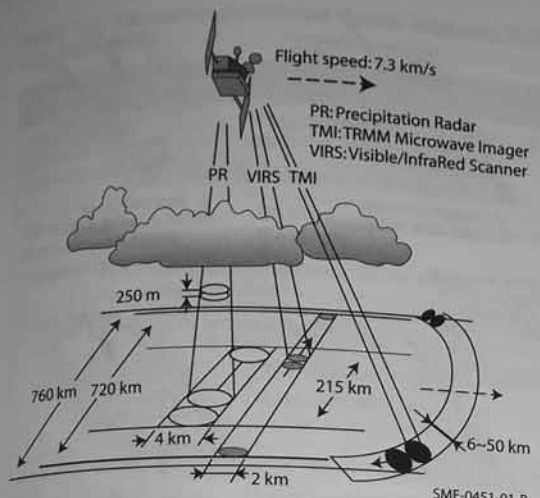


Fig. 17-1. The TRMM PR observed rainfall observation in conjunction with a passive microwave imager, the TMI, and a visible-infrared imager called VIRS. While the PR and VIRS made cross-track scans, the TMI made a conical scan, similar to the SSM/I and SSMI/S.

the relatively relaxed tolerances at these long RF wavelengths for coherent phase and Doppler shift measurements of a radar return signal

- Synthetic Aperture Radar (SAR) processes coherent measurements of phase and Doppler shift across the wide beamwidth of a RF system to synthesize a virtual aperture with a size equivalent to the distance the physical antenna moves while the scene location remains illuminated by the beam. The corresponding spatial resolution in the along spacecraft track direction approaches resolutions achievable with visible and infrared systems.

- SEASAT launched in 1978 was the first known SAR in space. It operated in L-band at a frequency of 1.275 GHz or 23.5 cm in wavelength. SEASAT operated for only about 105 days, failing when a short circuit in the satellite's electrical system ended the mission. Nevertheless, the mission was very successful and laid the processing foundation for following systems

- Since SEASAT, SARs have been developed and flown by Canada, Japan, Europe, and Russia. TerraSAR-X, an X-band (9.65 GHz) SAR developed by the German Aerospace Center (DLR) and launched in 2007 has produced remarkable imagery that is especially useful for urban landscapes. Along with its partner satellite, TanDEM-X, launched in 2010, high resolution digital elevation maps are being produced for flood plain maps and traffic monitoring among many other applications

To determine the radiometric performance of an optical instrument, we must consider the observation geometry, effective aperture, integration time, detector sensitivity, spectral bandwidth, ground pixel size, and the transmission through the atmosphere. For passive Earth observations, we define three basic categories, depending on the spectral range

Wavelengths covered in the first case range from 0.3 to 2.5 μm, and they include the ultraviolet, visible, near and short wavelength infrared. The instrument receives radiation reflected from the atmosphere, as well as from the surface of the Earth, when illuminated by the Sun. We will normally neglect thermal radiation emitted by the Earth in this case

The second case applies to wavelengths ranging from the long infrared region (8–14 μm) through the microwave and RF. In these wavelengths, instruments receive thermal radiation emitted from the atmosphere and the surface of the Earth when reflected radiation from the Sun is negligible

The mid-wavelength infrared spectral region (3–8 μm) is our final case, where contributions from both emissive and reflected sources must be considered. The radiance from emissive and reflected sources is shown in Fig. 17-2. It can be seen that the thermal emitted radiance of the Earth increases with wavelength, corresponding to Planck's law, and that the reflected radiance from the Earth's surface decreases with wavelength

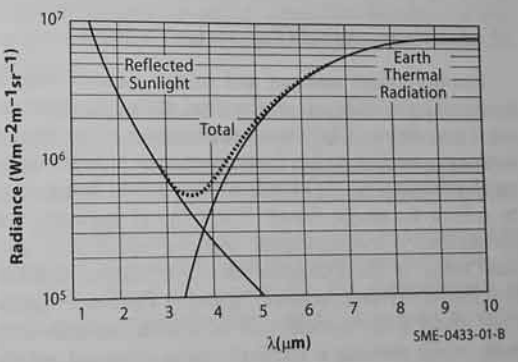


Fig. 17-2. Radiance from Direct and Reflected Sources. Radiance contribution in Watts per square meter per meter (wavelength) per unit solid angle of reflected sunlight from the Earth and emitted radiation from the Earth as a function of wavelength. The sum is shown as a dashed line. The Sun is modeled as a blackbody with a temperature of 6,000 K, the reflection coefficient of the Earth's surface and the transmission of the atmosphere are modeled as constants for clarity.

In the ultraviolet through SWIR bands, we resolve images produced by energy (chiefly from the Sun) reflected from the target scene rather than energy from the relatively low self-emission that occurs in the reflectance bands. But in the infrared, we see things almost entirely by their self-emission, with very little energy being reflected, particularly at night

We may use the same optical train elements —lenses, prisms, mirrors, and filters—to collect

Table 17-1◀, Fig. 17-2, Eq. 17-0

infrared energy as for visible and UV, but we must apply them differently. For example, ordinary glass is opaque to IR beyond 3 μm, whereas germanium, which is opaque in the visible band, is transparent in the 1.8 to 25 μm region

We must consider atmospheric scattering caused by molecules, aerosols and other particles in the air. The amount of scattered radiation from atmospheric molecules is a function of the inverse fourth power of the wavelength

For haze, the situation is more complex, but the scattering extinction coefficient goes roughly as the inverse first to second power of the wavelength. Thus, IR penetrates haze and dust much better than visible radiation because the IR wavelengths are four or more times those in the visible spectrum. The same phenomena explain the reddish color of the sky near dawn and sunset. At these times, shorter green, blue, indigo, and violet wavelength signals are greatly attenuated as they travel farther through the atmosphere than when the Sun is overhead

The following sections provide more detailed information on the best known observation payloads— those that use electro-optical techniques in the solar reflectance and emissive spectral regions to observe and measure physical properties of scenes in the ultraviolet through submillimeter spectral region, and those that use radio frequency technology.

17.1.1 Electro-optical Observation Payloads

Electro-optical imagers and remote sensors scan a scene using mechanical or electrical means. Geostationary Earth Orbit (GEO) spacecraft appear to be almost stationary relative to the Earth, so that an optical instrument operating in GEO needs to scan in two dimensions to collect an image frame. The original approach for geostationary environmental imaging spacecraft, still used today in the European Meteosat Second Generation (MSG) system, involves spinning the entire spacecraft about a north-south axis to enable the east-west scan, while moving a relatively large elliptical mirror (~80 cm along the major axis) in front of the imager optics to perform the north-south scan. Three axis stabilized spacecraft in geosynchronous orbit like the US GOES environmental satellite system have imagers that use a two axis scanner in front of the imager collection optics to scan the scene in two dimensions. Alternatively, large format two dimensional focal plane array and optical technologies for advanced missile warning sensors have been used to create a full-Earth staring imager that requires no mechanical scanner to sample the entire Earth scene.

The sub-satellite point for a spacecraft in low Earth orbit (LEO) moves with respect to the scene on the ground at a speed of approximately 7 km/s (See Chap. 8). In most LEO systems, the motion of the spacecraft is used to eliminate scanning in the *ground track* direction (except for spacecraft motion and image rotation corrections), so that the scanning system of the optical instrument need move only in the *cross-track* direction. Several methods for doing so are described below. Illustrated in Fig. 17-3A, *whiskbroom sensors* scan a single detector element on the ground in the cross-track direction, thereby creating a single scanline of pixels in the final image or other data product. Whiskbroom scanners can also use multiple detectors, to create multiple scanlines of pixels simultaneously (Fig. 17-3B). This technique reduces the required scan rate compared to a system with a single detector. The required scan rate is reduced so that the dwell time per detector sample is multiplied by the number of detector elements used. *Push broom scanners* cover a full swath width using a linear arrangement of detector elements called a linear array, line array, or scanning array. As illustrated in Fig. 17-3C, the read-out process delivers one line after another in the ground track direction, like pushing a broom, hence the name. The integration time is defined by the velocity of the sub-satellite point and the size of the pixels on the ground. To improve signal-to-noise ratio of the image, at the expense of area coverage rate, push broom imagers or their host spacecraft can compensate for motion of the sub-spacecraft point along the Earth's surface to increase effective dwell time.

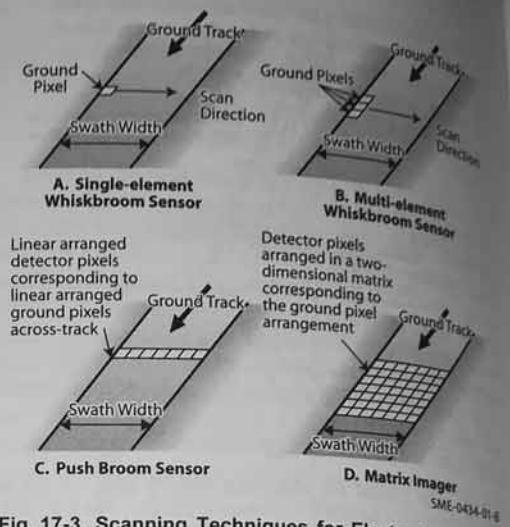


Fig. 17-3. Scanning Techniques for Electro-Optical Instruments. See text for discussion.

Step-and-stare scanners use a 2-dimensional array of detector elements to collect data over an entire region of interest by staring at part of the scene and then stepping to another location in the scene in a predetermined pattern of steps and staring again, until the region is completely covered. Each detector element corresponds to a spatial resolution element on the ground. These systems operate in two basic modes in LEO. In the first mode, integration times are determined by velocity of the sub-satellite point and size of the ground pixel, as in the pushbroom sensor. No advantage is achieved with

Table 17-2. Comparison of Optical Sensor Scanning Methods. We list relative advantages and disadvantages of different scanning mechanisms

Scanning Technique	Advantages	Disadvantages
Whiskbroom Scanner—Single Detector Element	High uniformity of the response function over the scene Relatively simple optics	Short dwell time per pixel High bandwidth requirement and time response of the detector
Whiskbroom Scanner—Multiple Detector Elements	Uniformity of the response function over the swath Relatively simple optics	Relatively high bandwidth and time response of the detector
Push Broom Sensor	Uniform response function in the along-track direction No mechanical scanner required Relatively long dwell time (equal to integration time)	High number of pixels per line imager required Relatively complex optics
Step-and-Stare Imager with Detector Array	Well defined geometry within the image Long integration time (if motion compensation is performed)	High number of pixels per array imager required Complex optics required to cover the full image size Calibration of fixed pattern noise for each pixel Highly complex scanner required if motion compensation is performed

respect to the integration time in the simplest case. A short integration time is needed to avoid image smear during read-out. The second mode allows a longer integration time. If we shift the image array in the focal plane or move the line of sight of the instrument, the image motion can be compensated to very low speeds relative to the ground. The instrument line of sight can be adjusted by moving a scan mirror or by tilting the spacecraft itself to compensate for the movement of the sub-spacecraft point. Step-and-stare sensors require relatively complex scanner optics or spacecraft operations to cover large areas at high signal to noise ratio from low Earth orbit, especially. The fixed pattern noise must be removed from the image because each pixel has a somewhat different responsiveness and dark signal, creating additional complexity. Table 17-2 summarizes the distinguishing features of optical scanning methods.

In GEO and other high altitude orbits, another approach to capturing a 2 dimensional scene can be used that involves positioning the scene with respect to the instrument. In this case, the instrument moves to the next part of the scene between integration periods either along-track or across track to image the entire scene. This scan motion can be continuous across a large swath, as in the geosynchronous GOES and MTSAT imagers. If this shift occurs in a series of discrete steps, this approach is referred to as a step-and-stare imager. This technique is used in geosynchronous orbit in the GOES Sounder, for example. If the instrument field of view only covers a part of the required scene, then moderately complex (and also moderately sized) optics are required. We can use highly agile and accurate pointing mirrors in front of the instrument's optics to adjust the line of sight.

For space missions, optical instruments often rely on existing designs. Custom tailoring of current optical and detector design approaches are common in optimizing the performance and cost of the instrument. Let's follow a scene photon through an observational payload by considering first the optical subsystem involving a telescope and a spectral separation approach and then the detectors.

Telescopes. This discussion considers the telescope optics used to collect and transform light from the scene onto a focal plane along with approaches used to separate the incoming light into different wavelength bands.

We treat reflective telescopes (telescopes using mirrors) and refractive telescopes (telescopes using lenses) together. There are advantages and disadvantages to both these telescopes that will be discussed in detail later.

There are several ways to describe a telescope. Parallel rays of light falling on a perfect lens will all converge at the focal point, whose distance from the lens is called the *focal length, f*. The focal length largely determines the length of the optical collection system and, for a single lens, it is related to the lens surface's radius of curvature. The focal length of a spherical reflecting surface is one-half its radius. For a parabolic reflector whose y-z surface is defined by the equation $z^2 = 4fy$, the quantity f is the focal length; it equals the distance from the focus to the nearest point on the reflecting surface.

In design practice, we normally determine the required focal length based on instantaneous field of view or angular spatial sampling size and the physical size of detector elements in the image plane. The plate scale, s, or length per field-of-view angle is given by:

Table 17-2, Fig. 17-3◀, Eq. 17-0

$$s = f \text{ (unit length/rad)}$$
$$= \left(\frac{\pi}{180}\right) f \qquad (17\text{-}1)$$
$$\approx 0.01745 f \text{ (unit length/rad)}$$

where s and f are in the same units. The image size is a function of s and the size of the detector— ranging from a single element to a large array—employed at the focal plane. As Fig. 17-4 shows, the focal length needed to record an object or scene of radius R, is given by the magnification, M,

$$M = \frac{f}{h} = \frac{r_d}{R} \qquad (17\text{-}2)$$

where h is the distance from the spacecraft to the object, r_d is the radius of the detector array in the image plane, and R is the radius of the object, with the image and object measured perpendicular to the line of sight. The magnification or scale, r_d/R, is the ratio of the image size to the object size. It is ordinarily a very small number for space-based observation payloads. We express the scale on the image plane as "1 μm equals × km on the ground."

We can also describe an optical element or system by its so-called infinity F-number or F-stop, often written as f/, F, F No., or F#. It is defined as f/D, where D is the effective entrance aperture, which is the effective diameter of the lens or primary telescope mirror. Image brightness is proportional to F^{-2}, so an $f/4$ lens gives an image four times brighter than an $f/8$ lens.

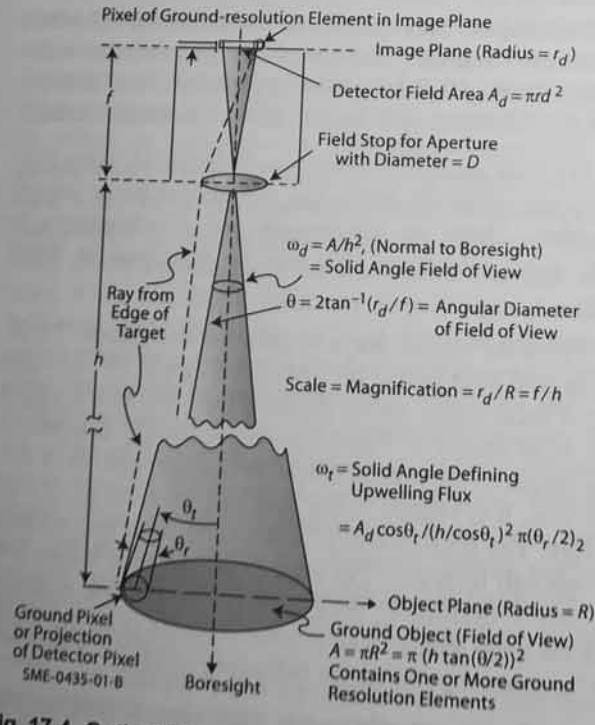


Fig. 17-4. Optical Characteristics of a Refractive System. Note one-to-one correspondence of the ground-resolution element's size to the pixel size at the image plane. The operating wavelength is λ. As resolution elements move away from nadir, flat-Earth approximations become less precise.

Table 17-3, Fig. 17-4, Eq. 17-4

The numerical aperture, NA, gives the same information in another way:

$$NA \equiv \frac{1}{2F\#} = \frac{D}{2f} \qquad (17\text{-}3)$$

or

$$F\# = \frac{1}{2NA} = \frac{f}{D} \qquad (17\text{-}4)$$

The largest numerical aperture for optics used in air is 1. Thus, the smallest F# is 0.5.

Table 17-3. Principal Aberrations in Optical Systems. See Table 17-4 for which of these are mitigated in various optical systems.

Chromatic Aberration is dispersion of the light due to the refractive index of a lens being a function of the wavelength. Causes different colors to focus at different distances.
Spherical Aberration is dispersion in which light from the periphery of a spherical lens or mirror is focused nearer the element than light from the center. Can be eliminated by making the optical surface parabolic, rather than spherical.
Coma is dispersion of off-axis portions of the image. (So named because in a telescope, off-axis star images look liketear drops, or the coma of a comet pointing toward the center of the image.)
Astigmatism is aberration in which the distorted image is asymmetric such as when light in a horizontal plane comes to a slightly different focus than light in a vertical plane. A common problem in human vision.
Distortion is when an otherwise sharp image is distorted in shape, such as when straight light on the surface being viewed appear curved on the focal plane. An uncorrectable distortion occurs when trying to image the celestial sphere onto a flat focal plane. (See Sec. 8.1.)
Curvature of Field is when a sharp image is formed on a focal surface which isn't flat. Can be corrected in film systems by using a slightly curved focal plane.

All telescopes experience *aberrations*, or imperfections in the quality of the image, in addition to diffraction which limits system resolution as discussed below. Principal optical aberrations are listed in Table 17-3. *Chromatic aberration* is wavelength or color dependent. This imperfection is due to various wavelengths being bent by different amounts when passing through the lens. As a result, only systems with at least one refractive element will suffer chromatic aberration. This is due to the fact that reflective surfaces treat all wavelengths the same, which is not absolutely true since some surfaces will reflect visible light, for example, but not X-rays. However, when reflection does occur, it is independent of wavelength to the first order.

Three basic types of telescopes are shown in Fig. 17-5. In each of these telescope approaches, there is a corresponding refractive and reflective design. Table 17-4 shows the aberrations that can be corrected in these telescopes. The lens doublet is the classic refractive telescope lens and can be designed to elimi-

Table 17-4. Aberrations that can be Corrected by the Three Basic Optical Systems. Checks indicate errors that are fully correctable and parenthetical checks indicates that corrections are possible only for dedicated design parameters. (See Table 17-3 for definitions.)

Image Error	Doublet/ Schmidt	TeleOptic/ Cassegrain	Triplet/ TMA
Lateral Chromatic Aberration	✓	✓	✓
Length Chromatic Aberration	✓	✓	✓
Spherical Aberration	✓	✓	✓
Coma	✓	✓	✓
Astigmatism		(✓)	✓
Distortion	(✓)	(✓)	✓
Curvature of Field		(✓)	✓

nate spherical aberrations, coma, distortion and chromatic aberrations. In tele-optic lens systems, the distance between the optical element and the focal plane is shorter than the focal length. Spherical aberrations, coma, astigmatism, curvature of field effects, and even chromatic aberrations can all be eliminated by tele-optic lenses. The simplest refractive (spherical) optical system that theoretically allows for correction of all distortions is the lens triplet. While this approach offers many important advantages, it can be relatively challenging to implement because each of the optical elements is highly sensitive to displacement or tilt. The ray traces in Fig. 17-5 show different locations of the image in the focal plane relating to various viewing angles. All five of the third-order aberrations as compensated by the lens triplet include: spherical aberrations, coma, astigmatism, curvature of field, and distortion. Similar to tele-optics, it is also free from chromatic aberrations. The same behaviors exist in the corresponding reflective systems as well. The Cassegrain telescope is a reflective implementation of a tele-optic lens, and the Schmidt Mirror System is an all-reflective doublet. The Three-Mirror Anastigmatic system is similar to the lens triplet with respect to all the aberration corrections, but with an all-reflective design. Generally, reflective optical systems are free from chromatic aberrations. However, reflective systems typically have a much smaller field of view than their refractive counterparts.

Space remote sensing telescopes tend to be more complex than ones used on the ground. This is due in part to effects such as thermal distortions and radiation which can alter performance of instruments and cause space-based telescopes to fall short of meeting performance requirements over the intended lifetime of the instrument. These environmental effects and the challenging image quality and spectral coverage requirements for many systems limit the technologies for manufacturing the lenses and mirrors. Even if operating temperature ranges are controlled to within a few deg, thermal distortions can still limit the performance of the optical system. Cosmic radiation effects can degrade the transparency of most optical glass over time. Figure 17-6 shows the cross sectional view of the high resolution optical lens system, a German-built Modular Optoelectronic Multispectral Scanner (MOMS 2P) designed to achieve a resolution of 6 meters on the ground.

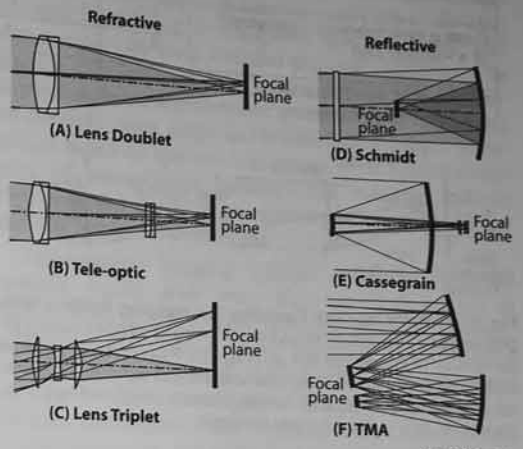


Fig. 17-5. Basic Configurations for Refractive and Reflective Optical Systems. Each of the reflective systems on the right is analogous to the corresponding refractive system on the left. TMA = Three-mirror anastigmatic.

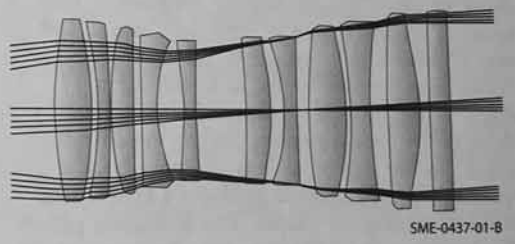


Fig. 17-6. Lens Cross Section of the Panchromatic Objective of the MOMS 2P Instrument. The sensor has a focal length of 0.66 m and an aperture size of 0.15 m. The complexity of this optical system is representative of sophisticated remote sensing payloads.

The angular resolution of an optical system determines its ability to enable recognition of fine spatial detail in Earth observation data. A telescope that can just barely resolve two close stars is said to have a resolving power equal to the angular separation of those two stars. We are more interested in the ability to distinguish fine detail on the surface for Earth observing systems. Thus, for these systems, spatial resolution is normally expressed as the smallest physical separation of two objects on the surface that can be resolved by the instrument.

Table 17-4, Fig. 17-6, Eq. 17-4◀

Live Calc

Table 17-5. Diffraction-Limited Resolution. Note that the Synthetic Aperture Radar provides resolutions similar to visual or IR systems, independent of range and wavelength for ranges up to the maximum signal-to-noise limit by synthesizing the required aperture.

		Ground Resolution = $h\lambda/D$		
	Aperture Size, D	Visible (λ = 0.5 μm)	IR (λ = 3 μm)	Passive Microwave (f = 10 GHz; λ = 3 cm)
From an orbiting spacecraft (h = 900 km)	1 m	0.45 m	2.70 m	27.0 km
	3 m	0.15 m	0.9 m	9 km
From a synchronous spacecraft (h = ~35,800 km)	1 m	17.9 m	107 m	1,074 km
	3 m	6.0 m	35.8 m	358 km
From SR-71 (h = 20 km or 70,000 ft)	0.3 m	0.033 m	0.200 m	2.00 km

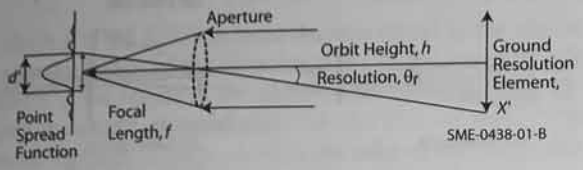


Fig. 17-7. Point Spread Function for Imaging System with Diffraction. The optical wave front from an ideal point source on the ground is imaged as the point spread function by the optical system. The diameter of the aperture and the wavelength determine the extent of the point spread function measured by the diameter, d', of the first intensity minimum.

For a perfectly designed and fabricated optical system, the theoretical limit to angular resolution performance is determined by diffraction, which is the apparent bending of light occurring whenever light encounters an obstacle, such as an element in a telescope or other optical system. Diffraction blurs the image of a point source of light, such as a distant star, into a pattern with elements that get progressively fainter away from the center, as shown in Fig. 17-7. The shape of the diffraction pattern is determined by the shape of the optical element causing the diffraction, so that a circular mirror or lens produces circular rings, whereas a square aperture produces a pattern with 2-D series of rectangular shapes. The diffraction pattern is also known as the diffraction disk, the Airy disk (for a circular aperture) or the point spread function. For a circular aperture, the angular distance, θ_r, from the maximum at the center of a point source image to the first dark interference ring, is called the Rayleigh limit and is given by

$$\theta_r = \frac{1.22\lambda}{D} = \frac{1.22}{f\,F_{cutoff}} \tag{17-5}$$

where λ is the wavelength, D is the aperture diameter of the optical instrument, f remains the focal length, and F_{cutoff} is the optical cutoff spatial frequency in units of inverse length (say m^{-1}) that limits the resolution of the image in the focal plane. θ_r is expressed in radians. The bright lobe at the center of this Airy disk diffraction pattern contains 84% of the total energy arriving at the focal plane from a point source, from its center out to the first minimum. For a satellite at altitude, h, the linear resolution or ground resolution limit, X', at nadir as defined by the cutoff frequency is

$$X' = \frac{h\lambda}{D} \tag{17-6}$$

A more general expression is to replace h with the slant range, R_s, from Eq. (8-31), to determine the resolution away from nadir (R_s here = D in Chap. 8). Keep in mind that this is the resolution perpendicular to the observation system line of sight and that actual spatial resolution on the surface of the Earth is degraded by the effect of geometric projection or 1/sin ε, where ε is the elevation angle at the specific spacecraft and Earth locations of interest, obtained from Eq. (8-28). Ground resolution at nadir for several representative system configurations is given in Table 17-5. Most space-based optical systems have the ability to produce images with angular resolution as good as the theoretical limit and are therefore described as being diffraction limited.

When we implement an optical system using a detector array, we add an additional design parameter, the quality factor, Q, defined as the ratio of the physical size in the focal plane corresponding to the spatial frequency cutoff to the detector element size, d, that is

$$Q = \frac{d'}{d} = \frac{X'}{X} \tag{17-7}$$

where d' is given by

$$d' = \frac{\lambda f}{D} \tag{17-8}$$

X is the ground pixel size, and X' is the ground resolution corresponding to d' in the focal plane (See Fig. 17-8). Q typically ranges from 0.4 to 2. For $Q > 1$, the detectors are smaller than the cutoff and resolution is limited by diffraction in the optics. This gives the best possible image resolution for a given aperture. For $Q < 1$, the resolution is limited by detector size. This design point may be chosen if image quality is less important than aperture size, as would be the case, for example, when increased light gathering power is required. A good starting point is $Q = 1$, which allows good image quality.

From the definition of the magnification, Eq. (17-2), we have:

$$\frac{d}{X} = \frac{d'}{X'} = \frac{f}{h} \tag{17-9}$$

Combining Eqs. (17-7) to (17-9), we obtain expressions for the detector size, d, in terms of the other basic system parameters:

$$d = \frac{d'X}{X'} = \frac{d'}{Q} = \frac{\left(\frac{\lambda f}{D}\right)}{Q} \tag{17-10}$$

where the parameters are defined above and, as usual, λ is the wavelength, f is the focal length, and D is the aperture diameter.

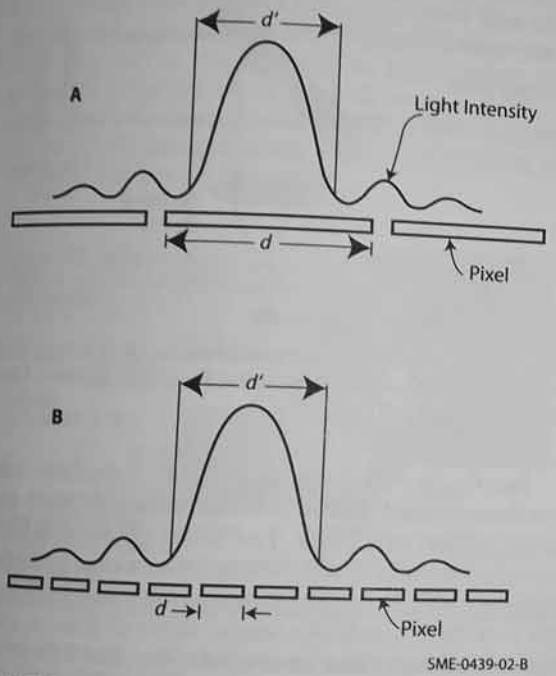


Fig. 17-8. Effect of Varying Quality Factor. Different sizing of the detector pixel with respect to the point spread function is shown by varying the Quality Factor, Q. A low quality factor results in the relative sizing in diagram (A) on top and a large quality factor results in the relative sizing in diagram (B) on bottom.

Spectral separation approaches. As noted above, space-based observation payloads operate with different spectral sampling approaches, depending on intended mission. These spectral sampling approaches fall into four broad categories summarized below called panchromatic, multispectral, hyperspectral and ultraspectral.

Panchromatic or PAN spectral sampling involves one band per broad spectral region (e.g., one band covering the entire visible part of the spectrum). PAN imaging is useful for collecting high SNR, high spatial resolution imagery where spectral content of a scene is considered less important than image sharpness and clarity. Well known systems using PAN imaging include the commercial imaging GeoEye and DigitalGlobe systems.

Multispectral Imaging (MSI) uses ~3–50 relatively wide ($\lambda/\delta\lambda$~10 typically) spectral bands to identify major features of a scene (rocks, trees, crops, water). Well known MSI examples include Landsat, AVHRR, GOES imager and sounder and MODIS.

Hyperspectral Imaging (HSI) systems collect ~30–300 moderately high spectral resolution ($\lambda/\delta\lambda$~100) spectral bands over a contiguous spectral region. HSI systems provide more detailed physical information on a scene than PAN and MSI system and are used to identify components within major scene features (e.g., types of minerals, trees, crops and health, bathymetry, composition) based on spectral signatures. Well known examples of space-based hyperspectral systems include, Hyperion and the TES instrument that mapped Mars at thermal infrared wavelengths.

Ultraspectral Imaging systems collect ~300–3,000 spectral bands with very high spectral resolution ($\lambda/\delta\lambda$~1,000) across a broad contiguous spectral region. Ultraspectral systems are used primarily in the infrared to collect detailed spectra of the Earth's atmosphere and surface for physical analysis of gas absorption features for trace gas detection and mapping and retrieval of vertical temperature structure and distribution of water vapor in the atmosphere. Well known ultraspectral systems include AIRS, MIPAS and IASI.

Spectral separation occurs in a device called a spectrometer that can use filters, dispersive elements or interferometers to isolate spectral ranges of light for remote sensing and imaging. PAN and MSI systems most often use filters to separate out desired ranges of wavelength for measurement whereas HSI and ultraspectral systems almost always use dispersive or interferometric approaches but can use filter spectrometers. Some systems sample the Earth with a hyperspectral approach and then process data onboard the spacecraft to MSI bands for transmission to Earth, as illustrated in Fig. 17-9.

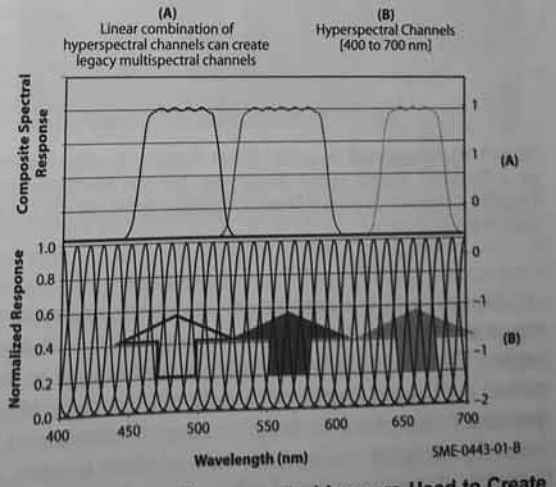


Fig. 17-9. Data from Hyperspectral Imagers Used to Create Virtual Multispectral Channels. (Courtesy of NASA)

Table 17-5◀, Fig. 17-9, Eq. 17-10

Filter spectrometers use optical bandpass filters to transmit narrow wavelength bands from the scene spectrum to an individual detector, or a 1-D or 2-D focal plane detector array. Early optical filters were based on stacks of colored glass that defined spectral bandpasses. Virtually all optical filters used today are based on an interferometric device called an etalon to define high tailored spectral responses precisely. Multispectral filter spectrometers can use discrete filters (one or more per focal plane), filter wheel mechanisms to rotate different filters into a detector line of sight or integrated spectral filter assemblies also known as butcher block assemblies (Fig. 17-10) to cover ~10 spectral bands in a single focal plane. Hyperspectral filter spectrometer approaches include tunable filters and spatially variable filters such as the wedge filter that can collect data as a pushbroom imager. Advantages of filter spectrometers include relatively high optical throughput and relatively easy implementation for multispectral applications, especially. One of the disadvantages of filter spectrometers is that high spectral resolution filters with predictable characteristics can be very expensive and difficult to produce consistently. Filter spectrometer examples include all of the best known MSI systems (e.g., MODIS, GOES, AVHRR) and the hyperspectral Linear Etalon Imaging Spectral Array (LEISA) on the New Horizons to Pluto.

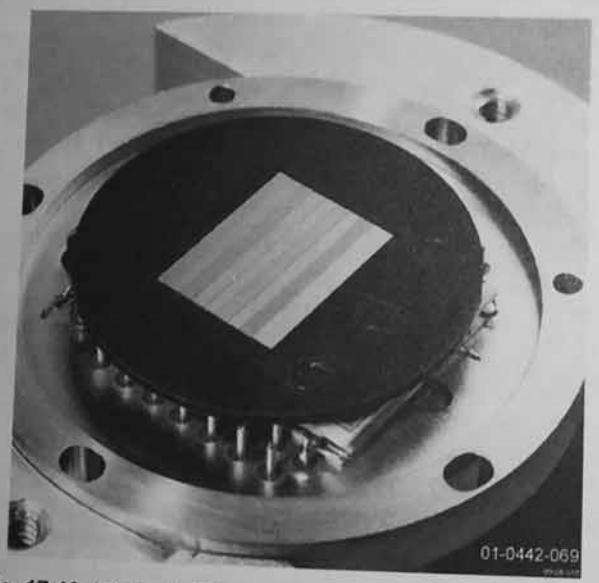


Fig. 17-10. Integrated Spectral Filter Assembly also Known as a Butcher Block Filter Assembly for the Multispectral THEMIS Imager On-board Mars Odyssey 2001. (Photo Courtesy of Raytheon)

Dispersive spectrometers use a grating or prism to separate light at different wavelengths into distinct angles. The spectrum of a single ground sample is dispersed and then focused onto different detectors along one dimension of a 2-D detector array. The spectrum is typically sampled at Nyquist rate so that number of detectors along the spectral axis is twice the number of points in reported spectrum. Advantages of dispersive spectrometers relative to Michelson interferometer approaches discussed in the next paragraph include superior performance in the SWIR and in that part of the MWIR where noise is photon limited and no moving parts in the spectrometer except possibly for a mechanical cooler. Disadvantages include sensitivity to temperature changes that can affect the focal length and therefore spectral calibration and radiometric accuracy and a tendency toward relatively large complex instruments especially for missions requiring high spectral resolution and wide field of view Earth coverage. Well known examples of space-based dispersive spectrometers include Galileo-NIMS, AIRS, and Hyperion.

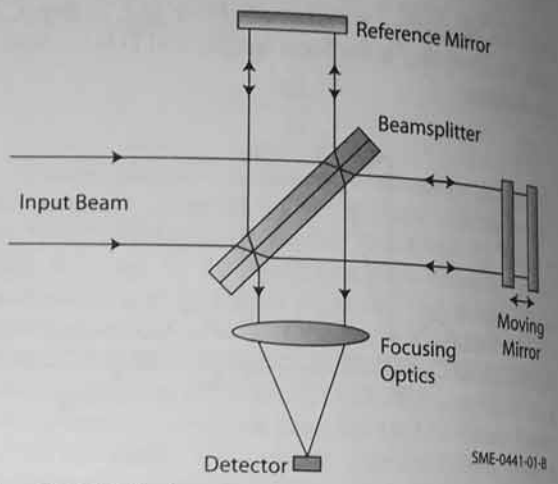


Fig. 17-11. Michelson Interferometer, the Most Widely Used Form for a Temporally Modulated Fourier Transform Spectrometer.

Interferometers achieve spectral separation using constructive and destructive interference of light from the same part of a scene. Two broad classes of interferometers flown in space include Fourier transform spectrometers (FTS) in which the optical path difference between two interfering beams varies in time or space and multibeam interferometers like the Fabry-Perot interferometer in which multiple beams from the same scene element interfere to select a series of narrowband transmission ranges from which one is selected using another etalon or blocking filter. Nearly all optical filters used in MSI systems are based on a special type of Fabry-Perot interferometer (FPI) called an etalon. The best known interferometer used in space is the temporally modulated FTS (TFTS) implemented frequently with a design approach similar to what is shown in Fig. 17-11. In principle, a temporally modulated FTS can measure all optical frequencies simultaneously. In reality, practical limits such as detector sensitivity cutoffs, optical coatings and electronics frequency ranges restrict spectral range. The moving element in the TFTS and the resulting variation in optical path difference modulates intensity of optical frequencies across a broad spectral range in front of a single detector for each pixel that converts received optical signal into an elec-

trical signal. The resulting plot of detected irradiance versus time (corresponding to changes in optical path difference) is called an interferogram. The interferogram is converted to an optical spectrum using a Fourier transform. General advantages of interferometer approaches include simplest and most compact instrument approach for high spectral resolution and coverage applications (especially for the FPI) and superior performance for detector noise limited TFTS instruments, because detector noise is multiplexed across the spectrum. Disadvantages for the TFTS include reliability concerns because of the moving elements in the interferometer and the so-called multiplex disadvantage for photon noise limited instruments in which photon noise from the broad spectral band observed by the TFTS in every interferogram sample is multiplexed into each spectral channel by the Fourier transform. FPI disadvantages include relatively low optical throughput, especially at high spectral resolution, which leads to larger collection optics than other spectrometers and limited spectral range for each etalon which can lead to requirements for many etalons in a system, to cover the broad spectral range required for some missions. Space-based interferometer examples include TFTS instruments like IASI, TE,S MIPAS, and FPI-based instruments like HRDI.

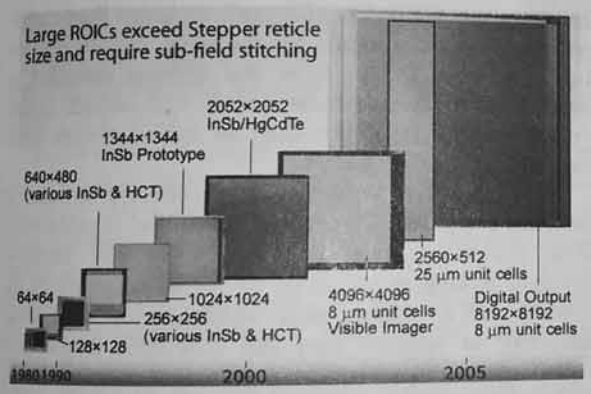


Fig. 17-12. Improvement in Detector Array Size and Readout Rate from 1980 to 2005.

Detectors. We make a distinction between discrete detectors consisting of one or a few light-sensitive elements with discrete electronics and detector arrays, which consist of a large number (hundreds to tens of millions) of tightly packed light-sensitive elements combined with a readout integrated circuit (ROIC) to recover the measured signal from each detector element with minimum complexity. As shown in Fig. 17-12, since the 1980s, detector technology has taken advantage of improvements in integrated circuit design and fabrication capabilities and advanced rapidly from relatively small format ~64 × 64 detector array designs to much larger format (~8,192 × 8,192) digital output approaches.

We must select the materials used for detector elements depending on the spectral range of the instrument being designed. The ability of photon detector elements to absorb photons relates directly to the energy of the incident photons (and consequently to the wavelength of the radiation as well) and the effective band gap of the material. Detector material generates random noise that reduces sensitivity of the instrument. This noise increases with operating temperature of the detector. Therefore, frequently, we must lower the temperature of the detector elements so that the self-generated noise does not degrade the signal-to-noise ratio of the instrument. This requirement becomes more stringent as the wavelength of the radiation being detected increases. With few exceptions, detectors have to be cooled for wavelengths in the short wave infrared (SWIR) and longer. All matter, at temperatures greater than absolute zero, generates thermal photons. Thus, frequently, for infrared imagers, the optics and interior structures of the instrument have to be cooled to reduce interference from light radiated by the instrument itself.

For the spectral range between 400 nm and 1,100 nm, silicon detectors and imagers are used most frequently. Silicon is attractive because it is possible to combine the detector elements and read-out electronics in a single monolithic chip. Furthermore, silicon substrates are available in larger size than other materials, which enables larger format devices. We can produce 1-dimensional line arrays or 2-D area arrays with a very large number of elements (e.g., 10,000 × 10,000 and greater) with silicon. Incident photons are converted to an electrical output signal in the array. For charge-coupled device (CCD) arrays with read-out electronics, the process begins when incident photons are converted by each detector element into electrons according to a conversion efficiency dictated by the characteristic spectral response. The electrons are then collected for the entire integration time for each detector element. The read-out of the charge generated—up to millions of electrons per detector element—is performed via an analog shift register across the arrays into a serial output port. 2-D CCDs may have more image smear relative to 1-D arrays during frame transfer. Also, in 2-D monolithic array designs, detector fill factor is generally lower than in 1-D arrays or in so-called hybrid arrays in which a detector array is bonded to a separate ROIC. Figure 17-13 shows a typical spectral response function for a silicon imager.

When radiometric performance of the optical instrument is paramount, we use time delay integration (TDI) methods. TDI describes an imaging principle that uses the image motion along the rows of a 2-D detector array to extend the integration time. Integration time is extended by electronically shifting the integrating pixel cell synchronously to the movement along the row. The signal-to-noise ratio of this concept is improved by the square root of the number of TDI stages. The primary advantage of TDI imager systems compared to line imagers is the improved signal-to-noise ratio. The disadvantage is the increased requirement for spacecraft attitude and orbit stability (due to the required synchronization of the shifting pixel). Some TDI sys-

Table 17-5◀, Fig. 17-12, Eq. 17-10◀

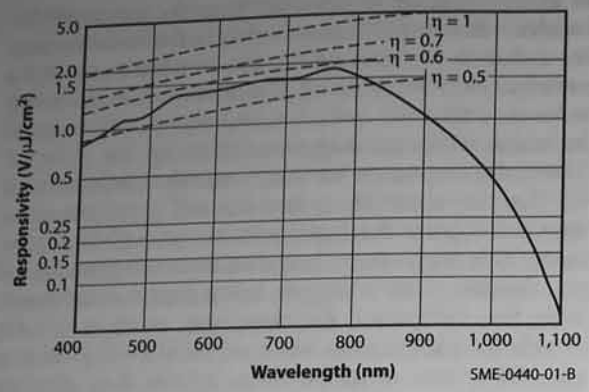


Fig. 17-13. Spectral Response Function of a Silicon Line Imager. The spectral response is shown in terms of the output voltage resulting from illumination by energy density vs. the wavelength from 400 to 1,100 nm. The imager reaches a quantum efficiency (generated electrons per incident photon), denoted by η, above 50% between 400–800 nm. Outside this wavelength region the quantum efficiency drops to small values making the imager less suitable above 900 nm.

tems perform the integration in software by adding together measurements from the different detectors that view the same part of the scene.

We classify and select infrared detectors according to their spectral band of operation and a figure of merit called specific detectivity for photon detectors. The operating temperature of the detector dictates the cooling requirements for the sensor focal plane. Infrared sensors often have non negligible time constants for response with respect to integration time. The size of infrared detector arrays is ultimately limited by the size of substrate material on which the detectors can be grown. Recent work at Raytheon has produced infrared arrays with format sizes up to 4,096 × 4,096. We detect infrared wavelengths with thermal detectors or photon detectors. Thermal detectors exploit the fact that absorbed heat raises the temperature of the detector, which changes its electrical characteristics. The advantage of thermal detectors is relatively uniform response with respect to wavelength. Thermal detectors can also be operated at ambient temperatures, although they have lower sensitivity and much slower response times, which can limit their applicability in whiskbroom scanners, for example. Photon detectors use absorbed photons to generate charge carriers. These systems offer the advantages of higher sensitivity and shorter time response, but they must be operated at low temperatures.

Infrared detectors are often rated by the specific detectivity, D^*, given by

$$D^* = \frac{\sqrt{A\Delta f}}{NEP} \qquad (17\text{-}11)$$

where A is the detector area, Δf is the noise equivalent bandwidth, and NEP is the noise equivalent power of the detector. The factor D^* is strongly wavelength dependent showing its peak value at the cut-off frequency. See the book website for a list of IR FPA manufacturers.

Get More

The selection of a detector or detector array is driven by several factors. The primary design issues center on maximizing detectivity in the spectral band of interest while operating at the highest possible temperature and a sufficiently small time constant. In addition, we must consider the geometry of the detector and the array as well as associated calibration issues.

17.1.2 Radio Frequency Observational Payloads

Radio frequency observation payloads can scan the Earth in ways that are analogous to electro-optical payloads. Likewise, systems for collecting or transmitting optical and RF signals operate exactly the same in theory with collecting, reflecting and focusing light—only the physical hardware is different. Thus, the primary mirror in a visible and infrared system and the large reflector of a dish antenna in passive microwave radiometry and radar, for example, are equivalent.

However, because of the much longer wavelengths at which RF systems operate compared with optical systems, the equations shown in the previous section for diffraction limited angular resolution and corresponding horizontal resolution on the surface lead to RF system apertures that are orders of magnitude larger than corresponding optical systems for the same surface resolution. Likewise, for the same real aperture diameter, a passive microwave or radar system will provide much coarser surface resolution than an optical system with the same aperture diameter.

Nearly all RF systems detect collected irradiance coherent using the superheterodyne method. Superheterodyne detection, invented in the early 20th century, involves mixing an incoming RF signal collected by an antenna with a coherent, phase stable and low noise local oscillator to produce sum and difference frequencies called intermediate frequencies (IFs) that are much lower in frequency than the incoming signal and therefore easier to process. The IF signals contain the same amplitude and phase information as the RF signal. The amplitude is scaled by a constant, measureable amount and the phase difference is a constant that can also be measured.

This ability of RF systems to measure both amplitude and phase of the received signal with a relatively high degree of accuracy enables a technique called synthetic aperture radar (SAR) that can provide radar imagery with spatial resolution in one dimension that is comparable to spatial resolutions obtained with high performance optical systems.

To understand the importance of SAR, consider first how real aperture radar works. Radar illuminates an area with pulses of microwave radiation. These pulses are reflected from the illuminated scene and collected and processed by a receiver. Radar determines the distance of the reflecting object by precisely measuring the time difference between transmission time of the outgoing

pulse and time the return pulse is received. This distance or range is the time difference times the speed of light divided by two. Spatial resolution along the radar line of sight is not range or directly wavelength dependent, as it is for horizontal resolution perpendicular to the line of sight. Instead, it is determined by the effective pulse-width, δt multiplied by the speed of light, c, divided by two, which is equivalent to the reciprocal of the effective pulse bandwidth, B times the speed of light. That is

$$\text{Range Resolution} = \frac{c\delta t}{2} = \frac{c}{B} \tag{17-12}$$

Pulse bandwidth and therefore data rate inside the instrument grow, as the range resolution improves. Many radars, including SARs, transmit a pulse using a linear frequency modulated (FM) "chirp" in which the frequency of the transmitted radar pulse varies linearly over a selected frequency range. This variation in frequency determines the RF bandwidth of the system. Chirp length and slope are constrained by range resolution for a system and depend on RF pulse power, pulse repetition frequency (PRF) and analog-to-digital converter (ADC) sampling rate and dynamic range. For example, the German TerraSAR-X system launched in 2007 and described by Werninghaus and Buckreuss, [2010], operates with a transmit bandwidth ranging from 5 to 300 MHz (with 300 MHz being used in an experimental mode). Consequently, TerraSAR-X is capable of operating with range resolutions extending from 1 m to 60 m.

In the direction perpendicular to the radar beam, the so-called cross range or azimuthal direction, spatial resolution of real aperture radar is limited by its physical size, just like a passive optical system. In this case, cross range spatial resolution is given by $1.22\,\lambda R_s/D$. From space, good range resolution can be achieved, but poor cross range horizontal spatial resolution, typically on the order of a kilometer even for practical but very large systems, cannot be avoided by real aperture radar at frequencies low enough to penetrate all weather conditions.

Interestingly enough, this significant disadvantage for radar, of large beam width or angular field of view, can be turned into an advantage, if the system has the ability to measure precisely, record and process the phase and Doppler shift of the radar return signal. A radar pointing toward the Earth's surface with a wide beam illuminates a scene location for an extended period of time (possibly up to seconds in duration) as the satellite passes overhead. If, during this time, the radar measures the phase and Doppler shift for each return, an aperture can be synthesized that is equivalent to the distance the real aperture antenna moves while the scene location is in the beam. This synthesized aperture can be orders of magnitude larger than the real radar antenna, which makes it possible to produce radar imagery with a meter of resolution. Longer wavelengths and smaller aperture diameters result in wider beam divergences, which can illuminate a scene location longer and therefore provide a longer baseline for creating a synthetic aperture.

Figure 17-14 illustrates a typical space-based SAR with co-located transmitter and receiver. This transmitter-receiver configuration is known as a monostatic configuration. As the satellite passes by a target location of interest, it maps out a strip along the Earth's surface. The dots in the figure represent a series of positions at which the SAR transmitter emitted a pulse. Each pulse travels to the Earth's surface and illuminates a scene location. A small fraction of that outgoing pulse returns to the receiver and is collected by the same antenna. The SAR records the phase histories of the responses at each scene location as the transmitter beam moves across the scene and then phase shifts and adds them together with appropriate weighting to minimize sidelobes in the synthesized beam to determine the range and scene brightness of one spatial resolution element at a time until the complete scene is described. SAR achieves a very high signal processing gain because this summation of the range correlated radar responses is done coherently with both amplitude and phase. In many cases, thousands of pulses are summed for each spatial resolution element, resulting in a signal processing gain of up to 10,000 or more for the TerraSAR-X compared with a single pulse.

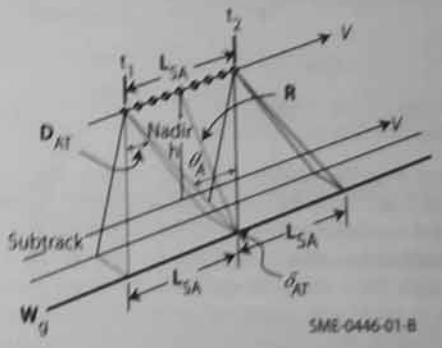


Fig. 17-14. Basic Principles of Synthetic Aperture Radar.

SAR performance is constrained by ambiguity relationships associated with the 2-D scene sampling that limit PRF and real aperture collector size. The basic idea is to collect data in the along track and in the range direction that can be processed into a unique image. In Fig. 17-14, L_{SA} is synthetic aperture length, and is the distance the satellite moves to illuminate the target. V is the satellite velocity, R is the target range, θ_A and D_{AT} are respectively the aperture azimuth beamwidth and the along-track antenna length of the real SAR antenna, δ_{AT} is the Earth target resolution element along-track dimension, and W_g is the satellite ground swath. The dwell or integration time, t_I, is given by:

$$t_I = \frac{L_{SA}}{V} = \frac{\lambda R}{V D_{AT}} \tag{17-13}$$

Table 17-5◀, Fig. 17-14, Eq. 17-13

The aperture azimuth beamwidth is given by:

$$\theta_A = \frac{\lambda}{D_{AT}} \tag{17-14}$$

Additionally, the following relations apply:

$$\mathbf{L_{SA}} = \theta_A \mathbf{R} = \frac{\lambda \mathbf{R}}{D_{AT}} \tag{17-15}$$

$$\delta_{AT} = \frac{\lambda}{2\mathbf{L_{SA}}}\mathbf{R} = \frac{\lambda \mathbf{R}}{2\lambda \mathbf{R}/\mathbf{D_{AT}}} = \frac{\mathbf{D_{AT}}}{2} \tag{17-16}$$

The achievable azimuthal (i.e., along track) resolution of a SAR is approximately one half the along track radar antenna size and this resolution is independent of radar frequency and range to first order and improves with smaller real antenna aperture. Based on that one equation alone, it appears that along track resolution could be improved simply by making physical size of the antenna length very small. However, achieving a given along track resolution also requires that the radar send out a pulse every time the radar moves half of the along track antenna size. That is

$$2 \times \frac{V}{D_{AT}} < \mathrm{PRF} \tag{17-17}$$

or

$$\mathrm{PRF} > \frac{V}{R_A} \tag{17-18}$$

where V is the satellite velocity, D_{AT} is the antenna length, and R_A is the azimuthal resolution. Better azimuthal resolution requires a higher transmitter PRF. This means that less time is available between pulses to collect data. Only one pulse can be in the target space at a time to avoid ambiguities in range that would prevent construction of a unique image solution. This condition sets another limit on PRF which is a function of swath width ($S = R_f - R_n$) as shown in Fig. 17-15 for the SEASAT SAR imaging geometry, and the pulse duration, T, resulting in

$$\mathrm{PRF} < \left(2T + \frac{2S}{c}\right)^{-1} \tag{17-19}$$

Not surprisingly, more spacing between pulses is required as the swath width or nadir angle increase. Thus, ultimately, a trade must be made in the design or specific system configuration between increased spatial coverage and improved along track spatial resolution, δ_{AT}. All of this reduces to a relationship between swath width, S, and along track spatial resolution for the single look case [Raney, 1998].

$$2 \times \frac{S}{\delta_{AT}} < \frac{\mathrm{c}}{V} \tag{17-20}$$

Table 17-5◀, Fig. 17-15, Eq. 17-23

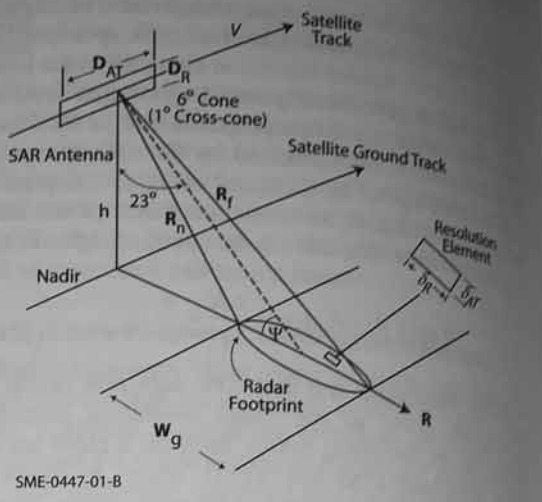


Fig. 17-15. SEASAT SAR Imaging Geometry.

In Fig. 17-15, the velocity, V is 7,450 m/s, the altitude, h is 800 km, and the ground swath, W_g is 100 km. $\mathbf{R}$ is the cross-track or range direction, Ψ is the grazing angle, and the satellite ground track is in the along track (AT) or azimuth direction. The following relations apply:

$$\delta_{AT} \geq \frac{\mathbf{D_{AT}}}{2} \tag{17-21}$$

$$\delta_R \geq \frac{\mathrm{c}}{2\beta} \tag{17-22}$$

$$\mathrm{W_g} = \frac{\mathbf{R_f} - \mathbf{R_n}}{\cos\Psi} = \frac{\lambda \mathbf{R}}{\mathbf{D_R} \sin\Psi} \tag{17-23}$$

Where c is the speed of light, and β is the transmitted pulse bandwidth.

17.2 Observation Payload Sizing

The sizing of an observational payload is an iterative process that involves trading and optimizing different payload approaches, frequently across a large trade space of candidate possibilities. In the early stages of this process, it is important to be able to examine different options without designing each one in detail. Initial assessment of design alternatives requires estimates of size, mass, power, data rate and constraints on spacecraft pointing control and stability among other interface characteristics to narrow down the possibilities for more thorough consideration. These assessments flow from detailed description and understanding of the mission to the operations concept and initial specifications for the observation payload that determine payload size, mass, and power, as illustrated in Table 17-6.

High-resolution wide field-of-view optical instruments, especially those with hyperspectral capability across a broad spectral region, typically generate data

Table 17-6. The Observation Payload Sizing Process. Note that Steps 2 and 3 are highly iterative and may well drive both the mission and payload design.

Step	Where Discussed
1. Define Mission Requirements and Objectives	Chaps. 3, 4, 6
2. Determine Mission Geometry and Orbit Parameters	Chaps. 8, 10
3. Define Payload Requirements (ITERATE WITH STEP 2)	Sec. 17.2.2
4. Define Payload Operations Concept	Sec. 17.2.1
5. Determine Spatial Sampling	Sec. 17.2.3
6. Determine On-board Signal Processing	Sec. 17.2.4
7. Determine Radiometric Sensitivity Performance	Sec. 17.2.5
8. Estimate Payload Size, Mass, and Power	Sec. 17.2.6
9. Determine Payload Figures of Merit	Sec. 17.2.7
10. Document and Iterate	

rates on the order of several hundred Mbps and above. To send this data stream to a ground station in real time, the system may need several parallel channels with capacities up to 300 Mbps each. If data collection is short compared to the available downlink time, the data can be stored on-board and transmitted to the ground over time to reduce the required downlink data rate.

Trends for future systems indicate a growing tendency toward more on-board processing to decrease downlink data rate requirements and to enable broadcast of information derived from observational payload data directly to users. Continuing improvements in powerful processing hardware and associated algorithms enable sophisticated on-board processing capability comparable to ground-based processing. Given the significant impact that high speed processing systems can have on payload size, mass, and power, selection of an on-board data processing approach is becoming an ever more important element in the observation payload design trade space. To maintain the flexibility required for continuous improvements in system performance, software uploads to the hardware must be possible to update and modify operational algorithms from the user community. The need to upload large quantities of software relatively quickly must be considered, when specifying uplink data rate requirements for an observation system.

This section discusses each of the steps in the observation payload sizing process. In addition, figures of merit for comparing different design approaches are considered along with the description of an observation payload that fulfills requirements for the FireSat II mission.

17.2.1 Payload Operations Concepts

The operations concept for using the observation payload must be considered early in the preliminary design process. The end-to-end mission need must be understood along with the physics, associated engineering and phenomenology of the collection and interpretation of data. The payload operations approach as driven by end data user requirements needs to be a realistic and cost-effective way to achieve mission objectives.

Specific collection operations involving tasking and scheduling of the sensor, processing of the mission data, and distribution of the data can dramatically increase complexity of the systems engineering challenge and decrease the final accuracy of the system, if not selected carefully and within context of quantitative figures of merit. Not only can physical effects such as atmospheric correction, calibration, and rectification degrade system performance, but technical effects such as quantization and data compression errors can decrease the resolution of the system from the perspective of the end user. For additional information about technical aspects of the end-to-end throughput or image chain problem for remote sensing and imagery collected from space, see Schott [2007].

The operations concept for a system like FireSat II must consider all aspects of the operational mission, including different mission phases and alternative operating modes (See Chap. 29). In preliminary mission planning, we should be aware of the expected sequence of events during each mission phase. For normal FireSat II mission operations, a sample timeline includes the following steps:

1. Fire starts somewhere on the Earth
2. Sensor field-of-view passes over the fire and collects data, as part of a global fire reconnaissance mission
3. Measurements of the fire are transmitted through the sensor data stream
4. Data is analyzed at the mission ground station or processed on board
5. Fire detection algorithm determines the probability of a fire at the analyzed location—this could be a multistep process with a rapid, but crude fire detection algorithm that triggers a more accurate algorithm or set of measurements
6. If fire is detected, system generates appropriate messages to Customer that indicates presence of fire at specific location
7. Customer organization reports fire and notifies responsible authorities and appropriate researchers
8. System continues to monitor the fire and region near the fire, in conjunction with its global reconnaissance mission to track progress of the detected fire and monitor nearby areas for new fires

To ensure mission success, an operations concept for each mission phase and each spacecraft operating mode—including contingency and failure modes—must be created within operating environment limitations (See Chap. 29).

Table 17-6, Fig. 17-15◀, Eq. 17-23◀

Table 17-7. Characteristics of Typical Payloads.

Purpose	Instrument Name	Size L × W × D (m)	Mass (kg)	Avg. Pwr. at 28 V (W)	Data Rate (Mbps)	Aperture (m)	Pointing Accuracy (deg)
Solar Physics	Lyman—Alpha Coronograph	2.8 × 0.88 × 0.73	250	87	13.5	—	0.003
	X-ray Telescope Spectrom.	2.7 × 1 dia.	465	30	0.4	—	0.003
	Solar Optical Telescope	7.3 × 3.8 dia.	6,600	2,000	50+	1.25	—
	Solar Magnetic Velocity Field	2 × 0.4 × 0.4	183	322	2+	—	0.003
	100 m Pinhole Camera	1 × 1 × 2	1,000	500	0.5	—	—
	Extreme UV Telescope	2.75 × 0.86 × 0.254	128	164	1.28	—	—
	Solar Gamma Ray Spectrom.	1 × 1 × 13	2,000	500	0.1	140 cm²	0.003
Space Plasma Physics	Ion Mass Spectrometer	0.5 × 0.5 × 0.4	80	334	0.01	—	1
	Beam Plasma	0.6 × 0.7 × 0.7+ two 0.7 dia. ant.	17	38	0.016	—	5
	Plasma Diagnostics	—	2,000	250	50	—	—
	Doppler Imaging Interferom.	$(0.25)^3$	100	620	0.2	—	—
	Proton (Ion) Accelerators	6.7 × 3.4 × 3.10	500	1,500	0.256 (4.2 TV)	—	—
High Energy Astrophysics	Gamma Ray Burst	2 × 4 dia.	1,000	120	0.01	3	1
	Cosmic Ray Transition	3.7 × 2.7 dia.	1,500	230	0.1	2.7	—
	X-ray Spectrom./Polarimeter	1.6 × 1.6 × 3	2,000	300	0.03	—	—
	Short X-ray	1 × 1 × 3	1,000	300	0.025	1 × 3	0.1
	Hi Energy Gamma Ray Tele.	3 dia. × 4	10,000	100	0.003	3	0.1
Resources	Gravity Gradiometer	0.23 m sphere	10	1			0.1
	Synthetic Aperture Radar	2.8 × 3.7 × 1.4	808	3,000	120	8 × 2.8	1–2
	Multi-Spectral Mid-IR	1.4 × 1.5 × 0.9	270	170	8	0.18	2.5
	Enhanced Thematic Mapper	2 × 0.7 × 1.1	400	600	75	0.406	0.1
Materials Processing	Materials Experiment Assem.	1 × 1 × 2	900	500	0	—	0.005
	Solidification Experiment		1,100	3,000	0.02	—	—
Environmental	Limb Scanning Radiometer	4.8 × 1.9 dia.	~800	125	0.52	—	—
	Microwave Radiometer	4 × 4 × 4	325	470	0.20	4	±0.1
	Dual Frequency Scatterom.	4.6 × 1.5 × 0.3	150	200	0.01	4.6 × 0.3	1
	Ocean SAR	20 × 2 × 0.2	250	300	120	20 × 2	0.1
	Solar Spectrum	0.4 × 0.3 × 0.6	16	60	Low	—	±3
	Doppler Imager	1.25 × 0.6 × 0.8	191	165	20	—	±3
	Imaging Spectoral	1.0 × 1.6 × 1.0	250	225	11	0.18	0.04
Planetary Science	Multi-Spectral Imager	0.54 × 0.37 × 0.29	13	12	2	0.12	—

17.2.2 Required Payload Capability

Often, mission requirements can be met in a number of different ways. It's not always obvious how to go through the various candidate approaches and select the best one for any given mission. The general approach outlined here is a simple, repeatable method for selecting a payload to satisfy an observation mission. After we select a physical phenomenology (for example, measuring thermal infrared radiance to detect the high brightness temperature that is characteristic of forest fires), three system characteristics need to be determined. The first is the spatial sampling approach needed to satisfy the mission information need, followed by the radiometric sensitivity levels needed to achieve fire detection and false alarm requirements. The third and frequently most important system characteristic from a mission implementation and cost perspective are the implications of spatial sampling and radiometric sensitivity characteristic on payload size, mass, and power.

Table 17-7, Fig. 17-15◀, Eq. 17-23◀

Table 17-7 provides a small sampling of remote sensing payloads and corresponding spacecraft missions. Categorizing remote sensing missions is complicated by the fact that sensors usually have multiple uses, and they can be categorized according to any number of different aspects, such as measurement technique (active or passive), event measured (such as fire or deforestation), and measurement resolution (spatial, spectral, radiometric, temporal). Frequently, these multiple uses lead to conflicting requirements that further complicate payload design and development. Remote sensing payload examples include imaging spectroradiometers that passively measure the intensity of scene spectral radiance across a broad range in wavelength and polarimeters that measure polarization state of a transmitted, scattered or reflected wave. Scatterometers are active RF instruments (radars) that measure surface roughness of the ocean by measuring radar return signals that depend on wind speed and direction.

Measurement procedures used in observation payloads and associated processing algorithms are customized to deliver information that can be retrieved from the measurements to understand the subject. We describe the fundamental measurement content provided by passive Earth observation instruments in terms of spatial and spectral resolution and radiometric sensitivity. We continue by discussing the basic types of detectors and collection techniques that are available in the design of a remote sensing instrument.

We frequently use imaging sensors, based on electro-optical devices, to measure and analyze spatial features. Visible wavelength imaging is a common approach for providing data needed for topographic mapping and photogrammetry. Altimeters are also especially valuable in providing fundamental spatial measurements. Kramer [2001] lists examples of these types of sensors.

As noted above, spatial resolution depends on the range from the sensor to the target, the aperture and focal length of the collection optics, detector element size, signal processing approach and the wavelength of the incident energy. We can characterize the spatial resolution by the smallest angular distance the sensor can resolve or directly as the ground range in units of length for the size of the smallest object that can be discriminated. Effective spatial resolution is a function of these many different design parameters in a remote sensing system and more, including the scene itself. Optical system and focal plane array characteristics determine theoretical limits to spatial resolution for any system geometry. In addition, these same characteristics define the ground sample distance (GSD), the distance on the ground (usually referenced to the sub-spacecraft point) at which the sensor samples the scene. Frequently, the effective spatial resolution is different from the GSD. Normally, the effective spatial resolution is about twice the GSD. However, for low contrast targets or target detections with low signal-to-noise ratio, effective spatial resolution can be coarser than twice the GSD. The smallest physical features that can be discriminated for specific target characteristics using any candidate sensor measurement approach determine the limits of spatial resolution in a remote sensing system. Information from spectral content of scenes can be used within context of plausible assumptions to derive spatial information that has finer resolution than the GSD.

Spectral content is provided by spectrometers that collect data that can be used to identify different components in a scene (e.g., rocks, trees, and water) and sometimes chemical composition of objects being measured. The spectral radiance received by a sensor is a composite of the spectral radiances emitted by all objects in the instantaneous field-of-view. The radiance reaching a possible multispectral FireSat II imager, for example, will include spectra from soil, vegetation, and cities in addition to any fire that might be there. The combined spectrum of all these scene elements may be very different from the spectrum of a forest fire by itself. Combining information from different spectral bands has been used often and successfully to isolate key scene elements from each other and improve the usefulness of a sensor.

Space-based spectrometers are frequently grouped into one of three categories: multispectral, hyperspectral and ultraspectral. Multispectral systems typically have tens of discrete spectral bands with relatively low ($\lambda/d\lambda$~10) spectral resolution. Hyperspectral and ultraspectral systems employ hundreds and thousands of spectral bands, respectively, across a contiguous spectral range with spectral resolutions that are typically moderate ($\lambda/d\lambda$~100) for hyperspectral systems and high ($\lambda/d\lambda$~1,000) for ultraspectral systems. Following Slater [1980], we consider a multispectral image of a scene with concrete, asphalt, soil, and grass. Figure 17-16 shows typical spectral reflectance for these four scene elements. If a candidate imager has only three bits of radiometric sensitivity (low, medium, and high), for example, then no single image, whether panchromatic or filtered to a particular band region could distinguish these four scene features. However, even a simple, but well calibrated two-band multispectral system can uniquely resolve these four scene elements using bands in the 600–700 nm and 700–900 nm wavelengths. Returns for each of these materials are shown in Table 17-8.

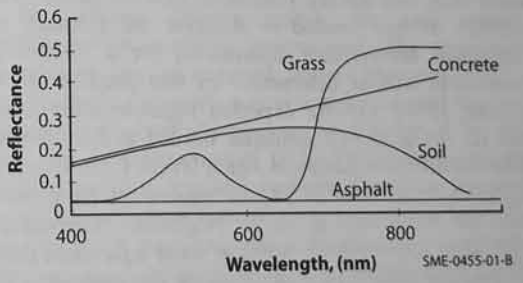


Fig. 17-16. Typical Spectral Reflectance Curves for Grass, Concrete, Soil, and Asphalt. A single-band low-resolution sensor can not distinguish all 4 types, but a 2-band, low-resolution sensor can, as shown in Table 17-8 [Slater, 1980].

Table 17-8. Spectral Resolution for Sample Two-Band Sensor. Combining low-resolution sensor returns for the two-band sensor can uniquely identify the four materials from Fig. 17-16.

Material	Sensor Reading 600–700 nm	Sensor Reading 700–900 nm
Concrete	Medium	High
Asphalt	Low	Low
Soil	Medium	Low
Grass	Low	High

Multispectral data with more spectral bands and especially hyperspectral and ultraspectral data contain much more physically meaningful detail, which can be used to recognize and quantify characteristics of com-

Table 17-8, Fig. 17-16, Eq. 17-23◀

plex scenes. However, the higher spectral resolution of hyperspectral and ultraspectral systems, especially, reduces measured signal-to-noise ratio (SNR)—all other system characteristics being equal—which can impact system design in many ways ranging from reducing spatial resolution to recover SNR to increasing aperture size to improve light gathering capability.

We can also use the measured spectral radiance level to extract useful information. For example, if we have an imager with four or more bits of sensitivity operating in the 700–900 nm wavelength, then a single spectral radiance measurement could be used to distinguish among all four candidate scene types, if we assume a pixel contains only one scene type and we understand the details of the solar illumination and viewing geometry precisely. The ability to measure spectral radiance precisely goes beyond detecting either the presence or absence of energy at a given wavelength. For example, if a FireSat II sensor were designed to detect fires based on measuring brightness temperatures of Earth's thermal emission, then a high enough brightness temperature would indicate presence of fire. But, the radiometric resolution of the instrument needs to be good enough to allow the system to discriminate between a forest fire and a campfire.

Thus, merely detecting the presence of thermal energy in a spectral band characteristic of burning biomass does not satisfy FireSat II mission needs. The number of bits needed to describe the intensity or brightness temperature information for a FireSat II instrument will be determined by two practical limits [Slater, 1980]: (1) the expected signal-to-noise ratio, and (2) the level of confidence needed to discriminate between different threshold signal levels. Consequently, signal-to-noise ratio (SNR) or equivalent parameters such as difference in scene brightness temperature equivalent to the system noise or noise equivalent delta temperature (NEDT) are performance discriminators for candidate observation payload designs.

17.2.3 Spatial Sampling

The spatial sampling of a scene by an optical instrument is determined by the scanning method used as described in Table 17-2, and by spacing of the detector elements in the focal plane, which can be constrained by optical system design characteristics such as operating wavelength, aperture diameter, and focal length. If the spatial sampling method is not selected with care, obvious image artifacts and not so obvious radiometric errors may be introduced into the data. Figure 17-17 shows the effect of sampling by a detector array observing three scenes with different spatial frequency structures. The response of the detectors to each scene pattern is shown as a gray level intensity in the three diagrams. Spatial frequency is defined as the inverse of the width of a black-and-white line pair, and is expressed in line pairs per meter or simply in units of inverse meters. The same detector array, shown to the right of each arrow, samples

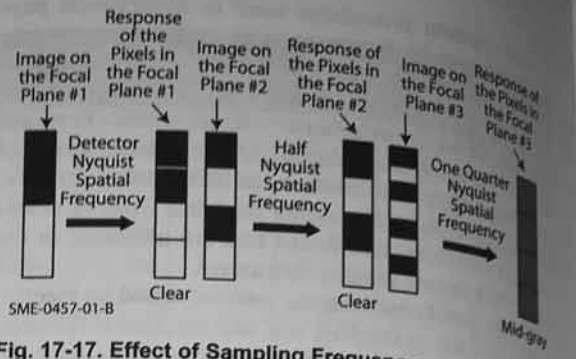


Fig. 17-17. Effect of Sampling Frequency on Image Quality. The three diagrams show illumination patterns having different spatial frequencies present in the scene (left of the arrows) sampled at the same sampling frequency, that is, the same detector array (right of the arrows). The response of the pixels is shown as intensity (gray level) in the three diagrams. From left to right: Sampling an illumination pattern with Nyquist frequency (the highest possible frequency that can be reconstructed without error). Sampling an illumination pattern with half the Nyquist frequency. Sampling an illumination pattern with one quarter the Nyquist frequency.

the scene in all three cases. That is, in these simple examples, the different scene structures are sampled at the same spatial sampling frequency, defined as the inverse of the distance between centers of the sampling detectors. The maximum spatial frequency which can be reconstructed without error after the sampling/restoration process is half of the sampling frequency or twice the sample distance in the focal plane. The left diagram in Fig. 17-17 illustrates this so-called detector Nyquist sampling case, where the scene feature as seen in the focal plane is twice the size of the detector. In the center, sampling of an illumination pattern occurs at half the Nyquist frequency of the scene structure, showing that reconstruction of the original illumination pattern is feasible without error even at half the Nyquist rate. However, if the sampling pattern is shifted by half a pixel with respect to the illumination pattern, then reconstruction fails. And on the right, sampling an illumination pattern with one quarter the Nyquist frequency of the scene structure results in a pixel response of mid-gray since every detector pixel is illuminated over half its area in the example. All samples in the third case show the same constant value which corresponds to a spatial frequency of zero, which was not present in the original scene. This creation of signal power at new frequencies is known as aliasing. Such frequency components cannot be removed by additional processing of the reconstructed image. To avoid these errors completely in an actual system, the scene must be sampled at twice the optical cutoff frequency described in Eq. (17-5). This Nyquist sampling of the optical cutoff frequency defines the lowest rate at which this band limited data must be sampled to avoid errors in scene reconstruction. Many systems do not sample the scene at the Nyquist rate of the cutoff frequency, because normally, very little power exists in the higher frequencies being aliased into the measurements and because radiometric sensitivity requirements typically determine aperture

Table 17-8◀, Fig. 17-17, Eq. 17-23◀

diameter and spatial sampling needs. Furthermore, it is not usually cost effective to sample an optical system at the Nyquist rate of the optical cutoff frequency because this approach results in a much larger number of detectors than would otherwise be required by alternative approaches, such as sampling at twice the Rayleigh criterion or at the spatial resolution required by the mission, which can significantly increase system cost relative to other approaches. The result is that we usually have a certain amount of unavoidable aliasing which degrades image quality and radiometric accuracy of such systems.

The Modulation Transfer Function (MTF) is the ratio of the measured contrast variation in the image to the actual contrast variation in the ground scene as a function of spatial frequency. MTF is zero at the optical cutoff frequency described in Eq. (17-5). Figure 17-18 shows the theoretical MTF of an optical system with and without a central obscuration such as an on-axis secondary mirror and support structure. The theoretical MTF can be approximated by a line starting at 1 when the spatial frequency is 0 and falling to 0 at the cut-off frequency, $F_{cut\text{-}off} = D/\lambda f$, where D is the aperture diameter, λ is the wavelength, and f is the focal length of the optical system. The MTF of a telescope is the autocorrelation function of the entrance pupil function. In optical terms, the MTF is the absolute value of the complex Optical Transfer Function (OTF) which describes how the complex amplitudes of the optical wave front are transferred by an optical system at different spatial frequencies.

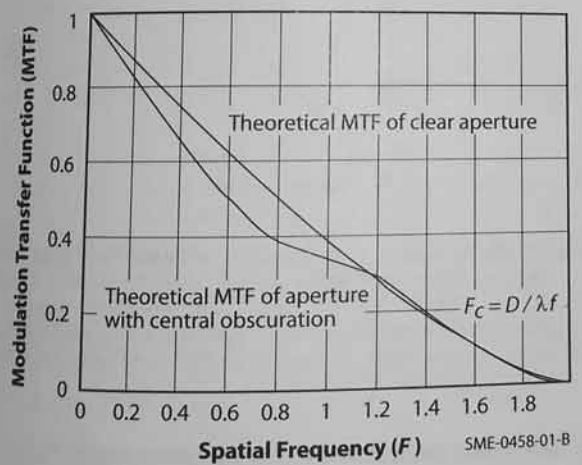


Fig. 17-18. Modulation Transfer Function of Circular Diffraction-Limited Optical System vs. Spatial Frequency. MTF curves for a clear circular optical system is compared with one having a central obscuration (as found in on-axis reflective telescopes). The MTF can be thought of as a function dependent on the spatial frequency, F, which describes the modulation (contrast) function through the optical system (analogous to the frequency dependent gain of an electrical transmission block). The MTF starts at 1 for spatial frequencies near 0 and drops to 0 at the cutoff frequency, $F_c = D/\lambda f$.

The MTF describes the ability of an imaging system to transfer contrast from a scene to an image in the focal plane as function of spatial frequency. It is a measure of the quality of an optical system. The point-spread function illustrated in Fig. 17-8 describes exactly the same properties by showing the two-dimensional intensity distribution in the focus of the optical system. The two functions are related by Fourier transform. System level MTF is also affected by spatial sampling in the focal plane by detector arrays and by signal processing of data collected with 2-D imaging detector arrays.

17.2.4 Signal Processing

Analog signal processing for visible wavelength detector arrays based on CCDs or Si:PIN diode arrays is very similar to processing used in infrared detector arrays with an integrated multiplexer. In all cases, weak analog signals need to be amplified and conditioned, maintaining high dynamic range and high processing speed. High speed electronic signal processing with high accuracy can become a cost and schedule driver in development of high-resolution wide field of view instruments. Concerns about electrical noise and other electromagnetic interference dictate that this processing be conducted as close to the focal plane assembly as possible—frequently on the chip itself. Simultaneously, careful thermal isolation between the detector arrays and the heat dissipating electronics must maintain a stable minimum operating temperature for the detector arrays.

For infrared detector arrays, the dark current increases with operating temperature. The exact rate depends on detector material, bias voltage and fabrication methods. However, across a wide range of designs, the dark noise typically goes up by a factor of 10 with each rise in temperature of 10 K. Infrared imagers are usually cooled to low temperatures; therefore, we must minimize each heat leak to their cryostat to keep the cooling power low. Frequently, the cooling requirement becomes more stringent as the wavelength of the radiation to be measured becomes longer. For wavelengths in the 100 μm range, detectors are typically cooled to 4 K. Even small heat leaks, such as those from the necessary electrical wiring and mechanical connection to the spacecraft, transport excess heat that increase required cooling power.

Figure 17-19 shows the typical functional blocks found in the electronics of an optical instrument. The signal flow through the electronics begins with a detector in the focal plane of the instrument. This detector converts photons to analog electrical signals, which are then amplified, conditioned and converted to a digital signal stream. Any additional signal processing is performed in the digital data processing block. The analog and digital signal processing blocks operate synchronously with the readout of the detector arrays. In most observation systems, a central timing subsystem supplies appropriate timing signals to all relevant electronic blocks to guarantee synchronous operation. Given electrical signals processed in the instrument electronics are weak, special attention to power conditioning and

Table 17-8◀, Fig. 17-18, Eq. 17-23◀

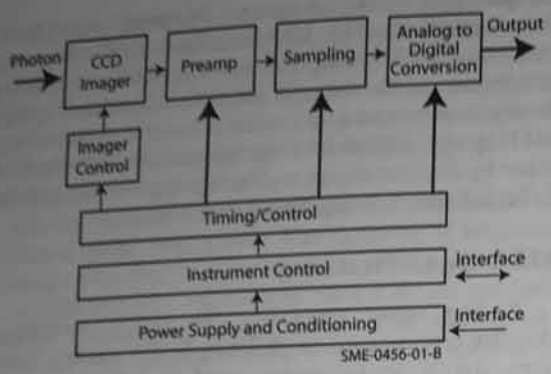


Fig. 17-19. Block Diagram of an Optical Instrument. Photons are collected by the Detector Array, then amplified and processed. The primary supporting functions of power and control are also depicted. The sampling process for charge coupled device signal processing typically uses correlated double sampling to eliminate reset noise.

grounding is required. Careful filtering and clean electrical grounding across a wide electrical bandwidth must be implemented to decouple digital and analog signals.

The instrument control processor manages the signal processing and timing functions performed by the instrument and it interfaces to the main spacecraft processor. Depending on the design, the spacecraft processor may transmit time-tagged commands that are executed by the instrument processor.

Analog signal processing for detector arrays frequently involves correlated double sampling. This technique takes two slightly time-shifted samples of the analog signal and subtracts one from the other to extract their image-related video signal. The adaptation of this video signal to the input range of the analog-to-digital converter requires setting gain and offset parameters for every detector in the system. Any non-uniformity in photo response and dark signal must be removed from each pixel. This task is frequently conducted on board the spacecraft because some of the most straightforward data compression algorithms become invalid if the detector samples are not normalized in this way. Furthermore, normalizing detector samples is required for most effective use of on-board calibration methods in the sensor.

The process of normalizing pixels is based on characteristics of the detector arrays. For arrays with a highly linear response, one-or two-point correction is generally sufficient. Highly nonlinear detectors, like some infrared detectors, require n-point correction techniques. Figure 17-20 illustrates an example of two-point correction. The offset (dark signal) and gain factor (response) are corrected by first subtracting the individual offset value and then by dividing by the individual response for each detector.

High-resolution wide field of view optical instruments typically generate data rates on the order of several hundred Mbps and above. To send this data stream to a ground station in real time, the system may need several parallel channels with capacities up to 300 Mbps each. If data collection is short compared to the available downlink time, a buffer memory can reduce the downlink data rate over a longer transmission time.

Trends for future systems indicate a growing tendency toward more on-board information processing to decrease communication requirements. Continuing improvements in powerful processing hardware and associated algorithms enable sophisticated on-board processing capability comparable to ground-based processing. To maintain the flexibility required for continuous improvements in system performance, software uploads to the hardware must be possible to update and modify operational algorithms from the user community. The need to upload relatively large quantities of software must be considered, when specifying uplink data rate requirements for an observation system.

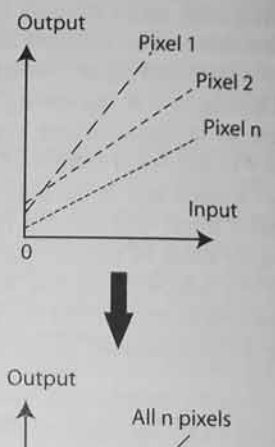


Fig. 17-20. Two-point Normalization of Pixel Response. The top diagram shows the original response functions of individual pixels. The bottom diagram shows the normalized response function of the same pixels, where the effects of the variations in offset values and responsivity values have been removed from the individual pixels.

17.2.5 Radiometric Sensitivity Performance

In order to describe the radiometric sensitivity performance of an observation payload, we must follow a scene photon through the entire end-to-end system. For a solar reflective system operating in the visible or near infrared, that means we start with the source of the reflected light, the Sun. Figure 17-21 shows the radiance distribution of the Sun versus wavelength.

The measured power density in a spectral band of interest is given by integration of the different components of spectral radiance input at the scene over solid angle and across the spectral bandwidth of the sensor. To first-order we assume that lambertian (ideal or dif-

Table 17-8◀, Fig. 17-20, Eq. 17-23◀

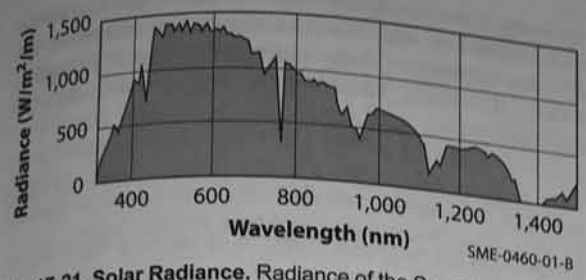


Fig. 17-21. Solar Radiance. Radiance of the Sun at sea level as a function of the wavelength.

fuse) reflection with a constant reflection coefficient occurs at the target scene (reasonably good approximation for small spectral bandwidth). The area of the ground sample radiating power to the instrument determines the power density per solid angle. The atmosphere attenuates this radiation by a constant transmission factor (again, invoking an approximation for small spectral bandwidth). Scattered light in the atmosphere contributes significantly to upwelling signal levels, especially over relatively low reflectance areas like the ocean. The entrance aperture of the payload at orbital altitude collects a very small fraction of the total power radiated by the scene. The collected signal power is attenuated further by transmission through the optics, during the integration time some amount of energy (power times integration time) is collected by each detector in the array. This collected energy divided by the photon energy (which is wavelength dependent, of course) leads to the number of available signal photons per detector. These signal photons are converted into electrons or pairs of electrons and holes inside the detector, depending on the detector material. The quantum efficiency of the detector multiplied by the number of photons is equal to the number of available electrons or electron/hole pairs. These charge carriers are collected by the detector junction and correspond to the output signal of the detector. This signal is expressed in units of electrons, whether electrons or holes are collected by the junction. For convenience, all noise components in the signal measurement process such as dark current, readout noise or instrument background for emissive infrared systems are stated in electrons.

Radiometric performance of an instrument is determined by the signal-to-noise ratio and dynamic range. The signal-to-noise ratio describes the image quality for any given set of measurement conditions, including sensor aperture diameter and instantaneous field of view and scene intensity. The number of noise electrons associated with the signal follows Poisson statistics and equals the square root of the number of signal electrons. Readout of the detector arrays results in additional noise electrons. The signal noise is added to the readout noise since they are statistically independent values, resulting in a total number of noise electrons in the denominator of the signal-to-noise ratio. For emissive infrared systems especially, background from structures inside the instrument that radiate spurious thermal photons create additional noise in the detector that has to be considered in the total number of noise electrons, too.

The required maximum dynamic range of the instrument is the ratio of the sum of the signal, instrument background, dark noise, and readout noise electrons the sensor sees between maximum possible brightness and completely dark scenes. In the solar reflectance part of the spectrum, the brightest scene is typically reflection from thick clouds or snow, and the darkest scene is deep space.

In order to estimate radiometric performance of instruments operating in the mid and long-wavelength emissive infrared spectral region, spectral emission of the surface and atmosphere of the Earth must be modeled. A reasonable approximation for much of the Earth's surface is a blackbody with an equivalent temperature of 290 K. The atmospheric transmission as a function of the wavelength is well understood for any given path orientation and set of atmospheric characteristics up to high zenith angles from the surface. The multiplication of the spectral radiance with the atmospheric transmission results in the upwelling radiance from the surface measured at the sensor. The atmosphere emits significant amounts of thermal radiation and can be the largest single source of emission, depending on wavelength. The integration of upwelling spectral radiance over the selected bandwidth and area of the spatial resolution element results in the power per solid angle. Input power at the sensor's entrance aperture is calculated, based on sensor instantaneous field of view and effective aperture of the receiving optics. This power is converted via the optical transmission factor to input power at detector (usually in the picoWatt range). As with the solar reflective sensors, during the integration time, a certain amount of energy is accumulated per detector. The division of that energy by the energy of one photon gives the number of photons available per detector which is converted by the quantum efficiency to the available number of electrons or electron/hole pairs per detector which corresponds to its output signal.

To characterize the radiometric performance of an instrument that measures thermal emission, a more commonly used figure of merit for radiometric sensitivity is the noise-equivalent delta temperature (NEDT) for the instrument at a given scene temperature. NEDT is given by the temperature difference at a specified scene temperature that is normally relevant to the mission, which generates a signal equivalent to the total number of noise electrons at that scene temperature. NEDT characterizes the instrument's ability to resolve temperature variations for a specified scene temperature.

17.2.6 Payload Size, Mass and Power

We need to be able to estimate the size, mass, power and other system level characteristics of an observation payload before completing a detailed design. We want to be able to compare system options without necessar-

Table 17-8◀, Fig. 17-21, Eq. 17-23◀

ily designing each in detail. This section describes methods to estimate the overall size and key parameters, including data rates. In Sec. 17.2.8, we will apply these methods to FireSat II.

Regardless of the nature of any particular payload, we would like to estimate its size, weight, and power even before we have done a detailed design. To do so, we can use three basic methods:

- Analogy with existing systems
- Scaling from existing systems
- Budgeting by components

The most straightforward approach is to use an analogy with existing systems. To do this, we turn to the list of existing payloads in Table 17-7 or other payloads that we may be aware of which have characteristics matching the mission we have in mind. Kramer [2001] offers a very thorough list of existing sensors. We look for payloads whose performance and complexity match what we are trying to achieve and make a first order estimate that our payload will have characteristics comparable to the previously designed, existing payload. While this approach is rough, it does provide a first estimate and some bounds to decide whether the approach we have in mind is reasonable.

A second approach, described in more detail below, is scaling the payload estimate from existing systems. This can provide moderately accurate estimates of reasonable accuracy if the scale of the proposed payload does not differ too greatly from current payloads. In addition, scaling provides an excellent check. Most existing payloads have been carefully designed and optimized. If our new payload is either too large or too small relative to prior ones, there should be some reason for this change in characteristics. If more detailed estimates based on detailed budgets don't scale from existing systems, we must understand why.

The most accurate process for first-order payload sizing is budgeting by components. Here, we develop a list of payload components such as detectors, optics, optical bench, and electronics. We then estimate the weight, power, and number of each. This is the best and most accurate approach but may be very difficult to apply at early mission stages because we simply don't have enough initial information. Ultimately, we will size the payload with budgeting by components. We will develop budgets as outlined in Chap. 14 for each payload instrument for weight, power, and any critical payload parameters. These budgets will then help us monitor the ongoing payload development. However, even with a detailed budget estimate, it is valuable to use scaling as a check on component budgeting. Again, we wish to understand whether the components scale from existing payloads and, if not, why not.

Scaling from Existing Systems

An excellent approach for early design concept work is to adjust the parameters in Table 17-7 to match the instrument we are designing. We will scale the instruments based on aperture—a main design parameter that we can determine from preliminary mission requirements. To scale, we compute the aperture ratio, R, defined by

$$R = \frac{A_i}{A_o} \tag{17-24}$$

where A_i is the required aperture of our new instrument, and A_o is the aperture of a similar instrument (Table 17-7). We then estimate the size, weight, and power based on ratios with the selected instrument from Table 17-7, using the following:

$$L_i \sim RL_o \quad L = \text{linear dimensions} \tag{17-25}$$

$$S_i \sim L_i^2 \quad S = \text{surface area} \tag{17-26}$$

$$V_i \sim L_i^3 \quad V = \text{volume} \tag{17-27}$$

$$W_i \sim KR^3 W_o \quad W = \text{weight} \tag{17-28}$$

$$P_i \sim KR^3 P_o \quad P = \text{power} \tag{17-29}$$

The factor K should be 2 when R is in the range of 0.5 or less and 1 otherwise. This reflects an additional factor of 2 in weight and power for increased margin when scaling the system down by a factor of more than 2. When the system grows, the R^3 term will directly add a level of margin. For instruments more than a factor of five smaller than those listed in Table 17-7, scaling becomes unreliable. We recommend assuming a mass density of 1 gm/cm^3 and power density of 0.005 W/m^3 for small instruments. An example of these computations for FireSat II is in Sec. 17.3.8.

17.2.7 Figures of Merit for Evaluating Candidate Payloads

Multi-parameter performance indices can be defined for comparing optical instruments with similar performance characteristics. For high spatial resolution imagers three basic figures of merit describe the quality or expected information content in the image. The three defining figures of merit are signal-to-noise ratio at zero spatial frequency, MTF of an instrument at the detector Nyquist frequency (corresponding to the leftmost diagram in Fig. 17-17), and the ground sample distance, GSD. We define a relative quality index (RQI) to allow relatively straightforward quantitative comparisons with a reference instrument denoted by the subscript ref.

$$\text{RQI} = \frac{\text{SNR}}{\text{SNR}_{ref}} \times \frac{\text{MTF}}{\text{SNR}_{ref}} \times \frac{\text{GSD}_{ref}}{\text{GSD}} \tag{17-30}$$

This relative quality index allows the designer to trade requirements with respect to each other. For example, a higher SNR can compensate for a lower MTF at the detector Nyquist frequency for a given GSD. Such

comparisons enable first-order insights into the relationships between complexity, performance, and cost of candidate imagers. For example, suppose we consider a well understood reference instrument with SNR of 512, MTF of 0.5 and GSD of 25 m. If we then compute design parameters for some specific mission, we can calculate a relative quality index, for this candidate design with respect to the known reference instrument. For instance, if our design choices lead to an instrument with SNR of 705.2, MTF of 0.47 and GSD of 30 m, then the RQI for this system will be 108%, suggesting the candidate instrument is slightly better than the reference instrument. This type of index offers a straightforward method for comparing several competing sensors across three (or potentially more) key performance figures of merit.

17.2.8 Example of Observation Payload Sizing

To illustrate the preliminary design process for payloads, we will estimate the basic parameters for the FireSat II payload developed throughout Chaps. 1–10. We cannot expect to carry out a detailed design without substantial input from an IR payload designer. Still, we would at least like to know whether the FireSat II payload is the size of a shoebox or the size of a truck.

Table 17-9 contains the details of the design analysis process for visible and infrared systems. We start with basic system parameters such as the orbital height, minimum observation angle and ground resolution. We then calculate quantities that describe instrument performance. In particular, we determine the image processing parameters and system data rate, size of the optics for a given spatial resolution, and the radiometric sensitivity performance of the sensor. Sample computations for the FireSat II imager are presented in the third column.

The FireSat II altitude trade led to a preliminary altitude, h = 700 km. From this, we can determine the apparent angular radius of the Earth, ρ:

$$\rho = \sin^{-1}\left(\frac{R_E}{R_E + h}\right) = 64.3\,\text{deg} \qquad (17\text{-}31)$$

A key parameter in the system design is the minimum elevation angle, ε, at which the system can work. We do not have an estimate of that yet, but we do know that IR payloads do not work well at small elevation angles. Therefore, we will tentatively assume a minimum elevation angle of 20 deg, recognizing that this may be a very critical trade at a later stage. With this assumption, we can compute the nadir angle range, η, the maximum ground-track angle or swath width, λ, and the maximum range to the target, D, from the formulas in Sec. 8.3:

$$\sin\eta = \cos\varepsilon \sin\rho = 57.9\,\text{deg} \qquad (17\text{-}32)$$

$$\pi = 90 - \eta - \varepsilon = 12.1\,\text{deg} \qquad (17\text{-}33)$$

$$D = R_E\left(\frac{\sin\lambda}{\sin\eta}\right) = 1{,}578\ \text{km} \qquad (17\text{-}34)$$

These equations imply that the sensor on board the spacecraft will have to swing back and forth through an angle of ±57.9 deg to cover the swath. The swath width on the ground will be 2 × 12.1 = 24.2 deg wide in Earth-central angle, with a maximum distance to the far edge of the swath of 1,578 km. Had we been able to work all the way to the true horizon (ε = 0), the maximum Earth central angle would be 90 − ρ = 27.5 deg, and the swath width would be 55 deg. Increasing the minimum elevation angle to 20 deg has very dramatically reduced the size of the available swath.

We next find the orbit period, P, and longitude shift per orbit, ΔL, (Secs. 10.2.1 and 10.2.1.6 web):

$$P = 1.659 \times 10^{-4} \times (6{,}378 + h)^{3/2} = 98.8\ \text{min} \qquad (17\text{-}35)$$

$$\Delta L = 98.8 \times (360/1440) = 24.7\,\text{deg} \qquad (17\text{-}36)$$

Therefore, at the equator, successive node crossings are 24.7 deg apart. Notice that this is slightly larger but very close to the 24.2 deg swath width which we computed above. This is an important characteristic for FireSat II. It would be extremely valuable to have the swaths overlap so that every FireSat II spacecraft can cover all locations on the Earth twice per day. Therefore, in designing the payload, we should work hard to maintain either the altitude or the minimum elevation angle to provide some swath overlap. Doing so could dramatically reduce the number of spacecraft required and therefore the cost of the system.



Fig. 17-22. Computation of FireSat II Ground-Track Parameters.

As Fig. 17-22 shows, the swath width does not need to be quite as large as the spacing between nodes along the equator. Even at the equator, it is enough to have a swath width equal to S, the perpendicular separation between the ground tracks. In Chap. 10, we selected an inclination for FireSat II of 55 deg to cover up to 65 deg latitude. Consequently, we can use the spherical triangle ABC shown in the figure to compute S as follows:

Table 17-8◀, Fig. 17-22, Eq. 17-36

Table 17-9. Parameter Calculations for a Passive Optical Sensor. IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
A. Orbit and Viewing Geometry			
Select altitude, h, and eccentricity, e	Table 10-1; Sec. 10.4	h =700 km, e = 0	See Sec. 10.6 for constellations
Compute orbit period, P	Tables 8-11, 8-12, IRC	P = 98.8 min	
Compute ground track velocity, V_g		V_g = 6.76 km/s	Varies with true anomaly
Compute node shift per orbit, ΔL	IRC	ΔL = 24.7 deg	= longitude shift at all latitudes
Compute Earth angular radius, ρ	Tables 8-11, 8-12, IRC	ρ = 64.3 deg	
Determine min working elevation angle, ε_{min}		ε_{min} = 20 deg	Major system trade; small elev angle provides more coverage, but makes it harder for payload
Compute max nadir angle, η_{max}	Tables 8-11, 8-12, IRC	η_{max} = 57.9 deg	Only slightly less than Earth ang. radius
Compute max Earth central angle, λ_{max}	Tables 8-11, 8-12, IRC	λ_{max} = 12.1 deg	
Compute slant range, R_s	Tables 8-11, 8-12, IRC	R_s = 1,584 km	min distance = altitude
Compute swath width, SW	= 2 λ_{max}	SW = 24.3 deg	
B. Pixel Parameters and Data Rate			
Specify maximum along-track ground sampling distance, Y_{max}	Design parameter	Y_{max} = 1.12 km	Based on spatial resolution requirements at λ_{max}
Determine along-track instantaneous field of view, IFOV	$IFOV_Y = \frac{Y_{max}}{R_s} \cdot \frac{180\text{deg}}{\pi}$	$IFOV_Y$ = 0.041 deg	One pixel width in along-track direction
Specify max cross-track ground sampling distance, X_{max}	Design parameter	X_{max} = 1.12 km	Based on spatial resolution requirements at λ_{max}
Specify number of cross-track detector samples at nadir in one pixel	Design parameter	N_{samp} = 3	Makes pixel same size in along-track and cross-track directions
Determine cross-track IFOV	$IFOV_X = \frac{X_{max}}{R_s} \cdot \frac{180\text{deg}}{\pi N_{samp}}$	$IFOV_X$ = 0.014 deg	
Determine cross-track ground pixel resolution, X, at nadir	$X = IFOV_X \cdot N_{samp} \cdot h\left(\frac{\pi}{180\text{deg}}\right)$	X = 495 m	Best cross-track resolution for this instrument
Determine along-track pixel resolution, Y, at nadir	$Y = IFOV_Y \cdot h\left(\frac{\pi}{180\text{deg}}\right)$	Y = 495 m	Best along-track resolution for this instrument
Determine number of cross-track pixels, Z_c	$Z_c = \left(\frac{2 \cdot \eta_{max}}{IFOV_X}\right)$	Z_c = 8,569	Ground pixel size varies along the swath
Find number of swaths recorded along-track in 1 sec, Z_a	$Z_a = \frac{V_g \cdot 1\,\text{sec}}{Y}$	Z_a = 13.7	Number of successive swaths without gaps at nadir
Find number of pixels recorded in 1 sec, Z	$Z = Z_c \cdot Z_a$	$Z = 1.12 \times 10^5$	
Specify number of bits used to encode each pixel, B	Design parameter	B = 14 bits	Based on radiometric resolution requirement and dynamic range
Specify number of spectral bands, N_{bands}	Design parameter	N_{bands} = 5	
Estimate data rate, DR	$DR = Z \cdot B \cdot N_{bands}$	DR = 8.2 Mbps	Large number may challenge downlink capacity
C. Define Sensor Integration Parameters			
Specify number of pixels for whiskbroom inst. N_m	Design parameter	N_m = 16	Must be large enough to allow sufficient integration time
Specify swath overlap, O	Design parameter	O = 1.185	
Find pixel integration period, T_i	$T_i = \frac{Y}{V_g} \cdot \frac{N_m}{O} \cdot \frac{X}{2\pi h}$	T_i = 111 μs	Integration time of each detector pixel at nadir
Find resulting pixel read-out frequency, F_p	$F_p = 1/T_i$	F_p = 9 kHz	

Table 17-9, Fig. 17-22◀, Eq. 17-36◀

Table 17-9. Parameter Calculations for a Passive Optical Sensor. (Continued) IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
D. Define Sensor Optics			
Specify width for cross-track detectors, d_x	Design parameter	$d_x = 50\ \mu m$	Typical for available infrared detectors
Specify quality factor for imaging, Q	Design parameter	$Q = 0.4$	$0.4 < Q < 2$ ($Q = 1.1$ for good image quality)
Specify operating wavelength, λ	Design parameter	$\lambda = 10.8\ \mu m$	Based on subject trades
Define focal length, f	$f = \frac{h \cdot d_x}{(X/N_{samp})}$	$f = 0.2$ m	Use altitude and Eq. (17-8)
Find diffraction-limited aperture, D	$D = \frac{\lambda f}{Q d_x}$	$D = 0.12$ m	Eq. 17-10
Compute F-number of optics, F#	$F\# = f/D$	$F\# = 1.85$	Typical range = 4–6
Compute along-track field of view of optical system, FOV	$FOV = IFOV_Y \cdot N_m$	$FOV = 0.65$ deg	FOV for the array of pixels
Determine cut-off frequency, F_c	$F_c = D/(\lambda f)$	$F_c = 51{,}000$/m	Referred to nadir
Determine cross-track Nyquist frequency, F_{nc}	$F_{nc} = 1/(2d_x)$	$F_{nc} = 10{,}000$/m	Referred to ground sample resolution at nadir
Determine along-track Nyquist frequency, F_{na}	$F_{na} = 1/(2d_x N_{samp})$	$F_{na} = 3{,}333$/m	Referred to ground sample resolution at nadir
Compute relative Nyquist frequencies, F_{qc} and F_{qa}	$F_{qc} = \frac{F_{nc}}{F_c}$; $F_{qa} = \frac{F_{na}}{F_c}$	$F_{qc} = 20\%$ $F_{qa} = 6.7\%$	% of the cutoff frequency used for this case
Find optics PSF as a function of distance, r, from center of detector	$PSF(r) = [\lambda J_1(Z)/Z]^2$ $Z \equiv \pi r D/\lambda f$		Use $-2d < r < 2d$ J_1 is the Bessel function of order 1
Find optical modulation transfer function (MTF_0) for clear circular optics	$MTF_0(F) = \frac{2C}{\pi}\left[\frac{\pi}{2} - \frac{C}{F_c\sqrt{1-C^2}} - a\sin C\right]$		Use $0 < F < F_c$ $C \equiv F / F_c$
Compute detector MTF cross-track, MTF_X, and along-track, MTF_Y	$MTF_X = [\sin(F_X)/F_X]^2$ $MTF_Y = \lvert\sin(F_Y)/F_Y\rvert$		Use $0 < F < F_c$ $F_X \equiv \pi X F$ $F_Y \equiv \pi Y F$
Compute system MTF cross-track, MTF_S	$MTF_S(F) = MTF_0(F) \cdot MTF_x(F)$		Let F range: $0 < F < F_c$
E. Estimate Sensor Radiometry (for Nadir Viewing)			
Define equivalent blackbody temperature, T	Design parameter	T = 290 K	Blackbody temperature of the Earth
Define the operating bandwidth, $\Delta\lambda$	Design parameter	$\Delta\lambda = 0.4$ μm	Based on subject trades
Determine blackbody spectral radiance, L_λ	$L_\lambda = E(\lambda)/\pi$ $E(\lambda)$ from Eq. (15-2)		Use range $\lambda \pm \Delta\lambda/2$
Look up transmissivity, $\tau(\lambda)$ of the atmosphere	See Fig. 15-5	See Fig 15.5	Evaluate operating bandwidth
Compute upwelling radiance, L_{upi}	$L_{upi}(\lambda) = L_\lambda\, \tau(\lambda)$		Total input radiance as a function of wavelength
Compute integrated upwelling radiance, L_{int}	$L_{int} = \sum L_{upi}(\lambda_i - \lambda_{i+1})$	$L_{int} = 3.31$ W/m²/sr	Evaluate over operating bandwidth
Compute radiated power, L, from a single detector sample at nadir	$L = L_{int} \cdot X \cdot Y/N_{samp}$	$L = 2.7 \times 10^5$ W/sr	Total power from the ground scene that arrives at the instrument
Compute input power, P_{in}, at sensor	$P_{in} = \frac{L}{h^2} \cdot \left(\frac{D}{2}\right)^2 \cdot \pi$	$P_{in} = 5.7 \times 10^{-9}$	P_{in} is power at the entrance to the optics
Define optical transmission factor τ_o	Design parameter	$\tau_0 = 0.80$	Typical value for optical systems
Find input power, P_D, at the detector pixel	$P_D = P_{in} \cdot \tau_0$	$P_D = 4.5 \times 10^{-9}$ W	Very little power arrives at each pixel
Determine available energy, E, after integration time at nadir	$E = P_D \cdot T_i / N_{samp}$	$E = 1.7 \times 10^{-13}$ J	Radiometric design challenge for FireSat II

Table 17-9. Parameter Calculations for a Passive Optical Sensor. (Continued) IRC = Inside Rear cover

Step	Calculation	FireSat II	Comments
E. Estimate Sensor Radiometry (for Nadir Viewing) (Continued)			
Find number of available photons, N_p	$N_P = E\,\lambda / hc$	$N_P = 9.2 \times 10^6$	h is Planck's constant, c is speed of light
Define quantum efficiency, QE, of detector at λ	Design parameter	$QE = 0.80$	Typical physical property of detector material
Compute number of photocarriers available, N_e	$N_e = N_p \cdot QE$	$N_e = 7.3 \times 10^6$	Evaluate for an ideal detector
Determine number of noise electrons, N_n	$N_n = \sqrt{N_e}$	$N_n = 2{,}708$	Considers only Shott noise
Determine number of read-out noise electrons, N_r	Design parameter	$N_r = 5{,}000$	Typical value
Determine total number of noise electrons, N_t	$N_t = \sqrt{N_e + N_r}$	$N_t = 2{,}709$	Assumes uncorrelated noise processes
Find signal-to-noise ratio of the image, SNR	$SNR = N_e / N_t$	$SNR = 2{,}707$	Assuming signal dominates background
Determine sensor dynamic range, DR	$DR = N_e / N_r$	$DR = 1{,}467$	With respect to cold space
F. Find the Noise-Equivalent Temperature Difference			
Compute noise-equivalent temperature difference, NEΔT	$NE\Delta T = \frac{\lambda T^2}{W\,SNR}\left[1 - \exp\left(\frac{W\Delta\lambda}{\lambda^2}\right)\right]$	$NE\Delta T = 0.02$ K	Temperature limit the instrument can resolve $W = 1.4388 \times 10^{-2}$

$$S = \sin^{-1}[\sin(24.8\text{ deg})\sin(55\text{ deg})] = 20.1\text{ deg} \quad (17\text{-}37)$$

The perpendicular separation between the orbits at the equator is 20.1 deg. Because the swath width is 24.2 deg, we now have some overlap margin even at the equator and substantial margin at higher latitudes, which are the primary areas of interest. We could, therefore, increase the minimum elevation angle to 25 deg. This would be a reasonable option. At present, we instead choose to hold ε at 20 deg and to provide some margin on altitude and elevation angle for later payload trades. We next compute the required resolution and data rates for FireSat II. From Table 3-4, we initially estimated the needed ground resolution as 500 m. Thus, the required resolution at nadir, θ_r from an altitude of 700 km is:

$$\theta_r = 0.500/700 = 7.1\times10^{-5}\text{ rad} = 0.041\text{ deg} \quad (17\text{-}38)$$

However, as shown in Fig. 17-23, the projected IFOV or Ground IFOV (GIFOV) at end of scan for a maximum slant range of 1,580 km grows to about 3.0 km in the cross track direction and 1.1 km in the along track dimension. This growth in GIFOV would cause the sensor to fall a factor of six short of the needed 500 m ground resolution in the worst case direction, and would probably prevent any further development of this system. We could mitigate this effect by designing in zoom optics that would reduce the IFOV with scan angle. However, we seek to develop a reliable, low cost system and introducing a complex zoom optics system would likely increase system cost while also reducing its reliability. A lower cost approach is to increase the format size of the focal plane assembly to sample at a higher spatial frequency to enable better ground resolution at end of scan. For example, increasing the spatial sampling rate by a factor of three in the cross track direction leads to the ground resolution versus nadir angle curve shown in the lower part of Fig. 17-23. In this case, GIFOV grows from 500 m × 500 m at nadir to 1.1 km × 1.1 km at end of scan. The price we pay in system performance for this approach is reduced sensitivity at end of scan. In the 3.8 μm band, NEDT at 300 K for this single +0.041deg^2 IFOV is close to 0.5 K. However, this is a fire detection and mapping system and while it might be useful to have better sensitivity at end of scan at room temperature, we are most interested in performance for fire signatures at temperatures close to 450 K and above. NEDT at 450 K for this system concept is about 0.2 K, which is more than adequate for distinguishing fire from other scene features.

Using this resolution, we can follow the procedure outlined in Table 17-9 to compute the data rate for FireSat II as 8 Mbps. This data rate is relatively high, but well within capabilities of existing technology. Nevertheless, to reduce cost, we should look at ways to reduce the downlink data rate. One approach is to process the data on board or more simply, turn off the imager over the oceans or other areas where fire detection is of limited utility or use well known and well understood data compression techniques such as Rice compression to reduce data rate by roughly a factor of two. For now, we will leave the value as computed, recognizing that this may need to be reduced later in the system design.

We next compute mapping and pointing budgets for FireSat II. We do not have a firm mapping requirement, but we do have some broad sense of what is needed. We begin, therefore, with a rough estimate of performance

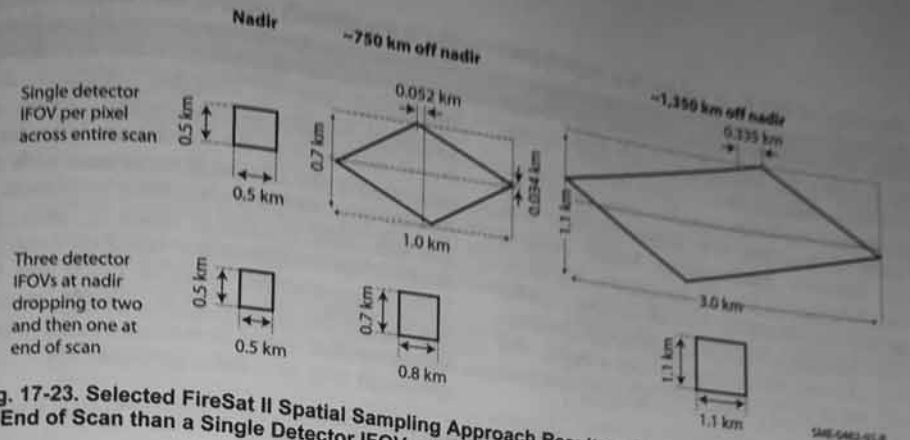
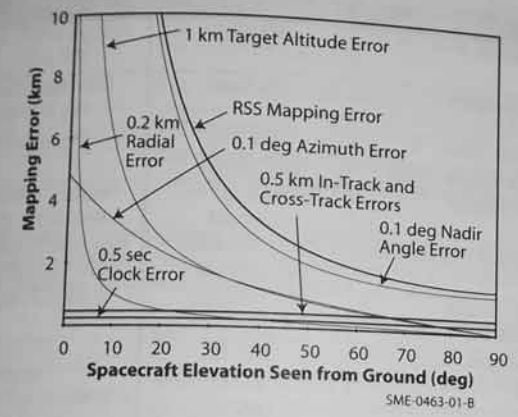
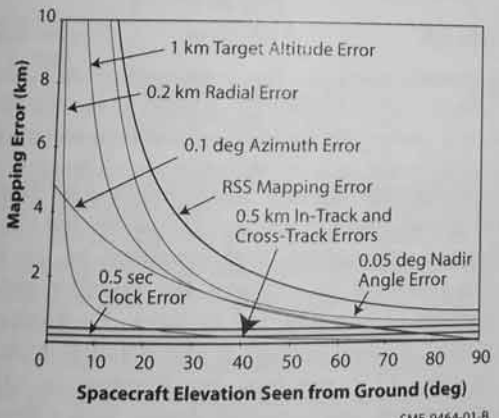


Fig. 17-23. Selected FireSat II Spatial Sampling Approach Results in Much Better Ground Resolution at End of Scan than a Single Detector IFOV per Pixel.



A. 0.1 deg Nadir Angle Error



B. 0.05 deg Nadir Angle Error

Fig. 17-24. FireSat II Mapping Budget. Reducing the nadir angle error below 0.05 deg will have relatively little impact on the overall mapping error because of the 1 km target altitude error. Compare with Fig. 8-50.

parameters and create the mapping error as a function of the elevation angle shown in Fig. 17-24. In this figure, we have used a 0.1 deg nadir angle and azimuth errors corresponding to a relatively inexpensive pointing system based on an Earth sensor. We know we can go to a more expensive system if necessary. In looking at Fig. 17-24A, we see that the mapping error at our chosen minimum elevation angle of 20 deg is between 6 and 8 km. While we are not certain what our mapping requirement is, we are reasonably sure that it is smaller than 6 km. We need to locate fires more accurately than this. Note also that the accuracy has been set almost entirely by our crude attitude number of 0.1 deg.

The next most critical parameter is the 1 km error in target altitude. This means that we assume we can determine the altitude of the fire above the Earth to 1 km—a reasonable accuracy with an oblate Earth model. But significantly improving this accuracy would require carrying a map of the altitudes of all of the regions of interest. That could be very difficult, particularly in mountainous areas, and would cost a lot more money. Therefore, it is of little value to drive the error in nadir angle down below approximately 0.05 deg because it would no longer be the dominant error source. Fig. 17-24B shows the curves that we would achieve with the error in nadir angle reduced to 0.05 deg and all of the other error sources remaining the same. Now the contribution of the errors in nadir angle and target altitude are comparable, so we will use this budget to establish a preliminary mapping requirement of 5.5 km at a 20 deg elevation angle, and 3.5 km at a 30 deg elevation angle. This may still be considerably cruder than we would like, so we may need to revisit this issue.

Now that we understand the mission geometry, we evaluate the optical, signal processing, and radiometric performance of the instrument using the process described in Table 17-9. The third column in that table summarizes results of computations for a whiskbroom sensor design for the FireSat II mission.

The example FireSat II design considers only initial feasibility of the instrument. Several challenges remain with this design and additional iterations are required within context of mission requirements and constraints to optimize the instrument design. For example, the computed data rate of 25.5 Mbps will present some design challenge for a very low cost system. A second challenge with the design is the relatively poor SNR for some of the bands at end of scan for lower temperature

Table 17-9◀, Fig. 17-24, Eq. 17-38◀

scenes in the range of 300 K or colder. We're prepared to accept this performance limitation, however, because our mission is to detect fire, not measure sea surface temperature or cloud top temperature with high spatial resolution across a wide swath.

We now select a "similar instrument" from Table 17-7 for our FireSat II example. We appear to have two options: the Planetary Multispectral Visible-Infrared Imager or the Multispectral Mid-IR instrument. We tentatively select the Multispectral Mid-IR as our similar instrument and will scale from its fundamental parameters of 1.4 m × 1.5 m × 0.9 m size, 270 kg weight, and 170 W power, for its 0.18 m aperture. We first compute the aperture ratio, R

$$R = 0.11/0.18 = 0.61 \quad (17\text{-}39)$$

The value 0.61 is close enough to 0.5 that we select K to be 2 to reflect our assessment that we are dealing with a significantly scaled down system that requires increased margin. With this fundamental ratio, we now estimate the FireSat II payload parameters as

$$\text{Size} = 0.9 \text{ m} \times 0.9 \text{ m} \times 0.5 \text{ m} \quad (17\text{-}40)$$

$$\text{Weight} = 2 \times 270 \times 0.61^3 = 123 \text{ kg} \quad (17\text{-}41)$$

$$\text{Power} = 2 \times 170 \times 0.61^3 = 77 \text{ W} \quad (17\text{-}42)$$

If we now scale from the Planetary Multispectral Visible-Infrared Imager, we derive a very different estimate for the FireSat II payload parameters. In this case,

$$R = 0.11/0.12 = 0.92 \quad (17\text{-}43)$$

We now estimate the FireSat II payload parameters as

$$\text{Size} = 0.50 \text{ m} \times 0.34 \text{ m} \times 0.27 \text{ m} \quad (17\text{-}44)$$

$$\text{Weight} = 12.8 \times 1.00^3 = 10.0 \text{ kg} \quad (17\text{-}45)$$

$$\text{Power} = 12 \times 1.00^3 = 9.3 \text{ W} \quad (17\text{-}46)$$

As described in Sec. 17.2.6, we have incorporated a factor of 2 to provide margin against scaling down the payload size for the Multispectral Mid-IR instrument significantly. The aperture diameter for the planetary instrument is the same as for this FireSat II payload. Therefore, we have not built margin into this scaling calculation.

Clearly, we find that simple scaling laws can give widely divergent estimates for size, mass, and power for concept payloads, depending on the reference payload. To resolve this difference, we need to look in more detail at the closest reference instrument, namely the Planetary Multispectral Visible-Infrared instrument and use a combination of the other methods described in Sec. 17.2.5, analogy with existing systems and budgeting by components to derive a higher confidence estimate.

The planetary instrument, while identical in aperture size with the FireSat II payload, is different at the next level of detail. It has two focal planes, versus the single focal plane for FireSat II. That difference allows us to reduce the dimensions of the instrument by a few centimeters in width and depth. Likewise, the planetary instrument has a long sunshield that is not needed for FireSat II, which further reduces the instrument length by tens of centimeters. However, FireSat II requires a scanner that adds back most of that length. A rough estimate of the scanner size is that it would be a few centimeters shorter than the sunshield so that a small reduction in instrument size is expected. Another difference is FireSat II requires active cooling, which adds to the power and mass of the instrument. Likewise, the higher data rate for the FireSat II instrument adds a substantial amount of power to the nominal instrument description. The size, mass, and power values in Table 17-10 account for all of these differences, based on existing hardware. Thus, as summarized in Table 17-10, our first guess is that the FireSat II payload is a simple, moderately sized instrument that would fit well on a small to medium-sized spacecraft with high data rate capability.

Table 17-10. Space Summary of FireSat II Initial Parameter Estimates.

Parameter	Nominal Value	Comments
Altitude, h	700 km	Range = 600 to 800 km
Inclination, i	55 deg	Coverage to 65 deg latitude
Swath width, $2\lambda_{max}$	24.2 deg	20.1 deg required
Nadir angle range, η	±57.9 deg	
Instrument	mid-range IR scanner	Payload needs expert input
Ground resolution	501 m at nadir (along track)	Key parameter—needs trade study
Instrument resolution	2.714 mrad	0.00409 deg
Aperture, A	0.11 m	
Size	0.48 × 0.22 × 0.18 cm^3	
Weight, W	20 kg	
Power, P	50 W	
Data rate, DR	8 Mbps/sec	May be limiting feature
Mapping	3.5 km @ ε=30 deg 5.5 km @ ε=20 deg	

Our initial analysis of a small, lightweight FireSat II payload shows that the mission is feasible but challenging. Several refinements and iterations on the design have the potential to improve the viability and cost-effectiveness of the payload concept. To illustrate the end point of a more complete and detailed design process, in Sec. 17.3, we turn our attention to VIIRS, a large instrument and a mature design with a fire detection mission among many other missions.

17.2.9 Life-cycle Cost and Operability of the Payload

Spacecraft designs are heavily driven by the need to minimize cost, while simultaneously meeting an acceptable level of performance. Over the years, a number of approaches have been proposed and implemented to treat cost as an independent variable. For systems where cost and performance may be traded, it is no longer sufficient to simply state mission requirements clearly and realistically. Rather, mission requirements may now become involved in the iterative process of design

Table 17-11. Impact of Remote Sensing Payloads on the Spacecraft Design.

Impact Area	Typical Requirements to Support Payload	Additional Considerations
Structure	Mount the optical instruments isostatically to the spacecraft bus	Carefully analyze aging of material (e.g., stress release in metal parts), humidity release, transition to microgravity, and acceleration forces
	Do not apply excessive forces or torques to the payload instrument	Typical stability requirements at critical locations within the optical instrument housing are in the μm and mdeg range
	Make the mounting structure or base plate for optical components stiff enough to prevent any misalignment when subjected to the forces and vibrations of launch	
Thermal	Make large opto-mechanical assemblies temperature stabilized or isothermal	Large reflective systems (which use Zerodur, Aluminum, or Beryllium or newly developed materials such as SiC or CsiC as materials for the mirrors) and mounting structures (which use composite materials) are temperature sensitive and may require semi-active temperature control of structure and mirrors
	Operate refractive optical systems typically within a specific temperature range to achieve required performance (frequently they employ semi-active temperature control)	Temperature gradients in optical components can severely degrade performance
	Make reflective systems entirely from the same material which leads to a compensation of thermal effects (typically done for cryogenic optical systems)	
External Alignment	Align the optical axis of the instrument and the line of sight of the pointing device with an external reference on the spacecraft. External alignments may need to be on the order of 1 arc sec	User reference cubes to achieve alignment
		External alignment requires a calibrated optical bench
Pointing	For monocular optical instruments, make the pointing requirements on the order of 0.1 to 0.01 of the swath width, typically	Mount attitude determination sensors (e.g., star sensors) to the instrument (not the bus) to minimize the effects of thermoelasticity
	For stereoscopic instruments, automated digital terrain mapping requires pointing knowledge of 1/5 of a pixel	Do pointing by maneuvering the spacecraft or by pointing devices (such as pointing mirrors for the front of the instrument or gimbals for the entire instrument)
Assembly Integration and Verification	Optical instruments require clean rooms and clean laminar air flow benches for all integration and verification activities	Cleaning optical surfaces is generally not possible
	Clean room requirements typically range from 100 to 100,000 ppm	During exposure to the environment, use cleanliness samples to verify the level of contamination
System Accommodation	Sensor must have an unobstructed field-of-view	Orient radiators and passive coolers for infrared systems to prevent interference with optical devices
	Sensor must have a guard cone to prevent performance degradation due to stray light	Calibration devices impose geometric constraints with respect to the optics of the system and the orbit
	Avoid pointing toward the Sun	

[Fleeter, 2000]. Several excellent examples of how to include cost as a system parameter rather than a given are described by Shishko, et al. [1996].

Working through trade-offs associated with cost, performance, and requirements in early stages of payload definition keeps payload designers focused on the best sensor characteristics to maximize mission performance and minimize cost. Designers sometimes want to perform a purely analytical evaluation of the costs and benefits of various design options. Unfortunately, relative benefits of different design features are difficult if not impossible to quantify in a clear and universally accepted way. Well understood and clearly stated analysis methods can be useful for providing a common framework for evaluating different designs. Ultimately, however, judgments about meeting mission goals within cost and schedule limitations rely on the insight of experienced designers and other analysts, adding to the challenges and significance of this portion of the payload definition process.

After we establish the final payload type and basic payload performance requirements, final design of the payload can begin. The final payload design could be as simple as an evaluation of existing payloads that are available or it could involve detailed design, fabrication, and testing of an entirely new instrument. The final step in the payload definition and sizing process is the decision by the responsible organization to buy or build the payload.

Integrating a payload into a spacecraft introduces a number of practical considerations for the spacecraft and other payload subsystems. These derived requirements can have a significant impact on other parts of the spacecraft. Table 17-11 contains an overview of some of the spacecraft accommodation considerations for a payload. Resolving the impact of these requirements on the spacecraft and other parts of the payload means we must assess the performance, cost, and technical risk for each subsystem to accommodate the payload.

17.3 Sample Mission—VIIRS

Spaceflight instruments that can automatically detect fires already exist. The remarkably versatile MODIS (Moderate-resolution Imaging Spectroradiometer), has been flying on board NASA Earth Observing System satellites since 1999. MODIS has provided an abundance of data in support of scientific investigations into Earth's atmosphere, oceans and land use that goes well beyond

Table 17-11, Fig. 17-24◀, Eq. 17-46◀

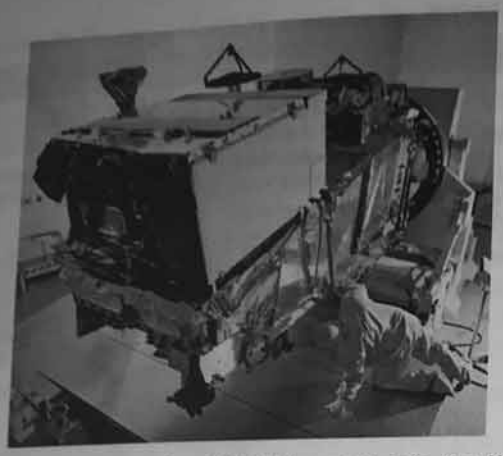


Fig. 17-25. Photograph of FU1 Integrated onto the NPP Spacecraft at Ball Aerospace. (Photo courtesy of Ball Aerospace)

fundamental needs for the FireSat II mission. MODIS is being replaced by the Visible/Infrared Imager Radiometer Suite (VIIRS), the next-generation imaging spectroradiometer for the NOAA/NASA Joint Polar Satellite System (JPSS) and the US Department of Defense Weather Satellite System (DWSS). VIIRS, described in Fig. 17-25, replaces and improves upon three different operational and research sensors with a single instrument. These three sensors are the MODIS, the Advanced Very High Resolution Radiometer (AVHRR) operating on board NOAA POES and EUMETSAT Polar System Metop since 1978 and the Operational Line Scanner (OLS) on board DMSP since 1976.

VIIRS is a whiskbroom scanner that monitors Earth from a 833 km polar Sun-synchronous orbits at visible-infrared wavelengths (0.4–12 μm) and provides well calibrated moderate (~km) spatial resolution measurements of light upwelling from Earth in support of a large number of high priority applications ranging from weather prediction for civilian and military needs to climate change monitoring, land usage, public health alerts and predictions of electrical power usage. Figure 17-26 shows that VIIRS offers significant improvements over the systems it replaces by providing 22 spectral bands with 4 times better spectral coverage than AVHRR, thereby enabling new agricultural, climate, disaster monitoring, public health and weather data products, 3 times better spatial resolution than AVHRR and MODIS at end of scan enabling sharper imagery over a much greater area and a fully calibrated Day/Night Band (DNB) that improves nighttime weather forecasting and military applications compared with OLS. VIIRS benefits from substantial US R&D investment in MODIS and other NASA Earth observing systems that led to a wide range of new environmental data products, including fire products, from well characterized instruments in polar sun synchronous orbits.

A key to the VIIRS architecture shown in Fig. 17-27 is a rotating telescope assembly that enables a flexible design needed to meet a remarkably diverse set of requirements for multispectral imaging spectroradiometry and low light level day/night imaging. Advantages of the rotating telescope design relative to scan mirror based systems like AVHRR and MODIS include better control of straylight, smaller range in angle of incidence of light on the fore optics to reduce striping and polarization sensitivity and better protection from contamination and degradation over time because all of the optical elements are placed deep inside the instrument housing,

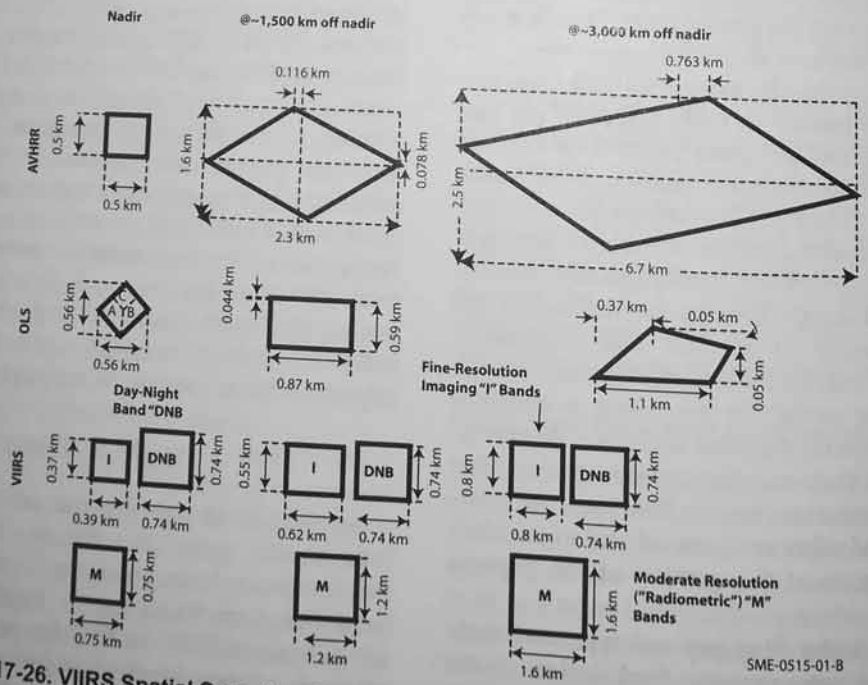


Fig. 17-26. VIIRS Spatial Sampling Approach Compared with the Legacy AVHRR and OLS Sensors. (Graphic courtesy of NASA)

Table 17-11◀, Fig. 17-26, Eq. 17-46◀

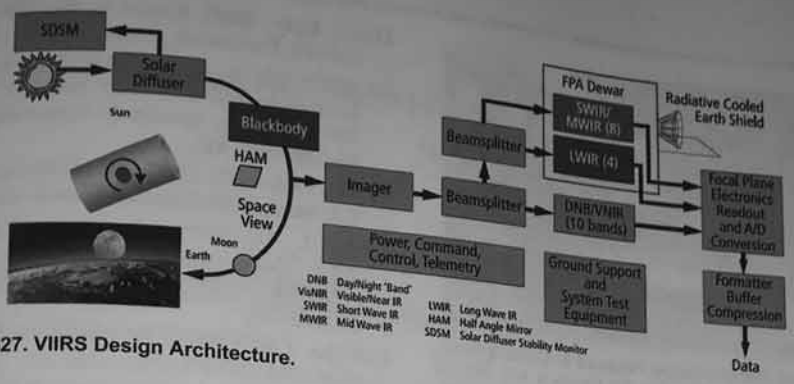


Fig. 17-27. VIIRS Design Architecture.

Many challenges were overcome in design and fabrication of the rotating telescope assembly to produce a well understood high performance subsystem. Significant US investment in the VIIRS architecture has resulted in an excellent imager that is ready to provide high fidelity data for the international science, weather and other environmental data product communities with unprecedented spatial resolution at end of scan.

The VIIRS fire detection algorithm is derived from the MODIS algorithm described by Kaufman and Justice [1998] and Giglio, et al. [2003]. The algorithm uses calibrated and geolocated data from five VIIRS spectral bands: M5 (0.672 μm), M7 (0.865 μm), M11 (2.25 μm), M13 (4.05 μm), M15 (10.763 μm), M16 (12.013 μm). In addition, the algorithm makes use of solar zenith and azimuth angles along with a global database describing whether a pixel is land, water or sometimes water. The algorithm follows these steps.

Identify potential fire pixels. The algorithm prescreens pixels by selecting only those clear sky pixels on land that have valid measurements. Then, the algorithm selects potential fire pixels by identifying pixels with spectral characteristics consistent with fire, based on temperature thresholds and differences in apparent brightness temperature between bands.

Generate statistics on pixel characteristics near potential fire pixels and separate potential fire pixels from the background. In this part of the process, the algorithm uses neighboring pixels to estimate the radiometric signal of a potential fire pixel in the absence of fire. Then, a series of threshold tests are used to perform relative fire detection. These tests look for the characteristic signature of an active fire in which both the M13 brightness and the M13 and M15 brightness temperature difference depart substantially from that of the non-fire background.

Make tentative fire detection based on absolute brightness temperature in M13. If the M13 brightness temperature is greater than 320 K at night, then fire detection is confirmed. During the day, the M13 brightness temperature in a pixel must be greater than 360 K for tentative fire detection.

Reject false alarms. Tentative fire detections made during the day are susceptible to false alarms because brightness temperature measurement can be contaminated with sunglint and other false signals resulting from background characterization errors in regions with abrupt boundaries between land and water or along the boundaries of deserts. These false alarms can be rejected with good reliability by using information derived from the solar reflective M5, M7 and M11 bands.

Export fire pixel locations with quality control information.

The VIIRS sensor and algorithm system illustrates many of the features of an automated fire detection system. In the context of the FireSat II mission, this example provides a point design with finalized tradeoffs in size, mass, power, resolution, and data processing that has already been built and tested.

17.4 The Evolution of Observation Payloads

In less than 60 years, space-based observation payloads have evolved from simple 35-mm film movie cameras on board converted World War II missiles providing imagery for just a few minutes to aggregations of widely different instruments covering much of the electromagnetic spectrum simultaneously from Earth orbit for more than 10 years. The very first space-based observation payload flew on board a German V-2 rocket captured by the United States near the end of World War II. This rocket was launched from White Sands Missile Range on October 24, 1946 and produced the image shown in Fig. 17-28. Clyde Holliday from Johns Hopkins University/Applied Physics Laboratory, developer of this first space-based imager, wrote: "Results of these tests now are pointing to a time when cameras may be mounted on guided missiles for scouting enemy territory in war, mapping inaccessible regions of the earth in peacetime, and even photographing cloud formations, storm fronts, and overcast areas over an entire continent in a few hours…the entire land area of the globe might be mapped in this way" [Holliday and Hamilton, 1950]. See the book website for a more extended discussion of how these payloads have evolved.



As new observational payloads are developed with ever increasing data collection rates, the driving technology for these systems is expected to shift from focal

Table 17-11◀, Fig. 17-27, Eq. 17-46◀



Fig. 17-28. The First Observational Payload to Collect Images from Space (>100 km alt.). Did So Using a V-2 Rocket Launched from White Sands Missile Range on October 24, 1946.

plane technology for visible-infrared systems toward on board and ground processing to handle the vast amounts of data generated by ever more capable systems. Future systems based on distributed architectures that are refreshed routinely may be able to keep pace with availability of improved technology better than today's systems which are refreshed on decade time scales.

Space-based observation systems are most valuable when they're effective, ubiquitous, affordable, reliable, flexible and rapidly replaceable. Readers should not confuse low system cost with low performance. The original Iridium system proved that it is possible to achieve both low cost and unprecedented performance. Unfortunately, the failure of the original Iridium business model prevented this production approach from being sustained in near term upgrades and in the building of similar expected systems. Any future observation system must meet its operational or research performance requirements. For instance, regardless of satellite constellation, space imagers must meet operational performance requirements for parameters such as spatial resolution, spectral coverage, area coverage rate, sensitivity and geolocation. Likewise, systems in any architecture must meet system level hardness, reliability and operational lifetime needed to fulfill customer mission needs. System-level trade studies early in program development need to be broadened to consider alternative system architectures that might be built at lower overall cost using highly engineered and reliable systems designed for affordable, efficient manufacturing and operationally effective performance and built using well understood and continuously improving processes. These early studies would trade satellite design life, technology improvement schedule, constellation refresh rate, survivability, and cost within context of firm system performance parameters.

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19 Spacecraft Subsystems II—Control Systems

19.1 Spacecraft Attitude Determination and Control Systems
Control Modes and Requirements; Quantify the Disturbance Environment; Selection of Spacecraft Control Methods; Selection and Sizing of ADCS Hardware; Define the Determination and Control Algorithms

19.2 Spacecraft Trajectory Navigation and Control Systems
System Definition Process; Navigation; Trajectory Control; Design Considerations

19.1 Spacecraft Attitude Determination and Control Systems

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In the year 1900, Galveston, Texas, was a bustling community of approximately 40,000 people. The former capital of the Republic of Texas remained a trade center for the state and was one of the largest cotton ports in the United States. On September 8 of that year, however, a powerful hurricane struck Galveston island, tearing the Weather Bureau wind gauge away as the winds exceeded 100 mph and bringing a storm surge that flooded the entire city. The worst natural disaster in United States' history—even today—the hurricane caused the deaths of between 6,000 and 8,000 people. Critical in the events that led to such a terrible loss of life was the lack of precise knowledge of the strength of the storm before it hit.

In 2008, Hurricane Ike, the third costliest hurricane ever to hit the United States' coast, traveled through the Gulf of Mexico. Ike was gigantic, and the devastation in its path included the Turk and Caicos Islands, Haiti, and huge swaths of the coast of the Gulf of Mexico. Once again, Galveston, now a city of nearly 60,000, took the direct hit as Ike came ashore. Almost 200 people in the Caribbean and the United States lost their lives; a tragedy to be sure, but far less deadly than the 1900 storm. This time, people were prepared, having received excellent warning from the GOES satellite network (Fig. 19-6). The GOES satellites have been a continuous monitor of the world's weather since 1975, and they have since been joined by other Earth-observing satellites. This weather surveillance, to which so many now owe their lives, is possible in part because of the ability to point accurately and steadily at the Earth below. The importance of accurately pointing spacecraft to our daily lives is pervasive, yet somehow escapes the notice of most people. But the example of the lives saved from Hurricane Ike as compared to the 1900 storm is something no one should ignore. In this section, we will summarize the processes and technologies used in designing and operating spacecraft pointing (i.e., attitude) systems.

Attitude is the three-dimensional orientation of a vehicle with respect to a specified reference frame. *Attitude systems* include the sensors, actuators, avionics, algorithms, software, and ground support equipment used to determine and control the attitude of a vehicle. Attitude

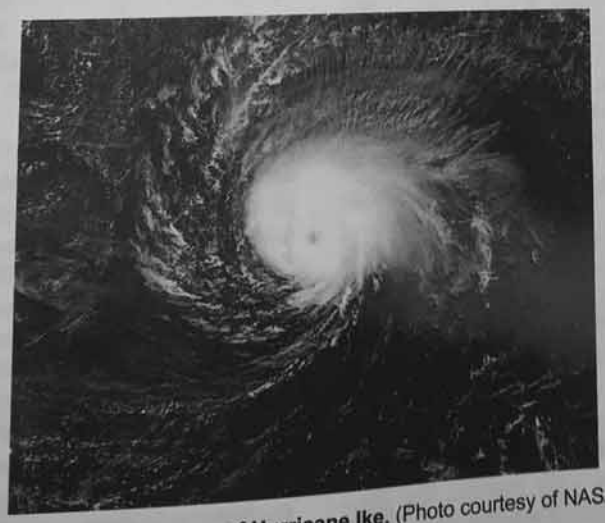


Fig. 19-1. Satellite Image of Hurricane Ike. (Photo courtesy of NASA)

systems can have a variety of names, such as attitude determination and control system (ADCS), attitude ground system (AGS), attitude and orbit control system (AOCS), guidance, navigation and control (GNC), or whatever other term describes the designers' focus in achieving the attitude needs of a particular mission. When we use an acronym in this section, we will use ADCS, but any given specialist may be more familiar with other terms.

Spacecraft attitude changes according to the fundamental equations of motion for rotational dynamics, the Euler equations, here expressed in vector form in the spacecraft's reference frame:

$$\dot{\mathbf{H}} = \mathbf{T} - \omega \times \mathbf{H} \tag{19-1}$$

This vector Eq. (19-1) represents the conservation equations for the *angular momentum*, which is denoted by **H**. Recall that linear momentum is the translational motion of a body that will remain constant unless a force acts to change it, and it is calculated (in Newtonian physics anyway) as mass times velocity. Analogously, angular momentum is the rotational motion of a body that will continue unless changed by a *torque*, and it is calculated as the body's *moment of inertia* times its angular velocity (or rotation rate). The moment of inertia is a 3-by-3 matrix of values that describe the distribution of mass in a body. There is always a coordinate frame, called the principal axis frame, for which the moment of inertia matrix is diagonal. The difference between the geometric and principal axis frames is of great interest to ADCS designers, as we will see later in this section.

Note that in the form above, the Euler equation makes it clear that the magnitude of angular momentum in a system can only be changed by applying external torques, T, because the change due to the term $\omega \times$ H can only change the direction of H, not the magnitude. The magnitude of a body's angular momentum will remain constant in the absence of external torques upon it, even when parts of the body can move with respect to other parts, such as a gyroscope spinning in its housing. If part of the body starts to spin in one direction in the absence of external torques, the rest of the body will have to spin in the opposite direction so that total angular momentum is conserved. With this in mind, we can relate the angular velocity of the spacecraft, ω, to H by the equation:

$$\mathbf{H} = \mathbf{I}\omega + \mathbf{h} \tag{19-2}$$

where I is the moment of inertia and h is the angular momentum stored by any rotating objects that are part of the spacecraft, such as momentum wheels or gyroscopes. So, by the product rule of calculus, the Euler equations can be rewritten as a matrix equation:

$$\dot{\mathbf{I}}\omega + \mathbf{I}\dot{\omega} + \dot{\mathbf{h}} = \mathbf{T} - \omega \times \mathbf{H} \tag{19-3}$$

or, after moving some terms around:

$$\mathbf{I}\dot{\omega} = \mathbf{T} - \mathbf{h} - \dot{\mathbf{I}}\omega - \omega \times \mathbf{H} \tag{19-4}$$

The form of Eq. (19-4) allows us to understand how attitude can change due to a variety of causes. The first term on the right-hand side represents the external torque's direct contribution to attitude dynamics; this term includes how some actuators can be used to control spacecraft attitude by creating external torques. The second term gives the relationship between changes in on-board rotating objects' speeds and changes in the spacecraft's rotational velocity; this term is where certain other control actuators enter into the dynamics as so-called internal torques. The third term shows how changes in the spacecraft moment of inertia (representing how mass is distributed in the spacecraft), such as by solar array articulation, can affect attitude dynamics; in the absence of changes in mass properties, the third term disappears. The fourth term is called the *gyroscopic torque*, and it shows how the angular momentum appears to change direction, but not magnitude, in the spacecraft's frame of reference when the spacecraft is rotating. All these effects combine to determine the rate of change of the angular velocity on the left-hand side.

Attitude determination is the process of combining available sensor inputs with knowledge of the spacecraft dynamics to provide an accurate and unique solution for the attitude state as a function of time, either onboard for immediate use, or after the fact (i.e., post-processing). With the powerful microprocessors now available for spaceflight, most attitude algorithms that formerly were performed as post-processing can now be programmed as onboard calculations. Therefore, though there are still good engineering reasons for certain processes to be performed only by ground-based attitude systems (see Chap. 28), it will be sufficient to focus our attitude determination discussions in this chapter on the design and implementation of onboard systems.

The product of attitude determination, the attitude *estimate* or *solution*, is attained by using sensors to relate information about external references, such as the stars, the Sun, the Earth, or other celestial bodies, to the orientation of the spacecraft. Frequently, any single sensor has a noise level or (see Sec. 16.3) other drawback that prevents it from providing a fully satisfactory attitude solution at all times. Therefore, more than one sensor is often required to meet all mission requirements for a given mission.

The combination of information from multiple sensors is a complex field of study. The possibilities for any given mission range from simple logical combination of sensors, depending on mode, to modern information filtering methods, such as Kalman filtering. Many methods require some prediction of a future attitude from current conditions. Because all spacecraft sensors must use the spacecraft's reference frame as a basis for attitude determination, the development of angular momentum according to the spacecraft's frame of reference can be important for some attitude determination algorithms. This is the reason for the spacecraft-referenced aspect of the Euler equations.

Attitude control is the combination of the prediction of and reaction to a vehicle's rotational dynamics. As

Table 19-1. Steps in Attitude System Design. An iterative process is used for designing the ADCS.

Step	Inputs	Outputs
1a) Define Control Modes *1b) Define or Derive System-Level Requirements by Control Mode*	Mission requirements, mission profile, type of insertion for launch vehicle	List of different control modes during mission. Requirements and constraints.
2) Quantify Disturbance Environment	Spacecraft geometry, orbit, solar/magnetic models, mission profile	Values for torques from external and internal sources
3) Select Type of Spacecraft Control by Attitude Control Mode	Payload, thermal & power needs Orbit, pointing direction Disturbance environment Accuracy requirements	Method for stabilization & control: 3-axis, spinning, gravity gradient, etc.
4) Select and Size ADCS Hardware	Spacecraft geometry and mass properties, required accuracy, orbit geometry, mission lifetime, space environment, pointing direction, slew rates. Failure detection and redundancy	Sensor suite: Earth, Sun, inertial, or other sensing devices. Control actuators: reaction wheels, thrusters, magnetic torquers, etc. Data processing avionics, if any, or processing requirements for other subsystems or ground computer.
5) Define Determination and Control Algorithms	Performance considerations (stabilization method(s), attitude knowledge & control accuracy, slew rates) balanced against system-level limitations (power and thermal needs, lifetime, jitter sensitivity, spacecraft processor capability)	Algorithms and parameters for each determination and control mode, and logic for changing from one mode to another.
6) Iterate and Document	All of above	Refined mission and subsystem requirements. More detailed ADCS design. Subsystem and component specifications.

discussed in Secs. 7.3 and 7.4, spacecraft exist in an environment of small and often highly predictable disturbances. Therefore, in certain cases they may be passively controlled. That is, a spacecraft may be designed in such a way that the environmental disturbances cause the spacecraft attitude to stabilize in the orientation needed to meet mission goals. Alternately, a spacecraft may include actuators that can be used to actively control the spacecraft orientation. These two general types of attitude control are not mutually exclusive. A spacecraft may be mostly or usually passively controlled and yet include actuators to adjust the attitude in small ways or to make attitude maneuvers (i.e., slews) to meet other objectives, such as targets of opportunity or communication needs.

As seen earlier, external torques change the total angular momentum, and internal torques exchange momentum between different rotating parts of the spacecraft. In this way *reaction wheels* or *control moment gyroscopes* may be used to change spacecraft pointing without affecting total angular momentum. Because environmental disturbances create external torques on the spacecraft, they also create angular momentum that must be either stored or removed by the attitude system. Small external torques that vary over the course of an orbit but have a mean of zero may be managed just through storage, but those torques that have a non-zero mean (secular torques) will cause a gradual increase in angular momentum, and this momentum build-up must eventually be removed with actuators that create external torques. Thrusters, magnetic torquers, or even solar trim tabs can be used to create controlled external torques on the spacecraft, thus controlling the total angular momentum.

Attitude system design is an iterative process. Table 19-1 lists typical steps in a design process and what inputs and outputs would be expected for each step. Figure 19-2 presents all the processes involved in attitude systems. The FireSat II spacecraft, shown in Fig. 19-3, and the Supplemental Communications System (SCS) constellation of spacecraft, shown in Fig. 19-4, will be used to illustrate this process.

19.1.1 Control Modes and Requirements

The first step of the attitude system design process is the definition of guiding requirements based ultimately on mission goals (Sec. 3.3). Since mission goals often require more than one mode of operating a spacecraft, the guiding requirements generally begin with a description of the control modes the ADCS is expected to execute to meet those goals. Tables 19-2 and 19-3 describe typical spacecraft control modes and requirements.

The final form of ADCS requirements and control modes will be the result of iteration; control modes are designed to achieve certain sets of requirements, and better understanding of the actual needs of the mission often results from having these modes of controlling the spacecraft well defined. This iteration takes place in a trade space where a single set of ADCS hardware must be used in different ways to meet different sets of requirements.

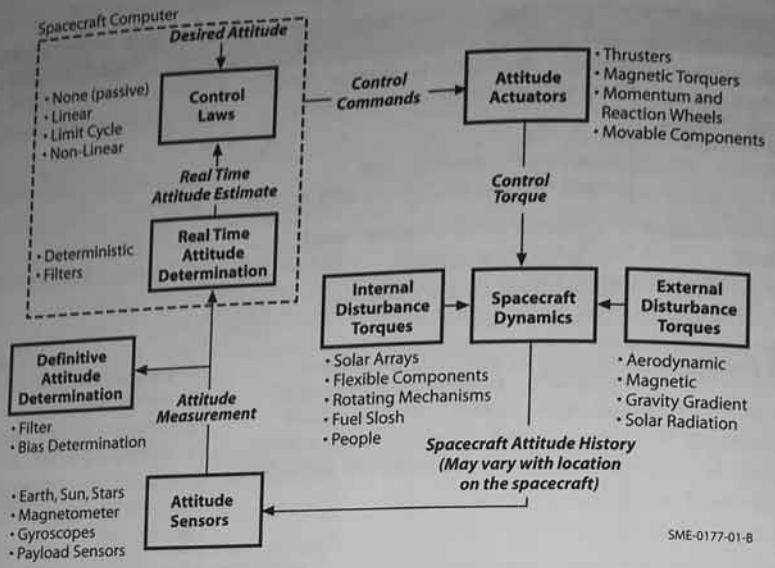


Fig. 19-2. Diagram of a Complete Attitude Determination and Control System. Definitive attitude determination usually occurs in ground processing of telemetry, whereas onboard, real-time determination design focuses on being extremely reliable and deterministic in its operation.

The ADCS will also be dependent on certain other subsystems, such as the power (Sec. 21.2) and structural (Sec. 22.1) subsystem. Attitude needs will also impose requirements on other subsystems, such as propulsion, thermal control, and structural stability. Figure 19-5 shows many of the complex interactions needed to bring the ADCS design in line with the needs of the whole mission (see Sec. 3.1).

Different mission needs may impose different requirements on the ADCS. One such critical need for many missions is orbit control. For many spacecraft the ADCS must control vehicle attitude during the firing of large liquid or solid rocket motors for orbit insertion or management. Large motors can create large disturbance torques, which can drive the design to larger actuators than may be needed for the rest of the mission.

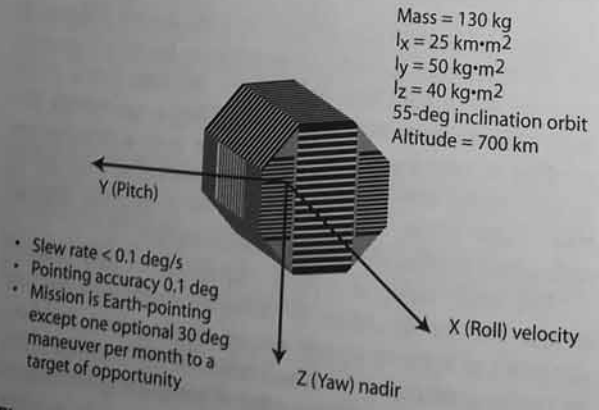


Fig. 19-3. Hypothetical FireSat II Spacecraft. We use this simplified example of a low-Earth orbiting satellite to discuss key concepts throughout the section.

Table 19-1◀, Fig. 19-4, Eq. 19-4◀

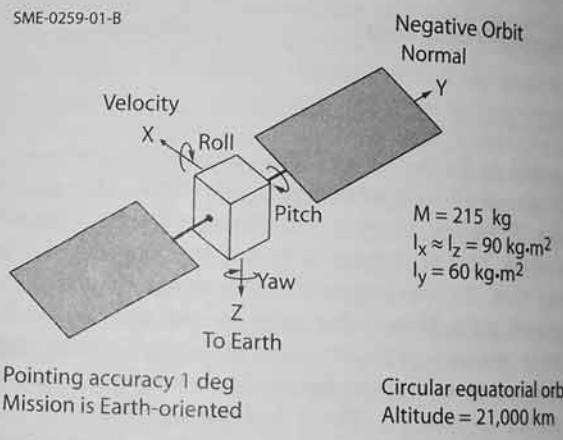


Fig. 19-4. One Member of the Hypothetical Supplemental Communications System (SCS) Constellation. We will also use this collection of three spacecraft in medium Earth orbit to illustrate attitude system design practices.

Once the spacecraft is on station, the payload pointing requirements usually dominate (see Sec. 8.6). These may require planet-relative or inertial attitudes and fixed or spinning fields of view. There is usually also a need for attitude slew maneuvers, and the frequency and speed of those maneuvers must be defined. Reasons for slews can include:

- Acquiring the desired spacecraft attitude initially or after a failure
- Repointing the payload's sensing systems to targets of opportunity or for calibration purposes
- Tracking stationary or moving targets, including communication stations
- Directing the vehicle's strongest motor to the proper direction relative to orbital motion

Table 19-2. Typical Attitude Control Modes. Performance requirements are frequently tailored to these different control operation modes.

Mode	Description
Acquisition	Initial determination of attitude and stabilization of vehicle for communication with ground and power generation just after launch. Also may be used to recover from power upsets or emergencies.
Orbit Insertion	Period during and after boost while spacecraft is brought to final orbit. Options include no spacecraft control, simple spin stabilization of solid rocket motor, and full spacecraft control using liquid propulsion system. May drive certain aspects of ADCS design.
Normal Mission, On-Station	Used for the vast majority of the mission. Requirements for this mode should drive system design.
Slew	Reorienting the vehicle when required.
Contingency or Safe	Used in emergencies if regular mode fails or is disabled. Will generally use less power or fewer components to meet minimal power and thermal needs.
Special	Requirements may be different for special targets or time periods, such as when the satellite passes through a celestial body's shadow, or *umbra*.

Table 19-3. Typical Attitude Determination and Control Performance Requirements. Requirements need to be specified for each mode. The following lists the performance criteria frequently specified.

Criterion	Definition*	Examples/Comments
Accuracy	Knowledge of and control over a vehicle's attitude relative to a target attitude as defined relative to an absolute reference	0.25 deg, 3σ, often includes determination errors along with control errors, or there may be separate requirements for determination and control, and even for different axes
Range	Range of angular motion over which determination & control performance must be met	Any attitude within 30 deg of nadir. Whenever rotational rates are less than 2 deg/sec.
Jitter	Specified bound on high-frequency angular motion	0.1 deg over 60 sec, 1 deg/s, 1 to 20 Hz; prevents excessive blurring of sensor data
Drift	Limit on slow, low-frequency angular motion	0.01 deg over 20 min, 0.05 deg max; used when vehicle may drift off target with infrequent command inputs
Transient Response	Allowed settling time or max attitude overshoot when acquiring new targets or recover from upsets	10% max overshoot, decaying to <0.1 deg in 1 min; may also limit excursions from a set path between targets

* Definitions vary with procuring and designing agencies, especially in details (e.g., 1σ or 3σ, amount of averaging or filtering allowed). It is always best to define exactly what is required.

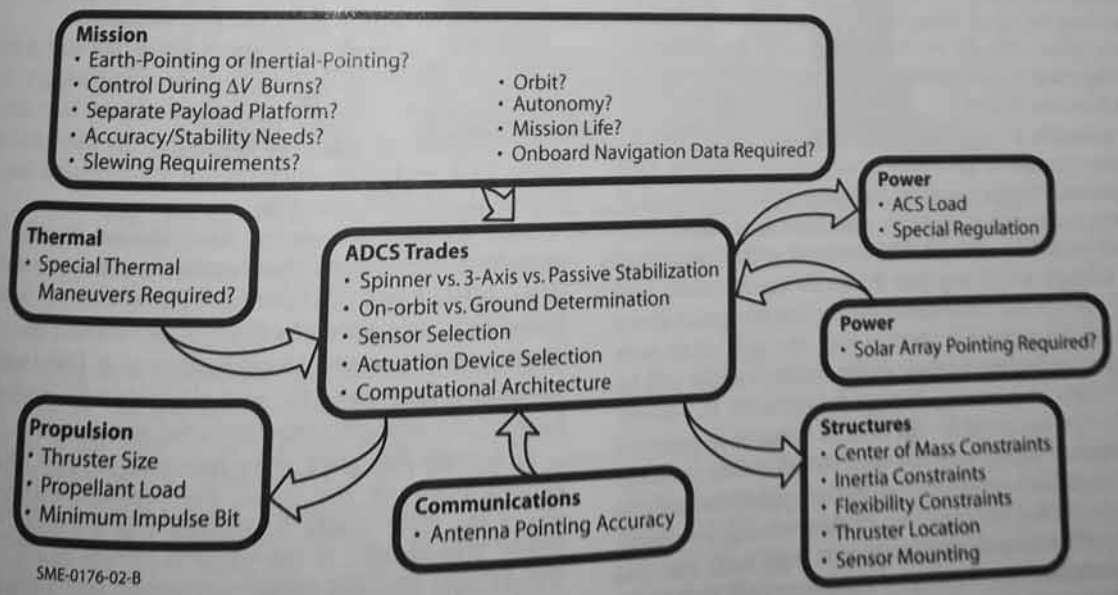


Fig. 19-5. The Impact of Mission Requirements and Other Subsystems on the ADCS. Direction of arrows shows requirements flow from one subsystem to another.

Table 19-3, Fig. 19-5, Eq. 19-4◀

19.1.2 Quantify the Disturbance Environment

The environment in which the spacecraft will operate constrains what types of control methods will be effective. For example, the relatively strong magnetic fields that occur in low Earth orbit (LEO) can create disturbance torques that need to be managed, but they also allow the use of magnetic torquers, a means of attitude control not available at much higher altitudes like geosynchronous orbit (GEO). Here, we will focus on the torque disturbance environment as the primary driver for control mode and hardware selection, but the sensitivity of the ADCS designer must be to more than just the external torque disturbances of the operational orbit. For example, some attitude sensors, such as star cameras that use charge-coupled devices (CCDs) for imaging, can be highly sensitive to the intense radiation in the Van Allen belts of the Earth's magnetosphere (see Sec. 7.3). Active Pixel Sensor (APS) cameras may be more robust. Depending on the specific model, the star camera may underperform or even provide no information at all when the spacecraft occupies these regions.

Only three or four sources of torque matter for the typical Earth-orbiting spacecraft: gravity-gradient effects, magnetic field torques on a magnetized vehicle (as most spacecraft will be), impingement by solar-radiation, and aerodynamic torques for LEO satellites. Figure 19-6 summarizes the relative effects of these disturbances for different flight regimes. Chapter 7 describes the Earth environment in detail, and Hughes [2004] provides a thorough treatment of disturbances.

Other external disturbances to the spacecraft are either small relative to the four main external disturbances, such as infrared emission pressure, or they are limited in time, such as outgassing. Occasionally, what is normally negligible can become surprisingly large, even exceeding the usual disturbance torque sources, but this is one of the reasons for maintenance of healthy engineering margins and operational plans that are adaptable to unforeseen events.

Centroids. Estimation of environmental torques often requires the use of geometrical averaging. Anyone with a technical education will be familiar with the centroid of an area, but it may have been some time since the reader encountered this concept. The centroid is the point in an area through which any line drawn in any direction will evenly divide moments about the line (or any point along the line). To express it another way, the sum of all area elements multiplied by their distances from a line will be zero for any line passing through the centroid. In a sense, it is the average point for the area. If a source of pressure were applied evenly over the area, the total solar pressure force could be represented as being applied entirely at the centroid for the purposes of determining moments, and therefore disturbance torques. A solid body can also have centroids. The *center of mass* (cm or CoM) is the point (usually inside) the body through which any plane will divide the mass moment evenly. By applying a force at or along the center of mass, no torques are created. This is why freely rotating bodies rotate about their centers of mass.

As a practical example, the point that may be regarded as the location of a body for purposes of gravitational forces is called the *center of gravity* (cg); i.e., all effects of gravity on the body can be considered to act at the cg. In the essentially uniform gravity that we humans occupy, the center of mass is usually indistinguishable from the center of gravity, but in the free-fall of a space orbit, the absence of direct gravitational forces and torques means that the change, or *gravity-gradient*, over the extent of a body can be important. For elongated or flattened objects in orbit, the cm may be offset from the cg, so that the gravitational force is effectively applied with an offset from the cm, creating a torque—this is the gravity gradient torque. Note that the cg is a function of the current attitude of the spacecraft, not just its mass configuration, which is critical in attitude analysis.

Other environmental effects can be understood in terms of offsets between centroids of different effects on a body. When the aerodynamic force centroid, which is at the centroid of the *ram area* (the area presented to the velocity direction), is not aligned with the cm, a torque is created. Solar radiation pressure is more intense on certain surfaces (reflective) than others (absorptive). The total pressure force over the Sun-pointing surface of a spacecraft can be considered to act through a *center of pressure* (cp) with an average reflectance, and the offset of that point from the cm results in solar radiation pressure torque. The location of this cp is a function of attitude as well as surface properties. Some modern surfaces can have their reflectance change with a change in applied voltage, usually for thermal reasons, but which results in a controlled change in cp location. So, in detailed modeling of spacecraft, the determination of the weighted averages of various forces is important to a good understanding of the torque environment.

Modeling Major Disturbances. Now, we will present the equations used to model major disturbances with some explanation and demonstration of how they can be used to design attitude systems. After the explanations, Table 19-4 will show disturbance calculations for the FireSat II and SCS examples.

Solar Radiation Pressure. As discussed in Sec. 18.5, sunlight has momentum, and therefore it exerts pressure on those objects it strikes. If an object absorbs all the sunlight falling on it, then it absorbs all of its momentum and experiences a certain pressure force because of it. If the sunlight is instead reflected exactly back along its path, such as by a mirror, the pressure force felt is twice as much.

If a sunlit flat plate were mirrored on one half and painted black on the other, the pressure distribution across the plate would be uneven and a torque would result. Alternately, if the plate were all black, but a weight were attached to one end in the plate's shadow, a torque would also result because the center of pressure would be in the center of the plate, but the center of mass would be closer to the weighted end. These phenomena are called *solar radiation pressure* (SRP) torques.

Table 19-3◀, Fig. 19-5◀, Eq. 19-4◀

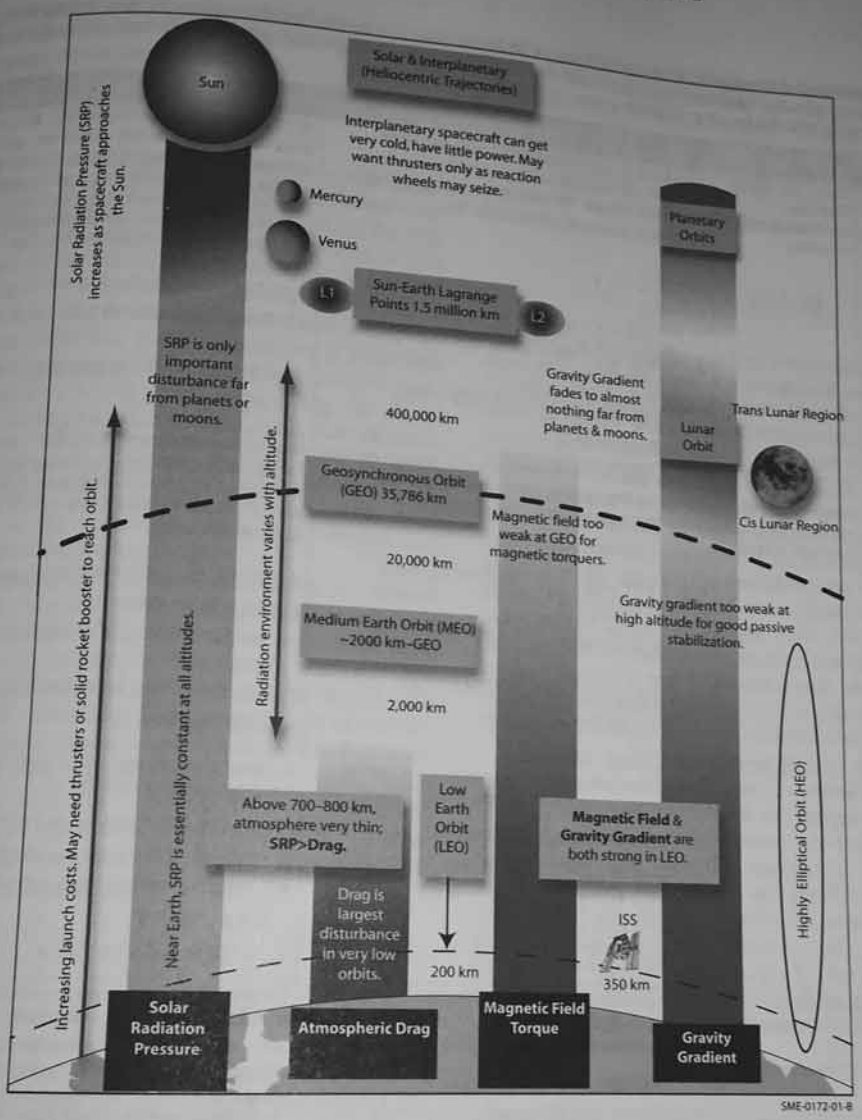


Fig. 19-6. Effects of Major Environmental Disturbances on Spacecraft Attitude System Design. The diagram has a roughly logarithmic scale of altitude. The columns represent the four major disturbance sources, with the intensity of color for each column indicating the strength of that disturbance in the various flight regimes.

Now imagine a spacecraft like FireSat II in sunlight. Some parts of the spacecraft stick out further from the center of mass than others. Some surfaces are more reflective than others; solar arrays would absorb more light than reflective metallic surfaces would. Also, surfaces that are angled with respect to the Sun would have less pressure on them than similar surfaces directly facing the Sun. All this goes to demonstrate that accurately predicting SRP torques is very tricky. That said, a good starting estimate can be gleaned by assuming a uniform reflectance and using the following equation:

$$T_s = \frac{\Phi}{c} A_s (1+q)(cp_s - cm)\cos\varphi \qquad (19\text{-}5)$$

where T_s is the SRP torque, Φ is the solar constant adjusted for actual distance from the Sun (average value: 1,366 W/m^2 at 1 AU), c is the speed of light (3×10^8 m/s), A_s is the sunlit surface area in m^2, q is the unitless reflectance factor (ranging from 0 for perfect absorption to 1 for perfect reflection), φ is the angle of incidence of the Sun, and cp_s and cm are the centers of solar radiation pressure and mass.

Table 19-3◀, Fig. 19-6, Eq. 19-5

Table 19-4. Disturbance Torque Summary and Sample Calculations. See text for detailed discussion and definition of symbols. FireSat II is mainly affected by magnetic torques, with the 30-degree offset attitude also affected by gravity gradient torques. SCS satellites are mainly affected by solar radiation pressure torques.

Disturbance	Type	FireSat II	SCS
Solar Radiation	Cyclic for Earth-oriented; constant for Sun-oriented	FireSat II is very small and Earth-oriented (though not Earth-pointing), and has body-mounted arrays. Therefore, its surface area is small and its cp_s is close to its *cm*. $A_s = 1.2\text{ m} \times 1.1\text{ m} = 1.3\text{ m}^2$; $q = 0.6$ $\varphi = 0$ deg; $cp_s - cm = 0.1$ m $T_s = (1{,}367)(1.3)(1 + 0.6)(0.1)/(3 \times 10^8)$ $= 9.6 \times 10^{-7}$ N·m	SCS is small and Earth-pointing, so the surface area will be fairly small, However, the need to balance the masses of three stowed satellites on the launch vehicle may reduce our control over mass placement and may cause the center of pressure to be considerably offset from the cm with respect to the Sun. $A_s = 2\text{ m} \times 1.5\text{ m} = 3\text{ m}^2$; $q = 0.6$ $\varphi = 0$ deg; $cp_s - cm = 0.3$ m $T_s = (1{,}367)(3.0)(1 + 0.6)(0.3)/(3 \times 10^8)$ $= 6.6 \times 10^{-6}$ N·m
Atmospheric Drag	Constant for Earth-oriented; variable for inertially oriented	Similar to SRP. The ram face will always be the same, so the *cp* and *cm* locations can be designed a bit better in this direction. $A_r = 1.3\text{ m}^2$; $cp_a - cm = 0.05$ m; $C_d = 2.0$ For 700 km orbit: $\rho \approx 10^{-13}$ kg/m^3; $V = 7{,}504$ m/s $T_a = (0.5)(10^{-13})(2.0)(1.3)(7504)^2(0.05)$ $= 3.7 \times 10^{-7}$ N·m	Similar assumptions as for SRP, except that being Earth-pointing, the same face will be presented to the ram direction all the time, so we can expect a little more control over the cp and *cm* locations. $A_r = 3\text{ m}^2$; $cp_a - cm = 0.2$ m; $C_d = 2.0$ For 21,000 km orbit: $\rho \approx 10^{-18}$ kg/m^3; V = 3,816 m/s $T_a = (0.5)(10^{-18})(2.0)(3)(3{,}816)^2(0.2)$ $= 8.7 \times 10^{-12}$ N·m
Magnetic Field	Cyclic	55 deg inclination orbit; assume 0.5 A·m^2 for a very small, uncompensated vehicle. $R = (6{,}378 + 700)$ km = 7,078 km $D = 0.5$ A·m^2; $\lambda = 1.9$ for 55 deg $B = (7.8 \times 10^{15})(1.9)/(7.078 \times 10^6)^3$ $= 4.2 \times 10^{-5}$ T $T_m = (0.5)B = 2.1 \times 10^{-5}$ N·m	Equatorial orbit; assume 1 A·m^2 for a small, uncompensated vehicle. $R = (6{,}378 + 21{,}000)$ km = 27,378 km $D = 1$ A·m^2; $\lambda \approx 1.2$ for equatorial orbit $B = (1)(7.8 \times 10^{15})(1.2)/(2.7378 \times 10^7)^3$ $= 4.6 \times 10^{-7}$ T $T_m = 1(B) = 4.6 \times 10^{-7}$ N·m
Gravity Gradient	Constant for Earth-oriented; cyclic for inertially oriented	With body-mounted arrays, FireSat II is fairly symmetric and small, so the moment of inertia can be balanced very well: We'll set $\theta = 1$ deg. $R = 7{,}078$ km $I = 25$ kg·m^2; $I_y = 50$ kg·m^2 Normal mode: $\theta = 1$ deg $T_g = (3)(3.986 \times 10^{14})\lvert 25 - 50\rvert \sin(2\text{ deg})$ $(2)(7.078 \times 10^6)^3$ $= 1.5 \times 10^{-6}$ N·m Target-of-opportunity: $\theta = 30$ deg $T_g = 3.7 \times 10^{-5}$ N·m	Deployed solar arrays dominate moment of inertia, so $I_x = I_z > I_y$. However, the ability to balance mass may be limited by the need to fit 3 satellites on the same launch vehicle, so we'll assume a large difference between the geometric and principal axes: $\theta = 10$ deg. $R = 27{,}378$ km $I_z = 90$ kg·m^2; $I_y = 60$ kg·m^2 $T_g = (3)(3.986 \times 10^{14})\lvert 90 - 60\rvert \sin(20\text{ deg})$ $(2)(2.7378 \times 10^7)^3$ $= 3.0 \times 10^{-7}$ N·m

Atmospheric Drag. In much the same way photons striking a spacecraft can exert pressure, so too can the rarified atmosphere that clings to Earth (and certain other planets) at the edge of space. The atmospheric density is roughly an exponentially decaying function of altitude, so that generally only spacecraft in low Earth orbit (LEO) encounter enough particles to cause noticeable disturbances. Those that do experience a pressure force known as *atmospheric* (or *aerodynamic*) *drag*. The atmospheric drag force itself is an important consideration for orbit planning (Chap. 9) and orbit prediction and tracking (Sec. 19.2). When the center of atmospheric pressure, determined by the spacecraft area exposed to the atmosphere in the direction of the orbital velocity

Table 19-4, Fig. 19-6◀, Eq. 19-5◀

(i.e., ram direction), is not aligned with the center of mass, a torque results. The atmospheric (or aerodynamic) torque can be estimated as

$$T_a = \frac{1}{2}\rho C_d A_r V^2 (cp_a - cm) \tag{19-6}$$

where T_a is the atmospheric drag torque, ρ is the atmospheric density in kg/m^3, C_d is the drag coefficient (usually between 2.0 and 2.5 for spacecraft), A_r is the ram area in m^2, V is the spacecraft's orbital velocity in m/s, and cp_a and cm are the centers of aerodynamic pressure and mass in m. Average atmospheric density and orbital velocity as functions of altitude are tabulated in App. I.

Magnetic Field. The Earth's liquid core is a dynamo that generates a magnetic field powerful enough to have important effects on the space surrounding the planet (see Sec. 7.3). Most spacecraft have some level of residual *magnetic moment*, meaning they have a weak magnetic field of their own. These residual moments can range anywhere from 0.1–20 A·m^2, or even more depending on the spacecraft's size and whether any onboard compensation is provided.

When a spacecraft's residual moment is not aligned with a local magnetic field, it experiences a *magnetic torque* that attempts to align the magnet to the local field, much like a compass needle. The Earth's magnetic field is complex, asymmetric, not aligned with the Earth's spin axis, and varies with both geographical movement of the dipole and changes in solar particle flux. However, for use in the ADCS design process, it is usually sufficient to model the Earth's magnetic field as a dipole and to determine the maximum possible value of the magnetic torque for a spacecraft's altitude. The following equation yields this maximum torque:

$$T_m = DB = D\left(\frac{M}{R^3}\lambda\right) \tag{19-7}$$

where T_m is the magnetic torque, D is the spacecraft's residual dipole moment in A·m^2, and B is the magnetic field strength in tesla (T). The magnetic field strength in turn is calculated from M, the magnetic moment of the Earth multiplied by the magnetic constant ($M = 7.8 \times 10^{15}$ tesla·m^3); R, the distance between the spacecraft and the Earth's center in m, and λ, which is a unitless function of the magnetic latitude that ranges from 1 at the magnetic equator to 2 at the magnetic poles. So, a polar orbit will see roughly twice the maximum magnetic torque of an equatorial orbit.

Gravity-Gradient. As described in the earlier subsection on centroids, *gravity-gradient* torques are caused when a spacecraft's center of gravity is not aligned with its center of mass with respect to the local vertical. Without getting into the math of the matter, the center of gravity of a spacecraft in orbit is dependent on its attitude relative to Earth (or whatever body the spacecraft is orbiting), and that cg is not, in general, the same as the center of mass. However, when one of the spacecraft's principal axes is aligned with the local vertical, the cg is always on that principal axis, and therefore there is no gravity-gradient torque. The gravity-gradient torque increases with the angle between the local vertical and the spacecraft's principal axes, always trying to align the minimum principal axis with the local vertical.

A simplified expression for the gravity-gradient torque for a spacecraft with the minimum principal axis in its Z direction is

$$T_g = \frac{3\mu}{2R^3}\left|I_z - I_y\right|\sin(2\theta) \tag{19-8}$$

Where T_g is the gravity-gradient torque about the X principal axis, μ is the Earth's gravitational constant (3.986×10^{14} m^3/s^2), R is the distance from the center of the Earth in meters, θ is the angle between the local vertical and the Z principal axis, and I_y and I_z are the moments of inertia about Y and Z in kg·m^2.

Remaining significant disturbances on the control system are internal to the spacecraft. Fortunately, we have some control over them. If we find that one is much larger than the rest, we can specify tighter values for that item. This change would reduce its significance but most likely add to its cost or weight. Table 19-5 summarizes the common internal disturbances. Misalignments in the center of gravity and in thrusters will show up during thrusting only and are corrected in a closed-loop control system and through on-orbit calibration of the thrusters.

Likewise, momentum wheel friction torques can be compensated in either a closed-loop or a compensatory fashion; some reaction wheels are designed with friction compensation included in some commanding modes. Liquid slosh and operating machinery torques are of greater concern but depend on specific hardware. If a spacecraft component has fluid tanks or rotating machinery, the system designer should investigate disturbance effects and ways to compensate for the disturbance, if required. Standard techniques include propellant management devices (e.g., slosh baffles) or counter-rotating elements.

19.1.3 Selection of Spacecraft Control Methods

Now that we understand the requirements on the control system and the environment in which it will operate, we can select one or more methods of controlling the spacecraft. Multiple methods may be indicated when different modes of operating the spacecraft have significantly different requirements or result in different disturbance profiles (as we will see in our FireSat II example). Table 19-6 lists several methods of control, along with typical characteristics of each. European Standard ECSS-E-60A, [ESA, 2004] for Spacecraft Control Engineering outlines their design and verification process.

Passive Control Techniques. *Gravity-gradient control* uses the inertial properties of a vehicle to keep it pointed toward the Earth. This relies on the fact that an

Table 19-5. Principal Internal Disturbance Torques. Spacecraft designers can minimize internal disturbances through careful planning and precise manufacturing, which may increase cost.

Disturbances	Effect on Vehicle	Typical Values
Uncertainty in Center of Gravity (cg)	Unbalanced torques during firing of couples thrusters Unwanted torques during translation thrusting	1–3 cm
Thruster Misalignment	Same as cg uncertainty	0.1–0.5 deg
Mismatch of Thruster Outputs	Similar to cg uncertainty	±5%
Reaction Wheel Friction and Electromotive Force (i.e., back EMF)	Resistance that opposes control torque effort. These torques are the limiting mechanism for wheels speed.	Roughly proportional to wheel speed, depending on model. At top speed, 100% of control torque (i.e., saturation)
Rotating Machinery (pumps, filter wheels)	Torques that perturb both stability and accuracy	Dependent on spacecraft design; may be compensated by counter-rotating elements
Liquid Slosh	Torques due to liquid dynamic pressure on tank walls, as well as changes in cg location.	Dependent on specific design; may be mitigated by bladders or baffles
Dynamics of Flexible Bodies	Oscillatory resonance at bending/twisting frequencies, limiting control bandwidth	Depends on spacecraft structure; flexible frequencies within the control bandwidth must be phase-stabilized, which may be undesirable.
Thermal Shocks ("snap") on Flexible Appendages	Attitude disturbances when entering/leaving umbra	Depends on spacecraft structure. Long inertia booms and large solar arrays can cause large disturbances.

Table 19-6. Attitude Control Methods and Their Capabilities. As requirements become tighter, more complex control systems become necessary.

Type	Pointing Options	Attitude Maneuverability	Typical Accuracy	Lifetime Limits
Gravity-Gradient	Earth local vertical only	Very limited	±5 deg (2 axes)	None
Gravity-Gradient + Momentum Bias	Earth local vertical only	Very limited	±5 deg (3 axes)	Life of wheel bearings
Passive Magnetic	North/South only	Very limited	±5 deg (2 axes)	None
Rate-Damping + Target Vector Acquisition	Usually Sun (power) or Earth (communication)	Generally used as robust safe mode.	±5–15 deg (2 axes)	None
Pure Spin Stabilization	Inertially fixed any direction	Repoint with precession maneuvers; very slow with torquers, faster with thrusters	±0.1 deg to ±1 deg in 2 axes (proportional to spin rate)	Thruster propellant (if applies)*
Dual-Spin Stabilization	Limited only by articulation on despun platform	Same as above	Same as above for spun section. Despun dictated by payload reference and pointing	Thruster propellant (if applies)* Despun section bearings
Bias Momentum (1 wheel)	Local vertical pointing or inertial targets	Fast maneuvers possible around momentum vector Repoint of momentum vector as with spin stabilized	±0.1 deg to ±1 deg	Propellant (if applies)* Life of sensor and wheel bearings
Active Magnetic with Filtering	Any, but may drift over short periods	Slow (several orbits to slew); faster at lower altitudes	±1 deg to ±5 deg (depends on sensors)	Life of sensors
Zero Momentum (thruster only)	No constraints	No constraints High rates possible	±0.1 deg to 5 deg	Propellant
Zero momentum (3 wheels)	No constraints	No constraints	±0.0001 deg to ±1 deg (determined by sensors and processor)	Propellant (if applies)* Life of sensors and wheel bearing
Zero Momentum (CMG)	No constraints Short CMG life may require high redundancy	No constraints High rates possible	±0.001 deg to ±1 deg	Propellant (if applies)* Life sensors and CMG bearings

* Thrusters may be used for slewing and momentum dumping at all altitudes, but propellant usage may be high. Magnetic torquers may be used from LEO to GEO.

elongated object in a gravity field tends to align its longitudinal axis through the Earth's center. The torques that cause this alignment decrease with the cube of the orbit radius and are symmetric around the nadir vector. Thus, the yaw of a spacecraft around the nadir vector is not controllable by this method. This technique is used on simple spacecraft in near-Earth orbits without yaw orientation requirements, often with deployable booms to achieve the desired inertias. Satellite geometry is shown in Fig. 19-4 and discussed in Sec. 8.1.2.

Frequently, we add dampers to gravity gradient satellites to reduce libration—small oscillations off of the nadir vector caused by other environmental disturbances. For example, long deployed booms are particularly susceptible to thermal shocks when entering or leaving umbra. These spacecraft also need a method of ensuring attitude capture with the correct end pointed at nadir—the gravity-gradient torques stabilize either end of the minimum inertia axis equally.

In the simplest gravity-gradient spacecraft, only two orientation axes are controlled; the orientation around the nadir vector is unconstrained. To control this third degree of freedom, a small constant-speed momentum wheel is sometimes added along the intended pitch axis. The momentum-biased wheel will be most stable when perpendicular to the nadir and velocity vectors, and therefore parallel to the orbital momentum vector. The stable state of the gravity-gradient plus momentum bias wheel establishes the desired attitude through small energy dissipations onboard without the need for active control.

A third type of purely passive control uses permanent magnets onboard to force alignment along the Earth's magnetic field. This is most effective in near-equatorial orbits where the North-South field orientation is reasonably constant for an Earth-referenced satellite.

Spin Control Techniques. *Spin stabilization* is a passive control technique in which the entire spacecraft rotates so that its angular momentum vector remains approximately fixed in inertial space. Spin-stabilized spacecraft (or *spinners*), employ the gyroscopic stability discussed earlier to passively resist disturbance torques about two axes. Additionally, spinners are generally designed to be either insensitive to disturbances around the third axis (the spin axis) or else have active means of correcting these disturbances.

The vehicle is stable (in its minimum energy state) if it is spinning about the principal axis with the largest moment of inertia. Energy dissipation mechanisms onboard, such as propellant slosh, structural damping, or electrical harness movement, will cause any vehicle to progress toward this state if uncontrolled. So, disk-shaped vehicles are passively stable whereas pencil-shaped vehicles are not. See Fig. 19-7. Spinners can be simple, survive for long periods without attention, provide a thermally benign environment for components (because of even heating), and provide a scanning or sweeping motion for sensors. The principal disadvantages of spin stabilization are that the vehicle mass properties must be carefully managed during vehicle design and assembly to ensure the desired spin direction and stability, and that the angular momentum vector that provides stability also limits maneuverability. More fuel is required to reorient the vehicle than a vehicle with no net angular momentum, making this technique less useful for payloads that must be repointed frequently.

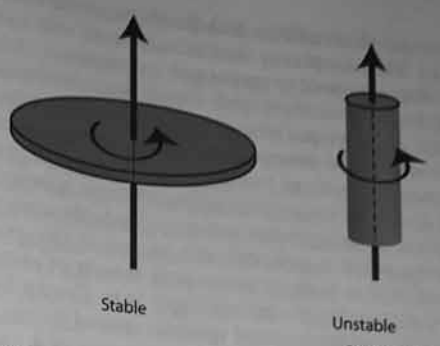


Fig. 19-7. Stable and Unstable Spinners.

A spinner requires extra fuel to reorient because of the gyroscopic stiffness, which also helps it resist disturbances. In reorienting a spinning body with angular momentum, **H**, a constant torque, **T**, will produce an angular velocity, ω, perpendicular to the applied torque and angular momentum vector, of magnitude $\omega = \mathbf{T}/\mathbf{H}$. (This follows from the earlier-introduced Euler equations.) Thus, the greater the stored momentum, the more torque must be applied for a given ω. For a maneuver through an angle θ, the torque-time product—an indication of fuel required for the maneuver—is a constant equal to $\mathbf{H}\cdot\theta$. Alternately, for a vehicle with no initial angular momentum, a small torque can be applied to start it rotating, with an opposite small torque to stop it when it has reached its new target. The fuel used for any angle maneuver can be arbitrarily small if a slow maneuver is acceptable. (Note that the spinner can only be maneuvered relatively slowly; a fast slew is usually not an option.)

A useful variation on spin control is called *dual-spin stabilization*, in which the spacecraft has two sections spinning at different rates about the same axis; this kind of spinner is also known as a *gyrostat*. Normally one section, the rotor, spins rapidly to provide gyroscopic stiffness, while the second section, the stator or platform, is despun to keep one axis pointed toward the Earth or Sun. By combining inertially fixed and rotating sections on the same vehicle, dual spinners can accommodate a variety of payloads in a simple vehicle. Also, by adding energy dissipation devices to the platform, a dual spinner can be passively stable spinning about the axis with the smallest moment of inertia, as long as the rotor is spinning about its own maximum moment of inertia. This permits more pencil-shaped spacecraft, which fit better in launch vehicle fairings and would not normally be stable spinning about their long axes. The disadvantage of dual-spin stabilization is the added complexity of the platform bearing and slip rings between the sections. (Slip rings permit

Table 19-6◀, Fig. 19-7, Eq. 19-8◀

power and electrical signals to flow between the two sections.) This complexity can increase cost and reduce reliability compared to simple spin stabilization.

Spinning spacecraft, both simple and dual, exhibit several distinct types of motion that are often confused. *Precession* is the motion of the angular momentum vector caused by external torques, including thruster firings used to correct environmental disturbances. *Coning* (or *wobbling*) is the apparent motion of the body when it is spinning about a principal axis of inertia that is not aligned with a body reference axis or axis of symmetry—for example, the intended spin axis. Coning looks like motion of the intended spin axis around the angular momentum vector at the spin rate. Figure 19-8 shows various natural rotations.

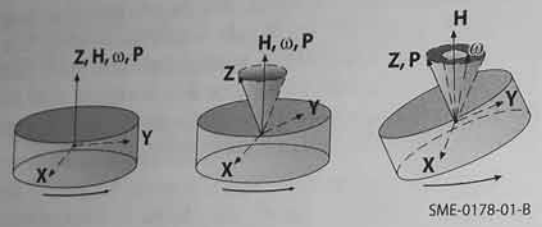


Fig. 19-8. Types of Rotational Motion. **H** = angular momentum vector; **P** = principal axis; ω = instantaneous rotation axis; **Z** = geometrical axis.

Nutation is the torque-free motion of a simple rigid body when the angular momentum vector is not perfectly aligned with a principal axis of inertia. For rod-shaped objects, this motion is a slow rotation (compared to spin rate) of the spin axis around the angular momentum vector. For these objects spinning about a minimum inertia axis, additional energy dissipation will cause increased nutation. For disk-shaped objects, spinning around a maximum inertia axis, nutation appears as a tumbling rotation faster than spin rate. Energy dissipation for these objects reduces nutation, resulting eventually in a clean spin. For these reasons, minimum-axis (or minor-axis) spinners are often concerned with minimizing energy dissipation, whereas maximum-axis (or major-axis) may actually include mechanisms, such as a passive nutation damper, to dissipate energy quickly.

Nutation is caused by disturbances such as thruster impulses, and can be seen as varying signals in body-mounted inertial and external sensors. Wobble is caused by imbalances and appears as constant offsets in body-mounted sensors. Such constant offsets are rarely discernible unless multiple sensors are available.

Spin stability normally requires active control, such as cold-gas thrusters or magnetic torquers, to periodically adjust the spacecraft's angular momentum vector to counteract disturbance torques. In addition, we may need to damp the nutation caused by disturbances or precession commands. Aggravating this nutation is the effect of energy dissipating phenomena like structural flexure and flexible harness or fluid motion, which are present in any spacecraft to some degree. Once the excitation stops, nutation decreases as these same factors dissipate the kinetic energy added by the control effort. However, this natural damping can take hours. We can neutralize this error source in minutes with nutation dampers. We can also reduce the amount of nutation from these sources by increasing the spin rate and thus the stiffness of the spinning vehicle. If the spin rate is 20 rpm and the nutation angle from a given disturbance is 3 deg, then nutation from the same disturbance would be reduced to 1 deg if the spin rate were 60 rpm. We seldom use spin rates above 90 rpm because of the large centripetal forces demanded of the structure and the consequent effects on design and weight. In thrusting and pointing applications, spin rates under 20 rpm may not be advisable as they may allow excessive nutation. However, applications unrelated to attitude control, such as thermal control or specialized payload requirements, are frequently less sensitive to nutation and may employ very low spin rates.

3-Axis Control Techniques. Spacecraft stabilized in all three axes are much more common today than those using spin or gravity-gradient stabilization. They can maneuver relatively easily and can be more stable and accurate, depending on their sensors and actuators, than more passive stabilization techniques. They are, however, more expensive and often more complex. Recent processor improvements have led to comparable or better total reliability, see Chap. 24.*

The control torques about the axes of 3-axis systems come from combinations of momentum wheels, reaction wheels, control moment gyros, thrusters, solar or aerodynamic control surfaces (e.g., tabs), or magnetic torquers. Broadly, however, these systems take two forms: one uses momentum bias by placing a momentum wheel along the pitch axis; the other is called *zero momentum* and does not use momentum bias at all—any momentum bias effects are generally regarded as disturbances. Either option usually needs some method of angular momentum management, such as thrusters or magnetic torquers, in addition to the primary attitude actuators.

In a *zero-momentum* system, actuators such as reaction wheels or thrusters respond to disturbances on the vehicle. For example, an attitude error in the vehicle results in a control signal that torques the wheel, creating a reaction torque in the vehicle that corrects the error. The torque on the wheel either speeds it up or slows it down; the aggregate effect is that all disturbance torques are absorbed over time by the reaction wheels, sometimes requiring the collected angular momentum to be removed. This momentum removal—called *desaturation*, *momentum dumping*, or *momentum management*—can be accomplished by thrusters or magnetic torquers acting automatically or by command from the ground.

When high torque is required for large vehicles or fast slews, a variation of 3-axis control using control

Table 19-6◀, Fig. 19-8, Eq. 19-8◀

* For critical space applications, there is no replacement for thorough risk and reliability assessment.

moment gyros, or CMGs, is available. These devices work like momentum wheels on gimbals. The control of CMGs is complex and their lifespan is limited, but their available torque for a given weight and power can make them attractive.

A very specialized form of zero momentum control, here called *active magnetic control*, can be attained from a combination of a magnetometer, a Global Positioning System (GPS) receiver (or other form or orbit information), and computationally intensive software filtering. The GPS feeds the spacecraft location to the onboard processor, which then determines the local magnetic field based on onboard models. The magnetometer data is filtered using the Euler equations to determine the attitude, and magnetic torquers make corrections in the two available directions at any given moment—corrections about the magnetic field vector are not possible. Active magnetic control can be an inexpensive backup control mode for a LEO satellite, or it can be a primary control mode for a satellite in a highly inclined orbit. (The highly inclined orbit has large changes in magnetic field direction, allowing the filtering algorithm to better determine a 3-axis attitude solution.) This attitude knowledge can also be combined with other sensors, such as Sun sensors, for more accuracy. Active magnetic control is an increasingly common control method, and we include it here as an example of how increased onboard computational power and the presence of new resources, such as the GPS constellation, have allowed completely new methods of attitude determination and control.

As a final demonstration of zero momentum 3-axis control, simple all-thruster systems are used for short durations when high torque is needed, such as during orbit insertion maneuvers or other orbit adjustments (delta V) from large motors discussed in Sec. 9.6. These thrusters then may be used for different purposes such as momentum dumping or provision of small delta Vs during other mission phases.

Momentum bias systems often have just one wheel with its spin axis mounted along the pitch axis, ideally normal to the orbit plane. The wheel is run at a nearly constant, high speed to provide gyroscopic stiffness to the vehicle, just as in spin stabilization, with similar nutation dynamics. Around the pitch axis, however, the spacecraft can control attitude by torquing the wheel, slightly increasing or decreasing its speed. Periodically, the momentum in the pitch wheel must be managed (i.e., brought back to its nominal speed), as in zero-momentum systems, using thrusters, magnets, or other means.

The dynamics of nadir-oriented momentum-bias vehicles exhibit a phenomenon known as *roll-yaw coupling*. To understand this coupling, consider an inertially fixed angular momentum vector at some error angle with respect to the orbit plane. If the angle is initially a positive roll error, then a quarter-orbit later it appears as a negative yaw error with no roll component remaining. As the vehicle continues around the orbit, the angle goes through negative roll and positive yaw before regaining its positive roll character. This coupling (or commutation) is due to the apparent motion of the Earth-fixed coordinate frame as seen from the spacecraft, and it can be exploited to control roll and yaw over a quarter orbit using only a roll (or only a yaw) sensor, instead of needing one sensor for each of the roll and yaw axes.

Effects of Requirements on Control Type. With the above knowledge of control techniques, we can proceed to select a control type that will best meet mission requirements in the expected operational environment. Tables 19-7 and 19-8 describe the effects of orbit insertion and payload slew requirements on the selection process. It is also useful here to once again reference Fig. 19-6 for information on how altitude can affect the space environment. Certain control types, such as gravity-gradient stabilization or active magnetic damping, are better in some orbits than in others.

A common control approach during orbit insertion is to use the short-term spin stability of the combination of spacecraft and orbit-insertion motor. Once on station the motor may be jettisoned, the spacecraft despun using thrusters or a yo-yo device, and a different control technique used from that point on.

Payload pointing will influence the attitude control method, the class of sensors, and the number and kind of actuation devices. Occasionally, pointing accuracies are so stringent that a separate, articulated platform is

Table 19-7. Orbit Transition Maneuvers and Their Effects. Using thrusters to change orbits creates special challenges for the ADCS.

Requirement	Effect on Spacecraft	Effect on ADCS
Large Impulse to Complete Orbit Insertion (thousands of m/s)	Solid motor or large bipropellant stage. Large thrusters or a gimbaled engine or spin stabilization for attitude control during burns.	Inertial measurement unit for accurate reference and velocity measurement. Different actuators, sensors, and control laws for burn vs. coasting phases. Need for navigation or guidance.
On-Orbit Plane Changes to Meet Payload Needs or Vehicle Operations (hundreds of m/s)	Large thrusters needed, but these thrusters may be needed for other reasons also, such as orbit insertion, coasting phase, or stationkeeping.	Separate control law for thrusting. Actuators sized for thrusting disturbances (possibly two sizes of thruster). Onboard attitude reference for thrusting phase.
Orbit Maintenance/Trim Maneuvers (<100 m/s)	One set of thrusters	Thrusting control law. Onboard attitude reference.

Table 19-8. Slewing Requirements That Affect Control Actuator Selection. Spacecraft slew agility can demand larger actuators for intermittent use.

Slewing	Effect on Spacecraft	Effect on ADCS
None or Time-Unconstrained	Spacecraft constrained to one attitude (highly improbable), or reorientations can take many hours.	Reaction wheels, if planned, can be smaller If magnetic torquers can dump momentum, reaction control thrusters may not be needed
Low Rates From 0.05 deg/s (orbital rate) to 0.5 deg/s	Minimal	Depending on spacecraft size, reaction wheels may be fully capable for slews If reaction wheels not capable, thrusters will be necessary Thrusters may be needed for other reasons; i.e., stationkeeping
High Rates >0.5 deg/s	– Structural impact on appendages – Weight and cost increase	– Control moment gyros or thrusters needed. If thrusters needed for other reasons, two thrust levels may be needed.

Table 19-9. Effects of Control Accuracy Requirements on Sensor Selection and ADCS Design. More accurate pointing requires better and more expensive sensors and actuators.

Required Accuracy (3σ)	Effect on Spacecraft	Effect on ADCS
>5 deg	• Permits major cost savings • Permits gravity-gradient (GG) stabilization	• Without attitude determination • No sensors required for GG stabilization • Boom motor, GG damper, and a bias momentum wheel are only required actuators • With attitude determination • Sun sensors and magnetometer adequate for attitude determination at ≥ 2 deg • Higher accuracies may require star trackers or horizon sensors
1 deg to 5 deg	• GG not feasible • Spin stabilization feasible if stiff, inertially fixed attitude is acceptable • Payload needs may require despun platform on spinner • 3-axis stabilization will work	• Sun sensors and horizon sensors may be adequate for sensors, especially a spinner • Accuracy for 3-axis stabilization can be met with RCS deadband control but reaction wheels will save propellant for long missions • Thrusters and damper adequate for spinner actuators • Magnetic torquers (and magnetometer) useful
0.1 deg to 1 deg	• 3-axis and momentum-bias stabilization feasible • Dual-spin stabilization also feasible	• Need for accurate attitude reference leads to star tracker or horizon sensors and possibly gyros • Reaction wheels typical with thrusters for momentum unloading and coarse control • Magnetic torquers feasible on light vehicles (magnetometer also required)
< 0.1 deg	• 3-axis stabilization is necessary • May require articulated and vibration-isolated payload platform with separate sensors	• Same as above for 0.1 deg to 1 deg but needs star sensor and better class of gyros • Control laws and computational needs are more complex • Flexible body performance very important

necessary. An articulated platform can perform scanning operations much more easily than the host vehicle and with better accuracy and stability. Trade studies on pointing requirements must consider accuracy in attitude determination and control, and the most stringent requirements will ultimately drive ADCS component selection. Table 19-9 summarizes the effects of accuracy requirements on the ADCS approach for the spacecraft. Section 14.5 discusses how to develop pointing budgets.

FireSat II Control Selection. For FireSat II, we consider two options for orbit insertion control. First, the launch vehicle may directly inject the spacecraft into its mission orbit. This common option simplifies the spacecraft design since no special insertion mode is needed. An alternate approach, useful for small spacecraft such as FireSat II, is to use a monopropellant propulsion system onboard the spacecraft to fly itself up from a low parking orbit to its final altitude (see Sec. 9.6). For small insertion motors, reaction wheel torque or momentum bias stabilization may be sufficient to control the vehicle during this burn. For larger motors, delta-V thruster modulation or dedicated attitude control thrusters become attractive.

Once on station the spacecraft must point its sensors at nadir most of the time and slightly off-nadir for brief periods. Since the payload needs to be despun and the spacecraft frequently reoriented, spin stabilization is not the best choice. Gravity-gradient and passive magnetic control cannot meet the 0.1 deg pointing requirement or the 30 deg slews. This leaves 3-axis control and momentum-bias stabilization as viable options for the on-station control.

Table 19-9, Fig. 19-8◀, Eq. 19-8◀

The periodic 180 deg yaw slews (i.e., slews around the nadir vector) would have to overcome any momentum bias in the Y axis (perpendicular to the orbit normal). When we do the example momentum bias sizing in Table 19-11 later, we will see that a bias of approximately 20 N·m·s would be required to maintain the required 0.1 deg accuracy. The magnetic torquers could not perform this slew in 45 minutes because of the bias to be overcome. Instead, we will use the propulsion system to perform the yaw slew.

For the optional off-nadir pointing requirement, 3-axis control with reaction wheels might be more appropriate. Mass and power will be especially precious for this very small satellite, so carrying and running 3 reaction wheels simultaneously may be more than the system budgets can handle. Since the SCS will provide an example of 3-axis control with reaction wheels, we will use the magnetic torque and momentum wheel combination for FireSat II, and for actuator sizing (Table 19-11) we will assume the 30 degree offset pointing is done about the pitch axis, so that the momentum wheel performs that slew also. If offset pointing is needed about roll, thrusters will have to be used; the thruster force sizing example in Table 19-12 shows how this might work with the momentum wheel stopped.

SCS Control Selection. For the Supplemental Communication System, we will focus on taking advantage of the gentle disturbance environment and 1 deg accuracy requirement to design a light, inexpensive attitude system that can be installed in all 3 satellites. At the SCS altitude of 21,000 km, with the configuration we've assumed, there are no good passive stabilization methods available to us. Reaction wheels will be needed to reject the disturbances, which are dominated by the solar radiation pressure torques (see Table 19-4). So, as long as the reaction wheels are there anyway, we might as well use them for 3-axis stabilization.

While the greater part of the SRP torques will be cyclic, some small part will be secular. Therefore, the momentum stored in the reaction wheels will gradually increase and will need to be removed periodically by thrusters. The use of 100 A·m^2 magnetic torquers for momentum removal is feasible from the point of view of the available magnetic field. However, the propulsion system has to be included anyway to separate the 3 satellites and establish the constellation, and constantly running torquers would be an additional power drain. Since power is a major challenge for this mission, we will use the thrusters for momentum unloading.

19.1.4 Selection and Sizing of ADCS Hardware

We are now ready to evaluate and select the individual ADCS components. For all ADCS hardware, we will determine the minimum performance level needed to meet requirements. Then, standard components available from manufacturers will be selected if possible, sometimes resulting in better performance than the minimum required. If standard components are not available, specialized components may be designed and built, but this can often be prohibitively costly for most customers, and so a revision of the requirements is more likely to be more in line with available hardware.

Actuators. Options for actuator selection are summarized in Table 19-10. First, we will discuss momentum-exchange devices, which conserve angular momentum in the spacecraft: reaction wheels, momentum wheels, and control moment gyros. Then, we will move on to external torque actuators, which change the angular momentum of the spacecraft when they are activated: magnetic torquers and thrusters (cold-gas, hot-gas and electric) are the most commonly used in this category.

Wheel control provides smooth changes in torque, allowing very accurate pointing of spacecraft. Some wheels can cause vibrations, or jitter, at high speeds, but this can often be mitigated with vibration isolators or changes in structural design. *Reaction wheels* are essentially torque motors with high-inertia rotors. They can spin in either direction and provide one axis of control for each wheel. *Momentum wheels* are reaction wheels with a nominal spin rate above zero to provide a nearly constant angular momentum. This momentum provides gyroscopic stiffness to two axes, and the motor torque may be controlled to change pointing around the spin axis. In sizing wheels we must always consider two performance quantities: angular momentum capacity, and torque authority.

To determine the necessary momentum capacity, we must distinguish between cyclic and secular disturbances in the spacecraft's environment. We typically size reac-

Table 19-10. Typical Attitude Actuators. Actuator weight and power usually scale with performance.

Actuator	Performance Range	Mass (kg)	Power (W)
Thrusters	Thrusters produce force; multiply by moment arm for torque	Mass and power vary (See Chap. 18 for details on propulsion systems)	
Hot Gas (Hydrazine)	0.5 to 9,000 N		
Cold Gas	< 5 N		
Reaction and Momentum Wheels	Maximum torques: 0.01 to 1 N·m Practical momentum storage capacity: 0.4 to 3000 N·m·s	2 to 20	Varies with speed: 10 to 100
Control Moment Gyros (CMG)	Max. torques: 25 to 500 N·m	> 10	90 to 150
Magnetic Torquers	1 to 4,000 A·m^2 (See Table 19-4 for torque examples)	0.4 to 50	0.6 to 16

tion wheels to be able to store the full cyclic component of momentum without the need for frequent momentum dumping. Therefore, the average disturbance torque for ¼ or ½ an orbit determines the minimum capacity of the wheels. The secular component of momentum will also need to be stored for the amount of time the spacecraft must be operational without a momentum dump being performed. This time may be determined by requirements on payload observation continuity, or it may be the amount of time the spacecraft must survive without ground intervention.

The necessary torque authority of the reaction wheels is defined either by slew requirements or the need for control authority to be well above the peak disturbance torque. If the peak disturbance is too close to the torque authority available, pointing accuracy may suffer.

For 3-axis control, at least three wheels with non-coplanar spin axes are required. Often, a fourth wheel is carried in case one of the primary wheels fails. If the wheels are not orthogonal (and a redundant wheel never is), additional torque authority or momentum capacity may be necessary to compensate for non-optimal geometry. However, the redundancy of the fourth wheel can have additional benefits, such as being able to avoid any wheel speed passing through zero (which can cause attitude error transients) or even as power storage, as driving a spinning wheel toward zero speed will provide power to the spacecraft.

For spin-stabilized or momentum-bias systems, the cyclic torques will cause cyclic variations of the attitude, whereas the secular torques cause gradual divergence of the attitude away from the ideal target. We typically design the stored angular momentum, determined by spin rate and inertia of the spinning body, to be large enough to keep the cyclic motion within accuracy requirements without active control. Periodic torquing will still be needed to counteract secular disturbances, and the cost of this torquing in propellant, if performed by thrusters, will constrain the maximum stored momentum.

For high-torque applications in which fine control is still needed, *control moment gyros* (CMG) may be used instead of reaction wheels. CMGs are single- or double-gimbaled wheels spinning at constant speed (and therefore providing momentum bias stiffness when not actuating). By turning the gimbal axis, we can obtain a high-output torque whose size depends on the speed of the rotor and the gimbal rate of rotation. Control systems with two or more CMGs can produce large torques about all three of the spacecraft's orthogonal axes, so we most often use them where agile (i.e., high angular rate) maneuvers are required. The use of CMGs requires complex control laws and careful momentum management to avoid wheel saturation. Also, because the CMG torque is created by twisting what is essentially a stiff gyroscope perpendicular to its spin axis, the bearings of the wheel suffer a great deal of wear and tear, causing most CMGs to have shorter lifetimes than other actuators. Because CMGs combine a short life with high cost, weight, and power needs, they are generally used only on very large spacecraft and only when absolutely necessary to achieve the mission goals.

Spacecraft also use *magnetic torquers* as actuation devices. These torquers are magnetic coils (electromagnets) designed to generate magnetic dipole moments of commanded magnitude. When three orthogonal torquers are mounted to a spacecraft, they can create a magnetic dipole of any direction and magnitude up to the strength of the torquers. Magnetic torquers can compensate for a spacecraft's residual magnetic field or attitude drift from minor disturbance torques. They also can be used to desaturate momentum-exchange systems, though they usually require much more time than thrusters. A magnetic torquer produces a torque proportional and perpendicular to the Earth's varying magnetic field; therefore, at any given moment, no torque can be provided about the Earth's magnetic field vector. However, as a spacecraft changes in latitude or altitude while following its orbit, different directions of the field become available.

Electromagnets have the advantage of no moving parts, requiring only a magnetometer for sensing and wire coiled around a metallic rod in each axis. (Some torquers are even internally redundant, as they have two coils around the same metallic core.) Because they use the Earth's natural magnetic field, and this field reduces in strength with the cube of distance from the Earth's center, magnetic torquers are less effective at higher orbits. We can specify the torquer's dipole strength in $A \cdot m^2$ and tailor it to any application. Table 19-11 describes sizing rules of thumb for wheels and magnetic torquers, and works through those rules for the FireSat II and SCS examples. SCS has no slewing requirements, so we size its reaction wheels based only on momentum storage needs.

Thrusters (i.e., rocket engines) are possibly the most frequently flown attitude actuator because of their dual use in adjusting orbital parameters. Almost every spacecraft that needs to perform orbital maneuvers (Sec. 9.6) will use thrusters to achieve that goal, and in many cases, some subset of the thrusters used is for attitude control. Thrusters produce a force on the spacecraft by expelling material, called propellant, at high velocity from their exit nozzles. Hot-gas propulsion systems include thrusters that chemically alter the propellant to extract the energy needed for rapid mass expulsion. These systems may be monopropellant, in which the propellant is catalyzed to break down chemically, or bipropellant, in which a fuel is mixed with an oxidizer to achieve combustion just prior to expulsion. Cold-gas systems include thrusters whose propellant is not altered chemically during propulsion. In a cold-gas system, the energy may come from phase change of the propellant, or simply from pre-pressurizing the propellant in its tank. A third type of thruster is electrical, which, due to the usually small forces involved, is only used for attitude control in special circumstances. Electrical propulsion (Sec. 18.4) is accomplished by using magnetic or electrostatic fields to eject plasma or magnetic fluid to achieve a reaction force on the spacecraft. Propulsion is discussed in detail in Chap. 18.

Table 19-10◀, Fig. 19-8◀, Eq. 19-8◀

Table 19-11. Simplified Equations for Sizing Reaction Wheels, Momentum Wheels, and Magnetic Torquers. The FireSat II momentum wheel is sized for the baseline pointing requirements and for the optional design with 30 deg slew requirements. SCS reaction wheels are sized for momentum storage capacity.

Parameter	Simplified Equations	Application to FireSat II and SCS Examples
Torque from Reaction Wheel for Disturbance Rejection	Control torque, whether from reaction wheels or magnetic torquers, must equal worst-case anticipated disturbance torque plus some margin: $T_C = (T_D)$(Margin Factor)	For the FireSat II spacecraft, magnetic torque dominates during regular observations: $T_D \approx T_m = 2.1 \times 10^{-5}$ N·m (see Table 19.4) This torque is manageable by most reaction wheels and magnetic torquers. Momentum requirements or slew torque, not disturbance rejection, will be used for actuator sizing.
Slew Torque for Reaction Wheels	For max-acceleration slews with no resisting momentum (½ distance in ½ time): $\theta/2 = \frac{1}{2}\,(T/I)(t/2)^2$ $T = 4\theta I/t^2$	FireSat II needs to make a 30-deg slew about the Y axis (50 kg·m^2) in 10 minutes using its momentum wheel: T = (4)(30 deg)(π/180 deg)(50 kg·m^2)/(600 sec)^2 $= 2.9 \times 10^{-4}$ N·m (This is a small value.) The total momentum change for the wheel during the slew is T·t/2 = 0.087 N·m·s.
Momentum Storage in Reaction Wheel	One approach to estimating wheel momentum, h, is to integrate the worst-case disturbance torque, T_D, over a full orbit, with period P. For gravity gradient, the maximum momentum accumulates in 1/4 of an orbit. A simplified expression for such a sinusoidal disturbance is: $h = T_D \cdot P \cdot (0.707)/4$ where 0.707 is the rms average of a sinusoidal function. For the SCS solar pressure torques, the torque buildup depends heavily on attitude and orbit geometry, but to keep this simple we will assume the maximum also accumulates in ¼ orbit.	For FireSat II, the maximum gravity gradient torque (during 30-deg off-pointing) is estimated at: $T_D = 3.7 \times 10^{-5}$ N·m (Table 19-4) and the 700 km orbital period is 5,926 sec: h = (3.7 × 10^{-5} N·m)(5,926 sec)(0.707/4) = 0.039 N·m·s If the magnetic torquers were not capable of handling this momentum, a small reaction wheel with storage of 0.4 N·m·s would be sufficient even if 30-deg off-pointing were held for an entire quarter-orbit. For SCS, $T_D = 6.6 \times 10^{-6}$ and P = 45,083 sec. SCS storage: $h = (6.6 \times 10^{-6})(45{,}083)(0.707/4) = 0.06$ N·m·s We will size 1.0 N·m·s wheels to minimize thruster burns and potential jitter from high wheel speeds.
Momentum Storage in Momentum Wheel	For momentum-wheel stabilization, roll and yaw accuracy depend on the wheel's momentum and the external disturbance torque. A simplified expression for the required momentum storage is: $h = (T/\theta_a)\cdot(P/4)$ where θ_a = allowable motion.	The value of h for the FireSat II yaw accuracy requirement of 0.1 deg (0.0017 rad) would be h = (2.1 × 10^{-5} N·m/0.0017 rad)·(5,926/4 sec) = 18.4 N·m·s For 1 deg accuracy, we would need only 1.8 N·m·s. The same results would hold for a spinner. Note that this capacity would need to be added to the momentum change needed for the 30 deg slews, which for FireSat II is less than 0.1 N·m·s.
Torque from Magnetic Torquers	Magnetic torquers use electrical current through the torquer to create a magnetic dipole (D). The Earth's magnetic field, B, acts on D, resulting in torque (T) on the vehicle: $D = T/B$ Magnets used for momentum dumping must equal the peak disturbance + margin to compensate for the lack of complete directional control. (Note that if magnetic torque is the dominant disturbance, the torquers need mainly to cancel the spacecraft's residual dipole.)	Table 19-4 estimates the maximum Earth field, B, for FireSat II to be 4.2×10^{-5} tesla. We calculate the torque rod's magnetic torquing ability (dipole) to counteract the worst-case gravity gradient disturbance, T_D, of 2.1×10^{-5} N·m·s as D = (2.1 × 10^{-5} N·m)/(4.2 × 10^{-5} tesla) = 0.5 A·m^2 which is a small actuator. The Earth's field is cyclic at twice orbital frequency; thus, maximum torque is available only twice per orbit. A torquer of 3 to 10 A·m^2 capability should provide sufficient margin, which also gives enough capacity to remove increased gravity gradient torques during offset pointing.

Note: For actuator sizing, the magnitude and direction of the disturbance torques must be considered. In particular, momentum accumulation in inertial coordinates must be mapped to body-fixed actuator axes.

Thrusters provide torque proportional to their moment arm, which is the amount that their line of force is offset from the vehicle's center of mass. When thrusters apply their force along a line that intersects the center of mass of the vehicle, there is no torque. So, while the amount of force available from a thruster may be large, the torque available is limited by the physical extent of the vehicle and how the thrusters are mounted on the vehicle. When mounted to maximize torque authority, thrusters have the advantage of being able to provide large, instantaneous control torques at any time in the orbit. The disadvantages of thrusters are that they use expendable propellant, they can disrupt orbit determination activities, and the plumes of expelled matter can impinge on the surface of the spacecraft, possibly heating or contaminating surfaces.

Attitude functions to which thrusters can be applied include controlling attitude, controlling nutation and spin rate, performing large, rapid slews, and managing angular momentum. In all of these functions, large torque authority can be helpful. To some extent, the ability to change angular momentum by relatively small amounts is also necessary. Another consideration in selection of thrusters is how fine and how precise the delivered torque impulse needs to be. Torque applied over a period of time creates a change in angular momentum, which has a corresponding change in the rotational rate of the spacecraft. As attitude changes with non-zero rate, attitude control accuracy is directly determined by the minimum thruster impulse available. If a 50 N thruster with a 1 meter moment arm must provide a change in angular momentum accurate to within 1 N·m·s, then it will be necessary to fire the thruster for a period of time no greater than 20 ms. The thruster valves must be capable of opening and closing that rapidly and precisely, and of propellant actually flowing into the thruster and out the nozzle, which takes a finite amount of time. This need for speed and precision in engine valve control also has implications in the design of the avionics that drive the thruster valves; often, special electronics cards must be developed for a given mission to meet thruster-based attitude control requirements.

FireSat II Actuators. The baseline FireSat II spacecraft will use magnetic torquers for momentum dumping, but thrusters will be needed for the 180 degree slews that must be done periodically to keep the Sun off the radiator. Because the slew must be about the yaw axis, the momentum bias in the wheel would resist the thruster torque, complicating the control design. Instead, we will use the thrusters to remove the momentum bias, rotate the spacecraft, and then re-establish the momentum bias. One way to remove the wheel momentum is to use thrusters for 3-axis control while driving the wheel to zero speed. As the wheel slows down, it causes the spacecraft to rotate, and the thrusters are fired to counteract the rotation; the same thing happens when the wheel is once again sped up to re-establish the momentum bias. The thrusters must provide much more torque than the wheel, but generally the attitude control accuracy requirements are loosened for this kind of maneuver to avoid firing the thrusters more than is necessary. Good accuracy is less important here because science data would not be collected anyway during the slew. The total fuel mass needed for the momentum changes and the slew depends on the specific impulse, I_{sp}, of the selected thruster/propellant combination; we will assume the I_{sp} is 218 sec.

SCS Actuators. The SCS satellites will need to use thrusters for orbit maintenance and momentum dumping. We will want to minimize the number and size of thrusters, since the spacecraft are so small. However, the lower limit will most likely come from the orbital maneuvering needs, and not the attitude control needs. We will suppose 4 1-lb thrusters are needed for orbital maneuvering and steering, and in Table 19-12, we will present and work through procedures and simplified equations for sizing thrusters using SCS as an example. SCS has no stated slew requirement, but for the purposes of the example, a slew requirement of 30 deg in 60 sec will be supposed. A thorough discussion on the topic of propulsion systems, including estimating propellant needs, can be found in Chap. 18.

Sensors. We complete this hardware unit by selecting the sensors needed for attitude determination. Consult Table 19-13 for a summary of typical devices as well as their performance and physical characteristics. Note, however, that sensor technology is changing rapidly, promising ever more accurate and lighter-weight sensors for future missions.

Sun sensors are visible-light or infrared detectors that measure one or two angles between their mounting base and incident sunlight. They are popular, accurate, and very reliable, but they require clear fields of view. They can be used as part of the normal attitude determination system, part of the initial acquisition or failure recovery system, or part of an independent solar array orientation system. Since most low-Earth orbits include eclipse periods, Sun-sensor-based attitude determination systems must provide some way of tolerating the regular loss of this data without violating pointing constraints (see Chap. 16).

Sun sensors can be quite accurate (<0.01 deg), but it is not always possible to take advantage of that feature. We usually mount Sun sensors near the ends of vehicles to obtain an unobstructed field view, so the Sun sensor accuracy can be limited by structural bending on large spacecraft. Spinning satellites use specially designed Sun sensors that measure the angle of the Sun with respect to the spin axis of the vehicle, and they issue a pulse correlated to the time the Sun crosses the sensor to provide spin-phase information. Also popular are coarse Sun sensors, which are simply small solar cells (see Sec. 21.2) that issue a current roughly proportional to the cosine of the Sun angle. These sensors are so small and inexpensive that it is often feasible to put several in many directions on a spacecraft, and then to estimate the Sun direction by solving the linear system equations that results. Because coarse Sun sensors use no power and almost never fail, they are often used in low-power acquisition and fault recovery modes.

Table 19-11◀, Fig. 19-8◀, Eq. 19-8◀

Table 19-12. Simplified Equations for Preliminary Sizing of Thruster Systems. SCS thruster requirements are small for this low-disturbance, minimal slew application. It is likely that the thrusters needed for orbit maintenance can also serve for momentum dumping. FireSat II thrusters are needed for 180 degree slews and for orbit insertion and maintenance.

Simplified Equations	Application to Examples
Thruster force level sizing for external disturbances: $F = T_D / L$ **where** **F is thruster force** **T_D is worst-case disturbance torque** **L is the thruster's moment arm**	For the SCS worst case T_D of 6.6×10^{-6} N·m (Table 19-4) and a thruster moment arm of 0.5 m $F = (6.6 \times 10^{-6}\ \text{N·m})/(0.5\ \text{m}) = 3.3 \times 10^{-6}$ N This small value indicates orbit maintenance and momentum dumping requirements, not disturbance torques, will determine thruster size. Also, using thrusters to fight cyclic disturbances uses precious propellant; it is generally better to store the momentum in wheels.
Sizing force level to meet slew rates: **Determine fastest slew rate, ω, required in the mission profile.** **Develop a slew profile that accelerates the vehicle quickly, coasts at that rate, and then decelerates quickly. The acceleration required, α, comes from equating these two torque definitions:** $T = F \cdot L = I \cdot \alpha$	Assume FireSat II does a 30-deg slew in less than 1 min (60 sec), accelerating for 5% of that time, coasting for 90%, and decelerating for 5%. $\omega = 30\ \text{deg} / 60\ \text{sec} = 0.5$ deg/sec To reach 0.5 deg/sec in 5% of 1 min, which is 3 sec, requires an acceleration $\alpha = \omega/t = (0.5\ \text{deg/sec})/(3\ \text{sec}) = 0.167\ \text{deg/sec}^2 = 0.003\ \text{rad/sec}^2$ $F = I\alpha/L = (25\ \text{kg·m}^2)(0.003\ \text{rad/sec}^2)/(0.5\ \text{m}) = 0.15$ N This is very small. It is certainly within the 5 N of thrust that the propulsion design selected. However, it is so much smaller that specialized circuitry would be needed to fire the thrusters for very brief amounts of time, which may impact thruster efficiency.
Determining fuel needed to complete FireSat II yaw slew: **Thrusters will need to remove the momentum bias from the wheel, rotate the satellite 180 deg, and then re-establish the momentum. Fuel mass, m, is estimated as:** $m = F \cdot t/(g \cdot I_{sp})$ **where** **t is the on-time of one thruster**	Changing momentum bias: The minimum thruster on-time needed to remove or create a given momentum, h, is $t = h/T = h/(F \cdot L)$. For FireSat II: $t_{bias} = (18.4\ \text{N·m·s})/(5\ \text{N} \cdot 0.5\ \text{m}) = 7.4$ sec We will bring this up to 10 seconds to account for any additional attitude correction firings needed during wheel torquing. Yaw slew: We'll plan on a slew rate of 1 deg/sec to finish the slew itself in under 5 minutes, leaving plenty of time for wheel stopping and starting. 1 deg/sec amounts to an angular momentum of: $h = (40\ \text{kg·m}^2)(1\ \text{deg/sec})(\pi\ \text{rad}/180\ \text{deg}) = 0.7$ N·m·s and therefore an on-time for both starting and stopping the slew of: $t_{accel} = t_{decel} = (0.7\ \text{N·m·s})/(5\ \text{N} \cdot 0.5\ \text{m}) = 0.3$ sec Total thruster on-time per slew: $t_{total} = 10 + 0.3 + 0.3 + 10 = 21$ seconds Fuel required per 180-degree slew: $m = (5\ \text{N})(21\ \text{s})/(9.8\ \text{m/s}^2 \cdot 218\ \text{s}) = 49$ g fuel
Sizing force level for momentum dumping: $F = h/(Lt)$ **where** **h is the stored wheel momentum** **L is the thruster moment arm** **t is the burn time**	For SCS with 1.0 N·m·s wheels and 1-sec burns, $F = (1.0\ \text{N·m·s})/(0.5\ \text{m} \times 1\ \text{sec})$ $= 2.0$ N This is still well within the range of 1-lb thrusters, which are commonly used for orbit maintenance on small spacecraft. Reaction wheels with even larger capacity might be desirable if it would further reduce the number of times the thrusters must be used.

Table 19-13. Typical ADCS Sensors. Sensors have continued to improve in performance while getting smaller and sometimes less expensive.

Sensor	Typical Performance Range	Mass (kg)	Power (W)
Gyroscopes	Drift Rate = 0.003 deg/hr to 1 deg/hr Drift rate stability varies widely	< 0.1 to 15	< 1 to 200
Sun Sensors	Accuracy = 0.005 deg to 3 deg	0.1 to 2	0 to 3
Star Sensors (Scanners & Cameras)	Accuracy = 1 arcsecond to 1 arcminute = 0.0003 deg to 0.01 deg	2 to 5	5 to 20
Horizon Sensors ***Scanner/Pipper*** ***Fixed Head (Static)***	Accuracy: 0.05 deg to 1 deg (0.1 deg is best for LEO) < 0.1 deg to 0.25 deg	1 to 4 0.5 to 3.5 0.3 to 1.2	5 to 10 0.3 to 5 < 1
Magnetometer	Accuracy = 0.5 deg to 3 deg		

Star sensors have improved rapidly in the past few years and represent the most common sensor for high-accuracy missions. Star sensors can be scanners or trackers. *Scanners* are used on spinning spacecraft. Light from different stars passes through multiple slits in the scanner's field of view. After several star crossings, we can derive the vehicle's attitude. We use *star trackers* on 3-axis stabilized spacecraft to track one or more stars to derive two- or 3-axis attitude information. The majority of star trackers used today work much like digital cameras (and many of these are increasingly called *star cameras*, rather than trackers), allowing starlight to fall on a CCD or an APS, to create an image of the star field. Then, internal processing determines a 3-axis attitude based on a star catalog. Many units are able to determine a very accurate attitude within seconds of being turned on.

While star sensors excel in accuracy, care is required in their specification and use. The most accurate star cameras are unable to determine attitude at all if the spacecraft is rotating too fast, and other star sensors must know roughly where they are pointing to make their data useful. Therefore, the vehicle must be stabilized to some extent before the trackers can operate effectively. This stabilization may require alternate sensors, which can increase total system cost. Also, old star sensors are susceptible to being blinded by the Sun, Moon, planets, or even high radiation levels, such as in the Van Allen belts, which is a disadvantage that must be accommodated in their application. Where the mission requires the highest accuracy and justifies a high cost, we often use a combination of star trackers and gyroscopes. The combination of these sensors is very effective: the gyros can be used for initial stabilization and during periods of inference in the star trackers, while the trackers can be used to provide a high-accuracy external reference unavailable to the gyros. As APS star trackers rapidly improve, an increasing number of spacecraft include the technology. They are often more robust than comparable CCD trackers and are able to ignore the moon and certain other celestial objects in the field of view. Some APS trackers are even able to track at full performance at rotational rates, much higher than that of a CCD.

Horizon sensors (often known as *Earth sensors*) are infrared devices that detect the contrast between the cold of deep space and the heat of the Earth's atmosphere (about 40 km above the surface in the sensed band). Simple narrow field-of-view fixed-head types (called *pippers* or *horizon crossing indicators*) are used on spinning spacecraft to measure Earth phase and chord angles, which, together with orbit and mounting geometry, define two angles to the Earth (nadir) vector. *Scanning horizon sensors* use a rotating mirror or lens to replace (or augment) the spinning spacecraft body. They are often used in pairs for improved performance and redundancy. Some nadir-pointing spacecraft use *staring sensors*, which view the entire Earth disk (from GEO) or a portion of the limb (from LEO). The sensor fields of view stay fixed with respect to the spacecraft. This type works best for circular orbits, as they are often tuned for a tight range of altitudes.

Table 19-13◀, Fig. 19-8◀, Eq. 19-8◀

Horizon sensors provide Earth-relative information directly for Earth-pointing spacecraft, which may simplify onboard processing. The scanning types require clear fields of view for their scan cones (typically 45, 60, or 90 deg half-angle). Typical accuracies for systems using horizon sensors are 0.1 to 0.25 deg, with some applications approaching 0.03 deg. For the highest accuracy in low-Earth orbit, it is necessary to correct the data for Earth oblateness and seasonal horizon variations.

Magnetometers are simple, reliable, lightweight sensors that measure both the direction and magnitude of the Earth's magnetic field. Magnetometer output helps us establish the spacecraft's attitude relative to the local magnetic field, which information then can be combined with magnetic field models and orbit information to determine attitude relative to the Earth and inertial reference frames. However, their accuracy is not as good as that of star or horizon sensors. The Earth's magnetic field can shift with time and is not known precisely in the first place. To improve accuracy, we often combine magnetometer data with data from Sun or horizon sensors. As a vehicle using magnetic torquers passes through changing magnetic fields during each orbit, we use a magnetometer to control the magnitude and direction of the torquers' output relative to the present magnetic field. In earlier spacecraft the torquers usually needed to be inactive while the magnetometer was sampled to avoid corrupting the measurement. However, improvements in onboard computing capability mean that coupling matrices can be used to extract the torquer inputs from the field measurement, allowing constant sampling even while torquing. Finally, good spacecraft ephemeris and magnetic field models can be used in place of magnetometers for some missions, but magnetometers will generally be more accurate.

GPS receivers are well known as high-accuracy navigation devices, but they can also be used for attitude determination. If a spacecraft is large enough to place multiple antennas with sufficient separation, attitude can be determined by employing the differential signals from the separate antennas. Such sensors offer the promise of low cost and weight for LEO missions. They can provide attitude knowledge accurate to 0.25–0.5 deg for antenna baselines on the order of 1 meter [Cohen 1996], and so are being used in low accuracy applications or as back-up sensors. Development continues to improve their accuracy, which is limited by the separation of the antennas, the ability to resolve small phase differences, the relatively long wavelength of the GPS signal, and multipath effects due to reflections off spacecraft components.

Gyroscopes are inertial sensors that measure the speed or angle of rotation from an initial reference, but without any knowledge of an external, absolute reference. We use gyros in spacecraft for precision attitude determination when combined with external references such as star or Sun sensors, or, for brief periods, for nutation damping, or attitude control during thruster firing. Manufacturers use a variety of physical phenomena, from simple spinning rotors to *ring lasers*, *hemispherical*

resonating surfaces, and *laser fiber optic bundles*. Gyros based on spinning rotors are called *mechanical gyros*, and they may be large *iron gyros* using ball or gas bearings, or may reach very small proportions in so-called *MEMS gyros*. (MEMS stands for *microelectromechanical systems*.) The gyro manufacturers, driven largely by aircraft markets, steadily improve accuracy while reducing size and mass.

Error models for gyroscopes vary with the technology, but characterize the deterioration of attitude knowledge with time. Some examples of model parameters are *drift bias*, which is simply an additional, false rate the sensor effectively adds to all rate measurements, and *drift bias stability*, which is a measure of how quickly the drift bias changes. When used with an accurate external reference, such as a star tracker, gyros can provide smoothing (filling in the gaps between tracker measurements) and higher frequency information (tens to hundreds of hertz), while the tracker provides lower frequency, absolutely referenced information whenever its field of view is clear. Individual gyros provide one or two axes of information and are often grouped together as an *inertial reference unit* (IRU) for three full axes and, sometimes, full redundancy. IRUs with accelerometers added for position and velocity sensing are called *inertial measurement units* (IMU).

Sensor Selection. Sensor selection is most directly influenced by the required orientation of the spacecraft (e.g., Earth-, Sun- or inertial-pointing) and its accuracy. Other influences include redundancy, fault tolerance, field of view requirements, and available data rates. Typically, we identify candidate sensor suites and conduct a trade study to determine the most cost-effective approach that meets the needs of the mission. In such studies the existence of off-the-shelf components and software can strongly affect the outcome. In this section we will only briefly describe some selection guidelines.

Full 3-axis knowledge requires at least two external, non-parallel vector measurements, although we use IRUs or spacecraft angular momentum (in spinners or momentum-biased systems) to hold or propagate the attitude between external measurements. In some cases, if attitude knowledge can be held for a fraction of an orbit, the external vectors (e.g., Earth or magnetic) will have moved enough to provide the necessary information. In 3-axis star trackers, each identified star acts as a reference vector, which allows a single piece of hardware to generate a full 3-axis attitude solution.

For Earth-pointing spacecraft, horizon sensors provide a direct measurement of pitch and roll axes, but require augmentation for yaw measurements. Depending on the accuracy required, we use Sun sensors, magnetometers, or momentum-bias control with its roll-yaw coupling for the third degree of freedom. For inertially pointing spacecraft, star and Sun sensors provide the most direct measurements, and IRUs are ideally suited. Frequently, only one measurement is made in the ideal coordinate frame (Earth or inertial), and the spacecraft orbit parameters are required to convert a second measurement or as an input to a magnetic field model. Either the orbit parameters are uplinked to the spacecraft from ground tracking and propagated by onboard processing, or they are obtained from onboard GPS antennas.

FireSat II Sensors. The external sensors for FireSat II could consist of any of the types identified. For the 0.1 deg Earth pointing requirement, however, horizon sensors are the most obvious choice since they directly measure the two axes we most need to control. The accuracy requirement makes a star sensor a strong candidate as well; its information would need to be transformed, probably using an onboard orbit ephemeris calculation, to Earth-relative for our use. The 0.1 deg accuracy is at the low end of horizon sensors' typical performance, so we need to be careful to get the most out of their data. We assume we also need a yaw sensor capable of 0.1 deg, and this choice is less obvious. Often, it is useful to question a tight yaw requirement. Many payloads, e.g., antennas, some cameras, and radars, are not sensitive to rotations around their pointing axis. For this discussion, we will assume this requirement is firm. We could use Sun sensors, but their data needs to be replaced during eclipses. Magnetometers don't have the necessary accuracy alone, but with our momentum-bias system, roll-yaw coupling, and some yaw data filtering, a magnetometer-Sun sensor system could work for normal operations. The magnetometers would also improve the control effectiveness of the magnetic torquers.

At this point we consider the value of an inertial reference package. Such packages, although heavy and expensive for high-accuracy equipment, provide a short-term attitude reference that would permit the Earth vector data to be used for full 3-axis knowledge over an orbit. A gyro package would also reduce the single measurement accuracy required of the horizon sensors, simplifying their selection and processing. Such packages are also useful to the control system if fast slews are required, and here is where FireSat II demands a gyro. We need to perform the 180 degree slews as quickly as we can, to avoid losing too much data. So, we will include a MEMS-based inertial package for FireSat II. They will be arranged perpendicular to one another, in a pyramid around the Z axis. This arrangement gives 3 axes of information with maximal redundancy around the slewing axis. Now that we have decided to include gyros, a careful trade study should be done to determine whether an inexpensive MEMS gyro package combined with just the Earth sensors eliminates our need to include Sun sensors and magnetometers. Leaving out some of these other sensors could give better reliability or lower total cost. We may also choose a slightly different arrangement of the sensors to improve accuracy in one direction at the expense of accuracy in another direction. We would need to do these kinds of detailed trade studies in later iterations of the design process.

Finally, we will want a simple, coarse control mode to initially point the arrays at the Sun and to protect the spacecraft in the event of an anomaly. By using 6 coarse Sun sensors pointing along the positive and negative of

Table 19-13◀, Fig. 19-8◀, Eq. 19-8◀

each axis, the spacecraft can derive the location of the Sun from any attitude and use magnetic torquers to rotate the spacecraft so the arrays are lit. Then, since the attitude relative to nadir will change as FireSat II follows its polar orbit, we can be sure to get a good communication signal at some point, so that we can receive telemetry and send commands.

SCS Sensors. For SCS attitude determination, low-power gyros can provide rate information. Accurate gyros can be heavy and often use a lot of power; we have neither high accuracy needs nor an excess of power or mass in our budgets. Therefore, we will use light and inexpensive MEMS gyros. We need a minimum of 3 MEMS gyros—one for each axis—but by employing 4–6 gyros we can cross-compare the gyro data and remove the larger bias errors that MEMS gyros normally have.

With rate information onboard, we only need an occasional update from an attitude sensor. If the magnetic field were stronger, we might be able to filter magnetometer data to get to 1 deg of accuracy, but it is doubtful at this high altitude. Star cameras are small and very accurate, but they are expensive. One useful rule of thumb is: If at all possible, sense the thing you need to point at. Sun pointers should have Sun sensors and Earth pointers should have Earth sensors. Because the satellites will have the same direction pointing toward the Earth throughout their mission, the best option appears to be an Earth sensor. We would need to select a sensor designed for high altitudes. However, we still have no yaw data, and since the satellites must point accurately at multiple targets simultaneously, yaw accuracy is critical.

Because the satellites will be communicating with each other, it is conceivable that the communication signal strength could be used as an attitude determination data source for yaw control. That is, a feedback loop would close around the communication system's own measure of its link margin; maximizing the link margin would provide the attitude goal we want. For this exercise we will not assume such an option is available. Instead, we will choose the star camera after all; a simple onboard ephemeris calculation will tell the spacecraft where its target satellite is in inertial space. There may be clever tricks that we could use with ground-based methods to avoid using star cameras, such as combining orbit tracking and attitude data. However, the complexity of operating three separate satellites that have to work together will likely prove more expensive in software development and operating costs than just buying three star cameras. At least we save money by not including horizon sensors. As for our rule of thumb of sensing what you're pointing at, we now see there can be situations for which this rule cannot be followed. Still, it's always a good place to start.

We will propose the same plan for an initial acquisition and safe control mode as for FireSat II. However, SCS will use reaction wheels for control, and will have the benefit of accurate rate sensors to improve performance. This is a good thing, since SCS satellites need more power than FireSat II, and so will probably not have as much time to acquire the Sun (i.e., get the solar arrays lit). Table 19-14 summarizes our FireSat II and SCS hardware selections.

Once the hardware selection is complete, as discussed in Sec. 6.4, it must be documented for use by other system and subsystem designers as follows.

- Specify the power levels and weights required for each assembly
- Establish the electrical interface to the rest of the spacecraft
- Describe requirements for mounting, alignment, or thermal control
- Determine what telemetry data we must process
- Document how much software we need to develop or purchase to support onboard calculation of attitude solutions

Specific numbers depend on the vendors selected. A typical list for FireSat II might look like Table 19-15, but the numbers could vary considerably with only slight changes in subsystem accuracies or slewing requirements.

19.1.5 Define the Determination and Control Algorithms

Finally, we must tie all of the control components together into a cohesive system. Generally, we begin with a single-axis control system design (See Fig. 19-2). As we refine the design, we add or modify feedback loops for rate and attitude data, define gains and constants, and enhance our representation of the system to include three axes of motion (though we may still treat these as decoupled for early design iterations). To confirm that our design will meet requirements, we need good mathematical simulations of the entire system, including sensor error models and internal and external disturbances. Usually, linear differential equations with constant coefficients describe the dynamics of a control system well enough to allow us to analyze its performance with the highly developed tools of linear servomechanism theory. With these same tools, we can easily design linear compensation to satisfy specifications for performance.

We typically apply linear theory only to preliminary analysis and design. We also maintain engineering margin against performance targets when using linear theory because, as the design matures, so does our understanding of the nonlinear effects in the system. Nonlinear effects may be inherent or intentionally introduced to improve the system's performance.

Another reason to maintain margin, especially in actuator sizing, is that while systems engineers always carefully budget mass, they cannot usually track the moment of inertia matrix as well. Moment of inertia is the most important quantity to ADCS designers, and a given mass may have a wide range of moment of inertia values depending on how the mass is distributed.

Table 19-13◀, Fig. 19-8◀, Eq. 19-8◀

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Table 19-14. FireSat II and SCS Spacecraft Control Component Selection.

Mission & Type	Components	Rationale
FireSat II Actuators	Momentum Wheel (1)	• Pitch axis torquing • Roll and yaw axis passive stability
	Magnetic Torquers (3)	• Roll and yaw control • Pitch wheel desaturation
	Thrusters (4)	• Rapid changes in pitch wheel momentum • 180-deg yaw slew maneuvers
FireSat II Sensors	Horizon Sensor (1)	• Provide basic pitch and roll reference • Can meet 0.1 deg accuracy
	Sun Sensors (6)	• Lower mass and cost than star sensors • Initially acquire vehicle attitude from unknown orientation • Coarse attitude data • Fine data for yaw
	Magnetometer (1)	• Coarse yaw data
	MEMS Gyros (3)	• Needed for more precise magnetic torquing • Needed for rotational rate data during thruster operations
SCS Actuators (for each satellite)	Reaction Wheels (3)	• 3-axis stabilization • Storing momentum from solar radiation torques
	Thrusters (4)	• Thrusting and steering during orbit maneuvers • Removing momentum from reaction wheels
SCS Sensors (for each satellite)	Star Camera (1)	• Determining absolute attitude
	MEMS Gyros (4)	• Determining rotational rates; having 4 gyros allows large biases to be canceled out • Propagating attitude solution when star sensor data unavailable
	Sun Sensors (6)	• Initially acquire vehicle attitude relative to Sun to get power quickly

Table 19-15. FireSat II Spacecraft Control Subsystem Summary. The baseline ADCS components satisfy all mission requirements, with thrusters required for orbit injection and 180 degree slews.

Components	Type	Weight (kg)	Power (W)	Mounting Considerations
Momentum Wheel	Mid-size, 20 N·m·s momentum	< 5 total, with drive electronics	10 to 20	Momentum vector on pitch axis
Electromagnets	3, 10 A·m^2	2, including current drive electronics	5 to 10	Orthogonal configuration best to reduce cross-coupling
Sun Sensors	6 wide-angle coarse Sun sensors providing 4π steradian coverage; 5-10 deg accuracy	< 1 total	0.0	Free of viewing obstructions and reflections
Horizon Sensors	Scanning type (2) plus electronics; 0.1 deg accuracy	5 total	10	Unobstructed view of Earth's horizon
Gyroscopes	MEMS (3) only low accuracy needed	< 1 total	0.1	Mounted perpendicular to each other, in a pyramid around Z axis
Thrusters	Hydrazine; 5 N thrust	Propellant weight depends on mission	N/A	Alignments and moment arm to center of gravity are critical
Magnetometer	3-axis	< 1	5	Need to isolate magnetometer from electromagnets, either physically or by duty-cycling the magnets

Feedback control systems are of two kinds based on the flow of their control signals. They are *continuous-data* systems when sensor data is electrically transformed directly into continuously flowing, uninterrupted control signals to the actuators. By contrast, *sampled-data* systems have sensor sampling at set intervals, and control signals are issued or updated at those intervals. Most modern spacecraft process data through digital computers and therefore use sampled-data control systems.

Although it is beyond the scope of this book to provide detailed design guidance on feedback control